SMC-BVS v3 — collected proofs of the prove-first list (2026-10-22)

Status: proved; per-theorem reviews (Astra x2 + Opus xhigh) and campaign review (Fable 5.1 + Astra xhigh): all ten correct; NOT ADOPTED.

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Overview

This document collects the ten finished proofs of the prove-first list of the SMC-BVS v3 theory architecture. It is assembled by copying: for each theorem, part (a) is the original conjecture text, part (b) is the `STATEMENT.md` that was proved, part (c) is the full final proof, and part (d) is the review record (statement-change and review sections of `FINAL_REPORT.md`, and every entry of `CLAIMS_OF_FATAL_FLAWS.md`). Markdown parts are typeset from the unchanged Markdown text; the proofs are the unchanged LaTeX bodies, with the three mechanical changes listed in `ASSEMBLY_NOTES.md` (chapter prefixes on labels and on theorem-environment names; no macro had to be renamed, because each proof's own macros are local to its chapter).

Status of every result in the table: proved, with the stage-A verdict of the lead and Astra #1 and the stage-B verdicts of Astra #2 and an Opus xhigh reviewer; the campaign review (stage C) is pending; nothing is adopted. Abbreviations: A, B = review stages; r1, r2, …= review rounds; "0/1/2 = fatal/gap/exposition findings.

Theorem and parts	Plain-language claim	Out-come	Statement changes	Hypotheses taken from other conjectures	Review history (Stage A, Stage B, judge)	Final proof file and SHA-256
T09 (T7); all parts (i)–(v)	The exact start sampler and every move kernel are valid; every round, in any order, mixes at rate at least ε_φ; a tiny ε_φ does not imply slow mixing.	Proved (Stage B done, settled); not adopted	None to hypotheses. Two informal sentences, (iii) and (v), written as precise statements; the lead's first reading of (v) was corrected after Astra #1 round 1.	None (self-contained; uses no P1, P6 or sibling v3 conjecture).	A: Astra #1 r1 not verified (1 fatal, the lead's own gloss of (v); 1 exposition), r2 verified. B: Astra #2 r1 and r2 verified; Opus xhigh r1 verified (7 exposition), r2 verified (3 exposition). Judge used: no.	`T09_T7_kernel_validity/proof/PROOF_v3.tex` bab55fdf1074bdb509f4409fc08d9b5ed342072da620a2df122bdc8d03276ba6
T11(iii) only	For a model S^* fixed by the data and the calibration, $P(S^*$ never among the final run's Q_f iid π_0 draws $\mid \mathcal{C}) = (1-\pi_0(S^*))^{Q_f} \le e^{-Q_f \pi_0(S^*)}$.	Proved (Stage B done, settled); not adopted	None to the claim. Setting (data fixed, launch event, EX-S/EX-A/CAL, final-run description) written out; the scoring corollary was restricted to independence proposals after Astra #1 round 2.	None (EX-A, EX-S, CAL only).	A: Astra #1 r1 verified (3 gaps, supplementary parts), r2 verified (1 gap, corollary), r3 verified (none). B: Astra #2 r1, r2, r3 verified; Opus xhigh r1 verified (1 gap F8, 7 exposition), r2 verified (4 exposition), r3 verified (1 note). Judge used: no.	`T11_LLED_scoring_ledger/proof/PROOF_v5.tex` 7f4e94c47415f2e4bb64c59038a4193bfbc4fcf71a774fd8c429a74184886260
T12 (T8); all parts (i)–(iv)	Given the frozen calibration, each island's $\widehat{Z}_r$ is exactly unbiased and the recorded RB PIPs satisfy $E[\widehat{Z}_r \widehat{\mathrm{PIP}}_{rj}] = Z\,\mathrm{PIP}_j$; pooled $0.05\widehat{Z} \le Z$ w.p. $\ge 95\%$; $\log \widehat{Z}$, normalised PIPs and conditioning on diagnostics are not covered (limits proved).	Proved (Stage B done, settled); not adopted	Made precise after an Astra #1 statement check (approved with changes): arbitrary integer parameters; START imported; negative parts of (iv) as component statements; (iv.d) restricted after the lead's over-extension; target formulas written out after Stage B. No conclusion changed in Stage B.	START (exact π_0 sampler and $\bar{Z}_0$), imported from T09(i) / project P1.	A: Astra #1 statement check approved with changes; review of v2 verified (0 fatal / 0 gap / 3 exposition). B: round 1 on v3: Astra #2 not verified (0/1/1), Opus xhigh not verified (0/2/6); round 2 on v4: both verified (0/0/2 each). Judge used: no.	`T12_T8_frozen_run_identities/proof/PROOF_v6.tex` d6b85fc0fea043a8df467aeb50c969a1b47dab1e29c42fcecff29a15496396f6
T13 (T10); all parts (i)–(iv)	The restricted-ESS test on a candidate set V controls the full-space $1+\chi^2$ once V carries enough mass at the evaluation points; coverage on $[0,1]$ alone is not enough.	Proved (Stage B done, settled); not adopted	Nine precisions, substance unchanged: $\varepsilon_C \in [0,1)$; (iv)(a) avoidance an explicit hypothesis; grid paired with its V; witnesses from real method data; coverage at v; all real u, v; positivity derived; one fixed implementation for (iv)(b); (ii)(d) strengthened. Three Astra #1 statement checks (one restart).	None from other conjectures (assumption code COVER only).	A: Astra #1 v1 not verified (0/1/2), v2 verified (0/0/1), v3 verified. B: v3: Astra #2 not verified (0/1/0), Opus xhigh not verified (0/4/5); v4: both verified (Opus 0/0/3). Judge used: no.	`T13_T10_restricted_ESS/proof/PROOF_v4.tex` 4049019c683731dd56e83c640696bd48fc41335df6bbbbf35c0fed019aa60f2d
T14(i) (L-FM, singleton case); part (i) only	$E\lvert\widehat{Z}_r/Z - 1\rvert \le \frac{2}{1-\varepsilon_*}\sum_t \eta_t$ for every particle count N, where η_t is the bridge mass outside a fixed model S°.	Proved (Stage B done, settled); not adopted	No mathematical change. Notation (numerical subscript = stage index) and the reading of "endpoint term is essential" as (i-d).	None (the T12 unbiasedness identity is re-proved inside the proof).	A: Astra #1 r1 verified (1 exposition), r2 verified. B: Astra #2 r1 and r2 verified; Opus xhigh r1 verified (3 exposition), r2 verified. Judge used: no.	`T14_LFM_first_moment_evidence/proof/PROOF_v3.tex` c12954d17b7f7383b026b13925c4558e917af95c14e40b844529f490397efdd4
T15 (L-OCC); all parts (i)–(iv)	The mean fraction of an island's particles outside a fixed set $\mathfrak{A}$ is at most $\pi(\mathfrak{A}^c)/c + P(\widehat{Z}_r/Z < c)$ (and a tail version); the lower tail must be islandwise, not pooled.	Proved (Stage B done, settled); not adopted	None to the inequalities. The T12 dependence made the explicit hypothesis H-T12; the pooled-tail sentence made precise as (iv); after Stage	H-T12: the conclusion of T12(i), $E_{\mathcal{C}}[\widehat{Z}_r \hat{q}_{r,\tau}(f)] = Z\pi(f)$.	A: Astra #1 r1 verified (1 exposition), r2 verified. B: Astra #2 r1 verified, r2 verified (2 exposition), r3 verified; Opus xhigh r1 verified (5 exposition), r2 verified (4), r3 verified (1 optional	`T15_LOCC_occupancy/proof/PROOF_v4.tex` 246f07076f0d71fd0f08f9191ae20e999b4370efe75cdfdf7c3e33837ead8b68

Theorem and parts	Plain-language claim	Out-come	Statement changes	Hypotheses taken from other conjectures	Review history (Stage A, Stage B, judge)	Final proof file and SHA-256
T20 (T15); clauses (i)–(ii)	`local_certified` means what it says: no scored neighbour is better; one scan from a neighbour of the true model certifies it on E_n, which holds w.p. $\to 1$ under H03.	Proved (Stage B done, settled); not adopted	No hypothesis changed or added. Method §6.10 formalised as an algorithm, H03 explicit, EXEC named, seven readings recorded; wording corrections in v2, v3, v3.1.	H03: the conclusion of T03(i), used only in (ii-C).	A: Astra #1 r1 not verified (1 "fatal" in an explanatory note, 1 gap, 1 exposition), r2 verified. B: Astra #2 verified (1 cosmetic remark); Opus xhigh verified (10 exposition or scope notes); ruling round: both verified. Judge used: no.	`T20_T15_local_search/proof/PROOF_v3.tex` 2e34a79120c1e92f6738acd97dbd54fb8051fbc47e38931a48d09667d2332b24
T22 (T16); parts (i), (ii), (iv)	Start-side bounds: $\pi_0(S^\circ) \geq (1-q)^{s+1}/(1+e)^s$; an exact product formula for $\pi_0(S_0)/\pi_0(S^\circ)$; $q_\varphi(S_0) \geq \min\{\pi_0(S_0), \pi(S_0)\}$ along the bridge.	Proved (Stage B done, settled); not adopted	Approved by Astra #1 (APPROVED WITH CORRECTIONS): C1 added hypothesis $S_0 \in \Omega$ for (ii) and (iv); C6 generalisation to any real X, y and block partition; C7 scope $u < 1$; readings (iv-c) conditional.	(iv-c) only: the conclusions of T03 and of T22(iii) as explicit hypotheses.	A: Astra #1 statement check approved with corrections; PROOF_v1 verified (1 exposition). B: Astra #2 verified (none); Opus xhigh verified (5 exposition); round 2: both verified (Opus 1 new cosmetic note). Judge used: no.	`T22_T16_start_aggregation/proof/PROOF_v3.tex` 08c3521c5ad28bc7d2be26f5093b191fcb0596d95eae811b850ed26922f95be1
T23 (T18); part (ii) only	Pathwise RB PIP error $\leq \bar{B}_r + u_\pi$ on every run, and under BASIN $\leq \delta + (1-a)^{\lfloor L/r_0 \rfloor} + (L+4)(\eta+\rho) + u_\pi$ in expectation; $P(\widehat{\mathrm{MPM}} \neq S_0) \leq 2\mathcal{B}_n$.	Proved (Stage B done, settled); not adopted	None. Five readings recorded in the statement (section 5); EX-S used in the form (F), which both Stage B reviewers ruled faithful.	None (no sibling conjecture; the BASIN conditions are hypotheses of the theorem itself).	A: Astra #1 on v1 verified (no findings). B: r1 Astra #2 verified, Opus xhigh verified (6 exposition); r2 both verified (7 further exposition); r3 Astra #2 verified, Opus verified (2 exposition); r4 both verified (Opus 1 cosmetic note). Judge used: no.	`T23_T18_basin_prefix/proof/PROOF_v5.tex` 36d38c99b7870cffb20d897f2f119c6ccb656730959df9e17b39332d63a564ff
T31 (T22); part (i) only	From unbiasedness alone, $P_c(Z \geq \alpha\widehat{Z}) \geq 1-\alpha$ and $E_c \log \widehat{Z} \leq \log Z$; this is not a relative-error interval, and conditioning on a final E2 flag can destroy coverage.	Proved (Stage B done, settled); not adopted	None to hypotheses or conclusions. "Under T12" made the explicit hypothesis H-T12; the two informal cautions written as non-implications (c), (d); explanatory wording revised twice after review.	H-T12: T12(iii), unbiasedness $E_c[\widehat{Z}] = Z$.	A: Astra #1 on v1 verified (none). B: r1 Astra #2 verified (none), Opus xhigh verified (11 exposition); r2 Astra #2 verified (2), Opus verified (5); r3 Astra #2 verified (none), Opus verified (3 cosmetic). Judge used: no.	`T31_T22_certificates/proof/PROOF_v3.tex` e140d9b81cd01b54b1b052011975d5d56a2db13dc64095d4fe7a658126e2d4cc

Dependencies between the proofs

The list below is taken from the `STATEMENT.md` files of the ten proofs (reproduced in each chapter, part (b)). "Hypothesis" means that the other result is used only as an explicit assumption of the statement being proved here; nothing about it is proved in the chapter that uses it.

- **T09**: no dependency (self-contained).
- **T11(iii)**: no dependency (EX-A, EX-S, CAL only).
- **T12**: hypothesis START (exact start sampler and exact normaliser $\bar{Z}_0$), imported from **T09(i)** (project P1). T12 does not otherwise use a sibling result.
- **T13**: no dependency on another result (assumption code COVER is a hypothesis of part (ii), not a result).
- **T14(i)**: no dependency (the T12 unbiasedness identity it builds on is re-proved inside its proof).
- **T15**: hypothesis H-T12, the conclusion of **T12(i)**: $E_{\mathcal{C}}[\widehat{Z}_r\, \hat{q}_{r,\tau}(f)] = Z\, \pi(f)$. Parts (i)–(ii) use only the instance $f = \mathbf{1}_{\mathfrak{A}^c}$; part (iv) uses no hypothesis.
- **T20**: hypothesis H03, the conclusion of **T03(i)** (not part of this collection), used only in (ii-C).
- **T22**: parts (i), (ii), (iv-a), (iv-b) are deterministic and use no other result; the conditional corollary (iv-c) takes the conclusions of **T03** (not part of this collection) and of **T22(iii)** (not part of this collection) as explicit hypotheses.
- **T23(ii)**: no dependency on another conjecture; the occupancy step is proved directly from the BASIN conditions and does not cite T15 or any part (i).

- **T31(i)**: hypothesis H-T12, taken from **T12(iii)**: unbiasedness $E_c[\widehat{Z}] = Z$ for every successful calibration value c.

Within this collection the dependency arrows are therefore: T09(i) $\rightarrow$ T12; T12(i) $\rightarrow$ T15; T12(iii) $\rightarrow$ T31(i). The conclusions of T03 and T22(iii) enter T20 and T22 only as hypotheses and are outside the collection.

Shared definitions

Source: `theorems/00_COMMON.md`, converted from Markdown; the text is not changed.

00_COMMON.md — shared definitions for every theorem file in this folder

Status of everything in this folder: **conjectured, not proven.** Methodology: SMC-BVS v3.0 of 2026-10-19, `C:\Users\felip\OneDrive\Documents\Local_to_global_Bayesian\01_Methodology_and_Theory\Methodology\SMC_BVS_v3_20261019\METHODOLOGY_v3.md`, SHA-256 `93043758058a6d422cca6ca4b6be8f3a9e8149b6f6de2e1bab1ce31c44b8019c`. The exact algorithm (blocks, tables, composite start, bridge, pilot, candidate grid, final run, local search, statuses) is written out in `..\THEORY_BRIEF.md` §2; the method file is authoritative. Each theorem file `T<k>_<name>.md` names the definitions and inventory codes it uses; give proof.md the theorem file, this file, THEORY_BRIEF.md and the method file (all by path).

Felipe's fixed decisions (no theorem changes them): the target (standardised X, y; $\beta_S \sim N(0, n\sigma^2(G_S+I)^{-1})$; $\sigma^2 \sim IG(0.01, 0.01)$; BB(1, p) capped at K = min(p, ⌊n/2⌋)); the HPM over all scored models with exact-equality ties; B = min(⌊√n⌋, ⌊√p⌋, 16); neighbour threshold $\rho_n = \sqrt{(2 \log p/n)}$; no secondary prior arm; the budgets of METHODOLOGY §3.

1. Data, target, decision objects

- Raw data (X^r, y^r), n ≥ 3, no constant column, non-constant y^r. $C = I - 11^T/n$; D_x = diag of retained raw column sample SDs. Standardised $X = CX^rD_x^{-1}$, $y = Cy^r/s_y$ with $s_y^2 = \|Cy^r\|^2/(n-1)$; so $\|x_j\|^2 = \|y\|^2 = n - 1$. Contrast coordinates: $U \in \mathbb{R}^{n\times(n-1)}$, $U^TU = I$, $UU^T = C$; $y_c = U^Ty^r$, $X_c = U^TX$.
- State space Ω = {S ⊆ [p] : |S| ≤ K}, K = min(p, ⌊n/2⌋). $G_S = X_S^TX_S = (n-1)R_S$; $z_S = X_S^Ty$; $H_S = (n+1)G_S + I$; $D_S = \det H_S/\det(G_S+I)$; $Q_S = y^Ty - n\, z_S^TH_S^{-1}z_S$; ν = 0.01 + n/2; $a_0 = b_0 = 0.01$.
- $m(S) = (2b_0)^{a_0}\Gamma(\nu)/(\pi^{n/2}\Gamma(a_0)) \cdot D_S^{-1/2}(2b_0 + Q_S)^{-\nu}$; w(k) = ½/C(2p−1, k); $W_K = \Sigma_{k\le K} C(p,k)w(k)$; ϖ(S) = w(|S|)/W_K.
- F(S) = log[ϖ(S)m(S)]; Z = Σ_Ω ϖ(S)m(S); π(S) = e^{F(S)}/Z; BF(S) = m(S)/m(∅).
- Global HPM S† = argmax_Ω F (ties: smallest size, then lexicographic). Reported HPM $\hat{S}$ = the same rule over $\mathcal{S}$, the set of all models ever scored (calibration, final run, rejected proposals, local scans); $\mathcal{S}_{cal} \subseteq \mathcal{S}$ is the calibration's scored set. PIP_j = π(j ∈ S); MPM = {j : PIP_j > ½}.
- Statistical model: $y^r = \alpha*1 + X^r\beta^* + \varepsilon$; $\mu = CX^r\beta^* = X\theta^*$, $\theta^* = D_x\beta^*$; $S_0 = \text{supp}\ \beta^*$, $s_0 = |S_0|$. Contrast-scale parameters $\theta = s_y\beta$, $v = s_y^2\sigma^2$. P_0 is over ε; "joint probability" adds the algorithm's randomness. $r_n := \sqrt{(\log n/n)}$. $m_n(\omega) := \lceil (s_0 \log(2p) + (s_0+1)(\log n)/2 + \omega)/\log 2 \rceil$, $m_n := m_n(½ \log n)$.

2. Sampler objects

- Blocks C_1, …, C_J (deterministic in X: complete linkage, top-down cut, next-fit packing), |C_b| ≤ B = min(⌊√n⌋, ⌊√p⌋, 16). Tables r_b(A) = BF(A) for A ⊆ C_b (full response).
- $\gamma_0(S) = \varpi(S)\prod_b r_b(S \cap C_b)$; $\pi_0 = \gamma_0/\bar{Z}_0$ ($\bar{Z}_0$ by exact convolution of block size profiles with the capped size prior; backward sampler gives exact iid π_0 draws). Start mode $\tilde{S}_0 := \operatorname{argmax} \pi_0$.
- Discrepancy d(S) = log BF(S) − Σ_b log r_b(S ∩ C_b). Bridge γ_φ = $\gamma_0 e^{\varphi d}$, q_φ = γ_φ/$\bar{Z}$_φ (defined for real φ; production φ ∈ [0,1]); $q_0 = \pi_0$, $q_1 = \pi$, Z = m(∅)$\bar{Z}_1$.
- Round K_φ: m = p + B + 2 labelled steps in a uniformly random order: p Gibbs flips (o_j(S, φ) = pre-draw inclusion probability of j at S; a_j(S,φ) = γ_φ(S+j)/γ_φ(S)), one uniform swap, B neighbour swaps (symmetric-union neighbour graph N(·) at threshold ρ_n), one independence proposal from π_0. Elementary kernels K_e; reflected kernels K_e^W(x,A) = K_e(x, A∩W) + K_e(x, W^c)1_A(x).
- Calibration $\boldsymbol{\mathcal{C}}$:= (grid $0 = \varphi_0 < \ldots < \varphi_T = 1$, T, L, N_e, tier e, candidate set V, every V-independent setting), measurable w.r.t. streams 0–9; final streams 10–13 are fresh; escalation is decided before final randomness. Δ_t = φ_t − φ_{t−1}; w_t = 2φ_t − φ_{t−1}; $\boldsymbol{\mathcal{E}}$_grid = {φ_0,…,φ_T, w_1,…,w_T}. Restricted ESS ρ_V(u,v) = $\bar{Z}_v(V)^2/(\bar{Z}_u(V)\bar{Z}_{2v-u}(V))$.
- Final run: 4 fresh islands × N_e iid π_0 draws; frozen grid; multinomial resampling after every reweighting; one round per stage; L burn-in rounds and 4 measured rounds at φ = 1; local search (≤ ℓ moves, scans score every feasible flip and swap of the centre). Outputs: $\hat{Z}$_r (island evidence), $\hat{Z}$ = ¼Σ_r$\hat{Z}$_r, bound 0.05$\hat{Z}$; PI$\hat{P}$_{rj} = Rao–Blackwell average of the pre-draw Gibbs conditionals over the 4 measured rounds; primary PI$\hat{P}$_j (equal weight), secondary ($\hat{Z}$_r-weighted); MP$\hat{M}$; $\hat{S}$; statuses {exact, completed, schedule_limit, numerical_failure, resource_limit, y_constant, no_predictors, invalid_input}; flags (E1–E3, low_empirical_ESS, local_certified, global_certified, …).
- Budgets: N_0 = max(2500, n, p); H_0 = ⌈$512N_0/2500$⌉; ℓ = ⌈$\log_2(n+p+1)$⌉; T_0 = max(1000, ℓ^2); (N_e, H_e, T_e) = 4^e(N_0, H_0, T_0), e ∈ {0,1}; L = max(4, ⌈$\log_2(n+1)$⌉). Q_f := 4N_e(T + L + 5) iid π_0 draws scored in the final run (initialisation + independence proposals); 4N_e(L+6) ≤ Q_f ≤ 4N_e(T_e + L + 5).
- Exact branch: Σ_{k≤K} C(p,k) ≤ ($5H_0 + 4N_0$)p; it precedes block/Gram construction and scores every feasible model.
- Stage notation: q_t = q_{φ_t}, G_t = e^{Δ_t d}, Z_t = m(∅)$\bar{Z}$_{φ_t}; $\hat{q}_t$ = empirical population after stage-t resampling and mutation; $\hat{Z}$_{r,t} = island r's evidence product through stage t. Reflected normalisation on W: q_t^W = q_t(·|W), g_t^W = G_t/[$\bar{Z}$_{φ_t}(W)/$\bar{Z}$_{φ_{t−1}}(W)], $\mathcal{Q}$^W_{s,t} = $\prod_{j=s+1}^t$ (g_j^W K_j^W); comparator mean with the original start normaliser Z·r_W, r_W = $q_1(W)/q_0(W)$.
- Leakage quantities: u_{T,l} = unit residual of x_l on X_T; δ_{T,l} = $u_{T,l}^T\mu$; Δ_T = $\|(I-P_T)\mu\|^2/(n-1)$; v_T = σ^{*2} + Δ_T; h_n = √(2(A+ε) log n); E_{T,l} = min{δ^2_{T,l}/(2Δ_T), δ^2_{T,l}/(2v_T) + |δ_{T,l}|σ*h_n/v_T} (E_{T,l} = 0 when δ_{T,l} = 0).
- Truth stability: $\delta_1 = 1 - \pi(S_0)$; $u_\pi = \delta_1/\pi(S_0)$; ρ_* = min{1, (B+3)u_π}.
- Regimes: R1: n → ∞, s_0 fixed, p ≤ n^A. R2: p fixed (SMC statements are component results; the exact branch eventually applies). R3: data and $\boldsymbol{\mathcal{C}}$ fixed, N or the island count varies (idealised). R4: actual budgets along R1. Fixed-data identities condition on a successful frozen $\boldsymbol{\mathcal{C}}$, never on passing later diagnostics.

3. Assumption inventory (codes used by the theorem files)

Code	Condition	Status
G	Fixed design; $\varepsilon \sim N_n(0, \sigma^{*2} I)$, $\sigma^{*2} > 0$ fixed; intercept removed by contrasts.	Standard/project (paper_draft §2); external precedent UNVERIFIED
R	s_0 fixed; $p \le n^A$, $A \ge 1$; $\|\mu\|^2/n \le M$; $\lambda_{\min}(R_{S_0}) \ge \varphi_0^2 > 0$; eventually $s_0 \le K$.	Project bundle (paper_draft T6); truth-dependent
SE	$\lambda_{\min}(R_T) \ge \varphi_0^2$ for every nonempty T with $\|T\| \le \min(p, m_n + s_0)$.	Project (proofs T7_T8 §1, T9_C9 §9); strong; not checkable exhaustively
BM-C	$\min_{j\in S_0} \theta^{*2}_j \ge M_* \sigma^{*2} \log n/(n\varphi_0^2)$, $M_* = 2L_*+1$, $L_* = \max\{4C_*, 32[s_0(A+1)+2]\}$, $C_* = 2s_0 + 3[Aj_* + s_0 + \log(j_*+1) + 1]$, $j_* = \lceil(2A+1)s_0\rceil + 1$.	New candidate fixed constant (J1 audit of the project proof; provisional); the proving run may replace M_*
BM-∞	Same inequality with $M_n \to \infty$ in place of M_*.	Project sufficient condition (v2's BM)
DX	min retained raw column SD $\ge d_- > 0$.	New; observable
GL	$R_{S_0} \to \Gamma_0 \succ 0$ with consistent support ordering.	Project limit form; generic BvM precedent UNVERIFIED
ORTH	$G = (n-1)I_p$ (so $p \le n-1$); for A4: fixed $1 \le s_0 < p$, equal magnitudes b_n on S_0, b_n^2/σ^{*2} bounded.	Project benchmark (P4/Pb1); observable
COH	$c_n = \max_{i\ne j}\|r_{ij}\|$, $c_n \log(p\vee n) \to 0$, $s_0 c_n \theta_{\max}/\theta_{\min} \to 0$.	New; optional
DUP	Exact signed duplicate columns; or two Gaussian truths with $\|\mu_1-\mu_2\|/\sigma^* \to 0$.	New alternative regimes
NIS	No distinct feasible $S \ne S'$ with $X_S X_S^T = X_{S'} X_{S'}^T$ (equal sizes then follow from traces).	New; decidable from X
VIS	For $U_b^* = S_0 \cap C_b \not\subset T \subseteq C_b$: $\|(P_{T\cup U_b^*} - P_T)\mu\|^2 \ge M_n^V v_T \log n$, $M_n^V \to \infty$.	New actual-block visibility; not implied by SE/BM
NL_avg(κ)	$(p-s_0)^{-1}\Sigma_{l\notin S_0} \max_{U^*_{b(l)}\subseteq T\subseteq C_{b(l)}\setminus l} e^{E_{T,l}} \le n^{½-κ}$, fixed $\kappa > 0$.	New; Gaussian-design prevalence UNVERIFIED
NL_η / TL	Sufficient cases: $\delta^2_{T,l} \le \eta\sigma^{*2} \log n$ with $\eta < (\sqrt{(2A+1)} - \sqrt{(2A)})^2$, over the block comparisons (NL_η) or a target-state family (TL).	New explicit threshold
TL_avg(κ; $\boldsymbol{\mathcal{R}}$)	For every $T \in \boldsymbol{\mathcal{R}}$, the average of $e^{E_{T,l}}$ over $l \notin S_0 \cup T$ is $\le n^{½-κ}$.	New; uniform before selection
EXP	Uniform actual-block expansion: $\log[r_b(T+l)/r_b(T)] \le -½ \log(n+1) + (\delta_{T,l} + \sigma^* Z_{T,l})^2/(2v_T) + R_n$, $Z_{T,l} = u_{T,l}^T \varepsilon/\sigma^*$; $R_n = O_P(1)$ (strict rate) or $o_P(\log n)$ (exponent rate); plus a uniform lower gain bound $c\cdot\text{gain}/v_T - CB \log n$ for visibility.	New explicit proof obligation (J5)
AGG	For feasible S°, $s = \|S^\circ\|$, $U_b^* = S^\circ \cap C_b$: $a_{bj} = \Sigma_{U\supseteq U_b^*, \|U\setminus U_b^*\|=j} r_b(U)/r_b(U_b^*)$; $D_b = \max_{j\ge1}(j! a_{bj})^{1/j}$; $q = (p-s)^{-1}\Sigma_b D_b$; $e = 2p \max_{b,U: a=\|U_b^*\setminus U\|>0}[r_b(U)/r_b(U\cup U_b^*)]^{1/a}$. Hypothesis $q < 1$. Conventions: empty maxima are 0; $q = 0$ if $s = p$; $e = 0$ if $s = 0$.	New deterministic table condition

Code	Condition	Status
BO	$X_{C_b}^T X_{C_b'} = 0$ for $b \neq b'$.	New benchmark; observable
ONE	$\tilde{S}_0$ and S_0 differ by one flip, or by a same-block proxy swap with $\lvert r_{ij'}\rvert > \rho_n$. $\mathfrak{P} = \{\tilde{S}_0, S_0\}$.	Project case definition
LK_d	$\boldsymbol{\mathcal{R}}_d = \{U \cup P : U \subseteq S_0, P \subseteq S_0^c, \lvert P\rvert \leq d\}$, fixed d, $s_0+d \leq K$; $\Lambda = \max_{T \in \boldsymbol{\mathcal{R}}_d} \Sigma_{l \notin S_0 \cup T}\, o_l(T,1)$; $(L+4)2^{s_0+d}\Lambda \to 0$.	New, truth-dependent
EXIT_d	ξ = max over starts in $\boldsymbol{\mathcal{R}}_d$ of the one-round probability that the first exit from $\boldsymbol{\mathcal{R}}_d$ is caused by a non-Gibbs label; $(L+4)\xi \to 0$.	New; does not bound internal non-Gibbs changes
GAIN_d	Uniform best missing-true addition gain $g^+ \to \infty$ and uniform best extra-deletion gain $g^- \to \infty$ from proper supersets, over $\boldsymbol{\mathcal{R}}_d$.	New; multi-extra deletion not derived
BASIN($\boldsymbol{\mathcal{R}}$, r_0, a, η, ρ, δ)	Given $\boldsymbol{\mathcal{C}}$: $\boldsymbol{\mathcal{R}} \ni S_0$; $\inf_{x \in \boldsymbol{\mathcal{R}}} K_1^{r_0}(x, S_0) \geq a$; one-round basin first exit ≤ η; truth prefix-departure ≤ ρ; expected stage-T outside fraction ≤ δ.	New abstract conditions
BOCC(d)	With probability → 1 all specified pilot/final elementary states have size ≤ d.	New; insufficient alone
EX-A	Exact scores, convolutions, probabilities, comparisons.	Declared idealisation
EX-S	Independent ideal uniform streams with the method's stream separation.	Declared idealisation
CAL	$\boldsymbol{\mathcal{C}}$ uses streams 0–9 only; streams 10–13 fresh; escalation before final randomness.	Method construction (M §§6.8–6.9)
SR	Each Gibbs update evaluates both feasible configurations (incl. the current one); both count as scored.	Scoring reading of M §§6.5, 6.10 (to be confirmed by Felipe)
COVER(W, ε_C)	$q_\psi(W) \geq 1 - \varepsilon_C$ at every $\psi \in \boldsymbol{\mathcal{E}}$_grid.	New full-space coverage
HIT	A specified scoring route succeeds with probability → 1 (e.g. $Q_f\pi_0(S_0) \to \infty$, table inclusion, scanned one-neighbour centre).	New conditional route
PX	On no prior discovery, ≥ $\omega_n k(p-k)$ uniform-swap executions at $\tilde{S}_0$ occur with probability → 1, $\omega_n \to \infty$.	New occupancy condition
CPL-E	Common-randomness coupling with the reflected particle system through all populations/ancestors used by the last weight evaluation, failure h_E; sufficient bound: initial mismatch + $\Sigma_e E[1\{\tau > e^-\}K_e(X_{\{e^-\}}, W^c)]$.	New particle-system obligation
CPL-H / RB	HPM transfer: couple all scored proposals and scans. PIP transfer: couple recorded rewards or bound the reward discrepancy.	New output-specific obligations
MISS-E	Data-measurable $\boldsymbol{\mathcal{M}}$, $\emptyset \notin \boldsymbol{\mathcal{M}}$; calibration scores avoid $\boldsymbol{\mathcal{M}}$; CPL-E for $W = \boldsymbol{\mathcal{M}}^c$ with $h_E \to 0$.	New conditional loss hypothesis
MISS-H	Calibration avoidance plus avoidance of every subsequent scoring opportunity, failure $h_H \to 0$; RB condition for PIP conclusions.	New
RET / OCC	RET: return visits from neighbours of $\mathfrak{P}$. OCC: expected outside fractions ≤ Cq_t or specified weighted slacks.	New first-moment conditions; not variance premises
BW2	$\int q^W_{t-1}(dx)[g_t^W(x)]^2 K_t^W(x,dy) K_t^W(x,dz) \leq C_t\, q_t^W(dy) q_t^W(dz)$.	New two-lineage domination
BW_W	$\sup_{s<t} \sup_x \mathcal{Q}^W_{s,t}1(x) / \inf_x \mathcal{Q}^W_{s,t}1(x) \leq C_0$.	New future-weight control
WS	Working set W; terminal initial law $\lambda \leq w\pi(\cdot\lvert W)$; reflected terminal round has absolute gap $g > 0$; $\sup_{s<t,x} \mathcal{Q}^W_{s,t}1(x) / q_s^W(\mathcal{Q}^W_{s,t}1) \leq a_n$.	New
PW2	With $W_t = \hat{Z}_t / Z_t$: $E[W^2_{t-1}\{(\hat{q}_{t-1} g_t)^2 - 1\}] \leq (a_t/N) E W^2_{t-1} + \nu_t$ with controlled slack.	New research obligation

Code	Condition	Status
OP-a / OP-s	Preflight passes under the allocation; / schedule failure probability $\to 0$ when final outputs are needed.	New operational conditions
NUM	Simultaneous error bound η for all compared log scores at fixed n.	New validation contract
NDR	No limiting finite bisection comparison equals the ESS threshold; no limiting pilot increment equals the floor.	New nondegeneracy
STIFF	Two-state restricted law, high-discrepancy mass a, gap D: $\rho(a, D(b-u)2^{-60}) < 0.9$, $\rho(a,x) = [a+(1-a)e^{-x}]^2/[a+(1-a)e^{-2x}]$.	Corrected mass-qualified obstruction
UB	Externally justified upper bound $U \geq Z$.	Optional external input

Retired as hypotheses: MASS; AGG-stat (a conclusion); ζ-EXIT; OP's numerics clause; V≤, OT-lb, NA (optional research ideas only). Regimes are not declared universally disjoint; exact duplicates conflict with SE and NIS; ORTH excludes p > n.

Chapter 1

T09 (T7): exact start, invariant kernels and whole-round minorisation

Final proof: `proofs/T09_T7_kernel_validity/proof/PROOF_v3.tex`. Sources of parts (a), (b), (d): `theorems/T09_T7_kernel_validity.md`, `STATEMENT.md`, `FINAL_REPORT.md`, `CLAIMS_OF_FATAL_FLAWS.md`.

(a) Original conjecture text (theorem file, verbatim)

T09 (T7): Exact start, invariant kernels and whole-round minorisation

Status: conjectured, not proven. Methodology SMC-BVS v3.0 (2026-10-19), SHA-256 93043758058a6d422cca6ca4b6be8f3a9e8149b6f6de2e1bab1ce31c44b8019c. Read with `00_COMMON.md`.

Category / rank: algorithmic validity and convergence; **core** (proposition).

Definitions used

Blocks, tables r_b, γ_0, $\bar{Z}_0$, π_0, backward sampler; d, γ_φ, q_φ, $\bar{Z}_\varphi$; round K_φ with m = p + B + 2 labelled steps (Gibbs, uniform swap, B neighbour swaps on the symmetric-union graph $N(\cdot)$, independence proposal from π_0), elementary kernels K_e, realised-order kernel K_σ; $\varepsilon_\varphi := \inf_S \pi_0(S)/q_\varphi(S)$ (00_COMMON §2).

Assumptions

EX-A, EX-S (declared idealisations); the actual tables, cap and move rules.

Precise statement

For every valid component input and every $\varphi \in [0, 1]$: (i) Convolving the actual block size profiles with the capped size prior gives $\bar{Z}_0$ exactly; the backward sampler returns exact iid π_0 draws. (ii) Each elementary kernel is q_φ-reversible, including the boundaries: addition at $|S| = K$ has Gibbs probability 0; the uniform swap self-loops at $|S| \in \{0, p\}$; the neighbour-swap reverse ratio reads 0 when the reverse insertion is not a neighbour; the stated Hastings ratio supplies the reverse proposal probability. (iii) $K_\varphi = (1/m!)\Sigma_\sigma K_{\sigma(1)}\cdots K_{\sigma(m)}$ is q_φ-invariant and self-adjoint on $L^2(q_\varphi)$. An individual realised order need not be reversible. (iv) Every realised order satisfies $K_\sigma(x, \cdot) \geq \varepsilon_\varphi$

q_φ(·) with ε_φ = ($\bar{Z}$_φ/$\bar{Z}_0$)e^{−φ max_S d(S)} > 0; hence ||K_φ^h(x,·) − q_φ||_TV ≤ (1 − ε_φ)^h and the averaged round's absolute L^2 gap is ≥ ε_φ. (v) ε_φ ≤ π_0(S)/q_φ(S) for every S. An exponentially small available minorisation constant does not establish exponentially slow mixing of the hybrid kernel; neither a switch nor a large discrepancy range alone supplies such a lower bound.

Dependencies

Builds on: method §§6.3–6.5; project P1/P6. Used by: T10–T19, T23, T25, T28, T29, T36.

Numerical evidence (provisional)

provisional evidence (Sol check J2): n = 6, p = 4, K = 3, B = 2: element and round detailed-balance residuals ≤ $2.2 \cdot 10^{-16}$ at φ ∈ {0, ½, 1}; minorisation constants 1, 0.786, 0.619 with nonnegative slack ($-1.1 \cdot 10^{-16}$ roundoff at φ = 0); addition ratios ≥ 1.09 × floor. Files: judge/sol2/J2_EXACT_IDENTITIES.md, J2_matrices.npz.

Proof-route hint (optional)

Sum block profiles; verify detailed balance including boundaries; pair each ordering with its reversal for self-adjointness; the independence kernel minorises q_φ (K_ind(S,S′) ≥ π_0(S′)min{1, w(S′)/w(S)} ≥ q_φ(S′)/sup w, w = q_φ/π_0) and its stochastic prefix and invariant suffix preserve the minorisation. Hardest step: boundary and Hastings bookkeeping. *When proof.md is run on this file, its lead considers the statement both with and without these hints; the hints never reach the reviewers.*

(b) STATEMENT.md (verbatim)

STATEMENT — T09 (T7): exact start, invariant kernels and whole-round minorisation

Source: `theorems/T09_T7_kernel_validity.md` (SMC-BVS v3.0, METHODOLOGY_v3.md SHA-256 93043758058a6d422cca6ca4b6be8f3a9e8149b6f6de2e1bab1ce31c44b8019c). Decision at step 1: **(a) correct as stated.** No hypothesis is added or changed. Two informal clauses (the last sentence of (iii) and the second sentence of (v)) are given a precise mathematical reading below; the original wording is quoted next to each reading.

Standing idealisations (declared in the source): **EX-A** exact scores, convolutions, probabilities and comparisons; **EX-S** independent ideal uniform random streams.

Change log. v2 (after review round 1): the last sentence of the reading of (v) said "(equivalently, no lower bound on its mixing time) that tends to 0", which is wrong as written (a mixing-time lower bound tending to 0 is trivially available). It now speaks of a lower bound tending to infinity and adds the explicit bound t_mix ≤ 55. The original wording, the hypotheses and the existence claim are unchanged.

1. Setting and definitions

Data. Integers n ≥ 3, p ≥ 1. X ∈ $\mathbb{R}$^{n×p} and y ∈ $\mathbb{R}^n$ are standardised data: every column x_j and y are centred, $\|x_j\|^2 = \|y\|^2 = n - 1$. (These are exactly the outputs of the method's preprocessing; a standardised pair is returned unchanged by it.) K = min(p, ⌊n/2⌋); Ω = {S ⊆ [p] : |S| ≤ K}.

Scores. For S ⊆ [p], k = |S|: $G_S = X_S^T X_S$, $z_S = X_S^T y$, $H_S = (n+1)G_S + I_k$, $D_S = \det H_S / \det(G_S +$

I_k), Q_S = $y^T y - n\, z_S^T H_S^{-1} z_S$, ν = 0.01 + n/2, b_0 = 0.01, A_S := $2b_0$ + Q_S, BF(S) = D_S^{−1/2} (A_∅ / A_S)^{ν} (so BF(∅) = 1; this is m(S)/m(∅)). Size prior: w(k) = ½ / C(2p−1, k), W_K = Σ_{k≤K} C(p,k) w(k), ϖ(S) = w(|S|)/W_K for S ∈ Ω.

Blocks and tables. C_1, …, C_J is the method's block partition of [p] (complete linkage, top-down cut, next-fit packing), |C_b| ≤ B = min(⌊√n⌋, ⌊√p⌋, 16). Tables r_b(U) = BF(U) for U ⊆ C_b. Block size profiles e_b(j) = Σ_{U⊆C_b, |U|=j} r_b(U), j = 0, …, |C_b|.

Start. $\gamma_0(S)$ = ϖ(S) ∏_b r_b(S ∩ C_b) for S ∈ Ω; $\bar{Z}_0$ = Σ_{S∈Ω} $\gamma_0(S)$; π_0 = $\gamma_0/\bar{Z}_0$. Convolution: c^{(0)} = δ_0 (c^{(0)}(0) = 1, else 0); c^{(b)}(t) = Σ_j e_b(j) c^{(b−1)}(t − j) (b = 1, …, J), c := c^{(J)}. **Backward sampler** (one draw, fresh uniforms): (1) draw the total size k ∈ {0, …, K} with probability w(k)c(k) / Σ_{k′≤K} w(k′)c(k′); (2) set t_J = k; for b = J, J−1, …, 1 draw j_b with probability e_b(j_b) c^{(b−1)}(t_b − j_b) / c^{(b)}(t_b) and set t_{b−1} = t_b − j_b; (3) for each b draw U_b ⊆ C_b with |U_b| = j_b with probability r_b(U_b)/e_b(j_b); return S = ∪_b U_b.

Bridge. d(S) = log BF(S) − Σ_b log r_b(S ∩ C_b); $\gamma_\varphi(S) = \gamma_0(S)\, e^{\varphi\, d(S)}$ for S ∈ Ω, $\gamma_\varphi(S) := 0$ for S ∉ Ω; $\bar{Z}_\varphi$ = Σ_Ω γ_φ; $q_\varphi = \gamma_\varphi/\bar{Z}_\varphi$ (a probability on Ω, positive everywhere on Ω).

Neighbour graph. N(j) ⊆ [p] \ {j} for each j: the method's symmetric-union graph (l ∈ N(j) ⇔ j ∈ N(l)). Statement (ii) is also asserted, with the stated reading convention, for an arbitrary family N(j) ⊆ [p] \ {j}.

Elementary kernels on Ω at temperature φ (all states, all moves in Ω):

- **Gibbs j** (j ∈ [p]): with S^+ = S ∪ {j}, S^- = S \ {j}: K_{G,j}(S, S^±) = $\gamma_\varphi(S^\pm)$ / ($\gamma_\varphi(S^+) + \gamma_\varphi(S^-)$). (When j ∉ S and |S| = K, $S^+ \notin \Omega$, so the addition has probability 0.)
- **Uniform swap**: if |S| ∈ {0, p}, stay. Otherwise pick j uniformly in S and l uniformly in [p] \ S, propose S′ = S \ {j} ∪ {l}, accept with probability $1 \wedge \gamma_\varphi(S')/\gamma_\varphi(S)$; else stay.
- **Neighbour swap**: if S = ∅, stay. Otherwise pick j uniformly in S; if N(j) \ S = ∅, stay; else pick l uniformly in N(j) \ S, propose S′ = S \ {j} ∪ {l}, accept with probability $1 \wedge ([\gamma_\varphi(S')/\gamma_\varphi(S)] \cdot |N(j) \setminus S| / |N(l) \setminus S'|)$, where this ratio is read as 0 when j ∉ N(l); else stay.
- **Independence**: draw S′ ∼ π_0 independently of S; accept with probability $1 \wedge e^{\varphi[d(S') - d(S)]}$; else stay.

Round. m = p + B + 2 labelled steps: Gibbs 1, …, Gibbs p, one uniform swap, B neighbour swaps, one independence step; their kernels K_1, …, K_m (as matrices on Ω). For a permutation σ of [m] (the realised order), K_σ := K_{σ(1)} K_{σ(2)} ⋯ K_{σ(m)} (step σ(1) is applied first). K_φ := (1/m!) Σ_σ K_σ (order uniform, independent of the state).

Analytic objects. $L^2(q_\varphi)$: functions on Ω with ⟨f, g⟩ = Σ q_φ f g. A kernel P is q-reversible if q(x)P(x,y) = q(y)P(y,x) for all x, y; P is self-adjoint on $L^2(q)$ if ⟨Pf, g⟩ = ⟨f, Pg⟩ for all f, g. Π_φ := the kernel $\Pi_\varphi(x, \cdot) = q_\varphi(\cdot)$. Absolute L^2 gap of P: gap*(P) := 1 − $\|P - \Pi_\varphi\|_{L^2(q_\varphi)\to L^2(q_\varphi)}$. $\|\mu - \nu\|_{TV} = \sup_A |\mu(A) - \nu(A)|$. $\varepsilon_\varphi := \min_{S\in\Omega} \pi_0(S)/q_\varphi(S)$. A **switch** means $\operatorname{argmax}_\Omega q_0 \neq \operatorname{argmax}_\Omega q_1$ (the bridge's mode changes between φ = 0 and φ = 1). The **discrepancy range** is max_Ω d − min_Ω d.

Valid component input: any (n, p, X, y) as above together with the method's blocks, tables and neighbour graph built from it. The kernels are component objects: they are defined for every such input, whether or not the method's exact-enumeration branch would be taken for that (n, p).

2. Claims (for every valid component input and every φ ∈ [0, 1], unless an existence claim says

otherwise)

(i) Σ_{k≤K} (w(k)/W_K) c(k) = $\bar{Z}_0$ exactly. Under EX-A and EX-S, the backward sampler is well defined (every conditional probability it uses has a positive denominator and sums to 1) and returns S with probability exactly $\pi_0(S)$; successive draws with fresh streams are iid π_0.

(ii) Each elementary kernel is a Markov kernel on Ω that is q_φ-reversible, including the boundary cases: the Gibbs addition at |S| = K has probability 0; the uniform swap self-loops at |S| ∈ {0, p}; the neighbour-swap ratio read as 0 when the reverse insertion is not a neighbour (j ∉ N(l)); and the stated Hastings ratios equal γ_φ(S′)q(S′, S) / (γ_φ(S)q(S, S′)) with q the actual proposal probabilities (so they supply the reverse proposal probability). For the neighbour swap this holds for any family N(j) ⊆ [p] \ {j}; for the method's symmetric graph the "read as 0" case never arises.

(iii) K_φ is a Markov kernel, q_φ-invariant, and self-adjoint on L^2(q_φ) (equivalently q_φ-reversible). *Original wording: "An individual realised order need not be reversible."* Reading: for every σ, K_σ is q_φ-invariant, and K_σ is q_φ-reversible if and only if K_σ = K_{$\bar{\sigma}$}, $\bar{\sigma}$ the reversed order; and there exist a valid component input, φ ∈ [0,1] and an order σ such that K_σ is not q_φ-reversible.

(iv) For every σ and every S ∈ Ω: K_σ(S, ·) ≥ ε_φ q_φ(·) (as measures), with ε_φ = ($\bar{Z}$_φ/$\bar{Z}_0$) e^{−φ max_Ω d} ∈ (0, 1]. Hence K_φ(S, ·) ≥ ε_φ q_φ(·), sup_S ‖K_φ^h(S, ·) − q_φ‖_TV ≤ (1 − ε_φ)^h for every integer h ≥ 0, and gap*(K_φ) ≥ ε_φ.

(v) ε_φ ≤ $\pi_0(S)$/q_φ(S) for every S ∈ Ω. *Original wording: "An exponentially small available minorisation constant does not establish exponentially slow mixing of the hybrid kernel; neither a switch nor a large discrepancy range alone supplies such a lower bound."* Reading: there exist constants c > 0, n_0 and, for every n ≥ n_0, a valid component input (with that n) such that at φ = 1 simultaneously: $\varepsilon_1 \le e^{-cn}$; the discrepancy range is ≥ cn; there is a switch; and gap*(K_1) ≥ 1/40 and sup_S ‖K_1^h(S, ·) − q_1‖_TV ≤ (39/40)^h for all h. Consequently, writing t_mix(K) := min{h ≥ 0 : sup_S ‖K^h(S, ·) − q_φ‖_TV ≤ 1/4}, the family has t_mix(K_1) ≤ 55 for every n ≥ n_0; so no upper bound on gap*(K_φ) that tends to 0, and no lower bound on t_mix(K_φ) that tends to infinity (in particular none exponentially large in n), can be derived from ε_φ being exponentially small, from a switch, or from a large discrepancy range, alone or together.

(c) Final proof, verbatim: body of PROOF_v3.tex

The proof below is the body of `PROOF_v3.tex` *from* `\begin{document}` *to* `\end{document}` *without the title block. Its preamble definitions are local to this chapter; labels carry the prefix* `T09:`.

All notation is that of STATEMENT.md, which is the statement proved here. Throughout, EX-A (exact arithmetic) and EX-S (ideal independent uniform streams) are in force. Kernels on the finite set Ω are identified with nonnegative $\Omega \times \Omega$ matrices; a measure is a row vector; $Pf(x) = \sum_y P(x,y)f(y)$ and $\mu P(y) = \sum_x \mu(x)P(x,y)$; $\mathbf{1}$ is the constant function 1. For P stochastic and q a probability on Ω, P is q-invariant if $qP = q$.

1 Preliminaries

Lemma 1 (Positivity)**.** *For every $S \subseteq [p]$: $Q_S \ge \|y\|^2/(n+1) > 0$, $D_S \in [1, \infty)$ is finite, $A_S = 2b_0 + Q_S > 0$ and $\mathrm{BF}(S) \in (0, \infty)$. Consequently $\gamma_\phi(S) \in (0, \infty)$ for all $S \in \Omega$ and all real ϕ, $\bar{Z}_\phi \in (0, \infty)$, and $q_\phi(S) > 0$ for every $S \in \Omega$.*

Proof. Let $k = |S|$. If $k = 0$, $Q_\emptyset = \|y\|^2$ and $D_\emptyset = 1$. Let $k \geq 1$ and write the spectral decomposition $G_S = \sum_{i=1}^k s_i b_i b_i^\top$ with $s_i \geq 0$ and (b_i) an orthonormal basis of $\mathbb{R}^k$. Then $H_S = \sum_i ((n+1)s_i + 1) b_i b_i^\top \succ 0$ and $G_S + I = \sum_i (s_i + 1) b_i b_i^\top \succ 0$, so $D_S = \prod_i \frac{(n+1)s_i+1}{s_i+1} \in [1, \infty)$. For $s_i > 0$ put $a_i = X_S b_i / \sqrt{s_i}$; the a_i are orthonormal, $X_S b_i = 0$ when $s_i = 0$, and $X_S H_S^{-1} X_S^\top = \sum_{i: s_i > 0} \frac{s_i}{(n+1)s_i+1} a_i a_i^\top$, whose eigenvalues are $< 1/(n+1)$. Hence $n z_S^\top H_S^{-1} z_S = n\, y^\top X_S H_S^{-1} X_S^\top y \leq \frac{n}{n+1}\|y\|^2$ and $Q_S \geq \|y\|^2/(n+1) = (n-1)/(n+1) > 0$. The remaining claims follow from $b_0 > 0$, the definitions, $\varpi(S) > 0$ on Ω, and $|\Omega| < \infty$. □

Lemma 2 (Metropolis–Hastings form)**.** *Let $q > 0$ be a probability on Ω, $\mathcal{Q}$ a proposal kernel on Ω, and $P(x,y) = \mathcal{Q}(x,y)\alpha(x,y)$ for $y \neq x$, $P(x,x) = 1 - \sum_{y \neq x} P(x,y)$, with $\alpha(x,y) \in [0,1]$. If, for all $x \neq y$ with $\mathcal{Q}(x,y) > 0$, $\alpha(x,y) = 1 \wedge r(x,y)$ with $r(x,y) = \dfrac{q(y)\mathcal{Q}(y,x)}{q(x)\mathcal{Q}(x,y)}$, then P is a Markov kernel and $q(x)P(x,y) = q(y)P(y,x)$ for all x, y.*

Proof. $P \geq 0$ and rows sum to one since $\sum_{y \neq x} \mathcal{Q}(x,y)\alpha(x,y) \leq 1$. For $x \neq y$: if $\mathcal{Q}(x,y) > 0$ and $\mathcal{Q}(y,x) > 0$ then $q(x)P(x,y) = \min\{q(x)\mathcal{Q}(x,y), q(y)\mathcal{Q}(y,x)\}$, which is symmetric. If exactly one of them is 0, say $\mathcal{Q}(y,x) = 0 < \mathcal{Q}(x,y)$, then $r(x,y) = 0$, so $P(x,y) = 0 = P(y,x)$. If both are 0, both sides are 0. The case $x = y$ is trivial. □

Lemma 3 (Adjoints)**.** *Let $q > 0$ be a probability on Ω. For a kernel P define $P^*(x,y) = q(y)P(y,x)/q(x)$. Then $\langle Pf, g\rangle_q = \langle f, P^* g\rangle_q$ for all f, g; $(PR)^* = R^* P^*$; P is self-adjoint on $L^2(q)$ iff $P = P^*$ iff P is q-reversible. If P is q-reversible and stochastic, then $qP = q$.*

Proof. $\langle Pf, g\rangle_q = \sum_{x,y} q(x)P(x,y)f(y)g(x) = \sum_y q(y)f(y) \sum_x P^*(y,x)g(x)$. The adjoint is unique (the form is nondegenerate because $q > 0$), which gives the product rule and the equivalences; $P = P^*$ is the detailed-balance identity. Finally $\sum_x q(x)P(x,y) = \sum_x q(y)P(y,x) = q(y)$. □

Lemma 4 (Doeblin contraction)**.** *Let P be a stochastic kernel on Ω with $qP = q$, $q > 0$ a probability, and suppose $P(x,\cdot) \geq c\,\nu(\cdot)$ for every x, for a probability ν and a constant $c \in (0,1]$. Then: (a) $\|\mu P\|_1 \leq (1-c)\|\mu\|_1$ for every signed measure μ with $\mu(\Omega) = 0$; (b) $\sup_x \|P^h(x,\cdot) - q\|_{\mathrm{TV}} \leq (1-c)^h$ for every integer $h \geq 0$; (c) if moreover P is self-adjoint on $L^2(q)$, then $\mathrm{gap}^*(P) := 1 - \|P - \Pi\|_{L^2(q) \to L^2(q)} \geq c$, where $\Pi(x,\cdot) = q(\cdot)$.*

Proof. (a) If $c = 1$ then $P = \mathbf{1}\nu$ and $\mu P = \mu(\Omega)\nu = 0$. If $c < 1$, $R := (P - c\mathbf{1}\nu)/(1-c)$ is nonnegative with unit row sums, and $\mu P = c\mu(\Omega)\nu + (1-c)\mu R = (1-c)\mu R$; also $\|\mu R\|_1 \leq \sum_{x,y} |\mu(x)| R(x,y) = \|\mu\|_1$. (b) Since $qP^h = q$, $P^h(x,\cdot) - q = (\delta_x - q)P^h$, and $(\delta_x - q)(\Omega) = 0$. By (a) iterated, $\|P^h(x,\cdot) - q\|_{\mathrm{TV}} = \frac{1}{2}\|(\delta_x - q)P^h\|_1 \leq (1-c)^h \frac{1}{2}\|\delta_x - q\|_1 = (1-c)^h(1 - q(x)) \leq (1-c)^h$. (c) Let $L_0^2 = \{f : \sum_x q(x)f(x) = 0\}$. Since $\sum_x q(x)(Pf)(x) = (qP)f = qf$, P maps L_0^2 into itself, and the restriction is self-adjoint; by the spectral theorem (finite dimension) $\|P|_{L_0^2}\| = \max|\lambda|$ over its eigenvalues. Let $Pf = \lambda f$ with $0 \neq f \in L_0^2$ and $\mu(x) := q(x)f(x)$. Then $\mu \neq 0$, $\mu(\Omega) = 0$, and by reversibility (Lemma 3) $\mu P(y) = \sum_x q(y)P(y,x)f(x) = q(y)\lambda f(y) = \lambda\mu(y)$. By (a), $|\lambda| \leq 1 - c$. Every $f \in L^2(q)$ splits orthogonally as $f = a\mathbf{1} + g$ with $g \in L_0^2$, and $(P - \Pi)f = Pg$ because $P\mathbf{1} = \Pi\mathbf{1} = \mathbf{1}$ and $\Pi g = 0$. Hence $\|(P - \Pi)f\| \leq (1-c)\|g\| \leq (1-c)\|f\|$. □

2 Part (i): exact normaliser and exact sampler

Put $e_b(j) = 0$ for $j \notin \{0, \ldots, |C_b|\}$, $c^{(b)}(t) = 0$ for $t < 0$, and $P_b = \sum_{i \leq b} |C_i|$. These are the conventions under which the recursion $c^{(b)}(t) = \sum_j e_b(j) c^{(b-1)}(t-j)$ and the draw of j_b in STATEMENT.md are read: j ranges over all integers, and only $0 \leq j \leq |C_b|$ with $t - j \geq 0$ contribute.

Lemma 5. *For $b = 0, \dots, J$ and every integer t, $c^{(b)}(t) = \sum \prod_{i \le b} r_i(U_i)$, the sum running over tuples $(U_1, \dots, U_b)$ with $U_i \subseteq C_i$ and $\sum_i |U_i| = t$. In particular $c^{(b)}(t) > 0$ iff $0 \le t \le P_b$.*

Proof. Induction on b. For $b = 0$ the only tuple is the empty one, with $t = 0$ and product 1. For the step, group the tuples by $j = |U_b|$: $\sum_j \big(\sum_{|U_b|=j} r_b(U_b)\big) c^{(b-1)}(t-j) = \sum_j e_b(j) c^{(b-1)}(t-j)$. Positivity: all $r_i > 0$ (Lemma 1) and a tuple with $\sum |U_i| = t$ exists iff $0 \le t \le P_b$. □

Proposition 6 (Part (i))**.** *$\sum_{k \le K} \frac{w(k)}{W_K} c(k) = \bar{Z}_0$. The backward sampler is well defined and returns S with probability $\pi_0(S)$ for every $S \in \Omega$; draws with fresh streams are iid π_0.*

Proof. Because $C_1, \dots, C_J$ partition $[p]$, the map $S \mapsto (S \cap C_1, \dots, S \cap C_J)$ is a bijection from subsets of $[p]$ onto tuples with $U_b \subseteq C_b$, and $|S| = \sum_b |S \cap C_b|$. Since $\varpi(S) = w(|S|)/W_K$ depends on S only through $|S|$,

$$\bar{Z}_0 = \sum_{k=0}^{K} \frac{w(k)}{W_K} \sum_{|S|=k} \prod_b r_b(S \cap C_b) = \sum_{k=0}^{K} \frac{w(k)}{W_K} c(k)$$

by Lemma 5 with $b = J$. (The values $c^{(b)}(t)$, $t \le K$, depend only on $c^{(b-1)}(t')$ with $t' \le t$, so computing every convolution only up to degree K is exact.)

Well-definedness. Step (1): the denominator $\sum_{k' \le K} w(k')c(k') \ge w(0)c(0) = \frac{1}{2} \prod_b r_b(\emptyset) = \frac{1}{2} > 0$, so the law of k is a probability, and k is drawn only where $c(k) > 0$. Step (2): we show by downward induction that $c^{(b)}(t_b) > 0$ whenever t_b is reached with positive probability. This holds for $b = J$. If $c^{(b)}(t_b) > 0$, the numbers $e_b(j)c^{(b-1)}(t_b - j)/c^{(b)}(t_b)$ are nonnegative and sum to 1 over j by the recursion, and j_b has positive probability only if $c^{(b-1)}(t_b - j_b) > 0$, i.e. $c^{(b-1)}(t_{b-1}) > 0$; also $e_b(j_b) > 0$, so $0 \le j_b \le |C_b|$. At the end $c^{(0)}(t_0) > 0$ forces $t_0 = 0$, so $\sum_b j_b = k$. Step (3): $e_b(j_b) > 0$ and $\{U \subseteq C_b : |U| = j_b\}$ has total weight $e_b(j_b)$, so the law of U_b is a probability. The output $S = \bigcup_b U_b$ has $|S| = k \le K$, so $S \in \Omega$.

Exactness. Fix $S \in \Omega$. The only run producing S is $k = |S|$, $j_b = |S \cap C_b|$, $t_b = \sum_{i \le b} j_i$, $U_b = S \cap C_b$; along it all quantities in the denominators are positive (Lemma 5, since $0 \le t_b \le P_b$). Its probability is

$$\frac{w(k)c(k)}{\sum_{k'} w(k')c(k')} \prod_{b=1}^{J} \frac{e_b(j_b)c^{(b-1)}(t_{b-1})}{c^{(b)}(t_b)} \prod_{b=1}^{J} \frac{r_b(U_b)}{e_b(j_b)} = \frac{w(k)c(k)}{\sum_{k'} w(k')c(k')} \cdot \frac{c^{(0)}(0)}{c^{(J)}(k)} \prod_b r_b(U_b) = \frac{\gamma_0(S) W_K}{W_K \bar{Z}_0} = \pi_0(S),$$

the middle product telescoping. Under EX-A each categorical draw from a finite law with an ideal uniform is exact; under EX-S the uniforms of different draws are independent, and each output is a fixed measurable function of its own uniforms with law π_0; hence the draws are iid π_0. □

3 Part (ii): reversibility of every elementary kernel

Fix ϕ and write $\gamma = \gamma_\phi$, $q = q_\phi$; recall $\gamma(S) = 0$ for $S \notin \Omega$ and $\gamma > 0$ on Ω (Lemma 1). For $S \ne S'$ of equal size with $|S \triangle S'| = 2$ write $j = j(S, S')$ for the element of $S \setminus S'$ and $l = l(S, S')$ for the element of $S' \setminus S$; the pair (j, l) determines $S' = S \setminus \{j\} \cup \{l\}$ and conversely.

Gibbs j. For $S \in \Omega$, $S^- = S \setminus \{j\} \in \Omega$, so the denominator $\gamma(S^+) + \gamma(S^-) \ge \gamma(S^-) > 0$ and $K_{G,j}(S, \cdot)$ is a probability on $\{S^+, S^-\} \cap \Omega$ (if $S^+ \notin \Omega$ its probability is $\gamma(S^+)/(\cdot) = 0$). This is the boundary case: $S^+ \notin \Omega$ iff $j \notin S$ and $|S| = K$, and then the addition has probability 0. Reversibility: let $T \ne S$ with $K_{G,j}(S, T) > 0$. Then T is the element of $\{S^+, S^-\}$ other than S,

so $T \in \Omega$ and $\{T^+, T^-\} = \{S^+, S^-\}$, and $q(S)K_{G,j}(S,T) = \gamma(S)\gamma(T)/(\bar{Z}_\phi(\gamma(S^+) + \gamma(S^-)))$ is symmetric in (S,T); in particular $K_{G,j}(T,S) > 0$ too. If $K_{G,j}(S,T) = 0$ and $T \neq S$ then either $T \notin \{S^+, S^-\}$ (so $S \notin \{T^+, T^-\}$ and $K_{G,j}(T,S) = 0$) or $T = S^+ \notin \Omega$, which is not a state. Hence detailed balance holds.

Uniform swap. If $|S| \in \{0, p\}$ the kernel is δ_S (boundary self-loop). If $1 \leq |S| \leq p-1$, the proposal is $\mathcal{Q}(S,S') = 1/(|S|(p-|S|))$ for each S' with $|S'| = |S|$, $|S \triangle S'| = 2$ (distinct pairs (j,l) give distinct S'), and $S' \in \Omega$. All proposals preserve size, so states of size 0 or p (each the unique set of its size) are never proposed from other states. For S' proposed from S, S is proposed from S' (delete l, add j) and $\mathcal{Q}(S',S) = 1/(|S'|(p-|S'|)) = \mathcal{Q}(S,S')$, so the Hastings ratio $\gamma(S')\mathcal{Q}(S',S)/(\gamma(S)\mathcal{Q}(S,S'))$ equals the stated $\gamma(S')/\gamma(S)$. Lemma 2 applies (with $\mathcal{Q}(S,\cdot) = \delta_S$ on the boundary states).

Neighbour swap (arbitrary $N(j) \subseteq [p] \setminus \{j\}$). If $S = \emptyset$ the kernel is $\delta_\emptyset$; all moves preserve size, so $\emptyset$ is never proposed from another state. For $S \neq \emptyset$ and $S' \neq S$, the move $S \to S'$ is proposed only by the pair $(j,l) = (j(S,S'), l(S,S'))$, so

$$\mathcal{Q}(S,S') = \begin{cases} \dfrac{1}{|S|\,|N(j) \setminus S|} & \text{if } l \in N(j), \\ 0 & \text{otherwise,} \end{cases} \qquad \mathcal{Q}(S',S) = \begin{cases} \dfrac{1}{|S'|\,|N(l) \setminus S'|} & \text{if } j \in N(l), \\ 0 & \text{otherwise,} \end{cases}$$

(here $j \in S$, $l \notin S$ by definition of $j(S,S')$, $l(S,S')$). The second formula holds because the reverse move deletes $l \in S'$ and must insert $j \notin S'$ from $N(l) \setminus S'$; when $j \in N(l)$ the set $N(l) \setminus S'$ contains j and is nonempty, and when $l \in N(j)$ the set $N(j) \setminus S$ contains l, so no denominator vanishes. (Proposals with $N(j) \setminus S = \emptyset$ are self-loops and do not affect Lemma 2.) Let $\mathcal{Q}(S,S') > 0$, so $|S'| = |S|$. If $j \in N(l)$, the Hastings ratio is

$$\frac{\gamma(S')\mathcal{Q}(S',S)}{\gamma(S)\mathcal{Q}(S,S')} = \frac{\gamma(S')}{\gamma(S)} \cdot \frac{|S|\,|N(j) \setminus S|}{|S'|\,|N(l) \setminus S'|} = \frac{\gamma(S')}{\gamma(S)} \cdot \frac{|N(j) \setminus S|}{|N(l) \setminus S'|},$$

the stated ratio. If $j \notin N(l)$, the Hastings ratio is 0, which is the stated ratio under the reading convention. So the stated acceptance is $1 \wedge r$ with the true Hastings ratio r in every case, and Lemma 2 gives reversibility. For the method's graph, $l \in N(j) \Rightarrow j \in N(l)$, so the convention never applies.

Independence. $\mathcal{Q}(S,S') = \pi_0(S') > 0$ for all $S, S' \in \Omega$. Since $q(S)/\pi_0(S) = (\bar{Z}_0/\bar{Z}_\phi)e^{\phi d(S)}$, $\dfrac{q(S')\pi_0(S)}{q(S)\pi_0(S')} = e^{\phi[d(S')-d(S)]}$, the stated ratio; Lemma 2 applies (a proposal $S' = S$ is a self-loop).

In each case the stated Hastings ratio equals $\gamma(S')\mathcal{Q}(S',S)/(\gamma(S)\mathcal{Q}(S,S'))$ with the actual proposal probabilities, i.e. it supplies the reverse proposal probability. This proves (ii).

4 Part (iii): the averaged round

By (ii) and Lemma 3, each K_e is stochastic, q-invariant and self-adjoint, $K_e^* = K_e$. Hence every K_σ is stochastic and $qK_\sigma = q$, and so is the average K_ϕ. By the product rule, $K_\sigma^* = K_{\sigma(m)}^* \cdots K_{\sigma(1)}^* = K_{\sigma(m)} \cdots K_{\sigma(1)} = K_{\bar{\sigma}}$, $\bar{\sigma}(i) = \sigma(m+1-i)$. Since $\sigma \mapsto \bar{\sigma}$ is a bijection of the symmetric group, $K_\phi^* = \frac{1}{m!}\sum_\sigma K_{\bar{\sigma}} = K_\phi$: K_ϕ is self-adjoint on $L^2(q_\phi)$, equivalently q_ϕ-reversible (Lemma 3). Also K_σ is q-reversible iff $K_\sigma = K_\sigma^* = K_{\bar{\sigma}}$.

That some realised order is not reversible is shown in Proposition 13 below, on the family of Section 6.

5 Part (iv): minorisation

Lemma 7. $\pi_0(S)/q_\phi(S) = (\bar{Z}_\phi/\bar{Z}_0)e^{-\phi d(S)}$, *so for* $\phi \geq 0$, $\varepsilon_\phi = \min_\Omega \pi_0/q_\phi = (\bar{Z}_\phi/\bar{Z}_0)e^{-\phi \max_\Omega d}$, *and* $0 < \varepsilon_\phi \leq 1$.

Proof. The identity is $\pi_0 = \gamma_0/\bar{Z}_0$ and $q_\phi = \gamma_0 e^{\phi d}/\bar{Z}_\phi$; for $\phi \geq 0$, $e^{-\phi d}$ is minimal where d is maximal. Positivity by Lemma 1. If $\pi_0(S) > q_\phi(S)$ for all S, summing gives $1 > 1$; so $\varepsilon_\phi \leq 1$. □

Proposition 8 (Part (iv))**.** *For every* σ *and* $x \in \Omega$, $K_\sigma(x,\cdot) \geq \varepsilon_\phi q_\phi(\cdot)$. *Hence* $K_\phi(x,\cdot) \geq \varepsilon_\phi q_\phi(\cdot)$, $\sup_x \|K_\phi^h(x,\cdot) - q_\phi\|_{\mathrm{TV}} \leq (1-\varepsilon_\phi)^h$ *for all* $h \geq 0$, *and* $\mathrm{gap}^*(K_\phi) \geq \varepsilon_\phi$.

Proof. Write $\varepsilon = \varepsilon_\phi$, $q = q_\phi$, K_{ind} for the independence kernel. For $u \neq v$, $K_{\mathrm{ind}}(u,v) = \min\{\pi_0(v),\, \pi_0(v)e^{\phi[d(v)-d(u)]}\}$. Here $\pi_0(v) \geq \varepsilon q(v)$, and by Lemma 7 $\pi_0(v)e^{\phi[d(v)-d(u)]} = q(v)\,\pi_0(u)/q(u) \geq \varepsilon q(v)$. For $v = u$, $K_{\mathrm{ind}}(u,u) \geq \pi_0(u) \geq \varepsilon q(u)$ (the proposal u is a self-loop). So $K_{\mathrm{ind}}(u,\cdot) \geq \varepsilon q(\cdot)$ for every u.

Let the independence step be at position i of σ, $A = K_{\sigma(1)} \cdots K_{\sigma(i-1)}$ (identity if $i = 1$) and $C = K_{\sigma(i+1)} \cdots K_{\sigma(m)}$ (identity if $i = m$). A is stochastic, $C \geq 0$ and $qC = q$ (part (iii)). Then

$$K_\sigma(x,y) = \sum_{u,v} A(x,u)K_{\mathrm{ind}}(u,v)C(v,y) \geq \varepsilon \sum_u A(x,u) \sum_v q(v)C(v,y) = \varepsilon q(y).$$

Averaging over σ gives $K_\phi(x,\cdot) \geq \varepsilon q$. Since K_ϕ is stochastic, q-invariant and self-adjoint (part (iii)), Lemma 4 with $\nu = q$, $c = \varepsilon \in (0,1]$ gives the TV bound and $\mathrm{gap}^*(K_\phi) \geq \varepsilon$. □

Remark 9. The same argument gives the TV bound for any product of h realised orders (each factor is q-invariant and $\geq \varepsilon q$), not only for the averaged kernel.

6 Part (v), and the non-reversible order of part (iii)

The first sentence of (v) is the definition of ε_ϕ as a minimum. For the second, the asymptotic parameter is the sample size n (with $p = 2$ fixed): "exponentially small" and "large range" refer to n; nothing is claimed about growth in p.

Notation for this section. The family has two predictors, and its states are written $\emptyset, \{1\}, \{2\}, \{1,2\}$; subscripts 1, 2, 12 on Q, D, A, t, R, κ refer to these states. The correlation parameter of the family is τ and the signal parameter ρ; the rate constant of the statement is κ. Throughout, $q_1 = \pi$ is the target at $\phi = 1$.

The family. Fix $\tau \in (0,1)$ and $\rho \in (0,1)$ (for instance $\tau = \frac{1}{2}$, $\rho = \frac{3}{5}$). For $n \geq 4$ the space $\mathbf{1}^\perp \subset \mathbb{R}^n$ has dimension $n-1 \geq 3$; fix orthonormal $h_1, h_2, h_3 \in \mathbf{1}^\perp$ and put $s = \sqrt{n-1}$,

$$x_1 = sh_1, \qquad x_2 = s(\tau h_1 + \sqrt{1-\tau^2}\, h_2), \qquad y = s(\rho h_1 + \sqrt{1-\rho^2}\, h_3).$$

Columns and y are centred with squared norm $n-1$, so this is standardised data in the sense of STATEMENT.md, Section 1. Here $p = 2$, $K = \min(2, \lfloor n/2 \rfloor) = 2$, $\Omega = \{\emptyset, \{1\}, \{2\}, \{1,2\}\}$, and $B = \min(\lfloor \sqrt{n} \rfloor, 1, 16) = 1$. Since every block has $|C_b| \leq B = 1$ and the blocks partition $\{1,2\}$, the blocks are $\{1\}$ and $\{2\}$ (in some order; γ_0 depends only on the partition). $m = p + B + 2 = 5$. The neighbour graph has $N(j) \subseteq \{1,2\} \setminus \{j\}$; no property of it beyond this is used. The prior: $w(0) = \frac{1}{2}$, $w(1) = w(2) = \frac{1}{6}$, so $w(1)/w(0) = \frac{1}{3}$ and $w(2)/w(1) = 1$.

Scores. $x_1^\top x_2 = (n-1)\tau$, $z_1 = x_1^\top y = (n-1)\rho$, $z_2 = (n-1)\tau\rho$. For a singleton, $G = n-1$, $H = (n+1)(n-1)+1 = n^2$, $D = n$, $Q = (n-1) - z^2/n$. So

$$Q_\emptyset = n-1, \quad Q_1 = (n-1)\Big(1-\rho^2\tfrac{n-1}{n}\Big), \quad Q_2 = (n-1)\Big(1-\tau^2\rho^2\tfrac{n-1}{n}\Big), \quad D_1 = D_2 = n.$$

For $\{1,2\}$, $G = (n-1)R$, $R = \begin{pmatrix} 1 & \tau \\ \tau & 1 \end{pmatrix}$, with eigenvectors $g_\pm = (1,\pm 1)/\sqrt{2}$ and eigenvalues $1\pm\tau$. Then H has eigenvalues $\alpha_\pm = (n^2-1)(1\pm\tau)+1$ and $G+I$ has $\beta_\pm = (n-1)(1\pm\tau)+1$; since $\alpha_\pm = (n+1)\beta_\pm - n$, $D_{12} = f(1+\tau)f(1-\tau)$ with $f(x) = n+1-\frac{n}{(n-1)x+1}$, increasing in $x>0$ with $f \le n+1$. With $z = (n-1)\rho(1,\tau)^\top$, $z^\top g_\pm = (n-1)\rho(1\pm\tau)/\sqrt{2}$ and, using $(1\pm\tau)^2/\alpha_\pm = \frac{1\pm\tau}{n^2-1}(1-1/\alpha_\pm)$,

$$t_{12} := nz^\top H^{-1} z = \frac{n(n-1)^2\rho^2}{2}\sum_\pm \frac{(1\pm\tau)^2}{\alpha_\pm} = \frac{n(n-1)\rho^2}{n+1}\Big(1-\frac{\delta}{2}\Big), \qquad \delta = \sum_\pm \frac{1\pm\tau}{\alpha_\pm} \in \Big(0, \frac{2}{n^2-1}\Big),$$

and $Q_{12} = (n-1) - t_{12}$.

Lemma 10. *(B1)* $|Q_{12}-Q_1| < 1/n$. *(B2)* $Q_{12} > (n-1)/(n+1) \ge \frac{1}{2}$. *(B3) If* $(n-1)(1-\tau) \ge 1$ *then* $n^2/4 \le D_{12} \le (n+1)^2$.

Proof. (B1) $t_{12} - \frac{(n-1)^2\rho^2}{n} = (n-1)\rho^2\big[\frac{n}{n+1} - \frac{n-1}{n}\big] - \frac{n(n-1)\rho^2\delta}{2(n+1)}$. The first term equals $\frac{(n-1)\rho^2}{n(n+1)} \in [0, 1/n)$; the second lies in $[0, \frac{n(n-1)\rho^2}{(n+1)(n^2-1)}) = [0, \frac{n\rho^2}{(n+1)^2}) \subset [0,1/n)$. A difference of two numbers in $[0,1/n)$ has absolute value $< 1/n$. (B2) $t_{12} < \frac{n(n-1)}{n+1}$, so $Q_{12} > \frac{n-1}{n+1}$, which is $\ge \frac{1}{2}$ for $n \ge 3$. (B3) $f \le n+1$ gives the upper bound. If $(n-1)x \ge 1$ then $f(x) \ge n+1-n/2 \ge n/2$; f increasing and $(n-1)(1-\tau) \ge 1$ give $f(1\pm\tau) \ge n/2$. □

Ratios at $\phi = 1$. $\gamma_1 = \varpi\mathrm{BF}$ and $\mathrm{BF}(S) = D_S^{-1/2}(A_\emptyset/A_S)^\nu$, $A_S = 0.02 + Q_S$, $\nu = 0.01 + n/2$. Define

$$R_1 = \frac{\gamma_1(\{1\})}{\gamma_1(\emptyset)} = \frac{\mathrm{BF}(\{1\})}{3}, \quad R_2 = \frac{\gamma_1(\{2\})}{\gamma_1(\emptyset)} = \frac{\mathrm{BF}(\{2\})}{3},$$

$$R_{12|1} = \frac{\gamma_1(\{1,2\})}{\gamma_1(\{1\})} = \Big(\frac{D_1}{D_{12}}\Big)^{1/2}\Big(\frac{A_1}{A_{12}}\Big)^\nu, \qquad R_{12|2} = \frac{\gamma_1(\{1,2\})}{\gamma_1(\{2\})} = \Big(\frac{D_2}{D_{12}}\Big)^{1/2}\Big(\frac{A_2}{A_{12}}\Big)^\nu,$$

using $w(2)/w(1) = 1$. Let $\kappa_1 = \frac{1}{2}\log\frac{1}{1-\rho^2}$ and $\kappa_2 = \frac{1}{2}\log\frac{1}{1-\tau^2\rho^2}$; then $\kappa_1 > \kappa_2 > 0$ because $0 < \tau < 1$, $\rho > 0$.

Lemma 11. *As* $n \to \infty$*:* $\frac{1}{n}\log R_1 \to \kappa_1$, $\frac{1}{n}\log R_2 \to \kappa_2$, $\frac{1}{n}\log R_{12|2} \to \kappa_1 - \kappa_2 > 0$, $\frac{1}{n}\log\frac{\gamma_1(\{2\})}{\gamma_1(\{1\})} \to -(\kappa_1-\kappa_2) < 0$, $\frac{1}{n}\log R_{12|1} \to 0$, *and* $R_{12|1} \le 2e^{1.01}n^{-1/2}$ *once* $(n-1)(1-\tau) \ge 1$.

Proof. $A_\emptyset/A_1 \to 1/(1-\rho^2)$, $A_\emptyset/A_2 \to 1/(1-\tau^2\rho^2)$ and, by (B1), $A_2/A_{12} \to (1-\tau^2\rho^2)/(1-\rho^2)$ and $A_1/A_2 \to (1-\rho^2)/(1-\tau^2\rho^2)$, all limits in $(0,\infty)$. Since $\nu/n \to \frac{1}{2}$, $\frac{1}{n}\log(\cdot)^\nu \to \frac{1}{2}\log(\lim)$. The remaining factors are: $\frac{1}{3}D_1^{-1/2} = \frac{1}{3}n^{-1/2}$ in R_1; $\frac{1}{3}D_2^{-1/2} = \frac{1}{3}n^{-1/2}$ in R_2; and, once $(n-1)(1-\tau) \ge 1$, $(D_2/D_{12})^{1/2} \in [n^{1/2}/(n+1),\, 2n^{-1/2}]$ in $R_{12|2}$ by (B3). Each lies in $[\frac{1}{3(n+1)},\, n^{1/2}]$, so contributes $O(\log n) = o(n)$ to the logarithm. This gives the first three limits. Next, $\gamma_1(\{2\})/\gamma_1(\{1\}) = (A_1/A_2)^\nu$ since $D_1 = D_2$ and $w(1) = w(1)$, which gives the fourth. Since $R_{12|1} = R_{12|2} \cdot \gamma_1(\{2\})/\gamma_1(\{1\})$, the fifth limit is $(\kappa_1-\kappa_2) - (\kappa_1-\kappa_2) = 0$. Finally, by (B1)–(B2), $A_1/A_{12} \le 1 + (1/n)/A_{12} \le 1 + 2/n$, so $(A_1/A_{12})^\nu \le e^{2\nu/n} = e^{1+0.02/n} \le e^{1.01}$; with (B3), $(D_1/D_{12})^{1/2} \le (4/n)^{1/2}$. □

Lemma 12 (Start, target, discrepancy)**.** *(a)* $\pi_0(\{1,2\}) \to 1$*, and for all large* n*,* $\arg\max_\Omega \pi_0 = \{1,2\}$*. (b)* $q_1(\{1\}) \to 1$*, and for all large* n*,* $\arg\max_\Omega q_1 = \{1\}$*. (c)* $d(\emptyset) = d(\{1\}) = d(\{2\}) = 0$ *and* $d(\{1,2\}) = \log R_{12|1} - \log(3R_2)$*, so* $\frac{1}{n} d(\{1,2\}) \to -\kappa_2$*; moreover* $-d(\{1,2\}) \geq \log(3R_2) - \log(2e^{1.01} n^{-1/2})$ *once* $(n-1)(1-\tau) \geq 1$*. (d)* $\varepsilon_1 \leq 1/(3R_2\, q_1(\{1\}))$*, so* $\limsup \frac{1}{n} \log \varepsilon_1 \leq -\kappa_2$*.*

Proof. (a) With blocks $\{1\}, \{2\}$, $\gamma_0(S) = \varpi(S)\mathrm{BF}(\{1\})^{\mathbf{1}[1\in S]}\mathrm{BF}(\{2\})^{\mathbf{1}[2\in S]}$. So $\gamma_0(\{1\})/\gamma_0(\{1,2\}) = 1/\mathrm{BF}(\{2\}) = 1/(3R_2)$, $\gamma_0(\{2\})/\gamma_0(\{1,2\}) = 1/(3R_1)$, $\gamma_0(\emptyset)/\gamma_0(\{1,2\}) = 3/(\mathrm{BF}(\{1\})\mathrm{BF}(\{2\})) = 1/(3R_1R_2)$; all $\to 0$ by Lemma 11. (b) $\gamma_1(\emptyset)/\gamma_1(\{1\}) = 1/R_1$, $\gamma_1(\{1,2\})/\gamma_1(\{1\}) = R_{12|1}$, and $\gamma_1(\{2\})/\gamma_1(\{1\})$; all $\to 0$. (c) Each singleton or the empty set lies in one block, where the table is the exact BF, so $d = 0$ there; and $\mathrm{BF}(\{1,2\})/\mathrm{BF}(\{1\}) = R_{12|1}\, w(1)/w(2) = R_{12|1}$, $\mathrm{BF}(\{2\}) = 3R_2$. The limit and the bound follow from Lemma 11. (d) $\varepsilon_1 \leq \pi_0(\{1\})/q_1(\{1\})$ and $\pi_0(\{1\}) \leq \gamma_0(\{1\})/\gamma_0(\{1,2\}) = 1/(3R_2)$; then use (b) and Lemma 11. □

Proposition 13 (Part (iii), second sentence)**.** *In the family, at* $\phi = 1$ *and for all large* n*, the realised order* $\sigma = ($*Gibbs* 2*, Gibbs* 1*, uniform swap, neighbour swap, independence*$)$ *(Gibbs 2 applied first) gives a kernel* K_σ *that is not* q_1*-reversible.*

Proof. By part (iii) it suffices to show $K_\sigma \neq K_{\bar\sigma}$. Lower bound along one path: $K_\sigma(\emptyset, \{1,2\}) \geq K_{G,2}(\emptyset, \{2\})\, K_{G,1}(\{2\}, \{1,2\})\, K_{\mathrm{us}}(\{1,2\}, \{1,2\})\, K_{\mathrm{ns}}(\{1,2\}, \{1,2\})\, K_{\mathrm{ind}}(\{1,2\}, \{1,2\})$. Here $K_{G,2}(\emptyset, \{2\}) = R_2/(1+R_2) \to 1$; $K_{G,1}(\{2\}, \{1,2\}) = R_{12|2}/(1+R_{12|2}) \to 1$; $K_{\mathrm{us}}(\{1,2\}, \{1,2\}) = 1$ ($|\{1,2\}| = p$); $K_{\mathrm{ns}}(\{1,2\}, \{1,2\}) = 1$ (for $j \in \{1,2\}$, $N(j) \setminus \{1,2\} = \emptyset$); $K_{\mathrm{ind}}(\{1,2\}, \{1,2\}) \geq \pi_0(\{1,2\}) \to 1$. So $K_\sigma(\emptyset, \{1,2\}) \to 1$. For $\bar\sigma =$ (independence, neighbour swap, uniform swap, Gibbs 1, Gibbs 2): $K_{\bar\sigma}(\emptyset, \{1\}) \geq K_{\mathrm{ind}}(\emptyset,\emptyset) K_{\mathrm{ns}}(\emptyset,\emptyset) K_{\mathrm{us}}(\emptyset,\emptyset) K_{G,1}(\emptyset, \{1\}) K_{G,2}(\{1\}, \{1\})$. From $\emptyset$ the independence step moves only if it proposes some $S' \neq \emptyset$ and accepts, with acceptance $1 \wedge e^{d(S')}$; so $K_{\mathrm{ind}}(\emptyset,\emptyset) \geq 1 - \pi_0(\{1\}) - \pi_0(\{2\}) - \pi_0(\{1,2\}) e^{d(\{1,2\})} \to 1$ by Lemma 12(a),(c). $K_{\mathrm{ns}}(\emptyset,\emptyset) = K_{\mathrm{us}}(\emptyset,\emptyset) = 1$; $K_{G,1}(\emptyset, \{1\}) = R_1/(1+R_1) \to 1$; $K_{G,2}(\{1\}, \{1\}) = 1/(1+R_{12|1}) \to 1$. Hence $K_{\bar\sigma}(\emptyset, \{1\}) \to 1$ and $K_{\bar\sigma}(\emptyset, \{1,2\}) \leq 1 - K_{\bar\sigma}(\emptyset, \{1\}) \to 0$. For large n, $K_\sigma(\emptyset, \{1,2\}) > \frac{1}{2} > K_{\bar\sigma}(\emptyset, \{1,2\})$. □

Proposition 14 (Part (v), second sentence)**.** *Let* $\kappa = \kappa_2/2$*. There is* n_0 *(depending on* τ, ρ*) such that for every* $n \geq n_0$ *the family's input satisfies, at* $\phi = 1$*:* $\varepsilon_1 \leq e^{-\kappa n}$*;* $\max_\Omega d - \min_\Omega d \geq \kappa n$*; a switch (*$\arg\max q_0 = \{1,2\} \neq \{1\} = \arg\max q_1$*);* $K_1(x, \{1\}) \geq \frac{1}{40}$ *for every* $x \in \Omega$*; hence* $\sup_x \|K_1^h(x,\cdot) - q_1\|_{\mathrm{TV}} \leq (39/40)^h$ *for all* $h \geq 0$*,* $\mathrm{gap}^*(K_1) \geq \frac{1}{40}$*, and* $t_{\mathrm{mix}}(K_1) \leq 55$*, where* $t_{\mathrm{mix}}(K) = \min\{h \geq 0 : \sup_x \|K^h(x,\cdot) - q_1\|_{\mathrm{TV}} \leq \frac{1}{4}\}$*.*

Proof. The first three properties hold for large n by Lemma 12: (d) gives $\frac{1}{n}\log\varepsilon_1 \leq -\kappa_2 + o(1)$; (c) gives $\max d - \min d \geq 0 - d(\{1,2\}) \geq \log(3R_2) - \log(2e^{1.01}n^{-1/2}) = \kappa_2 n + o(n)$; (a),(b) give the switch.

Mixing. Consider the $3! = 6$ orders whose last two steps are Gibbs 1 then Gibbs 2 (a fraction $6/120 = 1/20$ of all orders). For such σ and any x, let x' be the state before Gibbs 1. Gibbs 1 puts 1 in with probability $R_1/(1+R_1)$ if $2 \notin x'$ and $R_{12|2}/(1+R_{12|2})$ if $2 \in x'$ (the conditional depends only on $x' \setminus \{1\}$); then, from a state containing 1, Gibbs 2 removes or keeps 2 out with probability $\gamma_1(\{1\})/(\gamma_1(\{1\}) + \gamma_1(\{1,2\})) = 1/(1+R_{12|1})$, ending at $\{1\}$. So $K_\sigma(x, \{1\}) \geq \theta_n := \min\{\frac{R_1}{1+R_1}, \frac{R_{12|2}}{1+R_{12|2}}\} \cdot \frac{1}{1+R_{12|1}}$, and $K_1(x, \{1\}) \geq \theta_n/20$, all other orders contributing ≥ 0. By Lemma 11, $\theta_n \to 1$, so $\theta_n \geq \frac{1}{2}$ for large n and $K_1(x,\cdot) \geq \frac{1}{40}\delta_{\{1\}}(\cdot)$. K_1 is stochastic, q_1-invariant and self-adjoint (part (iii)); Lemma 4 with $\nu = \delta_{\{1\}}$ and constant $\frac{1}{40}$ gives the TV bound and $\mathrm{gap}^*(K_1) \geq \frac{1}{40}$. Since $(39/40)^{55} = \exp(-55\log(40/39)) < \exp(-1.39) < \frac{1}{4}$ (as $\log(40/39) > 0.0253$ and $55 \cdot 0.0253 > 1.39 > \log 4$), $t_{\mathrm{mix}}(K_1) \leq 55$. Take n_0 as the largest of the finitely many thresholds used. □

Remark 15 (Any prescribed exponential rate). Every argument of this section holds for every fixed $\tau, \rho \in (0,1)$, and $\kappa_2 = \frac{1}{2}\log\frac{1}{1-\tau^2\rho^2} \to \infty$ as $\tau\rho \uparrow 1$. So for any prescribed $K_0 > 0$, choosing τ, ρ with $\kappa_2 \geq 2K_0$ gives a family with $\varepsilon_1 \leq e^{-K_0 n}$ and discrepancy range $\geq K_0 n$ for all $n \geq n_0(\tau, \rho)$, a switch, and still $\mathrm{gap}^*(K_1) \geq \frac{1}{40}$ and $t_{\mathrm{mix}}(K_1) \leq 55$. Hence even a hypothesis that fixes the exponential rate ("$\varepsilon_\phi \leq e^{-K_0 n}$" or "range $\geq K_0 n$" for a given K_0) does not imply slow mixing.

Thus, for the hybrid kernel, an exponentially small minorisation constant ε_ϕ, a switch and a discrepancy range linear in n can all hold, at any prescribed exponential rate (Remark 15), while the absolute gap stays $\geq 1/40$ and the mixing time stays ≤ 55, for every $n \geq n_0$. Hence none of them, alone or together, implies an upper bound on $\mathrm{gap}^*(K_\phi)$ tending to 0 or a lower bound on $t_{\mathrm{mix}}(K_\phi)$ tending to infinity (in particular none exponentially large in n): any such implication would fail on this family. This proves the reading of (v). Together with Sections 2–5 all claims of STATEMENT.md are proved. □

(d) Review record

FINAL_REPORT.md: statement changes

2. Statement changes

No hypothesis was added or changed. Two informal sentences of the theorem file were written as precise mathematics in STATEMENT.md. The original wording is quoted next to each.

1. (iii) "An individual realised order need not be reversible" became: a fixed-order round is reversible exactly when it equals its reversed-order round, and an explicit instance exists where it is not.
2. (v) "does not establish exponentially slow mixing ..." became an existence claim. It is a family (n → ∞, p = 2) with ε₁ ≤ e^{−κn}, discrepancy range ≥ κn and a switch, yet gap ≥ 1/40 and mixing time ≤ 55.

One correction to my own reading was made after Astra #1's round-1 review. My first version said "no lower bound on mixing time that tends to 0", which is wrong; the corrected text says "tends to infinity". The correction is recorded in the STATEMENT.md change log and in CLAIMS_OF_FATAL_FLAWS.md. All three reviewers judged the final readings faithful to the original wording.

Dependencies: none were used. The proof is self-contained: it does not assume the project results P1 or P6 or any sibling v3 conjecture. Neither EXP nor EXP′ is needed.

FINAL_REPORT.md: review history

4. Review history

Stage	Reviewer	Version	Verdict	Findings
A, round 1	Astra #1 (gpt-6-astra high, session 01a16420-941b-7872-810a-426ec03c2ee9)	v1	not verified	1 fatal (my wrong gloss in the reading of (v)); 1 exposition (0/0 notation). Both accepted and fixed in v2.

Stage	Reviewer	Ver-sion	Verdict	Findings
A, round 2	Astra #1 (same session)	v2	**verified**	both findings ruled resolved; no new findings
B, round 1	Astra #2 (gpt-6-astra high, new session 01a16428-59e8-7843-847b-ceb1857aa7cb)	v2	**verified**	none
B, round 1	Opus 5.5 xhigh (native agent, Claude Personal, read-only)	v2	**verified**	7 exposition (F1 graded "borderline gap": prescribed rates). All accepted; unified revision v3.
B, round 2	Astra #2	v3	**verified**	agrees with all answers; 1 new exposition finding (argmax braces)
B, round 2	Opus 5.5 xhigh	v3	**verified**	agrees with all answers; 3 new exposition findings (notation clashes N1–N3)

- CLAIMS_OF_FATAL_FLAWS.md holds the one fatal finding verbatim with my answer, and the borderline F1. Every reviewer who saw the log agreed with the answers and with the fixes.
- **Astra judge: not used.** No dispute remained.
- **Open exposition notes on PROOF_v3:** all notation only; no reviewer found any effect on the mathematics. They are not applied, so that the final text is exactly the one all reviewers checked.
  - The rate symbol K_0 clashes with the cap K and the kernels K_φ.
  - The vectors h_1, h_2, h_3 clash with the step count h.
  - Subscript 1 means both the state {1} and φ = 1.
  - R means both the correlation matrix and the ratios R_1, R_2.
  - "w(1) = w(1)" should say "both singletons have the same prior weight".
  - "argmax = {1,2}" should read "the unique maximiser is {1,2}".

  Fixing them is a five-minute copy-edit for the campaign document.

CLAIMS_OF_FATAL_FLAWS.md (every entry, verbatim)

CLAIMS_OF_FATAL_FLAWS — T09 (T7)

Every fatal or gap finding, verbatim, with the lead's answer.

Stage A, Astra #1 (session 01a16420-941b-7872-810a-426ec03c2ee9), review of PROOF_v1, finding 1 (fatal)

Location: STATEMENT.md, claim (v), final sentence; PROOF_v1.tex, Section 6, closing paragraph (line 199).

Severity: fatal (the literal concluding claim is false).

Claim: The statement asserts that no upper bound on the absolute gap "(equivalently, no lower bound on its mixing time) that tends to 0" can follow from the listed conditions. The parenthetical

does not give an equivalent assertion: a mixing-time lower bound tending to zero is compatible with uniformly fast mixing and is available here.

Evidence: Define the usual worst-case mixing time $t_{mix}(1/4) = \min\{h \geq 0 : \sup_x \|K_\varphi^h(x,\cdot) - q_\varphi\|_{TV} \leq 1/4\}$. Every permitted state space contains at least $\emptyset$ and $\{1\}$. Therefore $\sup_x \|\delta_x - q_\varphi\|_{TV} = 1 - \min_x q_\varphi(x) \geq \frac{1}{2}$, so $t_{mix}(1/4) \geq 1$. In particular, $t_{mix}(1/4) \geq 1/n$, a valid lower bound tending to zero, including on the constructed family satisfying all three conditions.

The quoted original wording concerns an **exponentially large** mixing-time lower bound. The explicit family faithfully demonstrates that such a conclusion need not follow: its geometric TV bound gives a mixing time bounded independently of n. The final parenthetical, read literally, does not faithfully express that conclusion. This finding concerns the concluding statement, not the validity of the constructed family.

Lead's answer: accepted and fixed. The parenthetical was a mis-stated gloss in my reading of the informal clause (a gap upper bound tending to 0 corresponds to a mixing-time lower bound tending to infinity, not to 0). Changes:

- STATEMENT.md, reading of (v): the consequence now reads "the family has $t_{mix}(K_1) \leq 55$ for every $n \geq n_0$; so no upper bound on $\text{gap}^*(K_\varphi)$ that tends to 0, and no lower bound on $t_{mix}(K_\varphi)$ that tends to infinity (in particular none exponentially large in n), can be derived from ...", with t_{mix} defined. Recorded in the STATEMENT.md change log. The original wording, the hypotheses and the existence claim are unchanged.
- PROOF_v2, Section 6: the proposition now also states $t_{mix}(K_1) \leq 55$, proved from $(39/40)^{55} < 1/4$; the closing paragraph is rewritten to match.

(Finding 2 of the same review is exposition: the 0/0 form of the reverse proposal probability for a non-symmetric neighbour family. Fixed in PROOF_v2 by writing both proposal probabilities by cases.)

Stage B (independent reviews of PROOF_v2)

- Astra #2 (session 01a16428-59e8-7843-847b-ceb1857aa7cb): no findings; verdict verified.
- Opus 5.5 xhigh reviewer: verdict verified; seven findings, all graded exposition (F1 "exposition (borderline gap)"). No fatal or gap finding, so no entry is required here. For the record, F1 is quoted verbatim with its answer, because it was graded borderline:

F1. Location: Section 6, closing paragraph ("any such implication would fail on this family"); STATEMENT (v), the "Consequently..." sentence. Severity: exposition (borderline gap). Claim: The proof shows the family has $\varepsilon_1 \leq e^{-\kappa n}$ and discrepancy range $\geq \kappa n$ with $\kappa = \kappa_b/2$, where $\kappa_b = \frac{1}{2}\log(1/(1-c^2\rho^2))$ for fixed correlation c and ρ. In fact $\varepsilon_1 \asymp 1/\mathrm{BF}(b) \asymp e^{-\kappa_b n}$. Take a would-be implication whose hypothesis fixes a rate, such as "$\varepsilon_\varphi \leq e^{-5n} \Rightarrow \text{gap} \to 0$". At the example values $c = \frac{1}{2}$, $\rho = 3/5$ we get $\kappa_b \approx 0.047$, so the family does not satisfy that hypothesis and does not refute the implication. Evidence: $\kappa_b = \frac{1}{2}\log(1/0.91) \approx 0.047$. The STATEMENT's own wording ("there exist constants $c > 0$") reads "exponentially small" and "large range" existentially, and under that reading the sentence is proved. The stronger fixed-rate version is easy to repair, but the proof never states the repair: $\kappa_b \to \infty$ as $c\rho \uparrow 1$, and every argument in Lemmas 10–12 and Propositions 13–14 holds for any fixed $c, \rho \in (0,1)$.

Lead's answer: accepted; added in PROOF_v3 as Remark "Any prescribed exponential rate"

(for any $K_0 > 0$ choose τ, ρ with $\kappa_2 \geq 2K_0$; all arguments hold for fixed τ, ρ; gap $\geq 1/40$ and $t_{mix} \leq 55$ unchanged). The statement is unchanged (its existential reading was already proved). The other exposition findings F2–F7 are all accepted and fixed in PROOF_v3 (see review/STAGE_B_CHANGES_v2_to_v3.md).

Chapter 2

T11(iii): the final run's iid π_0 draws

Final proof: `proofs/T11_LLED_scoring_ledger/proof/PROOF_v5.tex`. *Sources of parts (a), (b), (d):* `theorems/T11_LLED_scoring_ledger.md`, `STATEMENT.md`, `FINAL_REPORT.md`, `CLAIMS_OF_FATAL_FLAWS.md`.

(a) Original conjecture text (theorem file, verbatim)

T11 (L-LED/T14-D): Unified scoring and discovery ledger

Status: conjectured, not proven. Methodology SMC-BVS v3.0 (2026-10-19), SHA-256 93043758058a6d422cca6ca4b6be8f3a9e8149b6f6de2e1bab1ce31c44b8019c. Read with `00_COMMON.md`.

Category / rank: algorithmic validity and discovery; **core** (lemma). Merges v2's L-LED and T14-D.

Definitions used

Scored set $\mathcal{S}$, $\mathcal{S}$_cal, reported HPM $\hat{S}$, global S†; the mechanisms (Gibbs flips, uniform swap, neighbour swaps on N(·), independence proposals and initial draws from π_0, block tables, local scans); execution-time state X_{pre-j} before coordinate j's step; Q_f = 4N_e(T+L+5); "opportunity" = a (particle, round, step) at which a mechanism could score the model; first-hit time τ; conditional hazard p_e (00_COMMON §2).

Assumptions

EX-A, EX-S, CAL; SR for Gibbs scoring. Component execution or launch conditions are explicit.

Precise statement

Fix S* ∈ Ω measurable w.r.t. data and $\mathcal{C}$ (before final randomness). (i) [per-mechanism rules] Gibbs coordinate j: under SR, S* is evaluated iff X_{pre-j} \ {j} = S* \ {j} (this includes the current-state case; before first discovery only the flip-neighbour case gives a new score). Uniform swap from a size-k exchange neighbour: the unique exchange producing S* is proposed with probability 1/[k(p−k)]. Neighbour swap: probability 1/[k|N(j)\S|] if the insertion is in N(j)\S, else 0 (degrees may exceed B). Independence proposal / initial draw: proposal probability π_0(S*), regardless of acceptance. Tables: every S* ⊆ C_b is scored when the tables execute. Local scan: given its centre, every feasible flip and

swap is scored; a random centre contributes a probability after averaging, not a deterministic unconditional indicator. (ii) [expected Gibbs evaluations] Σ_{r,i,ρ,j} P(X^{r,i,ρ}_{pre-j} \ {j} = S* \ {j}); if a round starts in stationarity every state-independent ordering prefix preserves q_φ. Actual particles are not assumed stationary. (iii) [iid draws] P(S* absent from the final iid draws | $\mathcal{C}$) = (1 − π_0(S*))^{Q_f} ≤ e^{−Q_fπ_0(S*)}. The adaptive calibration draw count is never a fixed exponent. (iv) [stopped opportunities] For chronological predictable opportunities with success probability ≥ b, with O the count up to first success or the horizon (including the successful one): P(no hit, O ≥ D) ≤ (1−b)^D; if P(no hit, O < D) ≤ δ then P(no hit) ≤ δ + (1−b)^D. An expected count alone does not supply this premise. Generally P(τ ≤ H) = Σ_e E[1{τ > e⁻}p_e], and P(hit) ≥ bEO ≥ 1 − E(1−b)^O (a weaker bound, not an independent-trials identity). (v) [set exposure] For a fixed set $\mathcal{M}$: P($\mathcal{M}$ scored) ≤ 1{$\mathcal{S}$_cal ∩ $\mathcal{M}$ ≠ ∅} + Q_fπ_0($\mathcal{M}$) + EΣ_e p_e^{noniid}, with future Gibbs, swap and local-scan opportunities in the last sum; clip at one. (vi) [launch] On the SMC branch the unconditional final-only miss bound is P(not launched, S* ∉ $\mathcal{S}$_cal) + E[1{launched, S* ∉ $\mathcal{S}$_cal}(1 − π_0(S*))^{Q_f}]. Enumeration scores every feasible model. (vii) If S† is scored, then $\hat{S}$ = S† (tie rule; NIS not needed).

Dependencies

Builds on: T09, T10, decision rule. Used by: T20, T21, T24, T27, T28, T29, T30, T31.

Numerical evidence (provisional)

provisional evidence (Sol check J8): exact stopped Bernoulli tree (b = 0.3): exact hit 0.51 = first-failure sum; the incorrect product expression gives 0.447; P(no hit, O ≥ 2) = 0.49 = bound. Literal Gibbs "iff" without the current-state case: FAIL. provisional evidence (Sol check J7): Q_f bounds at n = 200, p = 500 tier 0: Q_min = 140,000, Q_max = 10,130,000.

Proof-route hint (optional)

Condition on each execution prefix; fresh proposal randomness; first-hit decomposition and sequential survival bounds; the adaptive union bound with deterministic ceilings for (v). Hardest step: ensuring a claimed opportunity lower bound holds on the no-hit event. *When proof.md is run on this file, its lead considers the statement both with and without these hints; the hints never reach the reviewers.*

(b) STATEMENT.md (verbatim)

STATEMENT — T11(iii): iid π_0 draws of the final run

Source: `theorems/T11_LLED_scoring_ledger.md`, part (iii) only. Method: SMC-BVS v3.0, METHODOLOGY_v3.md (SHA-256 93043758058a6d422cca6ca4b6be8f3a9e8149b6f6de2e1bab1ce31c44b8019c). Shared definitions: `theorems/00_COMMON.md`.

Original wording (the file's preamble to all parts): "Fix S* ∈ Ω measurable w.r.t. data and $\mathcal{C}$ (before final randomness)." Original wording of (iii): "P(S* absent from the final iid draws | $\mathcal{C}$) = (1 − π_0(S*))^{Q_f} ≤ e^{−Q_fπ_0(S*)}. The adaptive calibration draw count is never a fixed exponent."

Statement changes: **none.** The items under "Setting" make explicit what the original wording leaves implicit; none adds a hypothesis beyond the file's assumptions (EX-A, EX-S, CAL) and the method's

own specification (item 5 transcribes the method's description of the final run; aborts imposed by the computing environment are outside the idealised model, as everywhere in this theory).

Setting

1. **Data.** The data (X, y) are fixed (all probabilities are over the algorithm's randomness given the data). p, n, K, the blocks C_1, …, C_J, the tables r_b, γ_0, $\bar{Z}_0$ and $\pi_0 = \gamma_0/\bar{Z}_0$ on $\Omega = \{S \subseteq [p] : |S| \le K\}$ are then fixed (they are functions of the data only; METHODOLOGY §6.3). The definitions (00_COMMON §§1–2, METHODOLOGY §5, §6.3):
   - K = min(p, ⌊n/2⌋); the blocks C_1, …, C_J partition [p];
   - size prior $\varpi(S) = w(|S|)/W_K$ for $|S| \le K$, with $w(k) = \tfrac{1}{2}/C(2p-1, k) > 0$ and $W_K = \Sigma_{k \le K}\, C(p,k)\, w(k)$;
   - tables $r_b(U) = m(U)/m(\emptyset)$ for $U \subseteq C_b$, where $m(\cdot) > 0$ is the marginal likelihood, so every $r_b(U) > 0$;
   - $\gamma_0(S) = \varpi(S) \prod_b r_b(S \cap C_b)$ for $S \in \Omega$, $\bar{Z}_0 = \Sigma_{S \in \Omega}\, \gamma_0(S)$, $\pi_0 = \gamma_0/\bar{Z}_0$. Hence $\pi_0(S) > 0$ for every $S \in \Omega$.
2. **Randomness (EX-S).** The algorithm's randomness is a countable family of independent streams, each an iid sequence of Uniform(0,1) variables, with the method's stream separation: the calibration uses streams 0–9 only and the final run uses streams 10–13 only. Every random choice of the algorithm is made by reading not-yet-read values from the streams; which value is read next is determined by the data and the values read so far. Random operations are executed one at a time (each, e.g. one call of the π_0 sampler, completes before the next begins). All maps are Borel measurable.
3. **Exactness (EX-A).** Scores, convolutions, probabilities and comparisons are exact. In particular, each call of the π_0 sampler (METHODOLOGY §6.3: total size, then block sizes backwards, then each block's sub-model within its size class) maps the values it reads to its output by a measurable map (which may depend on the run's state when the call starts), and, when the values it reads are iid Uniform(0,1), its output has law exactly π_0.
4. **Calibration output (CAL).** $\boldsymbol{\mathcal{C}}$:= (grid $0 = \varphi_0 < \ldots < \varphi_T = 1$, T, L, N_e, tier e, candidate set V, every V-independent setting) is a measurable function of the data and streams 0–9; it includes the indicator of the event Λ = {the final run is launched} (SMC branch, preflight passed, a grid was produced; all decided before any final randomness). On Λ, $T \ge 1$ is finite.
5. **Final run** (on Λ; METHODOLOGY §6.8, §6.5). 4 islands, each with N_e particles started from π_0 draws; the frozen grid; one move round per stage t = 1, …, T, then L burn-in rounds and 4 measured rounds at $\varphi = 1$; so each particle of each island goes through T + L + 4 rounds, and each round contains exactly one independence step, whose proposal is a π_0 draw (accepted or not). The **final iid draws** are these 4N_e initial draws and 4N_e(T + L + 4) independence proposals, in total Q_f := 4N_e(T + L + 5), a function of $\boldsymbol{\mathcal{C}}$. A launched final run executes all these rounds (no early stop; aborts imposed by the computing environment, such as wall time, are outside the idealised model).
6. **Target model.** S* is an Ω-valued random variable that is a measurable function of the data and $\boldsymbol{\mathcal{C}}$ (the file's "measurable w.r.t. data and $\boldsymbol{\mathcal{C}}$ (before final randomness)").

Claim (iii)

On Λ, almost surely,

P(S* is not among the Q_f final iid draws | $\mathcal{C}$) = (1 − π_0(S*))^{Q_f} ≤ exp(−Q_f π_0(S*)).

Scope note (the clause "the adaptive calibration draw count is never a fixed exponent")

The identity is asserted only for the final draws, with the $\mathcal{C}$-measurable exponent Q_f. Nothing is asserted about the calibration's π_0 draws: their number is adaptive (it depends on the calibration's own randomness, e.g. through the rebuild loop and the escalation decision), so the identity is **not** claimed with the calibration draw count in place of Q_f.

Method facts quoted verbatim (METHODOLOGY_v3.md), used only to justify Setting item 5 and the scoring corollary

- §2, statuses: "`numerical_failure` (scoring failed after the fallback; estimates unavailable; the best scored model returned, labelled incomplete) · `resource_limit` (preflight predicts the memory or time allocation will be exceeded; estimates of the inputs returned)"; "`schedule_limit` (no admissible schedule within the stage ceiling at the highest tier; the best scored model is returned, evidence and PIPs unavailable)".
- §7 pseudocode: "preflight (§8.6) → maybe return"; "if grid construction failed at e = 1: return schedule_limit (best scored model)"; then "islands r = 1..4 (streams 10..13): FINAL_ISLAND(grid, N_e) # bridge + L burn-in + 4 measured rounds".
- §6.13: "On a Cholesky failure, refactor from scratch, then use an augmented QR on X_S; if that fails, stop with `numerical_failure`".
- §6.5: "Independence proposal: S′ ∼ π_0 (exact draw), accepted with 1 ∧ e^{φ[d(S′) − d(S)]}. Every proposal is scored, accepted or not."

(c) Final proof, verbatim: body of PROOF_v5.tex

The proof below is the body of `PROOF_v5.tex` from `\begin{document}` to `\end{document}` without the title block. Its preamble definitions are local to this chapter; labels carry the prefix `T11:`.

Notation and the setting are those of `STATEMENT.md` (items 1–6). Throughout, the data (X, y) are fixed; $\Omega = \{S \subseteq [p] : |S| \le K\}$; $\pi_0 = \gamma_0/\bar{Z}_0$ is a fixed probability on Ω with $\pi_0(S) > 0$ for all $S \in \Omega$; Λ is the event that the final run is launched; and $Q_f = 4N_e(T + L + 5)$.

1 Two facts about reading fresh uniforms

Let $(U_\iota)_{\iota \in I}$ be a countable family of independent Unif$(0,1)$ variables. A *reading scheme* is a sequence of measurable maps $f_1, f_2, \ldots$ with $f_1 \in I$ a constant and $f_k : [0,1]^{k-1} \to I$, such that the indices

$$\iota_1 = f_1, \qquad \iota_k = f_k(V_1, \ldots, V_{k-1}), \qquad V_k := U_{\iota_k} \quad (k \ge 1)$$

are pairwise distinct for every realisation (each value is read at most once).

Lemma 1 (reading lemma)**.** *For every reading scheme,* $(V_k)_{k \ge 1}$ *is an iid* Unif$(0,1)$ *sequence.*

Proof. It suffices to show, for every $k \geq 1$ and Borel sets $A_1, \dots, A_k \subseteq [0,1]$, that $P(V_1 \in A_1, \dots, V_k \in A_k) = \prod_{i \leq k} \lambda(A_i)$, with λ Lebesgue measure. We induct on k; the case $k = 1$ holds because $V_1 = U_{f_1}$. Let $k \geq 2$ and $E = \{V_1 \in A_1, \dots, V_{k-1} \in A_{k-1}\}$. Then

$$P(E \cap \{V_k \in A_k\}) = \sum_{\iota \in I} P\big(E \cap \{\iota_k = \iota\} \cap \{U_\iota \in A_k\}\big).$$

Fix ι. Let $U^{(\iota)}$ be the family U with U_ι replaced by the constant $\frac{1}{2}$, and let (ι'_j, V'_j) be the indices and values the same scheme produces from $U^{(\iota)}$; these are functions of $(U_\kappa)_{\kappa \neq \iota}$. On the event that $\iota \notin \{\iota_1, \dots, \iota_{k-1}\}$, an induction on $j \leq k-1$ gives $\iota'_j = \iota_j$ and $V'_j = V_j$ (each index is computed from equal earlier values, and the value read there is not U_ι), and hence $\iota'_k = \iota_k$. Symmetrically, on the event $\iota \notin \{\iota'_1, \dots, \iota'_{k-1}\}$ the same equalities hold. Because indices are never repeated, $\{\iota_k = \iota\} \subseteq \{\iota \notin \{\iota_1, \dots, \iota_{k-1}\}\}$ and likewise for the primed scheme, so

$$E \cap \{\iota_k = \iota\} = E' \cap \{\iota'_k = \iota\}, \qquad E' := \{V'_1 \in A_1, \dots, V'_{k-1} \in A_{k-1}\}.$$

The right side is $\sigma(U_\kappa : \kappa \neq \iota)$-measurable, hence independent of U_ι. Therefore $P(E \cap \{\iota_k = \iota\} \cap \{U_\iota \in A_k\}) = P(E \cap \{\iota_k = \iota\})\, \lambda(A_k)$. Summing over ι gives $P(E \cap \{V_k \in A_k\}) = P(E)\lambda(A_k)$, and the induction hypothesis completes the step. □

Let $\mathcal{F}_t := \sigma(V_1, \dots, V_t)$ ($\mathcal{F}_0$ trivial). A random variable τ with values in $\{0, 1, 2, \dots\}$ is a *stopping time* if $\{\tau = t\} \in \mathcal{F}_t$ for every t.

Lemma 2 (fresh block)**.** *Let (V_k) be iid* Unif$(0,1)$ *and τ an a.s. finite stopping time (on the null set $\{\tau = \infty\}$ put $V_{\tau+j} := \frac{1}{2}$). Then $(V_{\tau+j})_{j \geq 1}$ is an iid* Unif$(0,1)$ *sequence independent of $\mathcal{F}_\tau := \{A \in \sigma(V_1, V_2, \dots) : A \cap \{\tau = t\} \in \mathcal{F}_t \ \forall t\}$.*

Proof. Let $A \in \mathcal{F}_\tau$ and let C be a measurable subset of $[0,1]^{\mathbb{N}}$. For each t, $A \cap \{\tau = t\} \in \mathcal{F}_t$ is independent of $(V_{t+j})_{j \geq 1}$, which is iid Unif$(0,1)$; hence $P(A \cap \{\tau = t\} \cap \{(V_{t+j})_j \in C\}) = P(A \cap \{\tau = t\})\, \mu(C)$, where μ is the law of an iid Unif$(0,1)$ sequence. Summing over t (using $\tau < \infty$ a.s.) gives $P(A \cap \{(V_{\tau+j})_j \in C\}) = P(A)\mu(C)$. Taking A to be the whole sample space identifies the law as μ, and the product form gives independence. □

2 The π_0 sampler

Setting item 3 (EX-A) states that one call of the π_0 sampler maps the values it reads to its output by a measurable map (possibly depending on the run's state when the call starts) and that, when these values are iid Unif$(0,1)$, its output has law exactly π_0. For completeness we verify this for the backward sampler of METHODOLOGY §6.3, written out as follows; only the conclusion of Lemma 3 is used later. The verification uses the definitions in Setting item 1: the blocks partition $[p]$, every $r_b(U) > 0$, $w(k) > 0$, and $\gamma_0(S) = \frac{w(|S|)}{W_K} \prod_b r_b(S \cap C_b)$ on Ω.

Let $e_b(j) = \sum_{U \subseteq C_b, |U| = j} r_b(U)$ for $0 \leq j \leq |C_b|$ (and $e_b(j) = 0$ otherwise), $c_k^{(0)} = \mathbf{1}\{k = 0\}$, and $c_k^{(b)} = \sum_j e_b(j) c_{k-j}^{(b-1)}$ for $b = 1, \dots, J$ and integers k (with $c_m^{(b-1)} = 0$ for $m < 0$). Since the blocks partition $[p]$, $c_k^{(J)} = \sum_{|S| = k} \prod_b r_b(S \cap C_b)$, so $W_K \bar{Z}_0 = \sum_{k \leq K} w(k) c_k^{(J)} =: M$. One call reads values $u_0, u_J, \dots, u_1, u'_J, \dots, u'_1$ (fresh) and:

(a) draws $k \in \{0, \dots, K\}$ with probability $w(k) c_k^{(J)} / M$ by the inverse distribution function at u_0;

(b) sets $k_J = k$ and, for $b = J, J-1, \dots, 1$, draws j_b with probability $e_b(j)c^{(b-1)}_{k_b-j}/c^{(b)}_{k_b}$ by the inverse distribution function at u_b, and sets $k_{b-1} = k_b - j_b$;

(c) for each b, draws $U_b \subseteq C_b$ with $|U_b| = j_b$ with probability $r_b(U_b)/e_b(j_b)$ by the inverse distribution function (in a fixed enumeration of the subsets) at u'_b; and returns $S = \bigcup_b U_b$.

Lemma 3. *Under exact arithmetic, if the values read are iid* $\mathrm{Unif}(0,1)$*, the returned S has law π_0.*

Proof. All tables are positive, so $e_b(j) > 0$ for $0 \le j \le |C_b|$ and $c^{(b)}_k > 0$ exactly for $0 \le k \le \sum_{b' \le b} |C_{b'}|$. The probabilities in (a)–(c) are nonnegative and sum to one (for (b) by the recursion defining $c^{(b)}$). A value of k_b with $c^{(b)}_{k_b} = 0$ is reached with probability 0: $k_J = k$ has $w(k)c^{(J)}_k > 0$, and j_b is drawn with positive probability only if $c^{(b-1)}_{k_b - j_b} > 0$. At $b = 1$ this forces $k_0 = 0$, so $\sum_b j_b = k$. An inverse distribution function evaluated at a $\mathrm{Unif}(0,1)$ value is an exact draw from the corresponding finite distribution, and the values are independent, so the probability of a path is the product of its factors. A set $S \in \Omega$ is produced by exactly one path of positive probability: $k = |S|$, $j_b = |S \cap C_b|$, $U_b = S \cap C_b$, with $k_b = \sum_{b' \le b} j_{b'}$. Its probability is

$$\frac{w(k)c^{(J)}_k}{M} \prod_{b=1}^{J} \frac{e_b(j_b)\, c^{(b-1)}_{k_{b-1}}}{c^{(b)}_{k_b}} \prod_{b=1}^{J} \frac{r_b(S \cap C_b)}{e_b(j_b)} = \frac{w(k)}{M}\, c^{(0)}_0 \prod_b r_b(S \cap C_b) = \frac{w(|S|)}{W_K} \frac{\prod_b r_b(S \cap C_b)}{\bar{Z}_0} = \pi_0(S),$$

the product of $c^{(b-1)}_{k_{b-1}}/c^{(b)}_{k_b}$ telescoping to $c^{(0)}_0/c^{(J)}_k$. Sets outside Ω are never returned since $k \le K$. □

3 The final run is a reading scheme

Remark 4 (the final run executes in full on Λ). The main proof uses Setting item 5 as stated. This remark only explains why item 5 is the method's specification, using the method facts quoted in `STATEMENT.md`. By the §7 pseudocode, once launched, the final run consists of `FINAL_ISLAND(grid,` N_e`)` for $r = 1, \dots, 4$ ("bridge + L burn-in + 4 measured rounds"). Of the statuses that end a run early, `resource_limit` is returned at the preflight and `schedule_limit` when grid construction fails at tier 1, both before the final run; `numerical_failure` means "scoring failed after the fallback" (§2, §6.13). We read EX-A ("scores are exact") as: every score computation returns $\log m(S)$ exactly, and $m(S)$ is a finite positive number for every S (Setting item 1); so scoring does not fail. Hence under EX-A no status interrupts a launched final run. Aborts imposed by the computing environment (wall time, memory) during the final run are outside the idealised model; Setting item 5 excludes them.

Write Υ for streams 0–9 and Ξ for streams 10–13. By EX-S, Υ and Ξ are independent. By CAL, $\mathcal{C}$ is a measurable function of Υ (the data being fixed), and so are Λ, T, N_e, Q_f and S^* (Setting items 4–6). The final run, including the order in which it performs its random operations, is a measurable function of (Υ, Ξ); it reads values only from Ξ.

Proposition 5. *Fix a value υ of Υ with $\upsilon \in \Lambda$, and run the final run with $\Upsilon = \upsilon$ and Ξ iid as in EX-S. Call a class of the run's π_0-sampler calls* designated *if whether a call belongs to it is determined by the run's state when the call starts, and suppose the run makes exactly Q designated calls, Q a constant (given υ). Number their outputs $D_1, \dots, D_Q$ in chronological order. Then $D_1, \dots, D_Q$ are iid with law π_0. In particular this holds for*

(a) the final iid draws of Setting item 5 (the $4N_e$ initial draws and the $4N_e(T+L+4)$ independence proposals), with $Q = Q_f$;

(b) the $4N_e(T+L+4)$ independence proposals alone, with $Q = Q_p := 4N_e(T+L+4)$.

Proof. With υ fixed, the final run reads not-yet-read values of Ξ, each choice of the next index being a measurable function of the values read so far (Setting item 2). After the run ends, continue reading the unread values of stream 10 in increasing position; this does not affect the run. This is a reading scheme, so by Lemma 1 the read sequence (V_k) is iid Unif$(0,1)$, and the whole run is a measurable function of (V_k).

The m-th designated call ($m \le Q$) starts after τ_m values have been read. The run's state after t reads is a function of $V_1, \dots, V_t$; whether the run starts a sampler call right after the t-th read, whether that call is designated, and how many designated calls preceded it are all determined by that state, so $\{\tau_m = t\} \in \mathcal{F}_t$ and τ_m is a stopping time. It is finite: the m-th designated call does occur (by hypothesis), and since operations are executed one at a time, each completing before the next begins (Setting item 2), only finitely many values are read before it starts. The call reads $V_{\tau_m+1}, V_{\tau_m+2}, \dots$ (the next values read by the run, in order, until the call returns), so $D_m = G_{Z_m}\big((V_{\tau_m+j})_{j\ge1}\big)$, where Z_m is the run's state when the call starts and, for each state z, G_z maps an input sequence to the sampler's output. Measurability: take the state to be the history $z = (t, v_1, \dots, v_t) \in \bigsqcup_{t\ge0}[0,1]^t$ (any coarser state is a measurable function of it). Then $Z_m = (\tau_m, V_1, \dots, V_{\tau_m})$ is $\mathcal{F}_{\tau_m}$-measurable, and, since the whole run is a Borel function of the values it reads (Setting item 2), $(z, w) \mapsto G_z(w)$ is jointly measurable on each piece $[0,1]^t \times [0,1]^{\mathbb{N}}$, hence on the disjoint union. By Setting item 3 (for the sampler of METHODOLOGY §6.3, Lemma 3, where G_z does not depend on z), G_z applied to an iid Unif$(0,1)$ sequence has law π_0 for every z. The calls do not overlap, so $D_1, \dots, D_{m-1}$ are functions of the values read before the m-th designated call starts, that is, they are $\mathcal{F}_{\tau_m}$-measurable. By Lemma 2, $W := (V_{\tau_m+j})_{j\ge1}$ is iid Unif$(0,1)$ and independent of $\mathcal{F}_{\tau_m}$; hence, for $A \in \mathcal{F}_{\tau_m}$ and $s \in \Omega$, Fubini's theorem gives $P(A \cap \{G_{Z_m}(W) = s\}) = E\big[\mathbf{1}_A\, P(G_z(W) = s)|_{z=Z_m}\big] = P(A)\,\pi_0(s)$. Taking $A = \{D_1 = s_1, \dots, D_{m-1} = s_{m-1}\}$, for all $s_1, \dots, s_Q \in \Omega$,

$$P(D_1 = s_1, \dots, D_m = s_m) = P(D_1 = s_1, \dots, D_{m-1} = s_{m-1})\,\pi_0(s_m), \qquad m = 1, \dots, Q,$$

and induction gives $P(D_1 = s_1, \dots, D_Q = s_Q) = \prod_m \pi_0(s_m)$.

For (a) and (b): whether a sampler call is an initial draw or the independence step of some particle's round is part of the run's state when the call starts, so both classes are designated; by Setting item 5 they contain exactly Q_f and Q_p calls respectively, constants given υ. □

4 The theorem

Theorem 6 (T11(iii))**.** *Under Setting items 1–6, on Λ, almost surely,*

$$P\big(S^* \notin \{D_1, \dots, D_{Q_f}\} \,\big|\, \mathcal{C}\big) = (1 - \pi_0(S^*))^{Q_f} \le \exp\{-Q_f \pi_0(S^*)\}.$$

Proof. Let $A = \Lambda \cap \{S^* \notin \{D_1, \dots, D_{Q_f}\}\}$. Its indicator is $\mathbf{1}_A = h(\Upsilon, \Xi)$ for a measurable h, and for fixed υ the map $\xi \mapsto h(\upsilon, \xi)$ is the indicator that the final run started from υ avoids $s^*(\upsilon)$ (the value of S^* at υ), or 0 if $\upsilon \notin \Lambda$. Because Υ and Ξ are independent, Fubini's theorem gives $E[h(\Upsilon, \Xi) \mid \Upsilon] = g(\Upsilon)$ a.s., with $g(\upsilon) := E\,h(\upsilon, \Xi)$. For $\upsilon \in \Lambda$, Proposition 5(a) gives

$$g(\upsilon) = P(D_m \ne s^*(\upsilon) \text{ for } m = 1, \dots, Q_f(\upsilon)) = (1 - \pi_0(s^*(\upsilon)))^{Q_f(\upsilon)},$$

and $g(\upsilon) = 0$ otherwise. So $g(\Upsilon) = \mathbf{1}_\Lambda (1 - \pi_0(S^*))^{Q_f}$, which is a function of $\mathcal{C}$ (Setting items 4–6). Since $\sigma(\mathcal{C}) \subseteq \sigma(\Upsilon)$, the tower property gives $P(A \mid \mathcal{C}) = E[g(\Upsilon) \mid \mathcal{C}] = \mathbf{1}_\Lambda (1 - \pi_0(S^*))^{Q_f}$ a.s. On $\Lambda \in \sigma(\mathcal{C})$, $P(S^* \notin \{D_m\} \mid \mathcal{C}) = P(A \mid \mathcal{C})$ a.s., which is the identity. The inequality follows from $0 \le 1 - x \le e^{-x}$ for $x = \pi_0(S^*) \in (0, 1]$, raised to the power $Q_f \ge 0$. □

5 Remarks

Remark 7 (scope: calibration draws). The argument uses two facts about the final draws that fail for the calibration's π_0 draws. (1) The exponent Q_f is fixed by $\mathcal{C}$ before any final value is read; the calibration's number of draws depends on its own randomness (rebuild loop, escalation). (2) The conditioning σ-field $\sigma(\mathcal{C})$ is independent of the final streams, but in general not of the calibration streams ($\mathcal{C}$ is computed from them), so given $\mathcal{C}$ the calibration draws need not be iid π_0. With a random count, $(1-\pi_0(S^*))^N$ is in general not the absence probability, even in expectation: with D_1, D_2 iid π_0, $x = \pi_0(s) \in (0,1)$, and $N = 1$ if $D_1 \neq s$, $N = 2$ otherwise, $P(s \notin \{D_1, \ldots, D_N\}) = 1 - x$ while $E(1-x)^N = (1-x)^2(1+x) < 1-x$. Accordingly, Theorem 6 is asserted only with the exponent Q_f.

Remark 8 (unconditional form). Taking expectations, $P(\Lambda,\ S^* \notin \{D_1, \ldots, D_{Q_f}\}) = E[\mathbf{1}_\Lambda(1 - \pi_0(S^*))^{Q_f}]$.

Corollary 9 (scoring; beyond the stated part)**.** *Read "scored" as "its score* $\log m(S)$ *is computed". By the method fact quoted in* `STATEMENT.md` *(§6.5: "Every proposal is scored, accepted or not"), each of the* $Q_p = 4N_e(T+L+4)$ *final independence proposals is scored. Applying the argument of Theorem 6 to the proposals alone (by Proposition 5(b) they are iid* π_0 *given* Υ*, and* Q_p *is a function of* $\mathcal{C}$*), on* Λ*,* $P(S^*$ *is not scored in the final run* $\mid \mathcal{C}) \leq (1-\pi_0(S^*))^{Q_p}$ *a.s. If the initial draws are also scored (the quoted text does not say so explicitly), the exponent* Q_p *may be replaced by* Q_f*.*

(d) Review record

FINAL_REPORT.md: statement changes

Statement changes

None to the claim. STATEMENT.md writes out what the original one-line wording left implicit:

- The data are fixed. The probability is over the algorithm's randomness.
- The formula holds on the event that the final run is launched.
- The stream model (EX-S), exact sampling (EX-A) and calibration-measurability (CAL) are spelled out.
- The final-run description of METHODOLOGY §6.5, §6.8 and §7 is written out. A launched run executes all its rounds; aborts by the computing environment are outside the idealised model.
- S* is "measurable w.r.t. data and $\mathcal{C}$ (before final randomness)", quoting the theorem file's own preamble.

During review, definitions (w, W_K, ϖ, the tables, γ_0) and short verbatim method quotes were added so that reviewers could check them. No new hypothesis was added. EXP / EXP′ is not needed.

FINAL_REPORT.md: review history

Review history

Stage A: lead with Astra #1 (GPT-6 Astra, high; session `01a1641a-4997-7370-a99a-43d4f2a4f245`).

Round	Proof	Verdict	Fatal	Gap	Outcome
1	v1	verified	0	3	All gaps were in supplementary parts, because the packet lacked definitions and method text. All three were accepted and fixed.
2	v2	verified	0	1	The gap concerned the scoring corollary, which is outside (iii): the quoted text does not show that initial draws are scored. Accepted; the corollary was restricted to proposals. Two exposition points were also fixed.
3	v3	verified	0	0	No findings. Astra #1 agreed with every answer. Stage A closed.

Stage B: two fresh independent reviews in parallel (amendment 1).

Round	Proof	Astra #2 (session `01a16421-b0ef-7281-984e-2af293f78cf1`)	Opus 5.5 xhigh (native agent, Claude Personal)
1	v3	verified, no findings	verified; 0 fatal; 1 gap (F8); 7 exposition
2	v4	verified; agreed with all answers	verified; agreed with all answers; 4 exposition notes
3	v5	verified, no findings	verified; 1 exposition note, no change required

- **Opus F8 (round 1)** was in Corollary 9, outside (iii). The proposals form a subfamily of the draws whose position in the chronological order may depend on the run, so the iid property of all draws did not directly transfer to them. It was accepted and fixed: Proposition 5 now covers any class of "designated" calls decided when each call starts.
- **Opus exposition notes (round 2)** were applied in v5.

Disputes: none. No Astra judge was needed.

The full log, including every fatal or gap claim verbatim with the lead's answer and the reviewers' rulings, is in `CLAIMS_OF_FATAL_FLAWS.md`.

CLAIMS_OF_FATAL_FLAWS.md (every entry, verbatim)

CLAIMS_OF_FATAL_FLAWS — T11(iii)

Every fatal or gap finding, copied verbatim, with the lead's answer.

Stage A, round 1 — Astra #1 (session 01a1641a-4997-7370-a99a-43d4f2a4f245), review of PROOF_v1 (verdict: verified)

A1-F1 (gap)

Finding 1

1. **Location:** Section 2, sampler definitions and final equality in Lemma `\ref{lem:sampler}`.
2. **Severity: gap** — confined to the supplementary sampler verification.
3. **Claim:** The identification of the displayed sampler distribution with π_0 is not established from the supplied setting.

4. **Evidence:** The setting does not define $w(k)$, W_K, or the product representation

$$\gamma_0(S) = \frac{w(|S|)}{W_K} \prod_b r_b(S \cap C_b).$$

Nor does it explicitly supply positivity of every table entry. The algebra is correct given these additional facts, but their applicability is unverified. This does not undermine Claim (iii), because EX-A already supplies sampler exactness.

Lead's answer: accepted and fixed. The packet's STATEMENT.md did not carry the definitions of w, W_K, ϖ, γ_0 and the positivity of the tables, which the sampler verification uses. Change: Setting item 1 of STATEMENT.md now lists these definitions (taken from 00_COMMON §§1–2 and METHODOLOGY §5, §6.3: w(k) = ½/C(2p−1,k) > 0, W_K = Σ_{k≤K}C(p,k)w(k), ϖ(S) = w(|S|)/W_K, r_b(U) = m(U)/m(∅) > 0, γ_0(S) = ϖ(S)∏_b r_b(S∩C_b)); these are definitions, not new hypotheses. PROOF_v2 §2 states that the verification uses exactly these. As the finding notes, the main claim uses only the exactness in Setting item 3.

A1-F2 (gap)

Finding 2

1. **Location:** Section 3, Remark `\ref{rem:full}`.
2. **Severity: gap** — confined to the supplementary implementation justification.
3. **Claim:** The assertions about all possible interruption statuses and all matrices factorised cannot be verified from the supplied files.
4. **Evidence:** The setting does not specify the cited failure mechanisms or matrix inventory. Positive definiteness follows for the displayed matrices, but their exhaustiveness is unsupported here. Full execution is already stipulated by Setting item 5, so the main proof does not require these assertions.

Lead's answer: accepted and fixed. STATEMENT.md now has a section "Method facts quoted verbatim" (statuses of §2, the §7 pseudocode, §6.13's failure path). Remark 4 of PROOF_v2 ("the final run executes in full") is rewritten: it says the main proof uses Setting item 5 as stated and only explains why item 5 is the method's specification. The argument no longer relies on an inventory of factorised matrices: `numerical_failure` is "scoring failed after the fallback", and under EX-A scores are computed exactly, so scoring does not fail; the positive-definiteness of H_S and G_S + I is kept only as an illustration.

A1-F3 (gap)

Finding 3

1. **Location:** Final corollary, “scoring; beyond the stated part.”
2. **Severity: gap**.
3. **Claim:** The scoring conclusion requires implementation facts absent from Setting items 1–6.
4. **Evidence:** Those items specify initial draws and independence proposals, but do not require every proposal’s score to be computed or specify the stage-1 weighting calculation. The corollary invokes unavailable methodology sections for those premises. Its set-inclusion

argument is valid if the premises hold; the premises remain unverified. This corollary is explicitly outside Claim (iii).

Lead's answer: accepted and fixed. The premises are now quoted verbatim in STATEMENT.md (§6.5: "Every proposal is scored, accepted or not."; §6.12: the evidence product, whose t = 1 factor contains d(S_i^{(0)}), which involves log m(S_i^{(0)})). The corollary in PROOF_v2 cites these quotes. It remains labelled as beyond the stated part (iii).

Stage A, round 2 — Astra #1 rulings on round-1 answers and review of PROOF_v2 (verdict: verified)

Rulings (verbatim):

Rulings on the three previous findings

1. **A1-F1 — sampler definitions: agree; the fix works.** Updated Setting item 1 supplies positivity of the tables, the size-prior normalisation, and the product formula for γ_0. These justify $M = W_K \bar{Z}_0$ and the sampler proof's final identification with π_0.

2. **A1-F2 — full execution: agree with the substantive answer; the fix works for the main argument.** The proof explicitly takes full execution from Setting item 5. The supplied quotations support the timing of preflight and schedule failures and identify the scoring-failure status. The argument no longer requires an exhaustive matrix inventory. One illustrative implementation assertion remains unsupported; see Finding 1 below.

3. **A1-F3 — scoring: partially agree; the fix is incomplete.** The new quotation establishes that every independence proposal is scored. However, the evidence-product formula does not, by itself, establish that each initial draw's $\log m(S)$ is computed during the final run. See Finding 2 below.

A1-R2-F2 (gap; outside Claim (iii))

Finding 2

1. **Location:** Final scoring corollary, assertion that each initial draw is scored "by the evidence product."
2. **Severity: gap** — outside Claim (iii).
3. **Claim:** An expression containing $d(S)$ does not necessarily require computing $\log m(S)$ during the final run.
4. **Evidence:** Consider the permitted case of one block, $C_1 = [p]$. Then

$$r_1(S) = m(S)/m(\emptyset), \qquad d(S) = \log\frac{m(S)}{m(\emptyset)} - \log r_1(S) = 0.$$

Thus every evidence-product factor can be evaluated as 1 without computing an initial model's score during that run. The quoted formula is compatible with this evaluation, while the explicit scoring instruction covers proposals only.

Consequently, the quotations do not establish that the event "S^* is not scored" is contained in the event "S^* is absent from all initial draws and proposals." The main absence theorem remains valid.

Lead's answer: accepted and fixed. The quoted method text guarantees scoring only for the independence proposals. PROOF_v3's corollary now uses the proposals alone: each of the Q_p = 4N_e(T+L+4) final proposals is scored (§6.5 quote), they are iid π_0 given the calibration streams (Proposition 5) and Q_p is $\mathcal{C}$-measurable, so P(S* not scored in the final run | $\mathcal{C}$) ≤ (1 − π_0(S*))^{Q_p} on Λ; the exponent Q_f is stated only conditionally on the initial draws also being scored. The §6.12 quote was removed from STATEMENT.md (no longer used). Claim (iii) itself is unaffected (Astra agrees).

Exposition findings of round 2 (not fatal/gap; handled in PROOF_v3): Finding 1 (matrix example in Remark 4 unsupported) — parenthetical removed. Finding 3 (σ($\mathcal{C}$) "not independent" of calibration streams too categorical) — reworded to "in general not ... ($\mathcal{C}$ is computed from them)".

Stage A, round 3 — Astra #1 rulings and review of PROOF_v3 (verdict: verified; no new findings). Stage A closed: lead and Astra #1 both hold PROOF_v3 correct.

Rulings on the author's responses

1. **Round-2 Finding 2 — scoring gap: agree; the fix works.** The revised corollary uses only the $Q_p = 4N_e(T + L + 4)$ independence proposals, which the quoted method text explicitly requires to be scored. Their absence probability gives the stated upper bound on the probability of not being scored. The stronger exponent Q_f is now explicitly conditional on the initial draws also being scored. Removing the unused §6.12 quotation is consistent with this change.

2. **Round-2 Finding 1 — unsupported matrix illustration: agree; the fix works.** The parenthetical implementation assertion has been removed. The main proof continues to use full execution as stipulated by Setting item 5.

3. **Round-2 Finding 3 — categorical non-independence: agree; the fix works.** "In general not" correctly allows degenerate cases, including constant calibration output, while explaining why independence from calibration randomness is not guaranteed.

Stage B, round 1 — Astra #2 (session 01a16421-b0ef-7281-984e-2af293f78cf1): verified, no findings. Opus 5.5 xhigh reviewer: verified, no fatal findings, one gap (F8, outside (iii)) and seven exposition findings. All addressed in PROOF_v4.

OB-F8 (gap; Corollary 9, outside Claim (iii)) — Opus reviewer

F8

- **Location:** Corollary 9, lines 211–218. This is outside claim (iii).
- **Severity:** gap (in the corollary only; it does not affect the verdict on (iii)).
- **Claim:** The step "by Proposition 5 they [the proposals alone] are iid π_0 given ϒ" does not follow from Proposition 5 as stated.
- **Evidence:**
  - Proposition 5 proves that the chronologically numbered D_1..D_{Q_f} are iid. The proposals are a sub-family whose positions in that order may depend on Ξ: item 2 lets the order of operations depend on the values read so far, for example through how the islands are

interleaved.

- A sub-family chosen with data-dependent positions is not iid in general. For example, with D_1, D_2 iid, choosing D_1 if D_1 = s and D_2 otherwise does not give a π_0-distributed result.
- The corollary can be repaired by rerunning Proposition 5's stopping-time argument over proposal calls only. Whether a call is a proposal is decided at its start, so that event is $\mathcal{F}$_{τ_m}-measurable. As written, though, the citation does not supply this.

Lead's answer: accepted and fixed. Proposition 5 is restated in PROOF_v4 for any class of *designated* sampler calls (membership decided by the run's state when the call starts) with a constant number Q of calls; its stopping-time argument is run over the designated calls only, exactly as the reviewer suggests ({m-th designated call starts after read t} ∈ $\mathcal{F}$_t). Parts (a) (all final draws, Q = Q_f) and (b) (independence proposals alone, Q = Q_p) are stated explicitly; Theorem 6 cites (a) and Corollary 9 cites (b).

Exposition findings (not fatal/gap), handled in PROOF_v4 / STATEMENT.md:

- F1 (constant 0 outside (0,1)): replacement constant is now 1/2.
- F2 ($\mathcal{F}$_τ, τ = ∞): $\mathcal{F}$_τ now requires A ∈ σ(V); V_{τ+j} := 1/2 on the null set {τ = ∞}.
- F3 (finiteness of τ_m): now justified by item 5 (the m-th designated call occurs) and item 2 (one operation at a time).
- F4 (other sampler calls): covered by the designated-call formulation.
- F5 (single map G): the proof now allows a state-dependent map G_{Z_m} with Z_m $\mathcal{F}$_{τ_m}-measurable, and Setting item 3 is worded accordingly (measurable map, possibly state-dependent; exact law π_0 on iid uniform input); the step uses Lemma 2 plus Fubini.
- F6: "exactly one path of positive probability".
- F7 (Remark 4): the remark now notes m(S) is finite and positive (item 1) and that environment-imposed aborts are outside the idealised model; Setting item 5 now says explicitly that a launched final run executes all its rounds, and the "Statement changes" paragraph says item 5 transcribes the method's specification.
- Setting item 6 faithfulness: STATEMENT.md now quotes the theorem file's own definition of S* ("Fix S* ∈ Ω measurable w.r.t. data and $\mathcal{C}$ (before final randomness)").

Stage B, round 2 — rulings on PROOF_v4

- Astra #2: agrees with every logged answer (A1-F1, A1-F2, A1-F3 as resolved by v4, A1-R2-F2, OB-F8) and with the handling of all exposition items; no new findings; **VERDICT: verified** (review/astra2_stageB_round2_review.md).
- Opus reviewer: agrees with every logged answer; OB-F8 closed by Proposition 5(b); F1–F7 and the item-6 concern adequately handled; four new exposition findings N1–N4, no fatal or gap; **VERDICT: verified** (review/opus_stageB_round2_review.md).
- No open fatal or gap claim. Exposition N1–N4 handled in PROOF_v5 (diff against v4 is exposition only): N1 "A = Ω" (clash with the model space) → "A the whole sample space"; N2 Section 2 paraphrase of item 3 updated to the reworded item 3; N3 joint measurability of (z, w) ↦ G_z(w) written out (state = history in ⊔_t [0,1]^t; the run is a Borel function of the values read, Setting item 2); N4 Remark 4 now states the reading of EX-A it uses ("every score computation returns

log m(S) exactly").

Stage B, round 3 — confirmation of PROOF_v5 (final)

- Astra #2: v5 differs from v4 only at N1–N4 (plus title); N3 measurability argument correct; no findings; **VERDICT: verified** (review/astra2_stageB_round3_review.md).
- Opus reviewer: same diff check; N3 correct step by step; one exposition note E1 ("for every z" is slightly broader than needed: G_z is only defined at histories where a call starts; harmless, "no change is required"); **VERDICT: verified** (review/opus_stageB_round3_review.md). Lead: E1 noted, no change (the Fubini step evaluates G_z only at Z_m, where it is defined).
- Final state: no open fatal or gap claim; all three reviewers (Astra #1, Astra #2, Opus xhigh) hold the proof correct; no Astra judge needed.

Chapter 3

T12 (T8): frozen-run evidence and recorded Rao–Blackwell identities

Final proof: `proofs/T12_T8_frozen_run_identities/proof/PROOF_v6.tex`. *Sources of parts (a), (b), (d):* `theorems/T12_T8_frozen_run_identities.md`, `STATEMENT.md`, `FINAL_REPORT.md`, `CLAIMS_OF_FATAL_FLAWS.md`.

(a) Original conjecture text (theorem file, verbatim)

T12 (T8): Frozen-run evidence and recorded-RB identities

Status: conjectured, not proven. Methodology SMC-BVS v3.0 (2026-10-19), SHA-256 93043758058a6d422cca6ca4b6be8f3a9e8149b6f6de2e1bab1ce31c44b8019c. Read with `00_COMMON.md`.

Category / rank: algorithmic validity; **core** (theorem).

Definitions used

Stage notation q_t, Z_t, q̂_{r,t}, Ẑ_{r,t}; the final run (4 islands × N_e, multinomial resampling after every reweighting, one round per stage, L burn-in + 4 measured rounds); PIP̂_{rj} = pre-draw Gibbs conditionals averaged over particles and the 4 measured rounds; pooled Ẑ = ¼Σ_rẐ_r; primary/secondary PIPs (00_COMMON §2).

Assumptions

EX-A, EX-S, CAL; prescribed multinomial resampling and independent particle mutations.

Precise statement

Conditional on a successful $\boldsymbol{\mathcal{C}}$, for every island r, N ≥ 1, stage t and bounded fixed f: (i) E[Ẑ_{r,t}q̂_{r,t}(f)] = Z_tq_t(f); in particular EẐ_r = Z. Appending terminal invariant rounds with unit potentials preserves the identity; it holds at fixed elementary prefixes and immediately before a specified Gibbs label in a state-independent random permutation; not at arbitrary state-dependent stopping times. (ii) For the actual pre-draw RB recordings: E[Ẑ_r PIP̂_{rj}] = Z·PIP_j. (iii) For the pooled evidence: EẐ = Z; E log Ẑ ≤ log Z; P(Z ≥ αẐ) ≥ 1 − α for 0 < α < 1 (Jensen also islandwise). (iv)

Primary and secondary normalised PIPs and log $\hat{Z}$ are not unbiased in general. Conditional island independence holds given $\mathcal{C}$, not necessarily after averaging over shared calibration. Averaging over a $\mathcal{C}$-measurable launch event preserves evidence unbiasedness; conditioning on favourable final diagnostics does not guarantee any of (i)–(iii). NUM alone supplies no unbiasedness or Markov guarantee for implemented floating-point evidence.

Dependencies

Builds on: T09; project P7/P11. Used by: T14, T15, T16, T17, T19, T25, T28, T31.

Numerical evidence (provisional)

provisional evidence (Sol check J2): exact weighted-law propagation (N = 1, T = 1, 2, 3, through terminal resampling, burn-in and measured rounds with pre-draw rewards): max relative residual of $E\hat{Z} = Z$ and $E[\hat{Z}\,\mathrm{PI}\hat{P}_j] = Z{\cdot}\mathrm{PIP}_j$ is $1.3{\cdot}10^{-15}$. Files: judge/sol2/J2_EXACT_IDENTITIES.md.

Proof-route hint (optional)

Induct on the unnormalised empirical measure through weighting, resampling and mutation (Feynman–Kac unbiasedness); append unit-potential invariant kernels; treat the pre-coordinate reward as a marked transition and replace the indicator by o_j (project P11); tower property. Hardest step: the recording position within a round; terminal indicators cannot replace recorded conditionals. *When proof.md is run on this file, its lead considers the statement both with and without these hints; the hints never reach the reviewers.*

(b) STATEMENT.md (verbatim)

STATEMENT — T12 (T8): frozen-run evidence and recorded Rao–Blackwell identities

Methodology: SMC-BVS v3.0 (METHODOLOGY_v3.md, SHA-256 93043758058a6d422cca6ca4b6be8f3a9e8149b6f6de2e1bab1ce31c44b8019c), §§6.3–6.5, 6.8, 6.11, 6.12. Source conjecture: `theorems/T12_T8_frozen_run_identities.md`. This file is the precise statement to be proved; nothing else is assumed.

0. Data and target (fixed, deterministic objects)

A **valid input** is (X^r, y^r) with finite entries, $n \geq 3$, at least one retained (non-constant) column, and y^r not constant. X and y are the standardised data (columns and y centred and scaled so that every squared norm is $n - 1$). $p \geq 1$ is the number of retained columns, $K = \min(p, \lfloor n/2 \rfloor) \geq 1$, and the state space is $\Omega = \{S \subseteq [p] : |S| \leq K\}$ (finite, $|\Omega| \geq 2$).

For $S \in \Omega$, with $k = |S|$, $G_S = X_S^{\mathrm{T}} X_S$, $z_S = X_S^{\mathrm{T}} y$, $a_0 = b_0 = 0.01$ and $\nu = a_0 + n/2$ (the v3 target, METHODOLOGY §5 and 00_COMMON §1):

- $H_S = (n + 1)G_S + I_k$, $D_S = \det H_S / \det(G_S + I_k)$, $Q_S = y^{\mathrm{T}}y - n\, z_S^{\mathrm{T}} H_S^{-1} z_S$ (for $S = \emptyset$: $D_\emptyset = 1$, $Q_\emptyset = y^{\mathrm{T}}y$);
- $m(S) = C_n\, D_S^{-1/2} (2b_0 + Q_S)^{-\nu}$ with $C_n = (2b_0)^{a_0}\Gamma(\nu)/(\pi^{n/2}\Gamma(a_0)) > 0$, so $BF(S) := m(S)/m(\emptyset) = D_S^{-1/2}[(2b_0 + Q_S)/(2b_0 + y^{\mathrm{T}}y)]^{-\nu}$;
- $\varpi(S) = w(k)/W_K$ with $w(k) = B(1 + k, 2p - k)/B(1, p) = \tfrac{1}{2}/C(2p - 1, k)$ and $W_K = \Sigma_{j \leq K} C(p,$

j)w(j) (the BB(1, p) prior capped at K);

- Z = Σ_{S∈Ω} ϖ(S)m(S), π(S) = ϖ(S)m(S)/Z, PIP_j = π({S : j ∈ S}).

m > 0 (Q_S > 0, as Q_S ≥ $y^T y/(n + 1)$) and ϖ > 0. Everything except the explicit instance of (iv.b)/(iv.e) uses only m > 0 and ϖ > 0. The instance uses the formulas above.

The following are deterministic functions of the data (method §§6.3–6.4):

- blocks C_1, …, C_J partitioning [p]; tables r_b(U) = BF(U) > 0 for U ⊆ C_b;
- $\gamma_0(S) = \varpi(S)\prod_b r_b(S \cap C_b)$, $\bar{Z}_0 = \Sigma_\Omega \gamma_0$, $\pi_0 = \gamma_0/\bar{Z}_0$ (so $\pi_0(S) > 0$ for every S ∈ Ω);
- the discrepancy d(S) = log BF(S) − Σ_b log r_b(S ∩ C_b) (so d(∅) = 0); for real φ, γ_φ = $\gamma_0 e^{\varphi d}$, $\bar{Z}_\varphi = \Sigma_\Omega \gamma_\varphi$, q_φ = γ_φ/$\bar{Z}_\varphi$; note $q_0 = \pi_0$ and $q_1 = \pi$, and m(∅)$\bar{Z}_1$ = Z;
- the neighbour graph N(·), symmetric: l ∈ N(j) ⇔ j ∈ N(l); B = min(⌊√n⌋, ⌊√p⌋, 16).

1. Elementary steps and rounds (method §6.5)

At temperature φ there are m = p + B + 2 labelled elementary steps, each a Markov kernel on Ω:

- **Gibbs coordinate j** (j ∈ [p]): at S, with $S^+ = S \cup \{j\}$ and $S^- = S \setminus \{j\}$: if $|S^+| \le K$, move to S^+ with probability o_j(S, φ) = γ_φ(S^+)/(γ_φ(S^+) + γ_φ(S^-)) and to S^- otherwise; if $|S^+| > K$, set o_j(S, φ) = 0 and move to S^- (= S). The value o_j(S, φ) is the **pre-draw recording** of the step.
- **Uniform swap:** at S with |S| ∈ {0, p}, stay. Otherwise choose j ∈ S and l ∈ [p] \ S uniformly and independently, propose S′ = S \ {j} ∪ {l}, accept with probability 1 ∧ γ_φ(S′)/γ_φ(S), else stay.
- **Neighbour swap** (B copies, each a separate label): at S = ∅, stay. Otherwise choose j ∈ S uniformly; if N(j) \ S = ∅, stay; else choose l uniformly in N(j) \ S, propose S′ = S \ {j} ∪ {l}, accept with probability 1 ∧ [γ_φ(S′)/γ_φ(S)]·|N(j) \ S|/|N(l) \ S′|, else stay.
- **Independence proposal:** draw S′ ∼ π_0, accept with probability 1 ∧ $e^{\varphi[d(S') - d(S)]}$, else stay.

A **round at φ** applied to a particle: draw an order σ (a bijection from positions {1, …, m} to the m labels) uniformly at random, independently of the particle's state, and apply the elementary steps σ(1), …, σ(m) in turn. The states X_{[0]} (round start), X_{[1]}, …, X_{[m]} (round end) form the particle's **within-round path**. K_φ denotes the resulting round kernel.

2. The frozen run with fixed parameters

Frozen parameters c consist of: a grid $0 = \varphi_0 < \varphi_1 < \ldots < \varphi_T = 1$ with integer T ≥ 1 (Δ_t = φ_t − φ_{t−1}, G_t = $e^{\Delta_t d}$); an integer particle number N ≥ 1; an integer number L′ ≥ 0 of rounds at φ = 1 run after the stage-T round; an integer number M ≥ 1 of measured rounds; and a flag saying whether a round at $\varphi_0 = 0$ is applied at stage 0. **The method** (METHODOLOGY §6.8: "one move round per stage, then L burn-in rounds and 4 measured rounds") is the case N = N_e, M = 4, L′ = L, flag unset. Other values are generalisations, covered by the same statements.

One island I(c): 1. Stage 0: X_0^i iid ∼ π_0 (i = 1, …, N); $\hat{Z}_0 = Z_0$:= m(∅)$\bar{Z}_0$; if the flag is set, each particle performs one round at φ_0. Call the population X^{(0)}. 2. Stage t = 1, …, T: weights w^i = G_t(X^{(t−1),i}); $\hat{Z}_t = \hat{Z}_{t-1}\cdot(1/N)\Sigma_i w^i$; multinomial resampling (ancestors A^1, …, A^N iid with P(A = k) = $w^k/\Sigma_l w^l$ given the past); each resampled particle X^{(t−1),A^i} performs one round at φ_t; call the population X^{(t)}. $\hat{q}_t := (1/N)\Sigma_i \delta_{X^{(t),i}}$; $\hat{q}_0 := (1/N)\Sigma_i \delta_{X^{(0),i}}$. 3. After stage T: L′ rounds at φ = 1, then M measured rounds at φ = 1. In measured round u, $Y^i_{u,j}$ is

particle i's state immediately before its Gibbs-j step. Island evidence $\hat{Z} := \hat{Z}_T$; island RB estimate $\widehat{PI}_j := (MN)^{-1}\Sigma_{u=1}^{M}\Sigma_{i=1}^{N} o_j(Y^i_{u,j}, 1)$.

Stage notation: $q_t = q_{\varphi_t}$, $Z_t = m(\emptyset)\bar{Z}_{\varphi_t}$; so $Z_T = Z$.

Randomness (from EX-S and the prescribed operations). (R1) the stage-0 draws are iid π_0; (R2) given everything before a resampling, the ancestors are iid with the stated law; (R3) given the σ-field H of everything before a round (including the states entering it), the particles' (order, within-round path) pairs are conditionally independent across particles, each order is uniform and independent of H, and given H and its order σ, a particle's within-round path is a Markov chain started at its entering state with kernels $K_{\sigma(1)}, \ldots, K_{\sigma(m)}$ (each step uses fresh randomness; the independence proposal is a fresh exact π_0 draw).

Hypotheses.

- **EX-A:** exact arithmetic: every score, table, $\bar{Z}_0$, probability and comparison is exact (no numerical-failure status occurs).
- **START** (imported dependency: T09(i) / project P1, not re-proved here): the start sampler returns exact iid π_0 draws, the reported $\bar{Z}_0$ equals $\Sigma_\Omega \gamma_0$, and the independence proposals use the same exact sampler. This is not an extra restriction on inputs: the definition of I(c) above already uses these properties, and START is what makes the method's final island an instance of I(c). EX-A alone does not give it.
- **EX-S:** independent ideal uniform streams with the method's stream separation; in particular (R1)–(R3). The stream model includes measurability: the map $(c, u) \mapsto \Phi(c, u)$ below, from (calibration value, stream) to the island record, is jointly measurable, and the record takes values in a Polish space. This holds for any algorithmic realisation from uniform streams.
- **CAL:** the calibration output $\boldsymbol{\mathcal{C}}$ (grid, T, L, N_e, tier, candidate set and every other setting the final run uses) is measurable with respect to the data and streams 0–9 only; the final islands r = 1, …, 4 use streams 10–13, which are independent of each other and of streams 0–9; escalation is decided before any final randomness.
- **Success:** $\mathcal{A} \in \sigma(\boldsymbol{\mathcal{C}})$ is the event that calibration succeeds (a grid is constructed at the final tier, so the final run executes). On $\mathcal{A}$ the final islands are $O_r = \Phi(\boldsymbol{\mathcal{C}}, U_r)$, r = 1, …, 4, where U_r is stream 9 + r and $\Phi(c, \cdot)$ realises I(c) with the method's parameters (N = N_e, M = 4, L′ = L, flag unset). "Given $\boldsymbol{\mathcal{C}}$" means conditional on $\sigma(\boldsymbol{\mathcal{C}})$ on the event $\mathcal{A}$. Island r's outputs are $\hat{Z}_r$, $\widehat{PI}_{rj}$, and its stage quantities $\hat{Z}_{r,t}$, $\hat{q}_{r,t}$.
- **Pooled estimators:** $\hat{Z} = \frac{1}{4}\Sigma_r \hat{Z}_r$; primary $\bar{P}_j = \frac{1}{4}\Sigma_r \widehat{PI}_{rj}$; secondary $\tilde{P}_j = \Sigma_r \hat{Z}_r \widehat{PI}_{rj} / \Sigma_r \hat{Z}_r$.

3. Conclusions

Let E_c denote expectation for the frozen run with fixed parameters c. For every valid input:

(i) Evidence identities.

- (i.a) For every $c, t \in \{0, \ldots, T\}$ and $f : \Omega \to \mathbb{R}$: $E_c[\hat{Z}_t \hat{q}_t(f)] = Z_t q_t(f)$. In particular $E_c[\hat{Z}] = Z$.
- (i.b) *Terminal invariant rounds with unit potentials.* After stage T, let any fixed finite sequence of operations be applied, each either a round at φ = 1 or a multinomial resampling with unit weights (which leaves $\hat{Z}$ unchanged). For the population $\hat{q}$ after any of these operations, $E_c[\hat{Z}\, \hat{q}(f)] = Z\, \pi(f)$.

- (i.c) *State-independent prefixes.* Consider any round of the run (a stage-t round, with $\hat{Z}_t$ and target q_t, or a post-terminal round, with $\hat{Z}_T$ and target π) and, for each particle i, a prefix length $\kappa_i = \kappa(\sigma_i) \in \{0, \ldots, m\}$ that is a fixed function of that particle's order σ_i alone (examples: a fixed k; $\kappa(\sigma) = \sigma^{-1}(\ell) - 1$, the position just before a specified label ℓ). Then $E_c[\hat{Z}_t \, (1/N)\Sigma_i \, f(X^i_{[\kappa_i]})] = Z_t \, q_t(f)$.
- (i.d) *Not at state-dependent stopping times.* For every c and every round of the run (with its $\hat{Z}_t$, q_t as in (i.c)), there are f and a stopping time τ_i of particle i's within-round path for which $E_c[\hat{Z}_t \, (1/N)\Sigma_i \, f(X^i_{[\tau_i]})] > Z_t \, q_t(f)$. Namely $\tau_i = \min\{k \in \{0, \ldots, m\} : X^i_{[k]} = \emptyset\} \wedge m$, with $\min \emptyset = \infty$, and $f = 1\{\emptyset\}$.

(ii) Recorded RB identity. For every c and $j \in [p]$: $E_c[\hat{Z} \cdot PI\hat{P}_j] = Z \cdot PIP_j$.

(iii) Pooled evidence. Under EX-A, START, EX-S, CAL, given $\boldsymbol{\mathcal{C}}$ on $\mathcal{A}$: for every r and every stage $t \le T$ of $\boldsymbol{\mathcal{C}}$'s grid, $E[\hat{Z}_{r,t}\hat{q}_{r,t}(f) \mid \boldsymbol{\mathcal{C}}] = Z_t q_t(f)$ (Z_t and q_t are those of $\boldsymbol{\mathcal{C}}$'s grid, so this right side is $\boldsymbol{\mathcal{C}}$-measurable) and $E[\hat{Z}_r PI\hat{P}_{rj} \mid \boldsymbol{\mathcal{C}}] = Z \cdot PIP_j$; $E[\hat{Z} \mid \boldsymbol{\mathcal{C}}] = Z$; $E[\log \hat{Z} \mid \boldsymbol{\mathcal{C}}] \le \log Z$ and $E[\log \hat{Z}_r \mid \boldsymbol{\mathcal{C}}] \le \log Z$; and for every $0 < \alpha < 1$, $P(Z \ge \alpha\hat{Z} \mid \boldsymbol{\mathcal{C}}) \ge 1 - \alpha$ and $P(Z \ge \alpha\hat{Z}_r \mid \boldsymbol{\mathcal{C}}) \ge 1 - \alpha$.

(iv) Limits.

- (iv.a) *log $\hat{Z}$.* For every c: if $d \equiv 0$ on Ω then $\hat{Z} = Z$ almost surely; otherwise $E_c[\log \hat{Z}] < \log Z$, and, for the four islands of the method given $\boldsymbol{\mathcal{C}}$, $E[\log \hat{Z} \mid \boldsymbol{\mathcal{C}}] < \log Z$ and $E[\log \hat{Z}_r \mid \boldsymbol{\mathcal{C}}] < \log Z$.
- (iv.b) *Normalised PIPs.* Given $\boldsymbol{\mathcal{C}}$, $E[\bar{P}_j \mid \boldsymbol{\mathcal{C}}] - PIP_j = -\mathrm{Cov}(PI\hat{P}_{1j}, \hat{Z}_1 \mid \boldsymbol{\mathcal{C}})/Z$. *Component example.* There is a valid input and frozen parameters c for which both normalised estimators are biased: $n = 20$, $p = 2$ with the data of the appendix, $T = 1$, $N = 1$, $M = 4$, $L' = L = \max(4, \lceil \log_2(n + 1) \rceil) = 5$, flag unset, four independent islands I(c), and $j = 1$. Then $E[\bar{P}_1] - PIP_1 < -0.17$ and $E[\tilde{P}_1] - PIP_1 < -0.15$. The same bounds hold with the flag set and/or $L' = L - 1$ (generalisations). Scope: at $p = 2$ the full method would take the exact branch, and $N = 1$ is not the production budget. So this shows bias of the frozen SMC component, not bias of the full method's output on this input.
- (iv.c) *Island independence.* Given $\boldsymbol{\mathcal{C}}$, the islands $O_1, \ldots, O_4$ are independent, each with the law of $I(\boldsymbol{\mathcal{C}})$. Assume $P(\mathcal{A}) > 0$. Under $P(\cdot \mid \mathcal{A})$, for every bounded measurable h of an island's outputs, $\mathrm{Cov}(h(O_1), h(O_2)) = \mathrm{Var}(E[h(O_1) \mid \boldsymbol{\mathcal{C}}])$. Hence the islands are independent under $P(\cdot \mid \mathcal{A})$ if and only if the conditional law of an island given $\boldsymbol{\mathcal{C}}$ is a.s. constant, which none of the hypotheses ensure. In particular the island law does depend on c: if d has a unique maximiser on Ω, the law of $\hat{Z}$ under (grid, N) differs from that under (grid, 4N). (Not claimed: that the actual calibration procedure produces, on some specific input, two values of $\boldsymbol{\mathcal{C}}$ with different island laws, each with positive probability.)
- (iv.d) *Launch events.* Let $A \in \sigma(\boldsymbol{\mathcal{C}})$ with $A \subseteq \mathcal{A}$ and $P(A) > 0$. The conclusions of (iii) whose right side does not depend on $\boldsymbol{\mathcal{C}}$ hold under $P(\cdot \mid A)$: $E[\hat{Z} \mid A] = E[\hat{Z}_r \mid A] = Z$, $E[\hat{Z}_r PI\hat{P}_{rj} \mid A] = Z \cdot PIP_j$, $E[\log \hat{Z} \mid A] \le \log Z$, $E[\log \hat{Z}_r \mid A] \le \log Z$, $P(Z \ge \alpha\hat{Z} \mid A) \ge 1 - \alpha$ and $P(Z \ge \alpha\hat{Z}_r \mid A) \ge 1 - \alpha$. For an intermediate stage t and $A \subseteq \{T \ge t\}$, the averaged form holds: $E[\hat{Z}_{r,t}\hat{q}_{r,t}(f) \mid A] = E[Z_t q_t(f) \mid A]$, where $Z_t q_t(f)$ is computed on $\boldsymbol{\mathcal{C}}$'s grid. The left side is a number, whereas $Z_t q_t(f)$ is a random variable, and it is non-constant when, for example, φ_t takes two values with positive probability under $P(\cdot \mid A)$ and $f = 1\{S\}$ with $d(S) \ne 0$.
- (iv.e) *Favourable final diagnostics.* Let F be the event that the final E2 check passes: the standard deviation of $\{\log \hat{Z}_r\}_{r=1}^{4}$ is ≤ 1 nat (either the 1/4 or the 1/3 normalisation). In the component example of (iv.b) (same scope), $P(F) > 0$, $E[\hat{Z}_r \mid F] < Z$, $E[\hat{Z} \mid F] < Z$ and $E[\hat{Z}_r \, PI\hat{P}_{r1} \mid F] \ne Z \cdot PIP_1$. So conditioning on a favourable final diagnostic breaks an identity in each of (i) ($E\,\hat{Z}_r = Z$), (ii) and (iii) ($E\,\hat{Z} = Z$). This is not a claim that the Jensen or Markov inequalities of (iii) fail

under F; in this example they still hold, because E[$\hat{Z}$ | F] < Z.

- (iv.f) *NUM alone.* A NUM-compatible perturbed-score model with bound η > 0 is the same algorithm in exact arithmetic with every evaluated log m(S) replaced by a value within η of it. (This does not assert that any particular floating-point implementation makes this error; it shows that NUM alone does not exclude it.) For every η > 0 there is such an implementation (log m(S) + η for every S) with E_c[$\hat{Z}$^impl] = e^ηZ ≠ Z for every c and valid input; and on every valid input with d ≡ 0 (for example p = 1), P(Z ≥ α$\hat{Z}$^impl) = 0 for every α ∈ (e^{−η}, 1).

Appendix: data for (iv.b) and (iv.e)

n = 20, p = 2, i = 0, ..., 19: x1_i = (7i mod 20) − 9.5; x2_i = x1_i + ((5i^2 mod 7) − 3); y_i = x1_i − x2_i + ((3i mod 5) − 2). (Raw data; standardised as in §0.) Then B = 1, the blocks are {1}, {2}, N(1) = {2}, N(2) = {1}, K = 2, m = 5, L = 5.

Statement history

- v1 (2026-10-22): written by the lead from the conjecture. The formalisation choices are listed in review/packets/STATEMENT_REPORT_A1_call1.md.
- v2 (2026-10-22): statement check by Astra #1 (review/A1_call1_statement_check_REPLY.md), ruling APPROVED WITH CHANGES. All four required changes adopted: 1. (iv.d) repaired. v1 claimed every conclusion of (iii) under P(· | A). That is false for intermediate stages, whose targets depend on the random grid; only the $\boldsymbol{\mathcal{C}}$-free conclusions survive, and intermediate stages hold in averaged form. 2. The method is identified as L′ = L, flag unset (METHODOLOGY §6.8); other values are labelled generalisations; count parameters are declared integers. 3. START is labelled an imported dependency (T09(i)), not a new restriction. 4. Component scope stated beside (iv.b)/(iv.e). (iv.e) is stated as breaking the identities, not the Jensen/Markov inequalities. Also, following Astra's item 5: min ∅ = ∞ and time 0 are made explicit in (i.d).
- v2.1 (2026-10-22): after Astra #1's proof review (verdict verified; three exposition findings), the assumption P($\mathcal{A}$) > 0 was made explicit in (iv.c). This is exposition only.
- v2.2 (2026-10-22): after the Stage B reviews (Astra #2 and Opus xhigh; both raised the same gap). §0 now writes out the target formulas (m(S), BF, ϖ, w, W_K, a_0 = b_0 = 0.01, ν = a_0 + n/2), copied from METHODOLOGY §5 / 00_COMMON §1, Felipe's fixed target. EX-S now states that joint measurability of Φ is part of the stream model. In (iv.d), 'only … in general random' is replaced by an exact remark. No conclusion changed.

(c) Final proof, verbatim: body of PROOF_v6.tex

The proof below is the body of `PROOF_v6.tex` *from* `\begin{document}` *to* `\end{document}` *without the title block. Its preamble definitions are local to this chapter; labels carry the prefix* `T12:`.

We prove every conclusion of `STATEMENT.md`, with its definitions, notation and hypotheses (EX-A, START, EX-S, CAL, the randomness model (R1)–(R3), the success event $\mathcal{A}$). Section numbers refer to this document. All functions on Ω are bounded because Ω is finite, and every random quantity below is bounded, so all expectations exist. For a function h on Ω and a Markov kernel P on Ω, $Ph(x) = \sum_y P(x,y)h(y)$; for a measure μ on Ω, $\mu(h) = \sum_x \mu(x)h(x)$ and $\mu P(y) = \sum_x \mu(x)P(x,y)$. We write $J_j = \{S \in \Omega : j \in S\}$, so $\mathrm{PIP}_j = \pi(J_j) = \pi(\mathbf{1}_{J_j})$. Parts (i), (ii) and (iv.a)–(iv.b) are proved for the frozen run $I(c)$ with arbitrary integer parameters. The method's final island is the

case $N = N_e$, $M = 4$, $L' = L$ with the stage-0 flag unset; other values are generalisations. START is an imported dependency (T09(i)): it is what makes the stage-0 draws and the independence proposals exact π_0 draws and $\bar{Z}_0$ exact, as the definition of $I(c)$ requires.

1 Deterministic facts

Lemma 1.1 (bridge identities)**.** *For every real φ, $\bar{Z}_\varphi \in (0,\infty)$ and $q_\varphi(S) > 0$ for every $S \in \Omega$. For $t = 1,\dots,T$ and every $h : \Omega \to \mathbb{R}$,*

$$Z_{t-1}\, q_{t-1}(G_t h) = Z_t\, q_t(h).$$

Moreover $m(\emptyset)\bar{Z}_1 = Z$, so $Z_T = Z$; and $Z_0 = m(\emptyset)\bar{Z}_0$.

Proof. $\gamma_\varphi(S) = \gamma_0(S)e^{\varphi d(S)} > 0$ because $\varpi > 0$ and $r_b > 0$; Ω is finite, so $\bar{Z}_\varphi \in (0,\infty)$. Since $\varphi_{t-1} + \Delta_t = \varphi_t$, $q_{t-1}(x)G_t(x) = \gamma_0(x)e^{\varphi_{t-1}d(x)}e^{\Delta_t d(x)}/\bar{Z}_{\varphi_{t-1}} = \gamma_{\varphi_t}(x)/\bar{Z}_{\varphi_{t-1}}$. Hence $Z_{t-1}q_{t-1}(G_t h) = m(\emptyset)\bar{Z}_{\varphi_{t-1}}\sum_x \gamma_{\varphi_t}(x)h(x)/\bar{Z}_{\varphi_{t-1}} = m(\emptyset)\bar{Z}_{\varphi_t}q_t(h) = Z_t q_t(h)$. Finally $\gamma_0(S)e^{d(S)} = \varpi(S)\prod_b r_b(S\cap C_b)\cdot \mathrm{BF}(S)/\prod_b r_b(S\cap C_b) = \varpi(S)m(S)/m(\emptyset)$, so $m(\emptyset)\bar{Z}_1 = \sum_S \varpi(S)m(S) = Z$. □

Lemma 1.2 (elementary steps are q_φ-reversible)**.** *Fix a real φ. Each elementary step of STATEMENT §1 (Gibbs coordinate j, uniform swap, neighbour swap, independence proposal) defines a Markov kernel K_e on Ω that satisfies detailed balance with respect to q_φ: $q_\varphi(S)K_e(S,S') = q_\varphi(S')K_e(S',S)$ for all $S, S' \in \Omega$. Consequently $q_\varphi K_e = q_\varphi$.*

Proof. Write $q = q_\varphi$, $\gamma = \gamma_\varphi$, $\bar{Z} = \bar{Z}_\varphi$. Detailed balance for $S = S'$ is trivial. Summing it over S gives $qK_e(S') = q(S')\sum_S K_e(S',S) = q(S')$. So it suffices to check $S \neq S'$, after checking that each step stays in Ω with total probability one.

Gibbs coordinate j. At S the step moves to $S^+ = S\cup\{j\}$ (only if $|S^+| \le K$) or to $S^- = S\setminus\{j\}$, with probabilities $o_j(S,\varphi)$ and $1 - o_j(S,\varphi)$. Both targets are in Ω whenever they are used, since $|S^-| \le |S| \le K$. The only transitions between distinct states are between some pair $\{U, U\cup\{j\}\}$ with $j \notin U$ and $|U| + 1 \le K$. From either member of such a pair, $S^+ = U\cup\{j\}$ and $S^- = U$, so $K_e(U, U\cup\{j\}) = \gamma(U\cup\{j\})/(\gamma(U\cup\{j\}) + \gamma(U))$ and $K_e(U\cup\{j\}, U) = \gamma(U)/(\gamma(U\cup\{j\}) + \gamma(U))$. Then $q(U)K_e(U, U\cup\{j\}) = \gamma(U)\gamma(U\cup\{j\})/(\bar{Z}(\gamma(U\cup\{j\}) + \gamma(U))) = q(U\cup\{j\})K_e(U\cup\{j\}, U)$. No other pair of distinct states communicates, and every other $K_e(S,S')$ with $S \neq S'$ is 0 in both directions.

Uniform swap. If $|S| \in \{0, p\}$ the step stays at S. Otherwise, with $k = |S| \in (0,p)$, the pair $(j,l) \in S \times ([p]\setminus S)$ has probability $1/(k(p-k))$. It proposes $S' = S\setminus\{j\}\cup\{l\}$, with $|S'| = k \le K$, so $S' \in \Omega$ and $S' \neq S$. The map $(j,l) \mapsto S'$ is injective (its inverse is $S' \mapsto (S\setminus S',\, S'\setminus S)$). Hence for $S' \neq S$, $K_e(S,S') = \frac{1}{k(p-k)}\min\{1, \gamma(S')/\gamma(S)\}$ if $|S \triangle S'| = 2$ and $|S'| = |S|$, and $K_e(S,S') = 0$ otherwise. This condition is symmetric in (S,S') and $|S'| = k$, so $q(S)K_e(S,S') = \frac{1}{\bar{Z}\,k(p-k)}\min\{\gamma(S),\gamma(S')\} = q(S')K_e(S',S)$.

Neighbour swap. At $S = \emptyset$ the step stays. Otherwise, with $k = |S| \ge 1$, choose $j \in S$ with probability $1/k$. Let $A = N(j)\setminus S$. If $A = \emptyset$, stay. Else choose $l \in A$ with probability $1/|A|$ and propose $S' = S\setminus\{j\}\cup\{l\}$; then $|S'| = k$, $S' \in \Omega$, $S' \neq S$. Put $A' = N(l)\setminus S'$. Since $l \in N(j)$, the symmetry of the graph gives $j \in N(l)$, and $j \notin S'$, so $j \in A'$ and $|A'| \ge 1$; the acceptance probability $\min\{1, \gamma(S')|A|/(\gamma(S)|A'|)\}$ is well defined. The map $(j,l) \mapsto S'$ is injective as before. Hence for $S' \neq S$, $K_e(S,S') = \frac{1}{k|A|}\min\{1, \frac{\gamma(S')|A|}{\gamma(S)|A'|}\}$ when $S' = S\setminus\{j\}\cup\{l\}$ with $j \in S$ and $l \in N(j)\setminus S$, and 0 otherwise. If S' arises from S through (j,l), then S arises from S' through (l,j): $l \in S'$,

$j \in N(l) \setminus S' = A'$, and the sets used in the reverse move are $N(l) \setminus S' = A'$ and $N(j) \setminus S = A$. Hence $K_e(S', S) = \frac{1}{k|A'|} \min\{1, \frac{\gamma(S)|A'|}{\gamma(S')|A|}\}$, and

$$q(S)K_e(S, S') = \frac{1}{\bar{Z}k} \min\left\{\frac{\gamma(S)}{|A|}, \frac{\gamma(S')}{|A'|}\right\} = q(S')K_e(S', S).$$

If S' does not arise from S, then S does not arise from S' (by the same reversal), and both sides are 0.

Independence proposal. π_0 is supported on Ω. For $S' \neq S$, $K_e(S, S') = \pi_0(S') \min\{1, e^{\varphi(d(S')-d(S))}\}$. Since $q(S) = \pi_0(S)e^{\varphi d(S)}\bar{Z}_0/\bar{Z}$, $q(S)K_e(S, S') = \frac{\bar{Z}_0}{\bar{Z}}\pi_0(S)\pi_0(S') \min\{e^{\varphi d(S)}, e^{\varphi d(S')}\}$, which is symmetric. □

Lemma 1.3 (state-independent prefixes; recordings)**.** *Fix a real φ, let Σ be the set of orders (bijections from positions $\{1, \dots, m\}$ to labels), and let $\kappa : \Sigma \to \{0, \dots, m\}$. Define*

$$P_\kappa = \frac{1}{m!} \sum_{\sigma \in \Sigma} K_{\sigma(1)}K_{\sigma(2)} \cdots K_{\sigma(\kappa(\sigma))} \qquad (\textit{empty product} = \textit{identity}).$$

Then $q_\varphi P_\kappa = q_\varphi$. In particular the round kernel $K_\varphi = P_m$ is q_φ-invariant. Moreover, if K_{g_j} is the Gibbs-j kernel, then $K_{g_j}\mathbf{1}_{J_j} = o_j(\cdot, \varphi)$ and $q_\varphi(o_j(\cdot, \varphi)) = q_\varphi(J_j)$.

Proof. By Lemma 1.2, every K_e preserves q_φ, so every finite product does, and so does every average of products. For the second claim, at S the Gibbs-j step ends in a state containing j exactly when it moves to S^+, which happens with probability $o_j(S, \varphi)$ (this is 0 when $|S^+| > K$, where the step moves to S^-). Hence $K_{g_j}\mathbf{1}_{J_j} = o_j(\cdot, \varphi)$, and $q_\varphi(o_j) = (q_\varphi K_{g_j})(\mathbf{1}_{J_j}) = q_\varphi(J_j)$. □

Lemma 1.4 (positivity)**.** *Fix a real φ and let $R_d = \max_\Omega d - \min_\Omega d$.*

(a) For every order σ in which the independence label is last, and all $x, S \in \Omega$, $K_{\sigma(1)} \cdots K_{\sigma(m)}(x, \{S\}) \geq \pi_0(S)e^{-|\varphi|R_d}$. Hence $K_\varphi(x, \{S\}) \geq \pi_0(S)e^{-|\varphi|R_d}/m$. In particular, for every $\varphi \in [0, 1]$ (all temperatures used by the run), $K_\varphi(x, \{S\}) \geq \varepsilon_S/m$ with the temperature-free constant $\varepsilon_S := \pi_0(S)e^{-R_d} > 0$.

(b) For every order σ with $\sigma(1) =$ Gibbs coordinate 1, $K_{\sigma(1)} \cdots K_{\sigma(m)}(\emptyset, \Omega \setminus \{\emptyset\}) \geq \delta > 0$ for a constant δ that depends only on φ and the input. Hence $K_\varphi(\emptyset, \Omega \setminus \{\emptyset\}) \geq \delta/m > 0$.

Proof. (a) Whatever the state y before the last step, the independence step proposes S with probability $\pi_0(S)$. If $S \neq y$ it accepts with probability $\min\{1, e^{\varphi(d(S)-d(y))}\} \geq e^{-|\varphi|R_d}$; if $S = y$ the state stays S whether or not it accepts. Hence $K_{\text{ind}}(y, \{S\}) \geq \pi_0(S)e^{-|\varphi|R_d}$ for every y, and the earlier steps are stochastic. For $\varphi \in [0, 1]$, $e^{-|\varphi|R_d} \geq e^{-R_d}$. Exactly $(m-1)!$ of the $m!$ orders have the independence label last.

(b) At $\emptyset$, Gibbs coordinate 1 moves to $\{1\}$, which is in Ω since $K \geq 1$, with probability $o_1(\emptyset, \varphi) = \gamma_\varphi(\{1\})/(\gamma_\varphi(\{1\}) + \gamma_\varphi(\emptyset)) > 0$. For every elementary step e and every non-empty y, $c_e(y) := K_e(y, \Omega \setminus \{\emptyset\}) > 0$. Gibbs j with $j \notin y$ ends in y or $y \cup \{j\}$, so $c_e(y) = 1$. Gibbs j with $j \in y$ stays at y with probability $o_j(y, \varphi) > 0$, because $y^+ = y \in \Omega$. Swaps preserve $|y|$ or stay, so $c_e(y) = 1$. The independence step stays at y with probability at least $\pi_0(y) > 0$. Let $c = \min_e \min_{y \neq \emptyset} c_e(y) > 0$ (a minimum over finitely many positive numbers). By the Markov property along the fixed order, the probability of being non-empty after step 1 and of remaining non-empty after each later step is at least $\delta := o_1(\emptyset, \varphi)\, c^{m-1}$. Exactly $(m-1)!$ orders have Gibbs coordinate 1 first. □

2 Conditional means of the island operations

All random variables of one island live on one probability space. For a population $(X^1, \dots, X^N)$ entering an operation, $\mathcal{F}$ denotes the σ-field of everything generated before the operation.

Lemma 2.1 (resampling)**.** *Let the population $(X^1, \dots, X^N)$ and weights $w^k = G(X^k) > 0$ be $\mathcal{F}$-measurable, and let the ancestors $A^1, \dots, A^N$ be conditionally iid given $\mathcal{F}$ with $\mathbb{P}(A^i = k \mid \mathcal{F}) = w^k / \sum_l w^l$ (R2). Then for every $h : \Omega \to \mathbb{R}$,*

$$\mathbb{E}\Big[\Big(\tfrac{1}{N}\textstyle\sum_l w^l\Big)\tfrac{1}{N}\sum_i h(X^{A^i}) \,\Big|\, \mathcal{F}\Big] = \tfrac{1}{N}\textstyle\sum_k G(X^k)h(X^k).$$

Proof. $\mathbb{E}[h(X^{A^i}) \mid \mathcal{F}] = \sum_k h(X^k)w^k / \sum_l w^l$ for each i. Multiply by the $\mathcal{F}$-measurable factor $\frac{1}{N}\sum_l w^l$ and average over i. □

Lemma 2.2 (one round, state-independent prefix)**.** *Let a population $(\tilde{X}^1, \dots, \tilde{X}^N)$ enter a round at temperature φ, let $\mathcal{H}$ be the σ-field of everything before the round, and let $X^i_{[k]}$ be particle i's within-round path. For $\kappa : \Sigma \to \{0, \dots, m\}$ and every $f : \Omega \to \mathbb{R}$, under (R3), $\mathbb{E}[f(X^i_{[\kappa(\sigma_i)]}) \mid \mathcal{H}] = (P_\kappa f)(\tilde{X}^i)$.*

Proof. By (R3), given $\mathcal{H}$ and $\sigma_i = \sigma$, the path is a Markov chain started at $X^i_{[0]} = \tilde{X}^i$ with kernels $K_{\sigma(1)}, \dots, K_{\sigma(m)}$. Hence $\mathbb{E}[f(X^i_{[k]}) \mid \mathcal{H}, \sigma_i = \sigma] = (K_{\sigma(1)} \cdots K_{\sigma(k)} f)(\tilde{X}^i)$ for each fixed k, and in particular for $k = \kappa(\sigma)$. Since σ_i is uniform on the finite set Σ and independent of $\mathcal{H}$, averaging over σ gives the claim. □

3 Proof of (i.a)–(i.c)

Write $X^{(t-1),A^i} =: \tilde{X}^{(t),i}$ for the resampled particles of stage $t \geq 1$, and $\tilde{X}^{(0),i} := X^i_0$.

Claim C_t ($t = 0, \dots, T$). For every $h : \Omega \to \mathbb{R}$: (a) $\mathbb{E}[\widehat{Z}_t \frac{1}{N}\sum_i h(\tilde{X}^{(t),i})] = Z_t q_t(h)$, and (b) $\mathbb{E}[\widehat{Z}_t \frac{1}{N}\sum_i h(X^{(t),i})] = Z_t q_t(h)$. Moreover (c), for every round of stage t (the stage-t round, or the stage-0 round if the flag is set) and every κ, $\mathbb{E}[\widehat{Z}_t \frac{1}{N}\sum_i f(X^i_{[\kappa(\sigma_i)]})] = Z_t q_t(f)$.

$t = 0$. $\widehat{Z}_0 = Z_0$ is deterministic and the X^i_0 are iid $\pi_0 = q_0$ by (R1) and START, so (a) holds. If no stage-0 round is applied, $X^{(0)} = X_0$ and (b) is (a). If a round at $\varphi_0 = 0$ is applied, then for its prefix with κ, Lemma 2.2 and (a) give $\mathbb{E}[Z_0 \frac{1}{N}\sum_i f(X^i_{[\kappa]})] = \mathbb{E}[Z_0 \frac{1}{N}\sum_i (P_\kappa f)(X^i_0)] = Z_0 q_0(P_\kappa f) = Z_0 q_0(f)$ by Lemma 1.3. This is (c), and $\kappa \equiv m$ gives (b).

$t - 1 \to t$. Let $\mathcal{F}$ be everything before the stage-t resampling. $\widehat{Z}_t = \widehat{Z}_{t-1}\frac{1}{N}\sum_l w^l$ with $w^l = G_t(X^{(t-1),l})$, and $\widehat{Z}_{t-1}$ is $\mathcal{F}$-measurable. By Lemma 2.1, $\mathbb{E}[\widehat{Z}_t \frac{1}{N}\sum_i h(\tilde{X}^{(t),i}) \mid \mathcal{F}] = \widehat{Z}_{t-1}\frac{1}{N}\sum_k (G_t h)(X^{(t-1),k})$. Taking expectations and applying C_{t-1}(b) to the function $G_t h$ gives $Z_{t-1}q_{t-1}(G_t h) = Z_t q_t(h)$ by Lemma 1.1. This is (a). For the stage-t round with prefix κ, $\widehat{Z}_t$ is measurable with respect to $\mathcal{H}$ (everything before the round), so Lemma 2.2 gives $\mathbb{E}[\widehat{Z}_t \frac{1}{N}\sum_i f(X^i_{[\kappa(\sigma_i)]}) \mid \mathcal{H}] = \widehat{Z}_t \frac{1}{N}\sum_i (P_\kappa f)(\tilde{X}^{(t),i})$. By (a) with $h = P_\kappa f$ and Lemma 1.3 (at φ_t), its expectation is $Z_t q_t(P_\kappa f) = Z_t q_t(f)$. This is (c), and $\kappa \equiv m$ gives (b). □

Claim C_t(b) is (i.a). With $t = T$ and $h \equiv 1$, $\mathbb{E}\widehat{Z} = Z_T q_T(1) = Z$ by Lemma 1.1. Claim C_t(c) is (i.c) for the rounds of stages $0, \dots, T$.

Post-terminal operations: (i.b) and (i.c). Let $\mathcal{O}_1, \dots, \mathcal{O}_k$ be the fixed sequence of operations after stage T. Each is a round at $\varphi = 1$ or a multinomial resampling with unit weights. Write $\widehat{q}^{(s)}$ for the population after $\mathcal{O}_s$ and $\widehat{q}^{(0)} = \widehat{q}_T$. We show by induction on s that $\mathbb{E}[\widehat{Z}\,\widehat{q}^{(s)}(h)] = Z\pi(h)$ for all h; for $s = 0$ this is (i.a) at $t = T$, since $q_T = \pi$. A unit-weight resampling leaves $\widehat{Z}$ unchanged (its factor is $\frac{1}{N}\sum_l 1 = 1$), and by Lemma 2.1 with $G \equiv 1$, $\mathbb{E}[\widehat{q}^{(s)}(h) \mid \mathcal{F}] = \widehat{q}^{(s-1)}(h)$, with $\widehat{Z}$ $\mathcal{F}$-measurable. A round at $\varphi = 1$ with prefix κ gives, by Lemma 2.2, $\mathbb{E}[\widehat{Z}\frac{1}{N}\sum_i f(X^i_{[\kappa(\sigma_i)]}) \mid \mathcal{H}] = \widehat{Z}\,\widehat{q}^{(s-1)}(P_\kappa f)$. By the induction hypothesis and Lemma 1.3 at $\varphi = 1$, its expectation is $Z\pi(P_\kappa f) = Z\pi(f)$. This is (i.c) for that round, and $\kappa \equiv m$ gives the step of the induction. This proves (i.b), and (i.c) for every post-terminal round. In particular it covers the L' rounds and the M measured rounds of $I(c)$, which are post-terminal rounds at $\varphi = 1$. □

4 Proof of (i.d): state-dependent stopping times

Fix c and a round ρ of the run, with prefactor $\widehat{Z}_\rho$ (that is, $\widehat{Z}_t$ for a stage-t round and $\widehat{Z}_T$ for a post-terminal round) and target q_ρ (that is, q_t or π). Let $\mathcal{H}$ be everything before ρ, $\tilde{X}^i$ the entering states, and $\tau_i = \min\{k \in \{0, \dots, m\} : X^i_{[k]} = \emptyset\} \wedge m$ (with $\min\emptyset = \infty$, so $\tau_i = m$ when the path never visits $\emptyset$). Then τ_i is a stopping time for $\mathcal{F}^i_k = \sigma(\mathcal{H}, \sigma_i, X^i_{[0]}, \dots, X^i_{[k]})$. Let $f = \mathbf{1}\{\emptyset\}$.

Step 1: $D_i := f(X^i_{[\tau_i]}) - f(X^i_{[m]}) \geq \mathbf{1}_{E_i} \geq 0$, where $E_i = \{X^i_{[0]} = \emptyset,\ X^i_{[m]} \neq \emptyset\}$. If the path visits $\emptyset$, then $f(X^i_{[\tau_i]}) = 1 \geq f(X^i_{[m]})$. If it does not, $\tau_i = m$ and $D_i = 0$. On E_i, $\tau_i = 0$, $f(X^i_{[0]}) = 1$ and $f(X^i_{[m]}) = 0$.

Step 2: $\mathbb{P}(\tilde{X}^1 = \emptyset) > 0$. Let V be the event that every particle is at $\emptyset$ after every operation of the run before ρ, and at stage 0. $\mathbb{P}(X^i_0 = \emptyset\ \forall i) = \pi_0(\emptyset)^N > 0$ by (R1). If all particles are at $\emptyset$, every resampled particle is at $\emptyset$. Given $\mathcal{H}'$ (everything before a round) on the event that all entering particles are at $\emptyset$, the particles end the round at $\emptyset$ with conditional probability $K_\varphi(\emptyset, \{\emptyset\})^N \geq (\varepsilon_\emptyset/m)^N > 0$, by the conditional independence in (R3) and Lemma 1.4(a). Finitely many operations precede ρ, so $\mathbb{P}(V) > 0$, and $V \subseteq \{\tilde{X}^1 = \emptyset\}$.

Step 3. By (R3) and Lemma 1.4(b) at the temperature of ρ, $\mathbb{P}(E_1 \mid \mathcal{H}) = \mathbf{1}\{\tilde{X}^1 = \emptyset\}K_\varphi(\emptyset, \Omega \setminus \{\emptyset\}) \geq \mathbf{1}\{\tilde{X}^1 = \emptyset\}\delta/m$. Since $\widehat{Z}_\rho > 0$ surely (it is a product of positive factors) and is $\mathcal{H}$-measurable,

$$\mathbb{E}\Big[\widehat{Z}_\rho \tfrac{1}{N}\textstyle\sum_i D_i\Big] \ \geq\ \tfrac{1}{N}\mathbb{E}\big[\widehat{Z}_\rho\, \mathbf{1}\{\tilde{X}^1 = \emptyset\}\big]\,\delta/m \ >\ 0.$$

By (i.c) with $\kappa \equiv m$, $\mathbb{E}[\widehat{Z}_\rho \frac{1}{N}\sum_i f(X^i_{[m]})] = Z_\rho q_\rho(f)$, where Z_ρ is Z_t or Z. Hence $\mathbb{E}[\widehat{Z}_\rho \frac{1}{N}\sum_i f(X^i_{[\tau_i]})] > Z_\rho q_\rho(f)$. □

5 Proof of (ii)

Measured round $u \in \{1, \dots, M\}$ is a post-terminal round at $\varphi = 1$. Let $\kappa_j(\sigma) = \sigma^{-1}(g_j) - 1$, where g_j is the Gibbs-j label. Then $Y^i_{u,j} = X^i_{[\kappa_j(\sigma_i)]}$ in that round. By (i.c) with $f = o_j(\cdot, 1)$, and then Lemma 1.3 at $\varphi = 1$ (where $q_1 = \pi$),

$$\mathbb{E}\Big[\widehat{Z}\ \tfrac{1}{N}\textstyle\sum_i o_j(Y^i_{u,j}, 1)\Big] = Z\,\pi(o_j(\cdot, 1)) = Z\,\pi(J_j) = Z\,\mathrm{PIP}_j.$$

Averaging over $u = 1, \dots, M$ gives $\mathbb{E}_c[\widehat{Z}\,\widehat{\mathrm{PIP}}_j] = Z\,\mathrm{PIP}_j$. □

Remark 5.1. Lemma 1.3 is used only through $\pi(o_j) = \pi(J_j)$. The recorded value is a conditional probability at a state-independent prefix. So the identity holds whatever the position of the Gibbs-j label in the order, because that position is a function of σ_i alone.

6 Embedding in the method: conditioning on $\mathcal{C}$

Lemma 6.1 (freezing)**.** *Let $\mathcal{C}$ be a random element of a measurable space $(E, \mathcal{E})$, and let $U = (U_1, \dots, U_4)$ be a random element of $(\Upsilon, \mathcal{U})$ independent of $\mathcal{C}$. For every bounded measurable $g : E \times \Upsilon \to \mathbb{R}$, $\mathbb{E}[g(\mathcal{C}, U) \mid \mathcal{C}] = \bar{g}(\mathcal{C})$ a.s., where $\bar{g}(c) = \mathbb{E}[g(c, U)]$ is measurable.*

Proof. For $g = \mathbf{1}_{A \times B}$ and $C \in \mathcal{E}$: $\mathbb{E}[g(\mathcal{C}, U)\mathbf{1}_C(\mathcal{C})] = \mathbb{P}(\mathcal{C} \in A \cap C)\mathbb{P}(U \in B) = \mathbb{E}[\bar{g}(\mathcal{C})\mathbf{1}_C(\mathcal{C})]$, by independence, with $\bar{g}(c) = \mathbf{1}_A(c)\mathbb{P}(U \in B)$. The class of bounded measurable g for which $\bar{g}$ is measurable and the identity holds for all C contains the indicators of the π-system of measurable rectangles. It is closed under linear combinations and bounded monotone limits (monotone convergence). By the monotone class theorem it contains every bounded $\mathcal{E} \otimes \mathcal{U}$-measurable g. □

By CAL and EX-S, $\mathcal{C}$ (with the event $\mathcal{A} \in \sigma(\mathcal{C})$ that the final run executes) is a function of the data and streams 0–9. The final streams U_r (stream $9 + r$) are independent of each other and of streams 0–9, and are identically distributed. On $\mathcal{A}$, island r's record is $O_r = \Phi(\mathcal{C}, U_r)$. Here $\Phi(c, \cdot)$ realises $I(c)$ with the method's parameters ($N = N_e$, $M = 4$, $L' = L$, flag unset), and Φ is jointly measurable with values in a Polish space (part of EX-S in STATEMENT.md). Applying Lemma 6.1 to $g\,\mathbf{1}_{\mathcal{A}}(\mathcal{C})$, for every bounded measurable function g of $(O_1, \dots, O_4)$,

$$\mathbb{E}[g(O_1, \dots, O_4) \mid \mathcal{C}] = \mathbb{E}_c\big[g(O_1, \dots, O_4)\big]\big|_{c=\mathcal{C}} \quad \text{a.s. on } \mathcal{A}, \tag{1}$$

where on the right the four islands are independent copies of $I(c)$. Every conclusion stated for fixed c and for the four independent islands of $I(c)$ therefore holds given $\mathcal{C}$ on $\mathcal{A}$. This includes the identities of (i) and (ii), which hold for every c, and so hold almost surely given $\mathcal{C}$.

7 Proof of (iii)

Fix c and four independent islands of $I(c)$. By (i.a) and (ii) for each island, $\mathbb{E}_c[\widehat{Z}_{r,t}\widehat{q}_{r,t}(f)] = Z_t q_t(f)$ and $\mathbb{E}_c[\widehat{Z}_r\widehat{\mathrm{PIP}}_{rj}] = Z\mathrm{PIP}_j$. Hence $\mathbb{E}_c\widehat{Z} = \frac{1}{4}\sum_r \mathbb{E}_c\widehat{Z}_r = Z$. Each $\widehat{Z}_r$ and $\widehat{Z}$ takes values in the compact interval $[Z_0 e^{\min d}, Z_0 e^{\max d}] \subset (0, \infty)$, because each factor $\frac{1}{N}\sum_i e^{\Delta_t d(\cdot)}$ lies between $e^{\Delta_t \min d}$ and $e^{\Delta_t \max d}$ and $\sum_t \Delta_t = 1$. So $\log\widehat{Z}$ and $\log\widehat{Z}_r$ are bounded, and Jensen's inequality for the concave log gives $\mathbb{E}_c \log\widehat{Z} \le \log \mathbb{E}_c\widehat{Z} = \log Z$, and likewise for $\widehat{Z}_r$. For $0 < \alpha < 1$, Markov's inequality for the non-negative $\widehat{Z}$ gives $\mathbb{P}_c(\alpha\widehat{Z} > Z) \le \alpha\mathbb{E}_c\widehat{Z}/Z = \alpha$, that is $\mathbb{P}_c(Z \ge \alpha\widehat{Z}) \ge 1 - \alpha$; likewise for $\widehat{Z}_r$. By (1) applied to the bounded functions involved (indicators, $\widehat{Z}_{r,t}\widehat{q}_{r,t}(f)$, $\widehat{Z}_r\widehat{\mathrm{PIP}}_{rj}$, $\widehat{Z}$, $\log\widehat{Z}$, $\log\widehat{Z}_r$), all of these hold given $\mathcal{C}$ on $\mathcal{A}$. For the stage identities, fix t and apply (1) on $\mathcal{A} \cap \{T(\mathcal{C}) \ge t\}$ (an event in $\sigma(\mathcal{C})$). This gives $\mathbb{E}[\widehat{Z}_{r,t}\widehat{q}_{r,t}(f) \mid \mathcal{C}] = Z_t q_t(f)$, with Z_t, q_t computed on $\mathcal{C}$'s grid; the right side is a $\sigma(\mathcal{C})$-measurable random variable. □

8 Proof of (iv)

8.1 (iv.a) $\log\widehat{Z}$

If $d \equiv 0$ on Ω, then $G_t \equiv 1$, every factor of $\widehat{Z}$ is 1, and $\widehat{Z} = Z_0 = m(\emptyset)\bar{Z}_0 = m(\emptyset)\bar{Z}_1 = Z$ (Lemma 1.1, since $\gamma_1 = \gamma_0$).

Suppose $d \not\equiv 0$; since $d(\emptyset) = 0$, pick S with $d(S) \neq 0$. For $U \in \{\emptyset, S\}$, let E_U be the event that all stage-0 draws equal U, and that after every round preceding the stage-T resampling (the stage-0 round if the flag is set, and the stage-t rounds for $t \le T - 1$) every particle is at U. As in Step 2 of §4 (using Lemma 1.4(a) for U and the fact that a population entirely at U resamples

to U), $\mathbb{P}_c(E_U) \geq \pi_0(U)^N(\varepsilon_U/m)^{NT} > 0$, with at most T rounds involved. On E_U, every weight at stage t equals $e^{\Delta_t d(U)}$, so $\widehat{Z} = Z_0 e^{d(U)\sum_t \Delta_t} = Z_0 e^{d(U)}$. Thus $\widehat{Z}$ takes the two distinct values Z_0 and $Z_0 e^{d(S)}$ with positive probability, so $\mathbb{P}_c(\widehat{Z} \neq Z) > 0$. For the bounded positive $W = \widehat{Z}$ with $\mathbb{E}_c W = Z$, the inequality $\log w \leq \log Z + (w - Z)/Z$ holds for all $w > 0$, with equality only at $w = Z$. Taking expectations, with strict inequality on an event of positive probability, gives $\mathbb{E}_c \log \widehat{Z} < \log Z$. For the method's four islands given $\mathcal{C}$: by (1) the islands are independent copies of $I(\mathcal{C})$. On the event that all four islands are in their copy of E_U (conditional probability $\mathbb{P}_{\mathcal{C}}(E_U)^4 > 0$), $\widehat{Z} = \frac{1}{4}\sum_r \widehat{Z}_r = Z_0 e^{d(U)}$. So the same argument gives $\mathbb{E}[\log \widehat{Z} \mid \mathcal{C}] < \log Z$, and also $\mathbb{E}[\log \widehat{Z}_r \mid \mathcal{C}] < \log Z$. □

8.2 (iv.b) normalised PIPs

Given $\mathcal{C}$, the islands are identically distributed (by (1)), so $\mathbb{E}[\bar{P}_j \mid \mathcal{C}] = \mathbb{E}[\widehat{\mathrm{PIP}}_{1j} \mid \mathcal{C}]$. By (ii) and (i.a), $\mathbb{E}[\widehat{Z}_1 \widehat{\mathrm{PIP}}_{1j} \mid \mathcal{C}] = Z\mathrm{PIP}_j$ and $\mathbb{E}[\widehat{Z}_1 \mid \mathcal{C}] = Z$. Hence $\mathrm{Cov}(\widehat{\mathrm{PIP}}_{1j}, \widehat{Z}_1 \mid \mathcal{C}) = Z\mathrm{PIP}_j - Z\,\mathbb{E}[\widehat{\mathrm{PIP}}_{1j} \mid \mathcal{C}]$, which is the stated formula.

The instance. Take the appendix data of STATEMENT.md: $n = 20$, $p = 2$. As recorded there, $K = \min(2, 10) = 2$, so Ω is all four subsets. Further $B = \min(4, 1, 16) = 1$, so the blocks are $\{1\}, \{2\}$ and $r_b = \mathrm{BF}$ on singletons. The neighbour graph has $N(1) = \{2\}$ and $N(2) = \{1\}$, and $m = 5$, $L = \max(4, \lceil \log_2 21 \rceil) = 5$. The prior of STATEMENT §0 gives $w(k) = \frac{1}{2}/\binom{3}{k}$, that is $w(0) = \frac{1}{2}$ and $w(1) = w(2) = \frac{1}{6}$, and $W_K = \binom{2}{0}\frac{1}{2} + \binom{2}{1}\frac{1}{6} + \binom{2}{2}\frac{1}{6} = 1$.

The scores. By STATEMENT §0, C_n cancels in $\mathrm{BF}(S) = D_S^{-1/2}\big[(2b_0 + Q_S)/(2b_0 + y^\top y)\big]^{-\nu}$, with $\nu = a_0 + n/2 = 10.01$ and $a_0 = b_0 = 0.01$.

- For $S = \{j\}$: $G = x_j^\top x_j$, $H = (n+1)G + 1$, $D_S = H/(G+1)$, $Q_S = y^\top y - n(x_j^\top y)^2/H$.
- For $S = \{1, 2\}$: write g_{ab} for the entries of G_S, $z = (z_1, z_2) = X_S^\top y$ and $h_{ab} = (n+1)g_{ab} + \mathbf{1}\{a = b\}$. Then $\det H_S = h_{11}h_{22} - h_{12}^2$, $\det(G_S + I) = (g_{11}+1)(g_{22}+1) - g_{12}^2$, and $z^\top H_S^{-1} z = (h_{22}z_1^2 - 2h_{12}z_1z_2 + h_{11}z_2^2)/\det H_S$ (the 2×2 inverse).

These are exactly the expressions in `log_bf` of Appendix A. The function `standardise` centres each raw column and y and divides by the sample standard deviation with denominator $n - 1$, as in STATEMENT §0. So $\gamma_0 = (\frac{1}{2}, \frac{b_1}{6}, \frac{b_2}{6}, \frac{b_1 b_2}{6})$ and $\gamma_1 = (\frac{1}{2}, \frac{b_1}{6}, \frac{b_2}{6}, \frac{b_{12}}{6})$ on $(\emptyset, \{1\}, \{2\}, \{1, 2\})$, with $b_S = \mathrm{BF}(S)$. Moreover d vanishes except at $\{1, 2\}$, where $d(\{1, 2\}) = D := \log(b_{12}/(b_1 b_2))$. With $T = 1$ and $N = 1$, $\widehat{Z} = Z_0 e^{D\,B}$ with $B = \mathbf{1}\{X^{(0)} = \{1, 2\}\}$. Resampling of a single particle is trivial. The particle then performs $1 + L'$ rounds at $\varphi = 1$ before the $M = 4$ measured rounds. The recording $o_1(Y, 1)$ equals $c_0 = \gamma_1(\{1\})/(\gamma_1(\{1\}) + \gamma_1(\emptyset))$ if $2 \notin Y$, and $c_1 = \gamma_1(\{1, 2\})/(\gamma_1(\{1, 2\}) + \gamma_1(\{2\}))$ if $2 \in Y$. Hence $\widehat{\mathrm{PIP}}_1 = c_0 + (c_1 - c_0)k/4$, where k is the number of measured rounds whose pre-Gibbs-1 state contains 2.

Under the stage-0 flag, $X^{(0)} = X_0 K_0$ has law $\pi_0 K_0 = \pi_0$ (Lemma 1.3). Everything after stage 0 depends on the past only through $X^{(0)}$. Hence the joint law of (B, k), and so of $(\widehat{Z}, \widehat{\mathrm{PIP}}_1)$, is the same under both stage-0 readings. It is computed by propagating a law on triples (bit b, state x, count κ):

(1) start from $\mu_0(b, x, 0) = \pi_0(x)\mathbf{1}\{b = \mathbf{1}\{x = \{1, 2\}\}\}$;

(2) apply the round kernel K_1 to the state coordinate $1 + L'$ times;

(3) apply four measured rounds, each mapping (b, x, κ) to $(b, z, \kappa + \mathbf{1}\{2 \in y\})$ with weight $\mathsf{J}(x, y, z) = \frac{1}{120}\sum_\sigma [K_{\sigma(1)} \cdots K_{\sigma(\kappa_1(\sigma))}](x, y)\,[K_{\sigma(\kappa_1(\sigma)+1)} \cdots K_{\sigma(5)}](y, z)$, where y is the pre-Gibbs-1 state;

(4) marginalise over the final state.

This is the law of (B, k) by (R3) and Lemma 2.2. Within a round, the path splits at the pre-Gibbs-1 position into the prefix product and the suffix product of the same order. The 4×4 elementary kernels at $\varphi = 1$ are given by STATEMENT §1. At $p = 2$, $B = 1$ the uniform and neighbour swaps coincide: both are the Metropolis swap $\{1\} \leftrightarrow \{2\}$ with acceptance $\min\{1, b_2/b_1\}$ (resp. $\min\{1, b_1/b_2\}$), and they stay at $\emptyset$ and $\{1, 2\}$. The expectations then are finite sums: $\mathbb{E}[\bar{P}_1] = \mathbb{E}[\widehat{\mathrm{PIP}}_1]$, and $\mathbb{E}[\tilde{P}_1] = \sum \prod_{r=1}^{4} \mathbb{P}(B_r, k_r) \cdot \frac{\sum_r e^{DB_r} \widehat{\mathrm{PIP}}(k_r)}{\sum_r e^{DB_r}}$ over the 10^4 joint values of four independent islands.

We evaluated every quantity in 60-digit interval arithmetic (mpmath `iv`). Every input (n, the data, $a_0 = b_0 = 0.01$) enters as an enclosing interval, every operation is rigorous interval arithmetic, and $\min\{1, x\}$ is enclosed by $[\min(1, \underline{x}), \min(1, \overline{x})]$. The listing is in Appendix A. Every enclosure has width below 10^{-52}: the last `CHECK` line of each block of the log (Appendix B) subtracts the two exact endpoints of each enclosure in ordinary `mpf` arithmetic and compares the largest difference with the floating-point constant `1e-52`; that subtraction rounds by at most one part in 2^{53}, against a margin of a factor five, and no conclusion of the proof uses this width. The five decisive `CHECK` lines compare an exact interval endpoint with a floating-point constant whose rounding cannot cross the threshold it stands for. Rounded to 12 digits, they are:

	$L' = L$ (method; $1 + L' = 6$)	$L' = L - 1$ (generalisation; $1 + L' = 5$)
D	8.14676240634	
$a = \pi_0(\{1, 2\})$	0.0279141173179	
$c_0,\ c_1$	0.0694711126908, 0.998708337452	
PIP_1	0.990618012243	
$(\mathbb{E}\widehat{Z} - Z)/Z_0$	interval $\ni 0$	interval $\ni 0$
$(\mathbb{E}[\widehat{Z}\widehat{\mathrm{PIP}}_1] - Z\mathrm{PIP}_1)/Z_0$	interval $\ni 0$	interval $\ni 0$
$\mathbb{E}[\bar{P}_1] - \mathrm{PIP}_1$	-0.173974177850	-0.212065526919
$\mathbb{E}[\tilde{P}_1] - \mathrm{PIP}_1$	-0.159677356166	-0.194638440548

The first two difference rows check (i.a) and (ii) on the instance; they are not used in the argument. In both columns and under both stage-0 readings, the biases are certified to be below -0.17 (primary) and -0.15 (secondary). *Scope.* This is a component example. At $p = 2$ the full method would take the exact branch, and $N = 1$ is not the production budget. It shows that the frozen SMC component's normalised PIP estimators are biased, not that the full method's output on this input is. □

8.3 (iv.c) island independence

By (1) with $g = \prod_r h_r(O_r)$, $\mathbb{E}[\prod_r h_r(O_r) \mid \mathcal{C}] = \prod_r \psi_{h_r}(\mathcal{C})$ on $\mathcal{A}$, where $\psi_h(c) = \mathbb{E}_c[h(O)]$ is the same function for every island. This is conditional independence with conditional law that of $I(\mathcal{C})$. Assume $\mathbb{P}(\mathcal{A}) > 0$, so that $\mathbb{P}(\cdot \mid \mathcal{A})$ is defined. Under $\mathbb{P}(\cdot \mid \mathcal{A})$ and for bounded h: $\mathbb{E}[h(O_1)h(O_2)] = \mathbb{E}[\psi_h(\mathcal{C})^2]$ and $\mathbb{E}h(O_r) = \mathbb{E}\psi_h(\mathcal{C})$, so $\mathrm{Cov}(h(O_1), h(O_2)) = \mathrm{Var}(\psi_h(\mathcal{C}))$.

Characterisation. The record O_r is a finite collection of real and finite-valued variables, so it takes values in a Polish space O. Choose a countable π-system $\{B_k\}$ generating its Borel σ-field. If O_1, O_2 are independent under $\mathbb{P}(\cdot \mid \mathcal{A})$, then $\mathrm{Var}(\psi_{\mathbf{1}_{B_k}}(\mathcal{C})) = \mathrm{Cov}(\mathbf{1}_{B_k}(O_1), \mathbf{1}_{B_k}(O_2)) = 0$ for every k. So, outside one null set, $\mathbb{P}(O_1 \in B_k \mid \mathcal{C}) = \mu(B_k)$ for all k, where μ is the law of O_1 under $\mathbb{P}(\cdot \mid \mathcal{A})$. Two probability measures that agree on a generating π-system are equal, so the conditional law is a.s. μ. Conversely, if the conditional law is a.s. a fixed μ, then $\mathbb{E}[h_1(O_1)h_2(O_2)] = \mathbb{E}[\psi_{h_1}\psi_{h_2}(\mathcal{C})] =$

$\mu(h_1)\mu(h_2) = \mathbb{E}h_1(O_1)\mathbb{E}h_2(O_2)$; so the islands are independent, and the same holds for all four by the same computation. None of the hypotheses restricts the law of $\mathcal{C}$ beyond its measurability in streams 0–9. So they do not exclude, for example, a calibration law that puts positive mass on two values $c \neq c'$ with different island laws, and then the islands are dependent under $\mathbb{P}(\cdot \mid \mathcal{A})$. The next paragraph shows such pairs exist: (grid, N) and $(\mathrm{grid}, 4N)$ on any input where d has a unique maximiser. The instance of (iv.b) is one, since there $d = (0, 0, 0, D)$ with $D > 0$. Whether the method's actual calibration procedure produces such a law on some input is not claimed.

The island law depends on c. Suppose d has a unique maximiser S^* on Ω, and let $d^* = d(S^*)$. Then $\widehat{Z}/Z_0 = \prod_{t=1}^{T} \frac{1}{N}\sum_i e^{\Delta_t d(X^{(t-1),i})} \leq e^{d^*}$, with equality iff $X^{(t-1),i} = S^*$ for all i and all $t = 1, \ldots, T$. This event has probability a^N, where $a = \pi_0(S^*)\prod_{t=1}^{T-1} K_{\varphi_t}(S^*, \{S^*\})$ does not depend on N. Here we used (R1)–(R3): $X^{(0),i}$ has law π_0 under either stage-0 reading (by Lemma 1.3), independently across i; a population at S^* resamples to S^*; and the rounds act independently across particles. Now $0 < a \leq \pi_0(S^*) < 1$ by Lemma 1.4(a) and $|\Omega| \geq 2$. So $\mathbb{P}_{(\mathrm{grid},N)}(\widehat{Z} = Z_0 e^{d^*}) = a^N \neq a^{4N} = \mathbb{P}_{(\mathrm{grid},4N)}(\widehat{Z} = Z_0 e^{d^*})$. □

8.4 (iv.d) launch events

Let $A \in \sigma(\mathcal{C})$, $A \subseteq \mathcal{A}$, $\mathbb{P}(A) > 0$. By the tower property, $\mathbb{E}[Y\mathbf{1}_A] = \mathbb{E}[\mathbb{E}[Y \mid \mathcal{C}]\mathbf{1}_A]$ for every bounded Y. For the conclusions of (iii) whose right side v is a constant (fixed by the data), this gives $\mathbb{E}[Y \mid A] = v$ for $Y = \widehat{Z}$, $Y = \widehat{Z}_r$ and $Y = \widehat{Z}_r\widehat{\mathrm{PIP}}_{rj}$ (with $v = Z$, Z, $Z\mathrm{PIP}_j$). Likewise $\mathbb{E}[\log\widehat{Z} \mid A] \leq \log Z$ and $\mathbb{E}[\log\widehat{Z}_r \mid A] \leq \log Z$. With $Y = \mathbf{1}\{Z \geq \alpha\widehat{Z}\}$ (resp. $\widehat{Z}_r$), $\mathbb{P}(Z \geq \alpha\widehat{Z} \mid A) = \mathbb{E}[\mathbb{P}(Z \geq \alpha\widehat{Z} \mid \mathcal{C})\mathbf{1}_A]/\mathbb{P}(A) \geq 1 - \alpha$. For an intermediate stage t and $A \subseteq \{T(\mathcal{C}) \geq t\}$, the right side $Z_t q_t(f)$ of (iii) is a $\sigma(\mathcal{C})$-measurable random variable. The same computation then gives only $\mathbb{E}[\widehat{Z}_{r,t}\widehat{q}_{r,t}(f)\mathbf{1}_A] = \mathbb{E}[Z_t q_t(f)\mathbf{1}_A]$, which is the averaged form. The left side divided by $\mathbb{P}(A)$ is a number. The right side's integrand $Z_t q_t(f)$ is non-constant when, for example, φ_t takes two values $u \neq u'$ with positive probability under $\mathbb{P}(\cdot \mid A)$ and $f = \mathbf{1}\{S\}$ with $d(S) \neq 0$. Indeed $Z_t q_t(\mathbf{1}\{S\}) = m(\emptyset)\gamma_0(S)e^{\varphi_t d(S)}$ by the definitions, and this takes different values at u and u'. □

8.5 (iv.e) favourable final diagnostics

In the instance of (iv.b), $\log\widehat{Z}_r = \log Z_0 + DB_r$ with B_r iid Bernoulli(a) across the four independent islands. If exactly $k \in \{1, 2, 3\}$ of the B_r equal 1, the four values DB_r have standard deviation $D\sqrt{k(4-k)}/4 \geq D\sqrt{3}/4$ (normalisation $1/4$) or $D\sqrt{k(4-k)/12} \geq D/2$ (normalisation $1/3$). Both exceed 1 because $D > 8 > 4/\sqrt{3}$. If $k \in \{0, 4\}$ the standard deviation is 0. Hence $F = \{B_1 = B_2 = B_3 = B_4\}$ under either convention, and $\mathbb{P}(F) = (1-a)^4 + a^4 > 0$. The four islands are iid and F is invariant under permuting them, so the pairs $(\widehat{Z}_r, \widehat{\mathrm{PIP}}_{r1})$ have the same law under $\mathbb{P}(\cdot \mid F)$ for every r. Hence $\mathbb{E}[\widehat{Z}_r \mid F] = \mathbb{E}[\widehat{Z} \mid F] = \mathbb{E}[\widehat{Z}_1 \mid F]$ and $\mathbb{E}[\widehat{Z}_r\widehat{\mathrm{PIP}}_{r1} \mid F] = \mathbb{E}[\widehat{Z}_1\widehat{\mathrm{PIP}}_{11} \mid F]$ for every r. Moreover

$$\mathbb{E}[\widehat{Z}_1 \mid F] = Z_0\,\frac{(1-a)^4 + a^4 e^D}{(1-a)^4 + a^4} = Z_0\big(1 + (e^D - 1)\theta\big), \qquad \theta = \frac{a^4}{(1-a)^4 + a^4}.$$

Since $a < \frac{1}{2}$, $\theta < a$ (because $a^4 < a(1-a)^4 + a^5 \iff a^3 < (1-a)^3$). Also $e^D > 1$ and $Z = \mathbb{E}\widehat{Z}_1 = Z_0(1 + (e^D - 1)a)$. Hence $\mathbb{E}[\widehat{Z}_1 \mid F] < Z$ and $\mathbb{E}[\widehat{Z} \mid F] < Z$. So the identity $\mathbb{E}\widehat{Z}_r = Z$ of (i) and the identity $\mathbb{E}\widehat{Z} = Z$ of (iii) fail under $\mathbb{P}(\cdot \mid F)$. For (ii),

$$\mathbb{E}[\widehat{Z}_1\widehat{\mathrm{PIP}}_1 \mid F] = \frac{\sum_{b,k}\mathbb{P}(B = b, k)\, Z_0 e^{Db}\,\widehat{\mathrm{PIP}}(k)\,\mathbb{P}(B = b)^3}{(1-a)^4 + a^4},$$

which the same interval computation evaluates as $(\mathbb{E}[\widehat{Z}_1\widehat{\mathrm{PIP}}_1 \mid F] - Z\mathrm{PIP}_1)/Z_0 = -95.609622724\ldots$ ($L' = L$) and $-95.648818292\ldots$ ($L' = L-1$), under either stage-0 reading; both are certified nonzero. This is a component calculation (scope as in (iv.b)). It breaks an identity in each of (i), (ii) and (iii). It does not break the Jensen or Markov inequalities of (iii): since $\mathbb{E}[\widehat{Z} \mid F] < Z$, they still hold under $\mathbb{P}(\cdot \mid F)$. □

8.6 (iv.f) NUM alone

Fix $\eta > 0$ and the NUM-compatible perturbed-score model in which every evaluated $\log m(S)$ is replaced by $\log m(S) + \eta$. (This is a model NUM does not exclude, not a claim about a particular floating-point implementation.) It satisfies the NUM bound with equality. Every quantity the run uses other than the prefactor $m(\emptyset)$ depends on scores only through ratios $m(S)/m(\emptyset)$ or $m(S)/m(S')$: the tables r_b, BF, d, γ_φ up to the common factor ϖ, π_0, $\bar{Z}_0$, all acceptance and Gibbs probabilities, and the weights. These are unchanged, so the run has the same law, and $\widehat{Z}^{\mathrm{impl}} = e^\eta m(\emptyset)\bar{Z}_0 \prod_t(\cdots) = e^\eta \widehat{Z}$. By (i.a), $\mathbb{E}_c\widehat{Z}^{\mathrm{impl}} = e^\eta Z \neq Z$. If $d \equiv 0$ (for example $p = 1$: a single block $\{1\}$, so $d(S) = \log \mathrm{BF}(S) - \log r_1(S) = 0$), then $\widehat{Z} = Z$ surely by (iv.a), so $\widehat{Z}^{\mathrm{impl}} = e^\eta Z$, and $\mathbb{P}(Z \geq \alpha\widehat{Z}^{\mathrm{impl}}) = \mathbf{1}\{\alpha e^\eta \leq 1\} = 0$ for $\alpha \in (e^{-\eta}, 1)$. So NUM alone gives neither unbiasedness nor the Markov bound. □

A Listing of the interval computation

The certified values of §8.2 and §8.5 are produced by `certify_instance.py iv`, which calls `p2_exact.py` with `IvArith(60)` (mpmath 1.3.0 interval arithmetic at 60 digits). The listings follow.

p2_exact.py

```
"""Exact law of one frozen-run island for a p = 2 component input, N = 1, T = 1.

Used for the explicit instances of Part (iv) of T12.  Arithmetic is generic:
pass `ar` = FloatArith() for exploration or IvArith() (mpmath interval
arithmetic) for the certified computation.  States are indexed
0 = {}, 1 = {1}, 2 = {2}, 3 = {1,2}.
"""
import itertools
import sys
import mpmath


class FloatArith:
    def c(self, x): # exact constant from a string or rational
        return float(mpmath.mpf(x))
    def log(self, x): return float(mpmath.log(x))
    def exp(self, x): return float(mpmath.exp(x))
    def sqrt(self, x): return float(mpmath.sqrt(x))
    def min1(self, x): return min(1.0, x)
    def contains_zero(self, x): return x == 0.0
    def upper(self, x): return x
    def lower(self, x): return x
    def show(self, x): return repr(x)


class IvArith:
    def __init__(self, dps=60):
        mpmath.iv.dps = dps
        self.iv = mpmath.iv
```

```
    def c(self, x): return self.iv.mpf(x)
    def log(self, x): return self.iv.log(x)
    def exp(self, x): return self.iv.exp(x)
    def sqrt(self, x): return self.iv.sqrt(x)
    def min1(self, x):
        # enclosure of min(1, x) for an interval x
        lo = min(mpmath.mpf(1), x.a)
        hi = min(mpmath.mpf(1), x.b)
        return self.iv.mpf([lo, hi])
    def ends(self, x):
        # exact endpoints (mpf) of the enclosure x
        return mpmath.mp.make_mpf(x._mpi_[0]), mpmath.mp.make_mpf(x._mpi_[1])
    def contains_zero(self, x):
        lo, hi = self.ends(x)
        return lo <= 0 <= hi
    def upper(self, x): return self.ends(x)[1]
    def lower(self, x): return self.ends(x)[0]
    def show(self, x):
        # the two endpoints of the enclosure, rounded to 25 significant digits for display
        lo, hi = self.ends(x)
        return "[" + mpmath.nstr(lo, 25) + ",␣" + mpmath.nstr(hi, 25) + "]"


def standardise(ar, cols):
    n = len(cols[0])
    out = []
    for col in cols:
        col = [ar.c(v) for v in col]
        mean = sum(col) / n
        cen = [v - mean for v in col]
        ss = sum(v * v for v in cen)
        sd = ar.sqrt(ss / (n - 1))
        out.append([v / sd for v in cen])
    return out


def dot(a, b):
    return sum(x * y for x, y in zip(a, b))


def log_bf(ar, n, X, y, S):
    """log␣BF(S)␣=␣log␣m(S)␣-␣log␣m({})␣for␣the␣v3␣target␣(C_n␣cancels)."""
    a0 = ar.c("0.01"); b0 = ar.c("0.01")
    nu = a0 + ar.c(n) / 2
    yy = dot(y, y)
    if not S:
        return ar.c(0)
    nn = ar.c(n)
    if len(S) == 1:
        j = S[0]
        G = dot(X[j], X[j]); z = dot(X[j], y)
        H = (nn + 1) * G + 1
        logD = ar.log(H) - ar.log(G + 1)
        t = nn * z * z / H
    else:
        i, j = S
        g11 = dot(X[i], X[i]); g22 = dot(X[j], X[j]); g12 = dot(X[i], X[j])
        z1 = dot(X[i], y); z2 = dot(X[j], y)
        h11 = (nn + 1) * g11 + 1; h22 = (nn + 1) * g22 + 1; h12 = (nn + 1) * g12
        detH = h11 * h22 - h12 * h12
        detG1 = (g11 + 1) * (g22 + 1) - g12 * g12
        logD = ar.log(detH) - ar.log(detG1)
        # z^T H^{-1} z for 2x2
        quad = (h22 * z1 * z1 - 2 * h12 * z1 * z2 + h11 * z2 * z2) / detH
        t = nn * quad
    Q = yy - t
    return -logD / 2 - nu * (ar.log(2 * b0 + Q) - ar.log(2 * b0 + yy))
```

```
STATES = [(), (0,), (1,), (0, 1)]
IDX = {s: k for k, s in enumerate(STATES)}


def setup(ar, n, xcols, ycol):
    X = standardise(ar, xcols)
    y = standardise(ar, [ycol])[0]
    lbf = [log_bf(ar, n, X, y, list(s)) for s in STATES]
    bf = [ar.exp(v) for v in lbf]
    # prior: p = 2, K = min(2, n//2) = 2 for n >= 4. w(k) = (1/2)/C(2p-1, k) = (1/2)/C(3, k),
    # i.e. w = (1/2, 1/6, 1/6) for k = 0, 1, 2, and W_K = sum_k C(2,k) w(k) = 1/2 + 2/6 + 1/6 = 1.
    assert n >= 4
    w = [ar.c(1) / 2, ar.c(1) / 6, ar.c(1) / 6, ar.c(1) / 6] # by state (sizes 0,1,1,2)
    # blocks {1},{2} (B = 1): r_1 = BF({1}), r_2 = BF({2})
    prod_r = [ar.c(1), bf[1], bf[2], bf[1] * bf[2]]
    d = [lbf[k] - ar.log(prod_r[k]) if k else ar.c(0) for k in range(4)]
    d[1] = ar.c(0); d[2] = ar.c(0) # single-block models: d = 0 exactly
    g0 = [w[k] * prod_r[k] for k in range(4)]
    return dict(n=n, lbf=lbf, bf=bf, w=w, d=d, g0=g0)


def gamma(ar, P, phi, k):
    return P["g0"][k] * ar.exp(phi * P["d"][k])


def zeros(ar):
    return [[ar.c(0)] * 4 for _ in range(4)]


def kernels(ar, P, phi):
    """Elementary␣kernels␣at␣temperature␣phi:␣G1,␣G2,␣US,␣NS,␣IND␣(p␣=␣2,␣K␣=␣2,␣B␣=␣1)."""
    gm = [gamma(ar, P, phi, k) for k in range(4)]
    Z0bar = sum(P["g0"])
    pi0 = [g / Z0bar for g in P["g0"]]
    Ks = {}
    for j in (0, 1): # Gibbs on coordinate j
        M = zeros(ar)
        for s in STATES:
            plus = tuple(sorted(set(s) | {j})); minus = tuple(sorted(set(s) - {j}))
            ip, im = IDX[plus], IDX[minus]
            o = gm[ip] / (gm[ip] + gm[im]) # |plus| <= 2 = K always feasible
            M[IDX[s]][ip] = M[IDX[s]][ip] + o
            M[IDX[s]][im] = M[IDX[s]][im] + (1 - o)
        Ks["G%d" % (j + 1)] = M
    # uniform swap: self-loop at |S| in {0, 2}; {1} <-> {2}
    # neighbour swap with N(1) = {2}, N(2) = {1}: from {j} propose {l}, ratio * 1/1;
    # at {1,2}: N(j)\S empty -> reject; at {}: self-loop. Same kernel as US here.
    for name in ("US", "NS"):
        M = zeros(ar)
        M[0][0] = ar.c(1); M[3][3] = ar.c(1)
        a12 = ar.min1(gm[2] / gm[1]); a21 = ar.min1(gm[1] / gm[2])
        M[1][2] = a12; M[1][1] = 1 - a12
        M[2][1] = a21; M[2][2] = 1 - a21
        Ks[name] = M
    M = zeros(ar)
    for x in range(4):
        rej = ar.c(0)
        for yv in range(4):
            if yv == x:
                continue
            acc = ar.min1(ar.exp(phi * (P["d"][yv] - P["d"][x])))
            M[x][yv] = pi0[yv] * acc
            rej = rej + pi0[yv] * (1 - acc)
        M[x][x] = pi0[x] + rej
    Ks["IND"] = M
    return Ks, pi0, gm
```

```
def matmul(ar, A, B):
    return [[sum((A[i][k] * B[k][j] for k in range(4)), ar.c(0)) for j in range(4)] for i in range(4)]


def eye(ar):
    return [[ar.c(1) if i == j else ar.c(0) for j in range(4)] for i in range(4)]


def round_objects(ar, Ks, label):
    """Round␣kernel␣K␣=␣avg␣over␣orders␣of␣products,␣and␣the␣joint␣kernel
␣␣␣␣J[x][y][z]␣=␣avg_sigma␣Prefix_sigma(x,y)␣*␣Suffix_sigma(y,z),␣where␣the␣prefix
␣␣␣␣stops␣immediately␣before␣the␣step␣`label`."""
    names = ["G1", "G2", "US", "NS", "IND"]
    perms = list(itertools.permutations(names))
    K = zeros(ar)
    J = [[[ar.c(0)] * 4 for _ in range(4)] for _ in range(4)]
    for perm in perms:
        pos = perm.index(label)
        pre = eye(ar)
        for nm in perm[:pos]:
            pre = matmul(ar, pre, Ks[nm])
        suf = eye(ar)
        for nm in perm[pos:]:
            suf = matmul(ar, suf, Ks[nm])
        full = matmul(ar, pre, suf)
        for i in range(4):
            for j in range(4):
                K[i][j] = K[i][j] + full[i][j]
                for k in range(4):
                    J[i][j][k] = J[i][j][k] + pre[i][j] * suf[j][k]
    m = len(perms)
    K = [[v / m for v in row] for row in K]
    J = [[[v / m for v in r2] for r2 in r1] for r1 in J]
    return K, J


def island_law(ar, P, Lprime, stage0_round, j=0, M_meas=4):
    """Joint␣law␣of␣(B,␣k)␣for␣one␣island,␣N␣=␣1,␣T␣=␣1,␣grid␣(0,␣1).
␣␣␣␣B␣=␣1{X^(0)␣=␣{1,2}}␣(so␣Zhat␣=␣m(0)␣Z0bar␣e^{D␣B});␣k␣=␣number␣of␣measured
␣␣␣␣rounds␣whose␣pre-Gibbs-j␣state␣has␣o_j␣=␣c1␣(i.e.␣contains␣the␣other␣variable).
␣␣␣␣Returns␣dict␣{(B,␣k):␣prob},␣c0,␣c1,␣pi0,␣pi."""
    K0s, pi0, _ = kernels(ar, P, ar.c(0))
    K1s, _, g1 = kernels(ar, P, ar.c(1))
    label = "G%d" % (j + 1)
    K1, J1 = round_objects(ar, K1s, label)
    other = 1 - j
    # o_j at state: depends on whether `other` in S
    def o_j(s):
        plus = tuple(sorted(set(s) | {j})); minus = tuple(sorted(set(s) - {j}))
        return g1[IDX[plus]] / (g1[IDX[plus]] + g1[IDX[minus]])
    c0 = o_j(()); c1 = o_j((other,))
    has_other = [other in s for s in STATES]
    # initial
    dist0 = list(pi0)
    if stage0_round:
        K0, _ = round_objects(ar, K0s, label)
        dist0 = [sum((dist0[x] * K0[x][z] for x in range(4)), ar.c(0)) for z in range(4)]
    # law over (B, state, k)
    law = {}
    for x in range(4):
        b = 1 if x == 3 else 0
        law[(b, x, 0)] = law.get((b, x, 0), ar.c(0)) + dist0[x]
    def apply_K(law, Kmat):
        new = {}
        for (b, x, k), pr in law.items():
            for z in range(4):
                key = (b, z, k)
```

```
                new[key] = new.get(key, ar.c(0)) + pr * Kmat[x][z]
        return new
    for _ in range(1 + Lprime): # stage-1 round + L' further rounds at phi = 1
        law = apply_K(law, K1)
    for _ in range(M_meas):
        new = {}
        for (b, x, k), pr in law.items():
            for yv in range(4):
                kk = k + (1 if has_other[yv] else 0)
                for z in range(4):
                    key = (b, z, kk)
                    new[key] = new.get(key, ar.c(0)) + pr * J1[x][yv][z]
        law = new
    out = {}
    for (b, x, k), pr in law.items():
        out[(b, k)] = out.get((b, k), ar.c(0)) + pr
    Z1 = sum(g1)
    pi = [g / Z1 for g in g1]
    return out, c0, c1, pi0, pi


def analyse(ar, n, xcols, ycol, Lprime, stage0_round, j=0):
    P = setup(ar, n, xcols, ycol)
    law, c0, c1, pi0, pi = island_law(ar, P, Lprime, stage0_round, j)
    D = P["d"][3]
    eD = ar.exp(D)
    # Zhat / Z0 = e^{D B}; Z / Z0 = Zbar_1 / Zbar_0
    Z_over_Z0 = sum(gamma(ar, P, ar.c(1), k) for k in range(4)) / sum(P["g0"])
    PIP = pi[IDX[(j,)]] + pi[3]
    def pip_hat(k): return (k * c1 + (4 - k) * c0) / 4
    tot = sum(law.values())
    EZP = sum((pr * (eD if b else 1) * pip_hat(k) for (b, k), pr in law.items()), ar.c(0))
    EZ = sum((pr * (eD if b else 1) for (b, k), pr in law.items()), ar.c(0))
    EP = sum((pr * pip_hat(k) for (b, k), pr in law.items()), ar.c(0))
    a = pi0[3]
    res = dict(D=D, a=a, c0=c0, c1=c1, PIP=PIP, total_prob=tot,
              identity_Z=EZ - Z_over_Z0, # should be 0 (part i)
              identity_ZP=EZP - Z_over_Z0 * PIP, # should be 0 (part ii)
              primary_bias=EP - PIP)
    # secondary over 4 iid islands: sum_r e^{D B_r} P_r / sum_r e^{D B_r}
    items = list(law.items())
    sec = ar.c(0)
    for combo in itertools.product(items, repeat=4):
        pr = combo[0][1] * combo[1][1] * combo[2][1] * combo[3][1]
        num = sum(((eD if b else 1) * pip_hat(k) for ((b, k), _) in combo), ar.c(0))
        den = sum(((eD if b else 1) for ((b, k), _) in combo), ar.c(0))
        sec = sec + pr * num / den
    res["secondary_bias"] = sec - PIP
    # conditioning on F = {B_1 = ... = B_4}: E[Zhat_1 P_1 | F], E[Zhat_1 | F]
    pB = {0: sum((pr for (b, k), pr in law.items() if b == 0), ar.c(0)),
          1: sum((pr for (b, k), pr in law.items() if b == 1), ar.c(0))}
    PF = pB[0] ** 4 + pB[1] ** 4
    EZ_F = (pB[0] ** 4 + pB[1] ** 4 * eD) / PF
    EZP_F = sum((pr * (eD if b else 1) * pip_hat(k) * pB[b] ** 3 for (b, k), pr in law.items()), ar.c(0)) / PF
    res["condF_Z_minus_Z"] = EZ_F - Z_over_Z0
    res["condF_ZP_minus_ZPIP"] = EZP_F - Z_over_Z0 * PIP
    # conditional Jensen / Markov under F
    res["condF_Elog_minus_logZ"] = (pB[1] ** 4 * D) / PF - ar.log(Z_over_Z0)
    return res


if __name__ == "__main__":
    print("Library␣module;␣run␣certify_instance.py␣(add␣the␣argument␣iv␣for␣the␣certified␣run).")
```

certify_instance.py

```
"""Certified␣(interval-arithmetic)␣evaluation␣of␣the␣T12␣Part␣(iv)␣instance.
Data:␣n␣=␣20,␣i␣=␣0..19,␣x1_i␣=␣(7i␣mod␣20)␣-␣9.5,␣x2_i␣=␣x1_i␣+␣((5␣i^2␣mod␣7)␣-␣3),
y_i␣=␣x1_i␣-␣x2_i␣+␣((3i␣mod␣5)␣-␣2).␣␣Variable␣j␣=␣1␣(index␣0)."""
import sys; sys.path.insert(0, '.')
import mpmath
from p2_exact import IvArith, FloatArith, analyse
from fractions import Fraction
n = 20
x1 = [str(Fraction(2*((7*i) % 20) - 19, 2)) for i in range(n)]
x2 = [str(Fraction(2*((7*i) % 20) - 19, 2) + ((5*i*i) % 7) - 3) for i in range(n)]
y = [str(Fraction(x1[i]) - Fraction(x2[i]) + ((3*i) % 5) - 2) for i in range(n)]
print("x1␣=", x1); print("x2␣=", x2); print("y␣␣=", y)
ar = IvArith(60) if (len(sys.argv) > 1 and sys.argv[1] == 'iv') else FloatArith()
for stage0 in (False, True):
    for Lp in (4, 5):
        r = analyse(ar, n, [x1, x2], y, Lp, stage0)
        print("stage0_round=%s␣rounds_before_measured=%d" % (stage0, 1 + Lp))
        for k, v in r.items():
            z = ar.contains_zero(v)
            print("␣␣␣␣%-24s␣%s%s" % (k, ar.show(v), "␣␣␣␣[interval␣contains␣0]" if z else ""))
        # certified checks used in the proof (exact endpoint comparisons)
        print("␣␣␣␣CHECK␣primary_bias␣upper␣endpoint␣<␣-0.17␣:", ar.upper(r["primary_bias"]) < mpmath.mpf("
            -0.1701")) # mpf("-0.1701") < -0.17 whatever its rounding
        print("␣␣␣␣CHECK␣secondary_bias␣upper␣endpoint␣<␣-0.15␣:", ar.upper(r["secondary_bias"]) < mpmath.mpf("
            -0.1501")) # mpf("-0.1501") < -0.15
        print("␣␣␣␣CHECK␣D␣lower␣endpoint␣>␣4/sqrt(3)␣(uses␣D␣>␣2.31)␣:", ar.lower(r["D"]) > mpmath.mpf("2.31"))
             # 2.31 > 4/sqrt(3) = 2.3094...
        print("␣␣␣␣CHECK␣a␣upper␣endpoint␣<␣1/2␣:", ar.upper(r["a"]) < mpmath.mpf("0.5"))
        print("␣␣␣␣CHECK␣condF_ZP_minus_ZPIP␣upper␣endpoint␣<␣0␣:", ar.upper(r["condF_ZP_minus_ZPIP"]) < 0)
        if not isinstance(ar, FloatArith):
            wmax = max(ar.upper(v) - ar.lower(v) for v in r.values())
            print("␣␣␣␣CHECK␣max␣enclosure␣width␣over␣all␣rows␣<␣1e-52␣:", wmax < mpmath.mpf("1e-52"), "(max␣
                width␣=␣%s)" % mpmath.nstr(wmax, 5))
```

B Output of the certified run

The output of `python -I certify_instance.py iv` (file `helpers/certify_iv.log`) follows. Each enclosure is printed as its two exact endpoints, rounded to 25 digits for display. The `CHECK` lines compare exact endpoints. The thresholds -0.1701 and -0.1501 lie below -0.17 and -0.15 whatever their binary rounding, and $2.31 > 4/\sqrt{3}$. Rows `identity_Z` and `identity_ZP` are $(\mathbb{E}\widehat{Z} - Z)/Z_0$ and $(\mathbb{E}[\widehat{Z}\widehat{\mathrm{PIP}}_1] - Z\mathrm{PIP}_1)/Z_0$. Rows `condF_Z_minus_Z` and `condF_ZP_minus_ZPIP` are $(\mathbb{E}[\widehat{Z}_1 \mid F] - Z)/Z_0$ and $(\mathbb{E}[\widehat{Z}_1\widehat{\mathrm{PIP}}_{11} \mid F] - Z\mathrm{PIP}_1)/Z_0$. Row `condF_Elog_minus_logZ` is the unscaled difference $\mathbb{E}[\log \widehat{Z}_1 \mid F] - \log Z$ (not used in the proof). The last `CHECK` line of each block computes, for every enclosure in that block, the difference of its two exact endpoints in ordinary `mpf` arithmetic and compares the largest with the floating-point constant `1e-52`; this is a display check and is not used by the proof. `rounds_before_measured`$= 1 + L'$.

```
x1 = ['-19/2', '-5/2', '9/2', '-17/2', '-3/2', '11/2', '-15/2', '-1/2', '13/2', '-13/2', '1/2', '15/2',
    '-11/2', '3/2', '17/2', '-9/2', '5/2', '19/2', '-7/2', '7/2']
x2 = ['-25/2', '-1/2', '15/2', '-17/2', '-3/2', '17/2', '-11/2', '-7/2', '17/2', '-7/2', '1/2', '15/2',
    '-5/2', '7/2', '11/2', '-5/2', '11/2', '19/2', '-7/2', '13/2']
y = ['1', '-1', '-4', '2', '0', '-5', '-1', '2', '0', '-3', '-2', '1', '-4', '0', '3', '-4', '-2', '-1', '2',
    '-3']
stage0_round=False rounds_before_measured=5
  D [8.1467624063406823962028 85, 8.1467624063406823962028 85]
  a [0.027914117317939631485254 63, 0.027914117317939631485254 63]
  c0 [0.069471112690807603572169 58, 0.069471112690807603572169 58]
  c1 [0.998708337451837812310299, 0.998708337451837812310299]
  PIP [0.990618012243358205054783 3, 0.990618012243358205054783 3]
  total_prob [1.0, 1.0]
```

```
   identity_Z [-8.662447223607153834171269e-54, 8.663393121929388727710754e-54] [interval contains 0]
   identity_ZP [-7.861659660827537541469239e-54, 7.862585645500883279355262e-54] [interval contains 0]
   primary_bias [-0.212065526918593961312 9795, -0.212065526918593961312 9795]
   secondary_bias [-0.194638440548470411723 3927, -0.194638440548470411723 3927]
   condF_Z_minus_Z [-96.3344135496253510853 4537, -96.3344135496253510853 4537]
   condF_ZP_minus_ZPIP [-95.6488182922841802955 4233, -95.6488182922841802955 4233]
   condF_Elog_minus_logZ [-4.578171180932416623428288, -4.578171180932416623428288]
   CHECK primary_bias upper endpoint < -0.17 : True
   CHECK secondary_bias upper endpoint < -0.15 : True
   CHECK D lower endpoint > 4/sqrt(3) (uses D > 2.31) : True
   CHECK a upper endpoint < 1/2 : True
   CHECK condF_ZP_minus_ZPIP upper endpoint < 0 : True
   CHECK max enclosure width over all rows < 1e-52 : True (max width = 1.7326e-53)
stage0_round=False rounds_before_measured=6
   D [8.146762406340682396202885, 8.146762406340682396202885]
   a [0.027914117317939631485254 63, 0.027914117317939631485254 63]
   c0 [0.069471112690807603572169 58, 0.069471112690807603572169 58]
   c1 [0.998708337451837812310299, 0.998708337451837812310299]
   PIP [0.990618012243358205054783 3, 0.990618012243358205054783 3]
   total_prob [1.0, 1.0]
   identity_Z [-9.598727253428585193034604e-54, 9.599792633644155020494868e-54] [interval contains 0]
   identity_ZP [-8.709373737214449131556501e-54, 8.710439117430018959016764e-54] [interval contains 0]
   primary_bias [-0.173974177849890505703 3109, -0.173974177849890505703 3109]
   secondary_bias [-0.159677356166443932157 8109, -0.159677356166443932157 8109]
   condF_Z_minus_Z [-96.3344135496253510853 4537, -96.3344135496253510853 4537]
   condF_ZP_minus_ZPIP [-95.6096227244769673011 5704, -95.6096227244769673011 5704]
   condF_Elog_minus_logZ [-4.578171180932416623428288, -4.578171180932416623428288]
   CHECK primary_bias upper endpoint < -0.17 : True
   CHECK secondary_bias upper endpoint < -0.15 : True
   CHECK D lower endpoint > 4/sqrt(3) (uses D > 2.31) : True
   CHECK a upper endpoint < 1/2 : True
   CHECK condF_ZP_minus_ZPIP upper endpoint < 0 : True
   CHECK max enclosure width over all rows < 1e-52 : True (max width = 1.9199e-53)
stage0_round=True rounds_before_measured=5
   D [8.146762406340682396202885, 8.146762406340682396202885]
   a [0.027914117317939631485254 63, 0.027914117317939631485254 63]
   c0 [0.069471112690807603572169 58, 0.069471112690807603572169 58]
   c1 [0.998708337451837812310299, 0.998708337451837812310299]
   PIP [0.990618012243358205054783 3, 0.990618012243358205054783 3]
   total_prob [1.0, 1.0]
   identity_Z [-8.679075120429598804812758e-54, 8.680001105102944542698781e-54] [interval contains 0]
   identity_ZP [-7.878158118932203000363219e-54, 7.879064189956659582595578e-54] [interval contains 0]
   primary_bias [-0.212065526918593961312 9795, -0.212065526918593961312 9795]
   secondary_bias [-0.194638440548470411723 3927, -0.194638440548470411723 3927]
   condF_Z_minus_Z [-96.3344135496253510853 4537, -96.3344135496253510853 4537]
   condF_ZP_minus_ZPIP [-95.6488182922841802955 4233, -95.6488182922841802955 4233]
   condF_Elog_minus_logZ [-4.578171180932416623428288, -4.578171180932416623428288]
   CHECK primary_bias upper endpoint < -0.17 : True
   CHECK secondary_bias upper endpoint < -0.15 : True
   CHECK D lower endpoint > 4/sqrt(3) (uses D > 2.31) : True
   CHECK a upper endpoint < 1/2 : True
   CHECK condF_ZP_minus_ZPIP upper endpoint < 0 : True
   CHECK max enclosure width over all rows < 1e-52 : True (max width = 1.7359e-53)
stage0_round=True rounds_before_measured=6
   D [8.146762406340682396202885, 8.146762406340682396202885]
   a [0.027914117317939631485254 63, 0.027914117317939631485254 63]
   c0 [0.069471112690807603572169 58, 0.069471112690807603572169 58]
   c1 [0.998708337451837812310299, 0.998708337451837812310299]
   PIP [0.990618012243358205054783 3, 0.990618012243358205054783 3]
   total_prob [1.0, 1.0]
   identity_Z [-9.615335236602141008022631e-54, 9.616400616817710835482894e-54] [interval contains 0]
   identity_ZP [-8.725842324845780856970287e-54, 8.726897748236906106603818e-54] [interval contains 0]
   primary_bias [-0.173974177849890505703 3109, -0.173974177849890505703 3109]
   secondary_bias [-0.159677356166443932157 8109, -0.159677356166443932157 8109]
   condF_Z_minus_Z [-96.3344135496253510853 4537, -96.3344135496253510853 4537]
   condF_ZP_minus_ZPIP [-95.6096227244769673011 5704, -95.6096227244769673011 5704]
   condF_Elog_minus_logZ [-4.578171180932416623428288, -4.578171180932416623428288]
   CHECK primary_bias upper endpoint < -0.17 : True
```

```
CHECK secondary_bias upper endpoint < -0.15 : True
CHECK D lower endpoint > 4/sqrt(3) (uses D > 2.31) : True
CHECK a upper endpoint < 1/2 : True
CHECK condF_ZP_minus_ZPIP upper endpoint < 0 : True
CHECK max enclosure width over all rows < 1e-52 : True (max width = 1.9232e-53)
```

(d) Review record

FINAL_REPORT.md: statement changes

Statement changes, and why

1. **Made precise (Astra #1 statement check: APPROVED WITH CHANGES).**
   - The frozen run is stated for arbitrary integer parameters. The method is the case N = N_e, M = 4, L′ = L, with no stage-0 round.
   - START, exactness of the π_0 sampler and of $\bar{Z}_0$, is an imported dependency (T09(i)), not an extra restriction.
   - The negative parts of (iv) are component statements, with the scope stated: the p = 2 example would take the exact branch in the full method, and N = 1 is not the production budget.
   - "Not necessarily independent after averaging over $\mathcal{C}$" is formalised as an exact characterisation: unconditional independence holds iff the island law given $\mathcal{C}$ is a.s. constant. No claim is made about the actual calibration law.
   - "Favourable diagnostics" is formalised as the final E2 check passing. "NUM alone" is formalised as a perturbed-score model that NUM does not exclude.
2. **One error of mine, caught by Astra #1.** My first draft extended (iv.d) to *every* conclusion of (iii) under a launch event. That is false for intermediate stages, whose targets depend on the random grid. It was repaired: only the $\mathcal{C}$-free conclusions are kept, and intermediate stages hold in averaged form. This was an over-extension beyond the conjecture, so it is not counted as a step-1 restart.
3. **Stage B round 1.** Both reviewers independently found that the statement pointed to 00_COMMON for the marginal-likelihood formula instead of writing it out. STATEMENT §0 now carries Felipe's fixed target formulas. Joint measurability of the stream map is now stated explicitly under EX-S. No conclusion changed.

FINAL_REPORT.md: review history

Review history

Stage / round	Reviewer	Object	Verdict	Fatal / gap / exposition
A, statement check	Astra #1 (high)	STATEMENT v1	APPROVED WITH CHANGES	1 false clause, (iv.d), plus 3 required clarifications; all adopted

Stage / round	Reviewer	Object	Verdict	Fatal / gap / exposition
A, review	Astra #1 (high), resumed	PROOF_v2	**verified**	0 / 0 / 3 (fixed in v3)
B, round 1	Astra #2 (high), new session	PROOF_v3	not verified	0 / 1 / 1
B, round 1	Opus 5.5 (xhigh), Claude Personal	PROOF_v3	not verified	0 / 2 / 6
B, round 2	Astra #2	PROOF_v4 + claims log	**verified**; agrees with all 3 answers	0 / 0 / 2 (fixed in v5)
B, round 2	Opus 5.5 (xhigh)	PROOF_v4 + claims log	**verified**; agrees with all 3 answers	0 / 0 / 2 (fixed in v5)

- CLAIMS_OF_FATAL_FLAWS.md has every gap finding verbatim, with my answer and both reviewers' rulings. No fatal finding was raised at any stage.
- The Astra judge was not needed.
- The independent checks of the numerical example agree to 12+ digits:
  - a Sonnet assistant wrote its own implementation (helpers/sonnet_check/);
  - the Opus reviewer re-ran the certificate, wrote its own float implementation, and ran a single-seed Monte Carlo of the literal algorithm with 200,000 islands, getting −0.1752 ± 0.0008 against the certified −0.17397.

CLAIMS_OF_FATAL_FLAWS.md (every entry, verbatim)

CLAIMS_OF_FATAL_FLAWS — T12 (T8)

Every fatal or gap finding, copied verbatim, with the lead's answer. Stage A (Astra #1 on PROOF_v2) had no fatal or gap findings, only 3 exposition findings, all fixed in PROOF_v3. The entries below are from Stage B round 1 on PROOF_v3. The fixes are in PROOF_v4 and STATEMENT v2.2.

Claim 1 (Astra #2, finding 1, gap)

Location: §8.2, `PROOF_v3.tex` lines 185–214; §8.5, lines 227–231; Appendix A, `helpers/p2_exact.py` lines 58–84.

Severity: gap.

Claim: The computational examples are not sufficiently linked to the marginal likelihood specified in the theorem.

Evidence: The helper's `log_bf` function asserts that it computes the v3 target's Bayes factors. It uses $a_0 = b_0 = 0.01$, $H = (n+1)X_S^\top X_S + I$, a determinant ratio involving $X_S^\top X_S + I$, and exponent $a_0 + n/2$. None of the permitted documents supplies the target formula or derives these expressions from it. `STATEMENT.md` refers to `00_COMMON §1`, whose contents are not supplied in the permitted materials, and says that only positivity of m and ϖ is used.

Positivity alone does not determine these Bayes factors or the numerical conclusions. In particular, the finite-state enumeration and saved intervals establish results for the scores

implemented by `log_bf`; they do not establish that those scores equal the theorem's $m(S)/m(\emptyset)$. Consequently, the bias bounds in **(iv.b)** and the instance-specific diagnostic conclusions in **(iv.e)** remain unverified against the stated target. This is not evidence that the implemented formula is false; its required identification is unsupported here.

Claim 2 (Opus xhigh reviewer, G1, gap). Same issue as Claim 1, found independently.

G1

- **Location:** §8.2, "The instance" and the interval-computation paragraph; §8.5; `helpers/p2_exact.py`, `log_bf` (lines 58–84) and `setup` (line 98).
- **Severity:** gap.
- **Claim:** The (iv.b) bias bounds (−0.17 and −0.15) and the (iv.e) result $E[\hat{Z}_r \cdot PI\hat{P}_{r1} \mid F] \neq Z \cdot PIP_1$ depend on the exact formula for m(S), on $a_0 = b_0 = 0.01$, and on the prior weights w(k). None of these is stated or justified in the proof or in STATEMENT.md, so I cannot verify that the computation is about the v3 target.
- **Evidence:**
  - The formula exists only in code, as $\log BF(S) = -\tfrac{1}{2} \log(\det H / \det(G+I)) - \nu[\log(2b_0 + y^T y - n\, z^T H^{-1} z) - \log(2b_0 + y^T y)]$, with $H = (n+1)G + I$ and $\nu = a_0 + n/2$. Its only justification is the comment "for the v3 target".
  - STATEMENT §0 defers the formula to 00_COMMON §1, which is not in the packet. It also says "only m > 0 and $\varpi > 0$ are used", which is not true for this instance.
  - The prior values w(0) = ½, w(1) = w(2) = 1/6, W_K = 1 match the standard BB(1,2) beta-binomial (I recomputed them), but that definition is also not in the packet.
  - The margins are small: at the method column, 0.0040 for the primary bound and 0.0097 for the secondary.
  - A diagnostic only: with $\nu = a_0 + (n-1)/2$ the bounds still hold (primary −0.179106, secondary −0.164769, D = 7.7534). So the conclusion may well survive, but I could not verify the formula itself.
  - The qualitative (iv.e) facts (P(F) > 0, $E[\hat{Z}_r \mid F] < Z$, $E[\hat{Z} \mid F] < Z$) need only $D > 4/\sqrt{3}$ and $a < \tfrac{1}{2}$. Both of those numbers come from the same unverified formula.

Lead's answer to Claims 1–2: accepted and fixed. The finding is right. STATEMENT.md delegated the target formulas to a file outside the packet, and its remark "only m > 0 and $\varpi > 0$ are used" was false for the instance. Fixes: 1. STATEMENT §0 (v2.2) now writes out the fixed v3 target, copied from METHODOLOGY_v3 §5 and 00_COMMON §1: H_S, D_S, Q_S, m(S) with C_n, $BF(S) = D_S^{\{-1/2\}}[(2b_0+Q_S)/(2b_0+y^Ty)]^{\{-\nu\}}$, $\nu = a_0 + n/2$, $a_0 = b_0 = 0.01$, $w(k) = B(1+k, 2p-k)/B(1,p) = \tfrac{1}{2}/C(2p-1,k)$, W_K. The positivity remark is restricted to everything except the instance. This adds no assumption and changes no conclusion: these are Felipe's fixed target, which the statement always referred to. 2. PROOF_v4 §8.2 has a new paragraph, "The scores". It derives the 1×1 and 2×2 expressions that `log_bf` computes from the STATEMENT §0 formula (C_n cancels; explicit 2×2 determinant and inverse; ν = 10.01), states that `standardise` implements the §0 standardisation, and computes the prior values from $w(k) = \tfrac{1}{2}/C(3,k)$. 3. The certificate was re-run after cosmetic helper changes (E6), with no change to any value. It now prints exact endpoint comparisons (CHECK lines), and its output is reproduced in a new Appendix B.

Claim 3 (Opus xhigh reviewer, G2, gap, minor)

G2

- **Location:** §6, the paragraph after Lemma 6.1 ("the joint measurability of Φ is part of the stream model").
- **Severity:** gap (minor).
- **Claim:** The proof assumes Φ is jointly measurable in (c, u). STATEMENT.md does not state this; EX-S, CAL and Success only say that Φ(c, ·) realises I(c).
- **Evidence:** Two steps need it. Lemma 6.1 is applied to $g(\Phi(c,u_1),\ldots,\Phi(c,u_4))\cdot 1_{\mathcal{A}}(c)$, which must be measurable on E×Y. The (iv.c) characterisation needs c ↦ law of I(c) to be a probability kernel. The assumption is standard and true for any algorithmic implementation, but it is imported rather than given.

Lead's answer: accepted and fixed. STATEMENT (v2.2) now says explicitly, under EX-S, that the stream model includes joint measurability of (c, u) ↦ Φ(c, u) with values in a Polish space; PROOF_v4 §6 cites it. This states what EX-S ("independent ideal uniform streams") already presupposes, since the island record is produced from uniform streams by finitely many measurable operations. It restricts no input and changes no conclusion.

Exposition findings (not fatal/gap; recorded for completeness)

- Astra #2 finding 2 = Opus E1 (broken \ref, log not in appendix): fixed. The reference is repaired and the log is reproduced in Appendix B.
- Opus E2 (ε_U temperature): fixed. Lemma 1.4(a) now defines the temperature-free ε_S = $\pi_0(S)e^{-R_d}$, valid for every φ ∈ [0, 1].
- Opus E3 ((iv.c) implicit step): fixed. Added the explicit sentence: the hypotheses admit a calibration law with mass on (grid, N) and (grid, 4N); the (iv.b) instance has a unique maximiser of d; no claim is made about the actual calibration procedure.
- Opus E4 ("only the averaged form"): fixed. STATEMENT (iv.d) now gives an exact remark instead of "only … in general random". PROOF_v4 §8.4 proves that $Z_{tq_t}(1\{S\}) = m(\emptyset)\gamma_0(S)e^{\varphi_t\, d(S)}$ is non-constant when φ_t takes two values.
- Opus E5 (block/graph facts rely on method rules): the proof now cites the STATEMENT appendix, where these facts are recorded.
- Opus E6 (helper cosmetics): fixed. Endpoint display, the prior comment and the `__main__` block are corrected; the certificate was re-run with identical values (certify_iv.log; previous log kept as helpers/certify_iv_v1.log).
- Opus, (iv.e) "extension from r = 1 to every r by exchangeability is implicit": made explicit in §8.5.

Rulings by the Stage B reviewers (round 2, on PROOF_v4 / STATEMENT v2.2)

- **Astra #2** (review/A2_call2_stageB_r2_ruling_REPLY.md): agrees with the answers to Claims 1, 2 and 3; each fix works. No remaining fatal or gap findings. Two exposition items: (a) the log displayed rounded endpoints, so the width claim was not visible; (b) the Appendix B description

wrongly called the log row condF_Elog "divided by Z_0". **Verdict: verified.**

- **Opus xhigh** (review/OPUS_stageB_r2_ruling.md): agrees with all three answers; each fix works, checked term by term and numerically. Two exposition items: X1, certify_iv_v1.log was not in its packet (packaging; the file is in the project's helpers/); X2, an unneeded remark about the method's max-abs scaling. **Verdict: verified.**
- Lead: all four exposition items were fixed in PROOF_v5 (exposition only, no mathematical change from the verified v4). The certificate now prints an exact width CHECK (max width $1.92 \cdot 10^{-53} < 10^{-52}$); the Appendix B row descriptions are corrected; the X2 remark is removed. No dispute remains, so the Astra judge was not needed.

Chapter 4

T13 (T10): restricted ESS and the omitted-mass identity

Final proof: `proofs/T13_T10_restricted_ESS/proof/PROOF_v4.tex`. *Sources of parts (a), (b), (d):* `theorems/T13_T10_restricted_ESS.md`, `STATEMENT.md`, `FINAL_REPORT.md`, `CLAIMS_OF_FATAL_FLAWS.md`.

(a) Original conjecture text (theorem file, verbatim)

T13 (T10): Restricted ESS and the omitted-mass identity

Status: conjectured, not proven. Methodology SMC-BVS v3.0 (2026-10-19), SHA-256 93043758058a6d422cca6ca4b6be8f3a9e8149b6f6de2e1bab1ce31c44b8019c. Read with `00_COMMON.md`.

Category / rank: algorithmic validity; **core** (lemma).

Definitions used

Bridge q_ψ for real ψ; candidate set V (frozen, nonempty); restricted ESS $\rho_V(u, v) = \bar{Z}_v(V)^2/(\bar{Z}_u(V)\bar{Z}_{2v-u}(V))$; $a(\psi) := q_\psi(V)$; for $W \subseteq V$, $a_W(\psi) := q_\psi(W)$, $q^W_\psi = q_\psi(\cdot|W)$; $\chi^2(q_v\|q_u) = \int(dq_v/dq_u)^2 dq_u - 1$; $\boldsymbol{\mathcal{E}}$_grid (00_COMMON §2).

Assumptions

EX-A; fixed V; COVER(V, ε_C) or COVER(W, ε_C) for the corollaries.

Precise statement

For nonempty frozen V, $u < v$, $w = 2v - u$: (i) [identity] $1 + \chi^2(q_v\|q_u) = \rho_V(u,v)^{-1}\cdot a(v)^2/(a(u)a(w))$. (ii) [COVER corollary] An accepted interval ($\rho_V \geq 0.9$) with $a(u), a(w) \geq 1 - \varepsilon_C$ satisfies $1 + \chi^2(q_v\|q_u) \leq 10/(9(1-\varepsilon_C)^2)$. COVER supplies this at every grid evaluation point (φ_{t-1}, φ_t, w_t); coverage on [0,1] alone does not supply it when $w_t > 1$; the largest w_t is not automatically the worst coverage point. (iii) [restricted working set] For nonempty $W \subseteq V$: $1 + \chi^2(q_v^W\|q_u^W) = \rho_V(u,v)^{-1}\cdot(a_V(v)^2/a_W(v)^2)\cdot(a_W(u)a_W(w))/(a_V(u)a_V(w))$; COVER($W$, ε_C) gives the same bound as (ii). (iv) [restricted grid] If $\emptyset \notin \boldsymbol{\mathcal{M}}$ and calibration never scores $\boldsymbol{\mathcal{M}}$, then $V \cap \boldsymbol{\mathcal{M}} = \emptyset$; with the same pilot temperatures and the same V, restricting γ_φ to $\boldsymbol{\mathcal{M}}^c$ leaves the grid construction

unchanged. No assertion derives particle mixing from ESS.

Dependencies

Builds on: bridge definitions; project L4/L7. Used by: T18, T25, T28, T31, T36.

Numerical evidence (provisional)

None specific (the identity is algebraic); provisional evidence (Sol check J6) uses it in the reflected-pair product check.

Proof-route hint (optional)

Substitute $\bar{Z}_\psi(V) = a(\psi)\bar{Z}_\psi$ into the two second-moment identities (project L4, L7; valid for every real ψ, so over-tempered points are covered); (iv): ρ_V depends only on γ_φ restricted to V. Hardest step: none for the identity; establishing coverage for the actual scored set is the substantive obstacle (T25). *When proof.md is run on this file, its lead considers the statement both with and without these hints; the hints never reach the reviewers.*

(b) STATEMENT.md (verbatim)

STATEMENT — T13 (T10): restricted ESS and the omitted-mass identity

Source: `Theory_architecture_v3/theorems/T13_T10_restricted_ESS.md` (all parts), read with `00_COMMON.md`, `THEORY_BRIEF.md` §2.10 and the method file `METHODOLOGY_v3.md` §6.7 (SHA-256 93043758058a6d422cca6ca4b6be8f3a9e8149b6f6de2e1bab1ce31c44b8019c). Version 3 (after statement checks 1 and 2). Changes against the source are listed in §2 with reasons.

0. Setting and definitions

- **Data and model objects** are those of 00_COMMON §1: standardised X, y; $\Omega = \{S \subseteq [p] : |S| \le K\}$, $K = \min(p, \lfloor n/2 \rfloor)$; m(S), $BF(S) = m(S)/m(\emptyset)$; $\varpi(S) = w(|S|)/W_K$ with $w(k) = 1/2/C(2p-1, k)$.
- **Bridge** (00_COMMON §2, METHODOLOGY §6.3): blocks $C_1, \ldots, C_J$; tables $r_b(U) = BF(U)$, $U \subseteq C_b$; $\gamma_0(S) = \varpi(S)\prod_b r_b(S \cap C_b)$; $d(S) = \log BF(S) - \Sigma_b \log r_b(S \cap C_b)$. For every real ψ: $\gamma_\psi = \gamma_0 e^{\psi d}$, $\bar{Z}_\psi = \Sigma_{S\in\Omega}\gamma_\psi(S)$, $q_\psi = \gamma_\psi/\bar{Z}_\psi$. (The proof shows from these definitions that $\gamma_0 > 0$ and d is finite on Ω, so every q_ψ has full support.) For nonempty $A \subseteq \Omega$: $\bar{Z}_\psi(A) = \Sigma_{S\in A}\gamma_\psi(S)$, $q_\psi(A) = \bar{Z}_\psi(A)/\bar{Z}_\psi$, $q^A_\psi = q_\psi(\cdot \mid A)$.
- **Candidate set.** V is a nonempty subset of Ω, held fixed. $a(\psi) = a_V(\psi) := q_\psi(V)$; for nonempty $W \subseteq V$, $a_W(\psi) := q_\psi(W)$.
- **χ^2 divergence.** For laws μ, λ on a finite set with λ > 0 wherever μ > 0: $\chi^2(\mu\|\lambda) = \Sigma_{S:\lambda(S)>0} \lambda(S)(\mu(S)/\lambda(S))^2 - 1$.
- **Restricted ESS.** For real u, v and w := 2v − u: $\rho_V(u, v) = \bar{Z}_v(V)^2/(\bar{Z}_u(V)\bar{Z}_w(V)) = \exp\{2F_V(v) - F_V(u) - F_V(w)\}$, $F_V(t) = \log \Sigma_{S\in V}\gamma_0(S)e^{td(S)}$. An interval (u, v), u < v, is **accepted** if $\rho_V(u, v) \ge 0.9$.
- **Grid.** $0 = \varphi_0 < \varphi_1 < \ldots < \varphi_T = 1$, $T \ge 1$; $w_t = 2\varphi_t - \varphi_{t-1}$; $\boldsymbol{\mathcal{E}}_{grid} = \{\varphi_0, \ldots, \varphi_T, w_1, \ldots, w_T\}$.
- **Candidate-grid construction** (METHODOLOGY §6.7). Inputs: an increasing list of pilot temperatures in (0, 1] ending at 1, the candidate set V, T_e. Start at u = 0; for the next pilot

temperature b: if ρ_V(u, b) ≥ 0.90 append b; else bisect on (u, b] with u fixed, at most 60 halvings, and append the largest tested endpoint that passes (ρ_V(u, ·) ≥ 0.90); continue from the appended point; validate strict progress and the inequality at every appended point; non-progress, failed validation or more than T_e intervals ⇒ failure. A **successful construction** outputs a grid. An **implementation** of the construction is a fixed deterministic procedure realising these rules (it fixes the bisection points, stopping and endpoint choices); it accesses the bridge only through evaluations of ρ_V.

- **Candidate set of the method's form.** For a set $\mathcal{P} \subseteq \Omega$ of scored models (the *pool*), V($\mathcal{P}$) := top-20p of $\mathcal{P}$ by log γ_0 ∪ top-20p of $\mathcal{P}$ by log γ_1 ∪ {∅} (ties lexicographic; if $|\mathcal{P}| \le 20p$ each top list is all of $\mathcal{P}$).
- **COVER(W, ε_C)**, for nonempty W ⊆ Ω, a grid, and **0 ≤ ε_C < 1**: q_ψ(W) ≥ 1 − ε_C for every ψ ∈ $\mathcal{E}$_grid.
- **EX-A**: exact scores, sums, probabilities and comparisons.
- **Method-class witness** (used in (ii)(c)–(d)): a raw data set (X^r, y^r) satisfying the method's input conditions (n ≥ 3, no constant column, y^r non-constant) on which the exact-branch condition $\Sigma_{k\le K} C(p, k) \le (5H_0 + 4N_0)p$ fails (so the SMC branch runs); the method's blocks, tables, γ_0 and d for these data; a candidate set V = V($\mathcal{P}$) of the method's form for some pool $\mathcal{P} \subseteq \Omega$; a tier e ∈ {0, 1} with T_e = 4^eT_0 computed from the data's (n, p) (00_COMMON §2 budgets); a fixed deterministic implementation of the construction as in this section; a list of pilot temperatures; and the grid that this implementation outputs, successfully and under EX-A, from V, that list and T_e. (No claim is made about the probability that the random pilot outputs that list, or that a calibration reaches the pool $\mathcal{P}$.)

1. Statement

Throughout, V ⊆ Ω is nonempty and fixed, u, v are real, w = 2v − u.

(i) Identity. For all real u, v (in particular u < v):
$1 + \chi^2(q_v \,\|\, q_u) = \rho_V(u, v)^{-1} \cdot a(v)^2/(a(u)a(w))$.

(ii) Coverage corollary. (a) Let 0 ≤ ε_C < 1. If (u, v) is accepted and a(u) ≥ 1 − ε_C and a(w) ≥ 1 − ε_C, then $1 + \chi^2(q_v \,\|\, q_u) \le 10/(9(1 - \varepsilon_C)^2)$. (b) Assume EX-A. Let $(\varphi_t)_{t=0}^T$ be the output of a successful candidate-grid construction run with candidate set V (in particular the frozen grid of a calibration, paired with the V of the construction that produced it). Then every interval $(\varphi_{t-1}, \varphi_t)$ is accepted; its evaluation points φ_{t-1}, φ_t, w_t lie in $\mathcal{E}$_grid; and if COVER(V, ε_C) holds, then $1 + \chi^2(q_{\varphi_t} \,\|\, q_{\varphi_{t-1}}) \le 10/(9(1 - \varepsilon_C)^2)$ for every t = 1, …, T. (c) Coverage on [0, 1] alone does not supply the hypothesis of (a) when $w_t > 1$: there is a method-class witness with grid T = 1 ($\varphi_0 = 0$, $\varphi_1 = 1$, $w_1 = 2$) such that, with ε_C = 0.1, the interval (0, 1) is accepted and q_ψ(V) ≥ 1 − ε_C for every ψ ∈ [0, 1], but $q_{w_1}(V) < 1 - \varepsilon_C$ and $1 + \chi^2(q_1 \,\|\, q_0) > 10/(9(1 - \varepsilon_C)^2)$. (d) The largest w_t is not automatically the worst coverage point: there is a method-class witness with grid T = 2 (both intervals accepted) such that the minimum of ψ ↦ q_ψ(V) over $\mathcal{E}$_grid is attained only at φ_0, not at max_t w_t, and the minimum over $\{w_1, \ldots, w_T\}$ is not attained at max_t w_t.

(iii) Restricted working set. For every nonempty W ⊆ V and all real u, v:
$1 + \chi^2(q^W_v \,\|\, q^W_u) = \rho_V(u, v)^{-1} \cdot (a_V(v)^2/a_W(v)^2) \cdot (a_W(u)a_W(w))/(a_V(u)a_V(w))$. In particular (W = V) $\rho_V(u, v) = 1/(1 + \chi^2(q^V_v \,\|\, q^V_u))$. Moreover, with 0 ≤ ε_C < 1, if (u, v) is accepted and a_W(v) ≥ 1 − ε_C, then $1 + \chi^2(q^W_v \,\|\, q^W_u) \le 10/(9(1 - \varepsilon_C)^2)$. Hence, for a grid as in (ii)(b), COVER(W, ε_C) gives at every interval t both $1 + \chi^2(q^W_{\varphi_t} \,\|\, q^W_{\varphi_{t-1}}) \le 10/(9(1 -$

ε_C)^2) and (because W ⊆ V, COVER(W, ε_C) implies COVER(V, ε_C)) the bound of (ii)(b).

(iv) Restricted grid. Let $\mathcal{M} \subseteq \Omega$ with $\emptyset \notin \mathcal{M}$. (a) Let V = V($\mathcal{P}$) be of the method's form. If $\mathcal{P} \cap \mathcal{M} = \emptyset$ (pool avoidance), then V ∩ $\mathcal{M}$ = ∅. Pool avoidance holds, in particular, when calibration never scores $\mathcal{M}$ and every member of $\mathcal{P}$ was scored by the calibration or belongs to an inherited set of scored models (tables, leaders carried across tiers) that is assumed to avoid $\mathcal{M}$. (b) Suppose V ∩ $\mathcal{M}$ = ∅. Let γ′_φ := γ_φ1_{$\mathcal{M}$^c} for real φ (the bridge on Ω′ = Ω \ $\mathcal{M}$ ∋ ∅ with γ_0, d restricted to Ω′; q′_φ = q_φ(· | $\mathcal{M}$^c)). Then F′_V = F_V and ρ′_V = ρ_V on all real arguments, and, under EX-A, one and the same implementation of the candidate-grid construction, run on the restricted and on the original bridge with the same pilot temperatures, the same V and the same T_e, has the same outcome (the same grid, or failure in both).

No assertion derives particle mixing from ESS.

2. Changes against the source statement, with reasons

1. **ε_C ∈ [0, 1)** added (approved in check 1): for ε_C ≥ 1 the bound is undefined or false.
2. **(iv)(a) pool avoidance** made the explicit hypothesis (modification of check 1): $\mathcal{P} \cap \mathcal{M} = \emptyset$ is sufficient for V ∩ $\mathcal{M}$ = ∅. The source's wording "calibration never scores $\mathcal{M}$" is kept as a sufficient condition, with the provenance condition that every pool member was scored by the calibration or comes from an inherited set assumed to avoid $\mathcal{M}$ (the method does not say whether tables or carried leaders feed V's top lists).
3. **(ii)(b) the V paired with a grid** is the candidate set of the construction that produced it (approved in check 1).
4. **(ii)(c)–(d) witnesses are method-class** (replaces the rejected abstract-bridge version): real data run through the method's blocks, tables, γ_0 and d, on the SMC branch; V of the method's form; a grid output by the construction. The pilot's random output is not claimed; the pilot temperatures enter as construction inputs, as they do in the source's (iv).
5. **(iii) the bound uses coverage at v** (a_W(v) ≥ 1 − ε_C), supplied by COVER(W, ε_C) because φ_t ∈ $\mathcal{E}$_grid; both the restricted and the full-space bound are stated (approved in check 1; the condition is sufficient, not claimed necessary).
6. **(i), (iii) for all real u, v** (approved in check 1).
7. **Positivity**: $\gamma_0 > 0$ and d finite on Ω are derived in the proof from the definitions of 00_COMMON §1 (ϖ > 0, m(S) ∈ (0, ∞)), not assumed.
8. **(iv)(b) implementation**: the "unchanged construction" conclusion is for one fixed deterministic implementation applied to both bridges (check 1: EX-A alone does not fix bisection and endpoint choices).
9. **(ii)(d) strengthened** (recorded at check 2): the witness must have the grid-evaluation minimum attained **only** at φ_0 (the source asks only that the largest w_t not be the worst point). This is a strengthening, not a weakening.
10. **Witness parameters** (modification of check 2): a witness names its tier e, uses T_e = 4^eT_0 from its own data, and fixes a deterministic implementation.

(c) Final proof, verbatim: body of PROOF_v4.tex

The proof below is the body of `PROOF_v4.tex` from `\begin{document}` to `\end{document}` without the title block. Its preamble definitions are local to this chapter; labels carry the prefix `T13:`.

All notation is that of STATEMENT.md (version 3). In particular $\Omega = \{S \subseteq [p] : |S| \le K\}$, $\gamma_\psi = \gamma_0 e^{\psi d}$, $\bar Z_\psi(A) = \sum_{S\in A}\gamma_\psi(S)$, $\bar Z_\psi = \bar Z_\psi(\Omega)$, $q_\psi = \gamma_\psi/\bar Z_\psi$, $q_\psi^A = q_\psi(\cdot \mid A)$, $a_W(\psi) = q_\psi(W)$, $a = a_V$, and $w = 2v - u$ whenever u, v are given. Throughout, $V \subseteq \Omega$ is nonempty and fixed.

STATEMENT.md defines the model and sampler objects by reference to `00_COMMON.md` and `METHODOLOGY_v3.md`. Every definition from those files that this proof uses is quoted verbatim in Appendix B (with the files' SHA-256 checksums); citations of the form [B: file, §, lines] point there.

1 Preliminaries

Lemma 1 (Positivity)**.** *For every $S \in \Omega$, $\varpi(S) > 0$ and $0 < m(S) < \infty$. Consequently $\gamma_0(S) \in (0,\infty)$ and $d(S) \in \mathbb{R}$ for all $S \in \Omega$; for every real ψ and nonempty $A \subseteq \Omega$, $\gamma_\psi > 0$ on Ω, $\bar Z_\psi(A) \in (0,\infty)$, and q_ψ and q_ψ^A give positive mass to every point of Ω and of A respectively.*

Proof. The definitions used are [B: 00_COMMON.md, §1, lines 10–11] (also [B: METHODOLOGY_v3.md, §5]): $G_S = X_S^\top X_S$, $z_S = X_S^\top y$, $H_S = (n+1)G_S + I$, $D_S = \det H_S/\det(G_S + I)$, $Q_S = y^\top y - n z_S^\top H_S^{-1} z_S$, $\nu = 0.01 + n/2$, $a_0 = b_0 = 0.01$, $m(S) = (2b_0)^{a_0}\Gamma(\nu)/(\pi^{n/2}\Gamma(a_0)) \cdot D_S^{-1/2}(2b_0 + Q_S)^{-\nu}$, $w(k) = \frac12/\binom{2p-1}{k}$, $W_K = \sum_{k\le K}\binom{p}{k}w(k)$, $\varpi(S) = w(|S|)/W_K$; and $\mathrm{BF}(S) = m(S)/m(\emptyset)$, $r_b(U) = \mathrm{BF}(U)$ for $U \subseteq C_b$, γ_0, d as in [B: 00_COMMON.md, §2, lines 18–20].

$\varpi(S) = w(|S|)/W_K$ with $w(k) = \frac12/\binom{2p-1}{k} > 0$ for $0 \le k \le K \le p \le 2p-1$, and $W_K = \sum_{k\le K}\binom{p}{k}w(k) \ge w(0) > 0$ is finite. For $m(S)$, and in fact for $m(U)$ with any $U \subseteq [p]$ (the argument does not use $|U| \le K$; we write it for S): $G_S = X_S^\top X_S$ is positive semidefinite, so $H_S = (n+1)G_S + I$ and $G_S + I$ are positive definite (for $S = \emptyset$ all matrices are 0×0, determinants are 1 and $Q_\emptyset = y^\top y$). Hence $D_S = \det H_S/\det(G_S+I) \in (0,\infty)$. With $\hat\beta = n H_S^{-1} z_S$,

$$\begin{aligned}\|y - X_S\hat\beta\|^2 + \tfrac1n\big(\|X_S\hat\beta\|^2 + \|\hat\beta\|^2\big) &= y^\top y - 2\hat\beta^\top z_S + \tfrac1n\hat\beta^\top\big((n+1)G_S + I\big)\hat\beta \\ &= y^\top y - 2n z_S^\top H_S^{-1} z_S + n z_S^\top H_S^{-1} z_S = Q_S,\end{aligned}$$

so $Q_S \ge 0$ and $2b_0 + Q_S \ge 0.02 > 0$. The constant $(2b_0)^{a_0}\Gamma(\nu)/(\pi^{n/2}\Gamma(a_0))$ is positive and finite. Hence $m(S) = \mathrm{const}\cdot D_S^{-1/2}(2b_0+Q_S)^{-\nu} \in (0,\infty)$, so $\mathrm{BF}(S) = m(S)/m(\emptyset) \in (0,\infty)$, every table entry $r_b(U) = \mathrm{BF}(U)$ ($U \subseteq C_b$) is in $(0,\infty)$ by the same argument applied to U, $\gamma_0(S) = \varpi(S)\prod_b r_b(S\cap C_b) \in (0,\infty)$, and $d(S) = \log\mathrm{BF}(S) - \sum_b \log r_b(S\cap C_b)$ is finite. The remaining claims follow because Ω is finite. □

Lemma 2 (Restricted second-moment identity)**.** *For every nonempty $A \subseteq \Omega$ and all real u, v,*

$$1 + \chi^2\big(q_v^A \,\big\|\, q_u^A\big) = \frac{\bar Z_u(A)\,\bar Z_w(A)}{\bar Z_v(A)^2}.$$

For $A = \Omega$ this reads $1 + \chi^2(q_v\|q_u) = \bar Z_u \bar Z_w/\bar Z_v^2$.

Proof. By Lemma 1, $q_u^A > 0$ exactly on A and q_v^A vanishes off A, so the χ^2 divergence is defined and its sum runs over A. For $S \in A$, $q_v^A(S)/q_u^A(S) = e^{(v-u)d(S)}\bar Z_u(A)/\bar Z_v(A)$. Hence

$$\sum_{S\in A} q_u^A(S)\Big(\frac{q_v^A(S)}{q_u^A(S)}\Big)^2 = \frac{\bar Z_u(A)^2}{\bar Z_v(A)^2}\sum_{S\in A}\frac{\gamma_0(S)e^{ud(S)}e^{2(v-u)d(S)}}{\bar Z_u(A)} = \frac{\bar Z_u(A)\bar Z_w(A)}{\bar Z_v(A)^2},$$

since $u + 2(v-u) = 2v - u = w$. Subtracting 1 gives the claim. For $A = \Omega$, $q_\psi^\Omega = q_\psi$. □

Lemma 3 (Omitted mass)**.** *For nonempty $A \subseteq \Omega$ and real ψ: $\bar{Z}_\psi(A) = a_A(\psi)\bar{Z}_\psi$ with $a_A(\psi) = q_\psi(A) \in (0,1]$. If $\emptyset \neq W \subseteq V$ then $\bar{Z}_\psi(W) = \bar{Z}_\psi(V)\, a_W(\psi)/a_V(\psi)$ and $a_W(\psi) \leq a_V(\psi)$.*

Proof. The first identity is the definition $q_\psi(A) = \bar{Z}_\psi(A)/\bar{Z}_\psi$; positivity is Lemma 1. The second follows by dividing the first for $A = W$ by the first for $A = V$. Monotonicity: $W \subseteq V$ and $\gamma_\psi \geq 0$. □

2 Part (i)

By Lemma 2 with $A = \Omega$ and Lemma 3 with $A = V$ (so $\bar{Z}_\psi = \bar{Z}_\psi(V)/a(\psi)$ for $\psi \in \{u, v, w\}$),

$$1 + \chi^2(q_v \| q_u) = \frac{\bar{Z}_u \bar{Z}_w}{\bar{Z}_v^2} = \frac{\bar{Z}_u(V)\bar{Z}_w(V)}{\bar{Z}_v(V)^2} \cdot \frac{a(v)^2}{a(u)a(w)} = \rho_V(u,v)^{-1} \frac{a(v)^2}{a(u)a(w)},$$

for all real u, v. All quantities are finite and positive by Lemma 1. □

3 Part (iii)

Let $\emptyset \neq W \subseteq V$. By Lemma 2 with $A = W$ and Lemma 3,

$$1 + \chi^2(q_v^W \| q_u^W) = \frac{\bar{Z}_u(W)\bar{Z}_w(W)}{\bar{Z}_v(W)^2} = \frac{\bar{Z}_u(V)\bar{Z}_w(V)}{\bar{Z}_v(V)^2} \cdot \frac{a_W(u)}{a_V(u)} \cdot \frac{a_W(w)}{a_V(w)} \cdot \frac{a_V(v)^2}{a_W(v)^2},$$

which is the identity of (iii), since the first factor is $\rho_V(u,v)^{-1}$. For $W = V$ the last three factors equal 1, so $\rho_V(u,v) = 1/(1 + \chi^2(q_v^V \| q_u^V))$.

Bound. Let $0 \leq \varepsilon_C < 1$, $\rho_V(u,v) \geq 0.9$ and $a_W(v) \geq 1 - \varepsilon_C > 0$. By Lemma 3, $a_W(u)/a_V(u) \leq 1$ and $a_W(w)/a_V(w) \leq 1$; also $a_V(v) \leq 1$. Hence $1 + \chi^2(q_v^W \| q_u^W) \leq (10/9) \cdot 1/(1 - \varepsilon_C)^2$.

Grid consequence. Let (φ_t) be as in (ii)(b) and assume COVER(W, ε_C). By (ii)(b) (proved below, without using (iii)) every interval $(\varphi_{t-1}, \varphi_t)$ is accepted, and $\varphi_t \in \mathcal{E}_{\text{grid}}$, so $a_W(\varphi_t) \geq 1 - \varepsilon_C$; the bound gives $1 + \chi^2(q_{\varphi_t}^W \| q_{\varphi_{t-1}}^W) \leq 10/(9(1 - \varepsilon_C)^2)$. Since $q_\psi(V) \geq q_\psi(W)$ for every ψ, COVER(W, ε_C) implies COVER(V, ε_C), and (ii)(b) gives the full-space bound. □

4 Part (ii)(a)–(b)

(a). By (i), $a(v) \leq 1$, $a(u)a(w) \geq (1 - \varepsilon_C)^2 > 0$ and $\rho_V(u,v)^{-1} \leq 10/9$: $1 + \chi^2(q_v \| q_u) \leq (10/9)(1 - \varepsilon_C)^{-2}$.

(b). Assume EX-A and let $0 = \varphi_0 < \cdots < \varphi_T = 1$ be the output of a successful construction with candidate set V. We read the construction of STATEMENT §0 (quoted from [B: METHODOLOGY_v3.md, §6.7]) as follows, which is the only reading consistent with "start at $u = 0$ … continue from the appended point" and with an output $0 = \varphi_0 < \cdots < \varphi_T = 1$: the current left point u is always the last appended point (initially $u = 0$), and the output grid is 0 followed by the appended points in the order appended. (The parenthetical case of the frozen grid of a calibration is the special case in which the frozen grid is such an output, paired with its V.) So φ_t was appended while $u = \varphi_{t-1}$. A point b' is appended only in two cases: $b' = b$ is the next pilot temperature and $\rho_V(u,b) \geq 0.90$; or b' is a tested bisection endpoint that passes, i.e. $\rho_V(u,b') \geq 0.90$. In both cases $\rho_V(\varphi_{t-1}, \varphi_t) \geq 0.9$, and under EX-A this comparison (and the validation of the inequality at every appended point) is exact. Strict progress gives $\varphi_{t-1} < \varphi_t$. Hence every interval is accepted. By definition of $\mathcal{E}_{\text{grid}}$, $\varphi_{t-1}, \varphi_t, w_t \in \mathcal{E}_{\text{grid}}$. Under COVER$(V, \varepsilon_C)$, $a(\varphi_{t-1}) \geq 1 - \varepsilon_C$ and $a(w_t) \geq 1 - \varepsilon_C$, and (a) with $(u, v, w) = (\varphi_{t-1}, \varphi_t, w_t)$ gives the bound for every t. □

5 Part (ii)(c)–(d): method-class witnesses

5.1 The data set

Let $n = 42$, $p = 40$. Let $e_1, \dots, e_{41}$ be an orthonormal basis of $\mathbf{1}^\perp = \{x \in \mathbb{R}^{42} : \sum_i x_i = 0\}$ (dimension 41). Put $x_j = \sqrt{41}\, e_j$ ($j \in [40]$), $X^r = X = (x_1, \dots, x_{40})$.

Blocks. We use three properties of the method's blocks, all from [B: METHODOLOGY_v3.md, §6.3] and [B: 00_COMMON.md, §2, line 18]. (a) *They depend on X only*: "Blocks $C_1, \dots, C_J$ (deterministic in X: complete linkage, top-down cut, next-fit packing)"; the clustering distance $1 - |r_{jl}|$ is "computed from X", and the cut and the packing use only the dendrogram, B and index order. So the blocks can be computed from X before y is chosen. (b) *They partition* $[p]$: the dendrogram's root contains all p retained columns; the cut repeatedly replaces a node by its two children (which partition it) until every retained cluster has at most B members, so the retained clusters partition $[p]$; packing places whole retained clusters ("a retained cluster is never split") into blocks, each cluster into exactly one block. Hence every column lies in exactly one block. (c) *Their size is capped*: $|C_b| \le B = \min(\lfloor\sqrt{n}\rfloor, \lfloor\sqrt{p}\rfloor, 16) = \min(6, 6, 16) = 6 < 40$. By (a) the blocks are fixed by X. Let $j_1 = 1$; by (c) the block of j_1 does not contain all 40 columns, so we can pick $j_2 \in [40]$ outside it; by (b) j_1 and j_2 lie in different blocks. Put

$$y^r = y = \sqrt{41}\,(c_1 e_{j_1} + c_2 e_{j_2} + c_3 e_{41}), \qquad c_1 = 0.97,\ c_2 = 0.10,\ c_3 = \sqrt{1 - c_1^2 - c_2^2} = \sqrt{0.0491}.$$

Input conditions and standardisation. All entries are finite, $n = 42 \ge 3$, $p = 40 \ge 1$; each x_j and y is centred and nonzero, hence non-constant, so no column is dropped and y is not constant [B: METHODOLOGY_v3.md, §4, items 1–3]. Both conventions in the sources use the sample SD with denominator $n - 1$: [B: 00_COMMON.md, §1, line 9] ("$s_y^2 = \|Cy^r\|^2/(n-1)$; so $\|x_j\|^2 = \|y\|^2 = n - 1$") and [B: METHODOLOGY_v3.md, §4, item 4] ("divide by its maximum absolute value, compute the mean, centre, then divide by the sample SD (denominator n − 1)"). Here $\|x_j\|^2 = \|y\|^2 = 41 = n - 1$ and the vectors are centred, so every sample SD equals 1. Under 00_COMMON's form, $D_x = I$, $s_y = 1$, $CX^r D_x^{-1} = X$ and $Cy^r/s_y = y$. Under the method's form, a column x is divided by $c = \max_i |x_i| > 0$, stays centred, and has sample SD $1/c$, so the final division returns x; the same holds for y. Either way the standardised data are X and y themselves. *SMC branch.* By [B: 00_COMMON.md, §2, lines 24–25] and [B: METHODOLOGY_v3.md, §3, §6.2]: $N_0 = \max(2500, n, p) = 2500$ and $H_0 = \lceil 512 N_0/2500 \rceil = \lceil 512 \cdot 2500/2500 \rceil = 512$, so $(5H_0 + 4N_0)p = 12560 \cdot 40 = 502400$; $K = \min(p, \lfloor n/2 \rfloor) = \min(40, 21) = 21$ and $\sum_{k \le 21} \binom{40}{k} \ge \binom{40}{20} = 137846528820 > 502400$. The exact branch is not taken. *Budget.* Same sources: $\ell = \lceil \log_2(n + p + 1) \rceil = \lceil \log_2 83 \rceil = 7$, $T_0 = \max(1000, \ell^2) = 1000$. Both witnesses use tier $e = 0$, $T_e = T_0 = 1000$.

Lemma 4 (Closed forms)**.** *Let* $\nu = 0.01 + n/2 = 21.01$, $t_i = (n-1)^2 c_i^2 / \big(n(n - 1 + 2b_0)\big) = 1681 c_i^2/(42 \cdot 41.02)$ *(*$i = 1, 2$*),* $g_i = (1 - t_i)^{-\nu}$*, and for* $S \in \Omega$ *let* $I(S) = \{i \in \{1, 2\} : j_i \in S\}$*. Then* $t_1 + t_2 < 1$ *and*

$$\mathrm{BF}(S) = n^{-|S|/2}\Big(1 - \sum_{i \in I(S)} t_i\Big)^{-\nu}, \qquad \gamma_0(S) = \varpi(S)\, n^{-|S|/2} \prod_{i \in I(S)} g_i, \qquad d(S) = D\, \mathbf{1}\{I(S) = \{1, 2\}\},$$

with $D = \nu \log \dfrac{(1 - t_1)(1 - t_2)}{1 - t_1 - t_2} > 0$*. These forms do not depend on the block partition beyond* j_1, j_2 *lying in different blocks.*

Proof. For $|S| = k$: $G_S = 41 I_k$ (orthogonality), so $H_S = (43 \cdot 41 + 1) I_k = n^2 I_k$, $G_S + I = n I_k$, $D_S = n^{2k}/n^k = n^k$. $z_j = x_j^\top y = 41 c_i$ if $j = j_i$ and 0 otherwise (as $e_{41} \perp x_j$). So $Q_S = y^\top y - n \sum_{j \in S} z_j^2 / n^2 = 41 - \sum_{i \in I(S)} 1681 c_i^2 / 42$, and $\mathrm{BF}(S) = D_S^{-1/2} \big((2b_0 + Q_S)/(2b_0 + Q_\emptyset) \big)^{-\nu} = n^{-k/2} (1 - \sum_{i \in I(S)} t_i)^{-\nu}$ since $2b_0 + Q_\emptyset = 41.02$. Next $t_1 + t_2 = 1681 \cdot 0.9509 / 1722.84 < 0.928 < 1$. A block C_b contains at most one of j_1, j_2, so $r_b(U) = n^{-|U|/2} \prod_{i \in I(U)} g_i$ for $U \subseteq C_b$; multiplying over blocks ($\sum_b |S \cap C_b| = |S|$, and each $j_i \in S$ lies in exactly one block, both by the partition property (b)) gives γ_0. Finally $d(S) = -\nu \log(1 - \sum_{I(S)} t_i) + \nu \sum_{I(S)} \log(1 - t_i)$, which is 0 when $|I(S)| \le 1$ and D when $I(S) = \{1, 2\}$; $D > 0$ because $(1 - t_1)(1 - t_2) = 1 - t_1 - t_2 + t_1 t_2 > 1 - t_1 - t_2 > 0$. □

Grouped sums. Let $N = [40] \setminus \{j_1, j_2\}$ ($|N| = 38$) and $\sigma(a) = \sum_{m=0}^{\min(38, K-a)} \binom{38}{m} \varpi(a + m) n^{-(a+m)/2}$. By Lemma 4, grouping S by $I(S)$ and $|S \cap N|$,

$$\bar{Z}_\psi = P + R\, e^{\psi D}, \qquad P = \sigma(0) + (g_1 + g_2)\sigma(1), \qquad R = g_1 g_2\, \sigma(2),$$

with P the γ_0-mass of models not containing both j_1, j_2 and R the γ_0-mass of the others. Write $r = R/P$.

Numerical values and their certification. Every number used below is a finite expression in exact rationals ($\varpi(k)$, t_i, ν, binomial coefficients) and the operations $+, -, \times, /, \sqrt{\cdot}, \log, \exp$. The script of Appendix A computes the rationals exactly (integer and fraction arithmetic) and evaluates each expression in interval arithmetic with outward rounding (`mpmath.iv`, 50 digits), so each internal output interval is guaranteed to contain the exact real value (for display, its endpoints are rounded outward, so the printed intervals contain it too). Every inequality between transcendental quantities used in §§5.2–5.3 is decided by comparing interval endpoints (lower endpoint of the larger side against upper endpoint of the smaller side); the script prints these seven checks ($D > 0$, $a_c(1) \ge 0.9$, $a_c(2) < 0.9$, $1 + \chi^2 > 1000/729$, $\beta > \alpha$, and the two acceptance inequalities), and all pass (output in Appendix A). The remaining facts used are exact: $t_1 + t_2 < 1$ (exact rational arithmetic, asserted in the script and shown by hand in Lemma 4); the set sizes $|V_c| = 117$, $|V_d| = 782 \le 800$, the counts $1, 2 \le T_e$ and the exact-branch inequality (integer arithmetic); $P > 0$ and $R > 0$ (Lemma 1). All enclosures have width below 10^{-20}. Below, "$x \approx c$" means that the certified enclosure of x lies within one unit of the last displayed digit of c (these displayed values are for orientation only; the proof uses only the seven certified checks):

$$t_1 \approx 0.91804979,\ t_2 \approx 0.0097571452 \qquad D \approx 2.457388918,\ e^D \approx 11.6742892$$
$$r = R/P \approx 5.27195546 \times 10^{-3} \qquad r e^D \approx 0.0615463326,\ r e^{2D} \approx 0.718509685$$

5.2 Witness for (ii)(c)

$V_c = \{\emptyset\} \cup \{\{j\} : j \in [40]\} \cup \{\{j_i, k\} : i = 1, 2,\ k \in N\}$, $|V_c| = 117$. All its members have size $\le 2 \le K$, so $V_c \subseteq \Omega$. With pool $\mathcal{P} = V_c$: $|\mathcal{P}| \le 20p = 800$, so both top-$20p$ lists equal $\mathcal{P}$, and $V(\mathcal{P}) = \mathcal{P} \cup \{\emptyset\} = V_c$; V_c is of the method's form. No member of V_c contains both j_1, j_2, so $d \equiv 0$ on V_c and $\bar{Z}_\psi(V_c) = \bar{Z}_0(V_c) =: Z_c$ for all ψ. Hence F_{V_c} is constant and $\rho_{V_c}(u, v) = 1$ for all u, v.
Grid. An implementation exists: e.g. test the pilot temperature b first; if it fails, test the midpoints of $(u, b]$ successively (halving towards u after a failure and towards b after a success, at most 60 halvings), append the largest tested point that passed, and fail if none passed. In both witnesses (§5.2 and §5.3) every pilot temperature passes on its first test, so no bisection, endpoint choice or stopping rule is ever used, and every implementation satisfying STATEMENT §0 gives the same output. Pilot list (1), tier 0: at $u = 0$, $b = 1$, $\rho_{V_c}(0, 1) = 1 \ge 0.90$, so b is appended (strict progress $1 > 0$; inequality validated; $1 \le T_e$ intervals); the list is exhausted and the output is the grid $(0, 1)$,

$T = 1$, $w_1 = 2$. The interval $(0, 1)$ is accepted. *Coverage.* $a_c(\psi) = Z_c/(P + Re^{\psi D})$ is strictly decreasing in ψ ($R > 0$, $D > 0$), where $Z_c = \varpi(0) + \varpi(1)n^{-1/2}(38 + g_1 + g_2) + 38\,\varpi(2)n^{-1}(g_1 + g_2)$, with certified $Z_c/P \approx 0.984162996$. Hence, by the certified checks,

$$\min_{\psi\in[0,1]} a_c(\psi) = a_c(1) = \frac{Z_c/P}{1 + re^{D}} \approx 0.92710319 \ \geq 0.9, \qquad a_c(2) = \frac{Z_c/P}{1 + re^{2D}} \approx 0.57268400 \ < 0.9.$$

Divergence. By Lemma 2 with $A = \Omega$,

$$1 + \chi^2(q_1 \| q_0) = \frac{\bar{Z}_0 \bar{Z}_2}{\bar{Z}_1^2} = \frac{(1 + r)(1 + re^{2D})}{(1 + re^{D})^2} \approx 1.53305467 \ > \ \frac{10}{9 \cdot 0.81} = \frac{1000}{729} \approx 1.37174211$$

(certified: the lower endpoint of the enclosure of the left side exceeds the upper endpoint of the enclosure of 1000/729). So with $\varepsilon_C = 0.1$ the interval is accepted and coverage holds on $[0, 1]$, while $q_{w_1}(V_c) < 1 - \varepsilon_C$ and the bound of (ii)(a) fails. This proves (ii)(c).

5.3 Witness for (ii)(d)

$V_d = \{\emptyset\} \cup \{\{j_1\}\} \cup \{\{j_1, k\} : k \in N\} \cup \{S : \{j_1, j_2\} \subseteq S,\ |S| \leq 4\}$, $|V_d| = 1 + 1 + 38 + (1 + 38 + 703) = 782 \leq 800$; sizes $\leq 4 \leq K$; as above $V(V_d) = V_d$ is of the method's form. By Lemma 4, $\bar{Z}_\psi(V_d) = A + Be^{\psi D}$ with

$$A = \varpi(0) + g_1\big(\varpi(1)n^{-1/2} + 38\varpi(2)n^{-1}\big), \qquad B = g_1 g_2\big(\varpi(2)n^{-1} + 38\varpi(3)n^{-3/2} + 703\varpi(4)n^{-2}\big).$$

Let $\alpha = A/P \approx 0.984162996$ (this agrees with Z_c/P to the digits shown but is a different number: $Z_c - A = \varpi(1)n^{-1/2}(38 + g_2) + 38\varpi(2)n^{-1}g_2 > 0$ is tiny relative to P; the certified enclosures in Appendix A are disjoint), $\beta = B/R \approx 0.996250785$ (certified: $\beta - \alpha > 0$, enclosure ≈ 0.0120877894) and $\lambda(\psi) = Re^{\psi D}/(P + Re^{\psi D})$, strictly increasing in ψ. Then $a_d(\psi) = (A + Be^{\psi D})/(P + Re^{\psi D}) = (1 - \lambda(\psi))\alpha + \lambda(\psi)\beta$, strictly increasing because $\beta > \alpha$. *Grid.* Pilot list $(\frac{1}{2}, 1)$, tier 0, any implementation. With $s = B/A \approx 5.33670722 \times 10^{-3}$,

$$\rho_{V_d}(u, v) = \frac{(1 + se^{vD})^2}{(1 + se^{uD})(1 + se^{(2v-u)D})}, \qquad \rho_{V_d}(0, \tfrac{1}{2}) \approx 0.97081344, \quad \rho_{V_d}(\tfrac{1}{2}, 1) \approx 0.91376273,$$

and the certified checks give $\rho_{V_d}(0, \frac{1}{2}) \geq 0.9$ and $\rho_{V_d}(\frac{1}{2}, 1) \geq 0.9$. So the construction appends $\frac{1}{2}$ and then 1 without bisection (strict progress, validated inequalities, $2 \leq T_e$ intervals) and outputs $(0, \frac{1}{2}, 1)$, $T = 2$, $w_1 = 1$, $w_2 = \frac{3}{2}$, $\mathcal{E}_{\text{grid}} = \{0, \frac{1}{2}, 1, \frac{3}{2}\}$; both intervals are accepted. *Coverage.* Since a_d is strictly increasing, the minimum over $\mathcal{E}_{\text{grid}}$ is attained only at $\varphi_0 = 0$, and the minimum over $\{w_1, w_2\}$ is attained at $w_1 = 1 < \max_t w_t = w_2$. (For illustration: $a_d(0) \approx 0.98422639$, $a_d(\frac{1}{2}) \approx 0.98437688$, $a_d(1) \approx 0.98486382$, $a_d(\frac{3}{2}) \approx 0.98626327$; COVER$(V_d, \varepsilon_C)$ holds with $\varepsilon_C = 0.016$, and the worst point is φ_0, not the largest w_t. Only $\beta > \alpha$ and the two acceptance inequalities are used.) This proves (ii)(d). □

6 Part (iv)

(a). Both top lists of $V(\mathcal{P})$ are subsets of $\mathcal{P}$, so $V \subseteq \mathcal{P} \cup \{\emptyset\}$ and $V \cap \mathcal{M} \subseteq (\mathcal{P} \cap \mathcal{M}) \cup (\{\emptyset\} \cap \mathcal{M}) = \emptyset$ under pool avoidance and $\emptyset \notin \mathcal{M}$. If calibration never scores a model of $\mathcal{M}$ and every member of $\mathcal{P}$ was scored by the calibration or belongs to an inherited set of scored models that avoids $\mathcal{M}$, then every member of $\mathcal{P}$ lies outside $\mathcal{M}$, i.e. $\mathcal{P} \cap \mathcal{M} = \emptyset$.

(b). Suppose $V \cap \mathcal{M} = \emptyset$ and let $\gamma'_\psi = \gamma_\psi \mathbf{1}_{\mathcal{M}^c}$. On $\Omega' = \Omega \setminus \mathcal{M}$ (nonempty, as $\emptyset \in \Omega'$), $\gamma'_\psi = \gamma_0|_{\Omega'} e^{\psi d|_{\Omega'}}$ with $\gamma_0|_{\Omega'} > 0$ (Lemma 1), so γ' is the bridge on Ω' with restricted γ_0, d, and $q'_\psi = \gamma_\psi \mathbf{1}_{\mathcal{M}^c}/\bar{Z}_\psi(\mathcal{M}^c) = q_\psi(\cdot \mid \mathcal{M}^c)$. Since $V \subseteq \mathcal{M}^c$, $\gamma'_0(S)e^{td'(S)} = \gamma_0(S)e^{td(S)}$ for every $S \in V$, hence $F'_V(t) = F_V(t)$ for all real t and $\rho'_V(u,v) = \exp\{2F'_V(v) - F'_V(u) - F'_V(2v-u)\} = \rho_V(u,v)$ for all real u, v. Fix one implementation $\mathcal{I}$ and the inputs (pilot list, V, T_e). By definition, $\mathcal{I}$ is deterministic and accesses the bridge only through evaluations of ρ_V; each of its steps (the next query point, a comparison with 0.90, a validation, appending, failing, stopping) is a function of the inputs and of the answers to the previous queries. Run $\mathcal{I}$ on both bridges. By induction on k: if the first $k-1$ queries and their answers coincide, then the k-th query point coincides (it is determined by the same inputs and answers), and its answer coincides because $\rho'_V = \rho_V$ at every real pair; under EX-A every comparison of the answer with 0.90 is exact, so the same branch is taken. Hence the whole transcripts coincide, and so do the outcomes: the same grid, or failure in both. □

Remark 5. Nothing above concerns particles: the statements are identities and inequalities between exact bridge quantities, and no assertion derives particle mixing from ESS.

A Certified computation of the witness inequalities

Script `witness_certified.py` (run with `python -I witness_certified.py`; Python 3 with mpmath):

```
"""Certified enclosures for the T13 witnesses (proof Section 5), by interval arithmetic.

All rational inputs (varpi(k), t_i, nu, binomial counts) are computed EXACTLY with Python integers
    / Fractions and
converted to intervals that contain them; every transcendental operation (sqrt, log, exp) uses
    mpmath.iv, whose
operations return intervals guaranteed to contain the exact result (outward rounding). Each
    internal interval
therefore contains the exact real value, and each inequality is decided by comparing internal
    interval endpoints.
For display, the internal binary endpoints are converted exactly to rationals and rounded OUTWARD
    to 22 decimals
(lower endpoint down, upper endpoint up), so every printed interval also contains the exact value.
Run: python -I witness_certified.py
"""
from fractions import Fraction as Fr
from math import comb
from mpmath import iv

iv.dps = 50
n, p = 42, 40
K = min(p, n // 2) # 21
nu = Fr(1, 100) + Fr(n, 2) # 21.01 = 2101/100
b0 = Fr(1, 100)
c1, c2 = Fr(97, 100), Fr(10, 100)
w = lambda k: Fr(1, 2) / comb(2 * p - 1, k)
WK = sum(comb(p, k) * w(k) for k in range(K + 1))
vp = lambda k: w(k) / WK # exact rationals
t = [Fr((n - 1) ** 2) * c * c / (n * (n - 1 + 2 * b0)) for c in (c1, c2)] # exact rationals
assert t[0] + t[1] < 1

def I(q): # interval containing the rational q
   return iv.mpf(q.numerator) / iv.mpf(q.denominator)
```

```
NU, N = I(nu), iv.mpf(n)
rs = 1 / iv.sqrt(N) # n^{-1/2}
g = [iv.exp(-NU * iv.log(I(1 - ti))) for ti in t] # (1 - t_i)^{-nu}
D = NU * iv.log(I((1 - t[0]) * (1 - t[1]) / (1 - t[0] - t[1]))) # D
eD = iv.exp(D)
Nn = p - 2
def sigma(a):
    tot = iv.mpf(0)
    for m in range(0, min(Nn, K - a) + 1):
        tot += iv.mpf(comb(Nn, m)) * I(vp(a + m)) * rs ** (a + m)
    return tot
s0, s1, s2 = sigma(0), sigma(1), sigma(2)
P = s0 + (g[0] + g[1]) * s1
R = g[0] * g[1] * s2
r = R / P
# witness (c)
Zc = I(vp(0)) + I(vp(1)) * rs * (Nn + g[0] + g[1]) + I(vp(2)) / N * Nn * (g[0] + g[1])
zc = Zc / P
ac1 = zc / (1 + r * eD)
ac2 = zc / (1 + r * eD * eD)
chi = (1 + r) * (1 + r * eD * eD) / (1 + r * eD) ** 2
bound = iv.mpf(1000) / iv.mpf(729) # 10/(9*0.81) = 1000/729 exactly
# witness (d)
A = I(vp(0)) + g[0] * (I(vp(1)) * rs + Nn * I(vp(2)) / N)
B = g[0] * g[1] * (I(vp(2)) / N + Nn * I(vp(3)) * rs ** 3 + comb(Nn, 2) * I(vp(4)) / N ** 2)
alpha, beta, s = A / P, B / R, B / A
def rho(u2, v2): # u = u2/2, v = v2/2 (half-integers); rho = (1+s e^{vD})^2 / ((1+s e^{uD})(1+s e
    ^{(2v-u)D}))
    E = lambda k2: iv.exp(D * iv.mpf(k2) / 2)
    return (1 + s * E(v2)) ** 2 / ((1 + s * E(u2)) * (1 + s * E(2 * v2 - u2)))
ad = lambda k2: (A + B * iv.exp(D * iv.mpf(k2) / 2)) / (P + R * iv.exp(D * iv.mpf(k2) / 2))

def exact_ends(x): # the interval's two binary endpoints as exact rationals
    def fr(t):
        sign, man, exp, _ = t
        q = Fr(man) * (Fr(2) ** exp)
        return -q if sign else q
    return fr(x._mpi_[0]), fr(x._mpi_[1])
def show(name, x, k=22): # outward-rounded decimal display: printed [lo, hi] contains the internal
    interval
    lo, hi = exact_ends(x)
    s = 10 ** k
    L = Fr((lo.numerator * s) // lo.denominator) # floor(lo * 10^k)
    H = Fr(-((-hi.numerator * s) // hi.denominator)) # ceil(hi * 10^k)
    fmt = lambda q: (str(q.numerator // s) if q >= 0 else "-" + str((-q.numerator) // s)) + "." +
        str(abs(q.numerator) % s).zfill(k)
    print(f"{name:18s} in [{fmt(L)}, {fmt(H)}] (decimals outward-rounded; width < 1e-20: {hi - lo <
         Fr(1, 10**20)})")
for nm, x in [("t1", I(t[0])), ("t2", I(t[1])), ("D", D), ("e^D", eD), ("r = R/P", r), ("r e^D", r
    * eD), ("r e^{2D}", r * eD * eD), ("Z_c/P", zc),
            ("a_c(1)", ac1), ("a_c(2)", ac2), ("1+chi2(q1||q0)", chi), ("1000/729", bound),
            ("alpha = A/P", alpha), ("beta = B/R", beta), ("beta - alpha", beta - alpha), ("s = B/A
                ", s),
            ("rho_Vd(0,1/2)", rho(0, 1)), ("rho_Vd(1/2,1)", rho(1, 2)),
            ("a_d(0)", ad(0)), ("a_d(1/2)", ad(1)), ("a_d(1)", ad(2)), ("a_d(3/2)", ad(3))]:
    show(nm, x)
checks = {
    "D > 0": D.a > 0,
```

```
    "a_c(1) >= 0.9": ac1.a >= (iv.mpf(9) / 10).b,
    "a_c(2) < 0.9": ac2.b < (iv.mpf(9) / 10).a,
    "1+chi2 > 1000/729": chi.a > bound.b,
    "beta > alpha": (beta - alpha).a > 0,
    "rho(0,1/2) >= 0.9": rho(0, 1).a >= (iv.mpf(9) / 10).b,
    "rho(1/2,1) >= 0.9": rho(1, 2).a >= (iv.mpf(9) / 10).b,
}
for k, v in checks.items(): print(f"CHECK {k:22s}: {v}")
print("ALL CHECKS PASS" if all(checks.values()) else "SOME CHECK FAILS")
```

Output (each printed interval is the internal enclosure rounded outward to 22 decimals, so it also contains the exact value):

```
t1 in [0.9180497898818230363817, 0.9180497898818230363818] (decimals outward-rounded; width < 1e
    -20: True)
t2 in [0.009757145178890668895, 0.009757145178890668896] (decimals outward-rounded; width < 1e
    -20: True)
D in [2.4573889178992225124231, 2.4573889178992225124232] (decimals outward-rounded; width < 1e-20:
     True)
e^D in [11.6742891816087945733827, 11.6742891816087945733828] (decimals outward-rounded; width < 1e
    -20: True)
r = R/P in [0.0052719554605095421403, 0.0052719554605095421404] (decimals outward-rounded; width <
    1e-20: True)
r e^D in [0.0615463325985499584319, 0.0615463325985499584320] (decimals outward-rounded; width < 1e
    -20: True)
r e^{2D} in [0.718509684822948469307 8, 0.718509684822948469308] (decimals outward-rounded; width <
     1e-20: True)
Z_c/P in [0.9841629956846136023762, 0.9841629956846136023763] (decimals outward-rounded; width < 1e
    -20: True)
a_c(1) in [0.9271031941446113197618, 0.9271031941446113197619] (decimals outward-rounded; width < 1
    e-20: True)
a_c(2) in [0.5726839972892024642265, 0.5726839972892024642266] (decimals outward-rounded; width < 1
    e-20: True)
1+chi2(q1||q0) in [1.5330546700532342666175, 1.5330546700532342666176] (decimals outward-rounded;
    width < 1e-20: True)
1000/729 in [1.3717421124828532235939, 1.3717421124828532235940] (decimals outward-rounded; width <
     1e-20: True)
alpha = A/P in [0.9841629956846136023757, 0.9841629956846136023758] (decimals outward-rounded;
    width < 1e-20: True)
beta = B/R in [0.9962507850715599508558, 0.9962507850715599508559] (decimals outward-rounded; width
     < 1e-20: True)
beta - alpha in [0.012087789386946348480 0, 0.012087789386946348481] (decimals outward-rounded;
    width < 1e-20: True)
s = B/A in [0.005336707221694864045 4, 0.005336707221694864045 5] (decimals outward-rounded; width <
    1e-20: True)
rho_Vd(0,1/2) in [0.9708134408716465566174, 0.9708134408716465566175] (decimals outward-rounded;
    width < 1e-20: True)
rho_Vd(1/2,1) in [0.9137627308499763841343, 0.9137627308499763841344] (decimals outward-rounded;
    width < 1e-20: True)
a_d(0) in [0.9842263877716183656109, 0.9842263877716183656110] (decimals outward-rounded; width < 1
    e-20: True)
a_d(1/2) in [0.9843768807694762001798, 0.9843768807694762001799] (decimals outward-rounded; width <
     1e-20: True)
a_d(1) in [0.9848638215300282733429, 0.9848638215300282733430] (decimals outward-rounded; width < 1
    e-20: True)
a_d(3/2) in [0.9862632653219948237377, 0.9862632653219948237378] (decimals outward-rounded; width <
     1e-20: True)
CHECK D > 0 : True
CHECK a_c(1) >= 0.9 : True
```

```
CHECK a_c(2) < 0.9 : True
CHECK 1+chi2 > 1000/729 : True
CHECK beta > alpha : True
CHECK rho(0,1/2) >= 0.9 : True
CHECK rho(1/2,1) >= 0.9 : True
ALL CHECKS PASS
```

B Verbatim quotations of the definitions used

The passages below are copied mechanically (by line number) from `00_COMMON.md` (SHA-256 `a7d05134350823efd62add9d66a8ec1e8ff78185f2efc78415a504cc819e023d`) and `METHODOLOGY_v3.md` (SHA-256 `93043758058a6d422cca6ca4b6be8f3a9e8149b6f6de2e1bab1ce31c44b8019c`), the files STATEMENT.md refers to. For display only, lines longer than 100 characters are wrapped at spaces, and continuation lines begin with " | "; removing these markers and rejoining recovers the source text (checked by the extraction script). Markdown markup is kept as in the source.

--- 00_COMMON.md, §1 Data, target (first three bullets), lines 9-11 ---
- Raw data (X^r, y^r), $n \geq 3$, no constant column, non-constant y^r. $C = I - 11^T/n$; D_x = diag of
  | retained raw column sample SDs. Standardised $X = CX^rD_x^{-1}$, $y = Cy^r/s_y$ with $s_y^2 = \|Cy^r\|^2/(n-1)$;
  | so $\|x_j\|^2 = \|y\|^2 = n - 1$. Contrast coordinates: $U \in \mathbb{R}^{n\times(n-1)}$, $U^TU = I$, $UU^T = C$; $y_c = U^Ty^r$, X_c
  | $= U^TX$.
- State space $\Omega = \{S \subseteq [p] : |S| \leq K\}$, $K = \min(p, \lfloor n/2 \rfloor)$. $G_S = X_S^TX_S = (n-1)R_S$; $z_S = X_S^Ty$; H_S
  | $= (n+1)G_S + I$; $D_S = \det H_S/\det(G_S+I)$; $Q_S = y^Ty - n\, z_S^TH_S^{-1}z_S$; $\nu = 0.01 + n/2$; $a_0 = b_0 =$
  | 0.01.
- $m(S) = (2b_0)^{a_0}\Gamma(\nu)/(\pi^{n/2}\Gamma(a_0)) \cdot D_S^{-1/2}(2b_0 + Q_S)^{-\nu}$; $w(k) = ½/C(2p-1, k)$; $W_K =$
  | $\Sigma_{k\leq K}\, C(p,k)w(k)$; $\varpi(S) = w(|S|)/W_K$.

--- 00_COMMON.md, §2 Sampler objects (blocks, start, bridge), lines 18-20 ---
- Blocks $C_1, \ldots, C_J$ (deterministic in X: complete linkage, top-down cut, next-fit packing), $|C_b| \leq$
  | $B = \min(\lfloor\sqrt{n}\rfloor, \lfloor\sqrt{p}\rfloor, 16)$. Tables $r_b(A) = BF(A)$ for $A \subseteq C_b$ (full response).
- $\gamma_0(S) = \varpi(S)\prod_b r_b(S \cap C_b)$; $\pi_0 = \gamma_0/\bar{Z}_0$ ($\bar{Z}_0$ by exact convolution of block size profiles with
  | the capped size prior; backward sampler gives exact iid π_0 draws). Start mode $\tilde{S}_0 := \operatorname{argmax} \pi_0$.
- Discrepancy $d(S) = \log BF(S) - \Sigma_b \log r_b(S \cap C_b)$. Bridge $\gamma_\varphi = \gamma_0 e^{\varphi d}$, $q_\varphi = \gamma_\varphi/\bar{Z}_\varphi$
  | (defined for real φ; production $\varphi \in [0,1]$); $q_0 = \pi_0$, $q_1 = \pi$, $Z = m(\emptyset)\bar{Z}_1$.

--- 00_COMMON.md, §2 Budgets and exact branch, lines 24-25 ---
- Budgets: $N_0 = \max(2500, n, p)$; $H_0 = \lceil 512N_0/2500 \rceil$; $\ell = \lceil \log_2(n+p+1) \rceil$; $T_0 = \max(1000, \ell^2)$; (N_e,
  | H_e, T_e) = 4^e(N_0, H_0, T_0), $e \in \{0,1\}$; $L = \max(4, \lceil \log_2(n+1) \rceil)$. $Q_f := 4N_e(T + L + 5)$ iid π_0 draws
  | scored in the final run (initialisation + independence proposals); $4N_e(L+6) \leq Q_f \leq 4N_e(T_e + L +$
  | 5).
- Exact branch: $\Sigma_{k\leq K}\, C(p,k) \leq (5H_0 + 4N_0)p$; it precedes block/Gram construction and scores every
  | feasible model.

--- METHODOLOGY_v3.md, §3 Notation and budgets (table rows), lines 62-73 ---
Symbol	Definition	n = 200, p = 500	n = 200, p = 1,000
p	number of retained (non-constant) predictors	500	1,000
K	size cap, $\min(p, \lfloor n/2 \rfloor)$	100	100
B	block size, $\min(\lfloor\sqrt{n}\rfloor, \lfloor\sqrt{p}\rfloor, B^*)$, $B^* = 16$ (implementation limit)	14	14
ℓ	$\lceil \log_2(n + p + 1) \rceil$ (local-search and rebuild budget)	10	11
L	$\max(4, \lceil \log_2(n + 1) \rceil)$ (burn-in rounds at $\varphi = 1$)	8	8
N_0	max(2,500, n, p) final particles per island	2,500	2,500
H_0	$\lceil 512 \cdot N_0/2{,}500 \rceil$ calibration particles (pilot and each diagnostic island)	512	512
T_0	$\max(1{,}000, \ell^2)$ stage ceiling	1,000	1,000
R	4 islands (final and diagnostic)	4	4
tier e	$e \in \{0, 1\}$; (N_e, H_e, T_e) = 4^e (N_0, H_0, T_0)		

--- METHODOLOGY_v3.md, §4 Preprocessing (items 1–4), lines 80-87 ---
1. Stop with `invalid_input` if any entry of X or y is not finite, or $n < 3$, or $p < 1$.
2. Drop column j only if all its entries are **exactly** equal; record it; it is never selected. p | := number kept. If $p = 0$, return $\hat{S} = \emptyset$, PIPs and evidence unavailable, status `no_predictors`.
3. If all entries of y are exactly equal, return $\hat{S} = \emptyset$ (the exact HPM: every $Q_S = 0$, $D_S \geq 1$ and w | is decreasing), PIPs and evidence unavailable, status `y_constant`. (The exact PIPs are not 0; they are not computed.)
4. Standardise stably: for each column, divide by its maximum absolute value, compute the mean, centre, then divide by the sample SD (denominator $n - 1$), so $\|x_j\|^2 = n - 1$. Same for y, recording $s_y^2 = \|Cy\|^2/(n-1)$ with $C = I - 11^T/n$. Original indices, means and scales are kept for reporting and back-transformation.

--- METHODOLOGY_v3.md, §5 Target posterior, lines 97-108 ---
On the standardised data (X, y), for $S \subseteq \{1, \ldots, p\}$ with $k = |S| \leq K$, $G_S = X_S^T X_S$, $z_S = X_S^T y$:

```text
y | β_S, σ², S ~ N_n(X_S β_S, σ² I_n)              (no intercept; y is centred)
β_S | σ², S  ~ N_k(0, n σ² (G_S + I_k)^{-1})      g = n (unit information), τ = 1 (one pseudo-observation)
σ²           ~ IG(a_0, b_0),  a_0 = b_0 = 0.01
ϖ(S)         = w(k) 1{k ≤ K} / W_K,   w(k) = B(1+k, 2p−k)/B(1, p) = ½ / C(2p−1, k),   W_K = Σ_{j≤K} C(p, j) w(j)

H_S = (n + 1) G_S + I_k,   Q_S = y^T y − n z_S^T H_S^{-1} z_S,
log m(S) = log C_n − ½ [log det H_S − log det(G_S + I_k)] − (a_0 + n/2) log(2b_0 + Q_S),
C_n = (2b_0)^{a_0} Γ(a_0 + n/2) / (π^{n/2} Γ(a_0)),
π(S | y) = ϖ(S) m(S) / Z,   Z = Σ_{|S|≤K} ϖ(S) m(S).
```

--- METHODOLOGY_v3.md, §6.2 Exact branch, lines 154-161 ---
6.2 Exact branch

If $E = \Sigma_{k \leq K} C(p, k) \leq (5H_0 + 4N_0) \cdot p$ (12,560·p at baseline; counted with saturating integer arithmetic), score every model with $|S| \leq K$ in a streaming pass: running best (strict order of §6.10), log-sum-exp accumulation of Z and of the PIP numerators.
No blocks are built. Status `exact`; `global_certified` when all comparisons are numerically validated. This applies for $p \leq 17$ for $n \geq 22$ ($p \leq 18$ at $n = 20, 21$), and for larger p when the cap binds (for example $p \leq 36$ at $n = 10$; all $p \leq 3{,}000$ at $n \leq 5$). Because the threshold grows with $N_0 \geq n$, every fixed-p problem is eventually enumerated as n grows.

--- METHODOLOGY_v3.md, §6.3 Blocks, tables and the composite start, lines 162-177 ---
6.3 Blocks, tables and the composite start

- **Clustering:** complete linkage on the distance $1 - |r_{jl}|$ computed from X; ties between candidate merges broken lexicographically on the sorted original-index lists.
- **Cut:** top-down from the root; any node with more than B members is split into its two children (visited smallest index first) until every retained cluster has at most B members.
- **Packing:** next-fit in dendrogram leaf order, whole clusters only (a retained cluster is never split), into blocks $C_1, \ldots, C_J$ of at most B members. $J \geq \lceil p/B \rceil$ (packing leaves gaps).
- **Tables:** for each block, every sub-model $U \subseteq C_b$ is scored against the full y; $r_b(U) = m(U)/m(\emptyset)$ (table exponent 1).

Since B ≤ K for n ≥ 3, no sub-model exceeds the cap. Every table score counts for the HPM. Table
| scores are not inserted in
the score cache.
- **Start:** $\gamma_0(S) = \varpi(S) \prod_b r_b(S \cap C_b)$, $\pi_0 = \gamma_0/\bar{Z}_0$. $\bar{Z}_0$ is computed exactly by convolving the
| block size profiles
$e_b(j) = \Sigma_{|U|=j} r_b(U)$ up to K with the size prior (valid because ϖ depends on S only through
| |S|). Exact iid draws:
draw the total size, then block sizes backwards, then each block's sub-model within its size
| class.
- **Discrepancy:** $d(S) = \log BF(S) - \Sigma_b \log r_b(S \cap C_b)$, $BF(S) = m(S)/m(\emptyset)$; bridge $\gamma_\varphi(S) = \gamma_0(S)$
| $e^{\varphi d(S)}$, $\varphi \in [0, 1]$;
$\gamma_1 \propto$ the posterior; $Z = m(\emptyset) \Sigma_S \gamma_1(S)$.

--- METHODOLOGY_v3.md, §6.7 Candidate set and candidate grid (first four bullets), lines 211-222 ---
6.7 Candidate set and candidate grid

- **V:** the union of the top 20p scored models by $\log \gamma_0$, the top 20p scored models by $\log \gamma_1$ (the
| target), and $\emptyset$;
deduplicated; list ties lexicographic; frozen during each construction.
- **Restricted log-normaliser:** $F_V(t) = \log \Sigma_{S \in V} \gamma_0(S) e^{t d(S)}$.
- **Restricted ESS** between temperatures $u < v$: $\rho_V(u, v) = \exp\{2F_V(v) - F_V(u) - F_V(2v - u)\}$,
| the population ESS fraction
of the weights $e^{(v-u)d}$ under the restricted law $q_u(\cdot \mid V)$ (evaluating F_V beyond 1 is a
| finite-sum moment, not a
production temperature).
- **Construction:** start at $u = 0$ with the pilot temperatures as targets. For the next pilot
| temperature b: if
$\rho_V(u, b) \geq 0.90$ append b; otherwise bisect on (u, b] with u fixed, at most 60 halvings, and
| append the largest tested
endpoint that passes; continue from the appended point. Validate strict numerical progress and the
| inequality at every
appended point. Non-progress, failed validation, or more than T_e intervals → the construction
| fails.

(d) Review record

FINAL_REPORT.md: statement changes

2. Changes to the statement, and why

All changes were checked by Astra #1 in three statement checks (`review\astra_01..03`). Check 1 rejected one reading, which counts as one restart of step 1 (the limit is 3). The final statement is `STATEMENT.md` §2.

1. **ε_C must be in [0, 1).** For $\varepsilon_C \geq 1$ the bound is undefined or false.
2. **(iv)(a): "calibration never scores $\mathcal{M}$" became an explicit hypothesis.** The hypothesis is that the scored pool V is drawn from avoids $\mathcal{M}$. The method does not say whether table scores or leaders carried across tiers feed V's top lists. The source wording is kept as a sufficient condition, with that provenance proviso.
3. **(ii)(b): each grid is paired with the V of the construction that produced it.**
4. **(ii)(c)–(d): the two cautionary examples must come from real data run through the method.** That means the method's blocks, tables, γ_0 and d, on the SMC branch, with a V of the method's form and a grid the construction actually outputs. My first version used abstract

weights; Astra rejected that as a weakening (restart 1). The witness also names its tier, its T_e and a deterministic implementation.

5. **(iii): the bound uses coverage at v.** COVER supplies it.
6. **(i), (iii) hold for all real u, v,** a free strengthening.
7. **Positivity is derived, not assumed.**
8. **(iv)(b) is for one fixed deterministic implementation.**
9. **(ii)(d) is strengthened:** the minimum is attained *only* at φ_0.

None of these changes the substance of the source theorem. EXP / EXP′ is not needed anywhere.

FINAL_REPORT.md: review history

4. Review history and my judgments

Stage	Reviewer	Version	Verdict	Findings (fatal / gap / exposition)	My answer
A	Astra #1	v1	not verified	0 / 1 / 2	Gap accepted: the witness inequalities had no certified error bound. Fixed by interval arithmetic (v2). Exposition fixed.
A	Astra #1	v2	verified	0 / 0 / 1	Printed endpoints were rounded to nearest; fixed with outward rounding (v3).
A	Astra #1	v3	verified	none	Stage A proof candidate.
B	Astra #2 (fresh)	v3	not verified	0 / 1 / 0	Gap accepted: the method definitions were not in the packet. Fixed by Appendix B (v4).
B	Opus xhigh (fresh)	v3	not verified	0 / 4 / 5	The four gaps are the same issue in detail (m(S) formula, n − 1 standardisation, block properties, budgets). All accepted and fixed in v4, with the exposition points.
B	Astra #2	v4	**verified**	none	Agrees with every logged answer.
B	Opus xhigh	v4	**verified**	0 / 0 / 3	The provenance note is closed by my mechanical check (9/9 quoted passages match the source lines; checksums match). The other two notes need no change (the reviewer agrees).

- No reviewer found a mathematical error at any stage. Every finding was about rigour or completeness of the record.
- Every gap finding, verbatim with my answer and the reviewers' rulings, is in `CLAIMS_OF_FATAL_FLAWS.md`.
- Raw replies are in `review\`: the Astra logs `astra_0*.log` and `stageB_astra2_0*.log`, and the Opus replies `stageB_opus_0*.md`.
- **Astra judge: not used.** There was no dispute: both Stage B reviewers verified v4 in their second round.

CLAIMS_OF_FATAL_FLAWS.md (every entry, verbatim)

CLAIMS OF FATAL FLAWS — T13 (T10) restricted ESS

Every fatal or gap finding, copied verbatim, with the lead's answer. Exposition findings are listed briefly at the end of each round.

Stage A, round 1 — Astra #1 review of PROOF_v1 (review\astra_05_proof_review_v1.log)

A1-1 (gap)

Location: §5.1, “Numerical values”; §§5.2–5.3, numerical comparisons; Appendix A, particularly `mp.dps = 40`.

Severity: gap.

Claim: The exact inequalities establishing the witnesses lack a justified numerical error bound.

Evidence: Appendix A evaluates noninteger powers, logarithms, exponentials, sums, and ratios using ordinary finite-precision `mpmath` arithmetic. It supplies neither certified enclosures nor an error analysis. Consequently, the displayed approximations alone do not rigorously establish $a_c(1) \geq 0.9$, $a_c(2) < 0.9$, the divergence-bound violation, $\beta > \alpha$, or the two acceptance inequalities. These comparisons are essential to parts (ii)(c)–(d). EX-A specifies the idealized construction’s arithmetic; it does not certify the appendix’s floating-point calculations. Independent agreement and the substantial numerical margins make this plausibly fixable, but do not supply the missing certification.

Lead's answer: accepted and fixed in PROOF_v2. All rational inputs (ϖ(k), t_i, ν, binomial counts) are now computed exactly with integer and fraction arithmetic and converted to enclosing intervals. Every transcendental step (√, log, exp) is evaluated in interval arithmetic with outward rounding (`mpmath.iv`, 50 digits), so each output interval contains the exact real value. The seven inequalities the witnesses use (D > 0, a_c(1) ≥ 0.9, a_c(2) < 0.9, $1+\chi^2 > 1000/729$, β > α, ρ_{V_d}(0,½) ≥ 0.9, ρ_{V_d}(½,1) ≥ 0.9) are decided by comparing interval endpoints, and all pass. All enclosures are narrower than 10^{-20}. The script `witness_certified.py` and its full output are in Appendix A of PROOF_v2. §5.1 now states the certification rule, and the displayed decimals are marked "≈" and described as orientation only.

Exposition findings, round 1 (not fatal or gap; handled in PROOF_v2)

- A1-2: the claimed independent validation (Cholesky check, reduced enumeration) could not be audited from the packet. **Fixed:** the claim is removed from the proof. The proof does not rely on it; the check itself is in `helpers\check_sonnet\`.
- A1-3: rounded values were written as exact equalities, and 10/(9·0.81) is exactly 1000/729. **Fixed:** "≈" with a stated meaning, and 1000/729 written exactly.

Stage A, round 2 — Astra #1 ruling on the log and review of PROOF_v2 (review\astra_06_ruling_v2.log)

Rulings: A1-1 agree, fix works; A1-2 agree, fix works; A1-3 agree for the main text, incomplete in Appendix A. Verdict on PROOF_v2: **verified**. No fatal or gap finding. One exposition finding: printed

interval endpoints were rounded to nearest, so a printed interval could fail to contain the exact value (e.g. the printed 1000/729 singleton). **Fixed in PROOF_v3:** the endpoints are converted exactly to rationals and rounded outward to 22 decimals, and the docstring and §5.1 now say so. The decisive checks were already on internal endpoints and are unchanged.

Stage A, round 3 — Astra #1 confirmation of PROOF_v3 (review/astra_07_confirm_v3.log)

Fix confirmed; no remaining findings; verdict **verified**. PROOF_v3 = Stage A proof candidate.

Stage B, round 1 — independent reviews of PROOF_v3

B-A2-1 (Astra #2, gap) — review/stageB_astra2_01_review_v3.log

- **Location:** §1, Lemma 1; §5.1, "The data set" and Lemma 4.
- **Severity:** gap.
- **Claim:** The identification of the proof's likelihood, block construction and budgets with the method incorporated into STATEMENT.md cannot be verified from the permitted files.
- **Evidence:** STATEMENT.md imports definitions from `00_COMMON.md` and `METHODOLOGY_v3.md` without reproducing the marginal-likelihood formula, hyperparameters, block-size rule, dependence of blocks on X alone, or budget formulas. The proof supplies these as facts. Their conformity with the imported method is essential: the likelihood determines positivity and the witness scores; the block rule permits choosing j_2 before defining y; and the budgets establish the SMC branch and allowable grids. The subsequent calculations are consistent, but the correspondence to the referenced method remains unchecked. This is a verification limitation, not evidence that those assertions are false.

Lead's answer: accepted and fixed in PROOF_v4. New Appendix B quotes verbatim, copied mechanically by line number with both files' SHA-256 checksums, every definition the proof uses: 00_COMMON §1 (data, standardisation, Ω, G_S, z_S, H_S, D_S, Q_S, ν, $a_0 = b_0 = 0.01$, m(S), w, W_K, ϖ), 00_COMMON §2 (blocks deterministic in X with $|C_b| \le B$, tables, γ_0, d, bridge; budgets; exact branch), and METHODOLOGY §3 (budget table), §4 items 1–4 (input checks, standardisation with the n − 1 SD), §5 (target), §6.2 (exact branch), §6.3 (clustering, cut, packing, tables, start, discrepancy) and §6.7 (V, F_V, ρ_V, construction). Lemma 1 now restates the definitions with citations. §5.1 now derives the three block properties (X-only, partition, size cap) and the standardisation and budget facts from the quotes.

B-O-1 (Opus, gap) — review/stageB_opus_01_review_v3.md, finding 1

> **Claim:** The proof uses a specific formula for m(S) that I cannot check against its source. STATEMENT.md defines m(S) only by reference to 00_COMMON §1, which is not in the packet. [...] The proof never states the definition of Q_S or the values of a_0 and b_0. [...] The only open question is whether this is the m(S) of 00_COMMON.

(Full text verbatim in review/stageB_opus_01_review_v3.md, finding 1.)

Lead's answer: accepted and fixed in PROOF_v4. The m(S) formula with $a_0 = b_0 = 0.01$, $\nu = 0.01 + n/2$, the definition of Q_S, H_S and D_S are now stated in the proof of Lemma 1, cited to Appendix B (00_COMMON §1 lines 10–11; METHODOLOGY §5). They match what v3 used.

B-O-2 (Opus, gap) — finding 2

> **Claim:** The step "$\|x_j\|^2 = 41 = n-1$, so every sample SD equals 1 … standardisation returns X and y unchanged" assumes 00_COMMON standardises with the n−1 denominator. That convention is not in the packet.

Lead's answer: accepted and fixed in PROOF_v4. Both sources use the n − 1 denominator: 00_COMMON §1 line 9 ("$s_y^2 = \|Cy^r\|^2/(n-1)$; so $\|x_j\|^2 = \|y\|^2 = n - 1$") and METHODOLOGY §4 item 4 ("divide by the sample SD (denominator n − 1)"). Both are quoted in Appendix B. §5.1 now shows that both forms of the standardisation (00_COMMON's $CX^rD_x^{-1}$, and the method's divide by max |x|, centre, divide by SD) return X and y unchanged. Lemma 4 is therefore correct as written.

B-O-3 (Opus, gap) — finding 3

> **Claim:** The proof uses three properties of the method's block construction that the packet does not contain. [...] (a) The blocks depend on X only, not on y. [...] (b) The blocks partition [p]. [...] (c) Block size is capped at $B = \min(\lfloor\sqrt{n}\rfloor, \lfloor\sqrt{p}\rfloor, 16)$.

Lead's answer: accepted and fixed in PROOF_v4. §5.1 "Blocks" derives the three properties from the quotes:

- (a) 00_COMMON §2 line 18 ("deterministic in X") and METHODOLOGY §6.3 (distance "computed from X"; the cut and packing use only the dendrogram, B and index order).
- (b) The root holds all p columns; the cut replaces a node by its two children until every retained cluster has at most B members; packing places whole clusters ("a retained cluster is never split") into blocks.
- (c) $|C_b| \le B = \min(\lfloor\sqrt{n}\rfloor, \lfloor\sqrt{p}\rfloor, 16)$.

Lemma 4's proof cites (b).

B-O-4 (Opus, gap, low risk) — finding 4

> **Claim:** The formulas $N_0 = \max(2500, n, p)$, $H_0 = \lceil 512 \cdot 2500/2500 \rceil$, $\ell = \lceil \log_2(n+p+1) \rceil$ and $T_0 = \max(1000, \ell^2)$, and the constants in the exact-branch condition, come from 00_COMMON §2 and cannot be checked in the packet. The H_0 expression as printed (2500/2500) does not show which of the two 2500s is N_0.

Lead's answer: accepted and fixed in PROOF_v4. The formulas are cited to 00_COMMON §2 lines 24–25 and METHODOLOGY §3 and §6.2, all quoted in Appendix B. H_0 is now written $\lceil 512N_0/2500 \rceil = \lceil 512 \cdot 2500/2500 \rceil$.

Exposition findings, Stage B round 1 (handled in PROOF_v4)

- B-O-5 (no implementation named): §5.2 "Grid" now names one (test b first, then midpoint bisection), and notes that no bisection is ever used in either witness, so every implementation gives the same output.
- B-O-6 (reading of the output grid; the frozen-grid parenthetical): §4 (ii)(b) now states the reading explicitly and why it is the only consistent one; the parenthetical is the special case.

- B-O-7 (a) (the seven checks do not cover every inequality): §5.1 now lists the exact integer and rational facts separately. (b) (re^D and re^{2D} had no printed enclosure): the script now prints certified enclosures for both.
- B-O-8 (Z_c/P and α display identically): §5.3 notes that they differ (Z_c − A > 0, about $5 \cdot 10^{-22}$ relative) and that their enclosures are disjoint.
- B-O-9 (B, C_b used before definition; |U| ≤ K step unneeded): Lemma 1 now argues m(U) ∈ (0, ∞) for every U ⊆ [p], without using the size cap.

Stage B, round 2 — rulings on this log and review of PROOF_v4

- **Astra #2** (review/stageB_astra2_02_ruling_v4.log): agrees with the lead's answer on every logged item (A1-1..3, Stage A round 2, B-A2-1, B-O-1..9), and every fix works. No new or remaining findings. **Verdict: verified.**
- **Opus reviewer** (review/stageB_opus_02_ruling_v4.md): agrees with the lead's answer on every logged item, and every fix works. No fatal or gap finding. **Verdict: verified.** Three exposition notes: 1. Provenance of the Appendix B quotes could not be checked from the packet. **Closed by the lead:** helpers/proof_build/check_appendix_B.py compares the quotes inside PROOF_v4.tex with the cited source lines. Result: 9/9 passages match, and both SHA-256 checksums in the proof equal those of the current files (output in helpers/proof_build/check_appendix_B_output.txt). 2. The example implementation in §5.2 is a sketch. **No change:** the reviewer agrees that existence is not in doubt, and that no unspecified branch is reached in either witness. 3. Lemma 1 extends the m formula to every U ⊆ [p], beyond the source's |S| ≤ K. **No change:** the reviewer rates it harmless; positivity is needed only where m is defined.

No fatal or gap finding is open. Astra judge not needed.

Chapter 5

T14(i) (L-FM, singleton case): first-moment evidence consistency

Final proof: `proofs/T14_LFM_first_moment_evidence/proof/PROOF_v3.tex`. *Sources of parts (a), (b), (d):* `theorems/T14_LFM_first_moment_evidence.md`, `STATEMENT.md`, `FINAL_REPORT.md`, `CLAIMS_OF_FATAL_FLAWS.md`.

(a) Original conjecture text (theorem file, verbatim)

T14 (L-FM): First-moment evidence consistency

Status: conjectured, not proven. Methodology SMC-BVS v3.0 (2026-10-19), SHA-256 93043758058a6d422cca6ca4b6be8f3a9e8149b6f6de2e1bab1ce31c44b8019c. Read with `00_COMMON.md`.

Category / rank: algorithmic convergence (consistency of the evidence); **core** (lemma).

Definitions used

Stage laws q_t, G_t = e^{Δ_td}, empirical population $\hat{q}_{t-1}$ reweighted at stage t, island evidence $\hat{Z}_r$ (00_COMMON §2). For a model S° fixed given $\mathcal{C}$: η_t = 1 − q_t(S°), ε_* = max_{0≤t≤T}η_t; R_t(S) = G_t(S)/G_t(S°); A_t = $\prod_{v=1}^{t} \hat{q}_{v-1}(R_v)$; b_t = q_0(S°)/q_t(S°). For a set $\mathfrak{A}$: ε_t = 1 − q_t($\mathfrak{A}$); c_t = q_{t−1}(G_t1_$\mathfrak{A}$)/q_{t−1}($\mathfrak{A}$); R_t = G_t/c_t; A_t, b_t = q_0($\mathfrak{A}$)/q_t($\mathfrak{A}$) analogously; Φ_t = E[A_{t−1}|$\hat{q}_{t-1}$((R_t − 1)1_$\mathfrak{A}$)|]/b_{t−1}.

Assumptions

EX-A, EX-S, CAL; exact start; prescribed multinomial resampling and q_φ-invariant rounds, one round per stage.

Precise statement

Conditional on a successful $\mathcal{C}$: (i) [singleton] $\hat{Z}_r/Z$ = A_T/b_T, and for every N ≥ 1 (ε_* < 1):

E|$\hat{Z}_r$/Z − 1| ≤ [Σ_{t=1}^T(b_{t−1}η_{t−1} + b_tη_t) + |1 − b_T|]/b_T ≤ (2/(1−ε_*))Σ_{t=0}^T η_t.

The endpoint term |1 − b_T| is essential. Convexity gives the same bound for the pooled $\hat{Z}$. (ii) [consequence] Along launched families, Σ_{t=0}^T η_t → 0 in data/calibration probability implies $\hat{Z}$/Z

→ 1 in joint probability. No over-tempered condition and no N^{−1/2} rate is asserted. (iii) [general set] $\hat{Z}_r/Z = A_T/b_T$ with the set definitions, and $E|\hat{Z}_r/Z - 1| \le [2\Sigma_{t=0}^{T} \varepsilon_t + \Sigma_{t=1}^{T} \Phi_t]/(1 - \max_t \varepsilon_t)$. The signed expression inside Φ_t's absolute value has weighted mean zero; $\Phi_t \ge 0$ need not vanish (it vanishes for a singleton); $\Phi_t \le q_{t-1}(|R_t - 1| 1_{\mathfrak{A}})$, which need not be summably small. Set mass alone does not control within-set weight dispersion; reflection is one possible further route, not an automatic transfer.

Dependencies

Builds on: T12. Used by: T15, T25 (first-moment alternative), T26.

Numerical evidence (provisional)

provisional evidence (Sol check J6): exact finite-state laws (N ≤ 2, T = 3): a rare state with singleton deficits (10^{-12}, $3\cdot10^{-8}$, 10^{-3}) has exact L^1 error 0.0020 ≤ telescoping-with-endpoint 0.00200 ≤ bound 0.00200, while $E(\hat{Z}/Z)^2 \approx 10^6$; a two-state set with full set mass has L^1 error 1.996 = Φ. Files: judge/sol2/J6_FIRST_MOMENTS.md.

Proof-route hint (optional)

Use $E[A_{t-1}\hat{q}_{t-1}(f)] = b_{t-1}q_{t-1}(f)$ (T12), the exact identity $q_{t-1}(R_t 1_{S^{\circ c}}) = (b_t/b_{t-1})q_t(S^{\circ c})$ and $R_t(S^\circ) = 1$ to get $E|A_t - A_{t-1}| \le b_{t-1}\eta_{t-1} + b_t\eta_t$; telescope A_t; retain $|1 - b_T|$; note $b_t/b_T = q_T(S^\circ)/q_t(S^\circ)$ and $|1 - b_T|/b_T = |q_T(S^\circ) - q_0(S^\circ)|/q_0(S^\circ)$. For sets, choose c_t so that R_t has $q_{t-1}(\cdot|\mathfrak{A})$-mean 1 and separate inside/outside contributions. Hardest step: Φ_t for larger sets. *When proof.md is run on this file, its lead considers the statement both with and without these hints; the hints never reach the reviewers.*

(b) STATEMENT.md (verbatim)

STATEMENT — T14(i) (L-FM, singleton case): first-moment evidence consistency

Source: `Theory_architecture_v3/theorems/T14_LFM_first_moment_evidence.md`, part (i) only. Method: SMC-BVS v3.0, METHODOLOGY_v3.md (SHA-256 93043758058a6d422cca6ca4b6be8f3a9e8149b6f6de2e1bab1ce31c44b8019c). Parts (ii) and (iii) of the file are not part of this run.

Statement analysis outcome: **(a) correct as stated.** No assumption was changed or added. EXP / EXP′ is not needed. No other v3 conjecture is used as a hypothesis (the unbiasedness identity of T12 that the theorem builds on is re-derived inside the proof).

1. Setting (from 00_COMMON.md §§1–2 and METHODOLOGY §§6.3, 6.5, 6.6, 6.8, 6.12)

- $\Omega = \{S \subseteq [p] : |S| \le K\}$ is a finite set. For $S \in \Omega$: $\gamma_0(S) = \varpi(S)\prod_b r_b(S \cap C_b) > 0$ (the size prior ϖ and the block tables r_b = BF are strictly positive), $d(S) \in \mathbb{R}$, and for real φ, $\gamma_\varphi(S) = \gamma_0(S)e^{\varphi d(S)}$, $\bar{Z}_\varphi = \Sigma_{S\in\Omega}\gamma_\varphi(S)$, $q_\varphi = \gamma_\varphi/\bar{Z}_\varphi$. Hence $q_\varphi(S) > 0$ for every $S \in \Omega$. At temperature φ = 0 the bridge is π_0, at φ = 1 it is π, and $Z = m(\emptyset)\cdot\bar{Z}_\varphi$ evaluated at φ = 1, with $m(\emptyset) > 0$. (Below, a numerical subscript on q or $\bar{Z}$ is a stage index; since $\varphi_T = 1$, $Z = m(\emptyset)\bar{Z}_T$ and $q_T = \pi$.)
- The calibration $\mathcal{C}$ fixes the grid $0 = \varphi_0 < \varphi_1 < \ldots < \varphi_T = 1$ (so T ≥ 1) and the per-island particle count N (= N_e). Write $\Delta_t = \varphi_t - \varphi_{t-1}$, $q_t = q_{\varphi_t}$, $\bar{Z}_t = \bar{Z}_{\varphi_t}$, $G_t(S) = e^{\Delta_t d(S)}$ (t = 1,

…, T).

- **Final island r** (one island; N ≥ 1 particles), given the data and $\mathcal{C}$: 1. ξ_0^1, …, ξ_0^N are iid with law q_0 (exact start). 2. For t = 1, …, T: (reweighting) particle i gets weight G_t(ξ_{t-1}^i); (multinomial resampling) ζ_t^1, …, ζ_t^N are drawn, conditionally on the past, iid from Σ_i G_t(ξ_{t-1}^i)$\delta_{\xi_{t-1}^i}$ / Σ_i G_t(ξ_{t-1}^i); (mutation, one round per stage) conditionally on the past and on ζ_t, the ξ_t^j are independent with $\xi_t^j \sim K_t(\zeta_t^j, \cdot)$, where $K_t = K_{\varphi_t}$ is a Markov kernel on Ω leaving q_t invariant ($q_t K_t = q_t$). 3. $\hat{q}_t = N^{-1}\Sigma_i \delta_{\xi_t^i}$ (empirical population after stage-t resampling and mutation; $\hat{q}_0$ = the initial draws). 4. Island evidence: $\hat{Z}_r = m(\emptyset)\bar{Z}_0 \prod_{t=1}^T \hat{q}_{t-1}(G_t)$, computed exactly. Pooled evidence: $\hat{Z} = \frac{1}{4}\Sigma_{r=1}^4 \hat{Z}_r$ over the four final islands. The burn-in rounds, measured rounds and local search that follow stage T do not enter $\hat{Z}_r$.
- **Reference model.** $S^\circ \in \Omega$ is fixed given $\mathcal{C}$: a function of the data and $\mathcal{C}$ only, not of the final-run randomness. Define
  - $\eta_t = 1 - q_t(S^\circ)$ (t = 0, …, T), $\varepsilon_* = \max_{0\le t\le T} \eta_t$;
  - $R_t(S) = G_t(S)/G_t(S^\circ)$ (t = 1, …, T), so $R_t(S^\circ) = 1$;
  - $A_t = \prod_{v=1}^t \hat{q}_{v-1}(R_v)$ ($A_0 = 1$);
  - $b_t = q_0(S^\circ)/q_t(S^\circ)$ ($b_0 = 1$).

2. Hypotheses

- **EX-A:** exact scores, normalisers ($\bar{Z}_0$ in particular), probabilities and arithmetic.
- **EX-S + CAL:** ideal independent uniform streams; $\mathcal{C}$ is measurable with respect to calibration streams 0–9 and the data; the final islands use streams 10–13, independent of $\mathcal{C}$. Consequently, conditionally on (data, $\mathcal{C}$), each final island evolves exactly as in §1 item 2.
- **Exact start:** the initial particles are exact iid $\pi_0 = q_0$ draws.
- **Prescribed resampling and moves:** multinomial resampling after every reweighting; one q_{φ_t}-invariant move round per stage, applied to each particle as in §1 item 2.
- **Successful $\mathcal{C}$:** the frozen grid exists (no `schedule_limit`); everything below is conditional on (data, $\mathcal{C}$). E denotes expectation over the final-run randomness given (data, $\mathcal{C}$).

3. Conclusion (to prove)

For every island r and every N ≥ 1 (note $\varepsilon_* < 1$ always, since $q_t(S^\circ) > 0$):

(i-a) $\hat{Z}_r/Z = A_T/b_T$, surely.

(i-b) $E|\hat{Z}_r/Z - 1| \le [\Sigma_{t=1}^T (b_{t-1}\eta_{t-1} + b_t\eta_t) + |1 - b_T|]/b_T \le (2/(1-\varepsilon_*))\,\Sigma_{t=0}^T \eta_t$.

(i-c) The same two bounds hold for the pooled $\hat{Z}$ in place of $\hat{Z}_r$.

(i-d) Reading of the file's sentence "the endpoint term |1 − b_T| is essential": the middle expression cannot in general be replaced by $\Sigma_{t=1}^T (b_{t-1}\eta_{t-1} + b_t\eta_t)/b_T$. Precisely: there is an instance of the abstract setting of §1 (finite Ω, positive γ_0, real d, a grid, q_t-invariant kernels K_t, N ≥ 1, $S^\circ \in \Omega$) for which $E|\hat{Z}_r/Z - 1| > \Sigma_{t=1}^T (b_{t-1}\eta_{t-1} + b_t\eta_t)/b_T$. (This sentence is an informal remark in the theorem file; this is the reading adopted here. It is not claimed that such an instance arises from an actual SMC-BVS data set.)

4. Change log

- 2026-10-22, after Astra review round 1 (exposition finding 1): notation only. "$q_1 = \pi$, $Z = m(\emptyset)\bar{Z}_1$" in §1 was rewritten so that a numerical subscript is always a stage index ($q_t = q_{\varphi_t}$); the endpoint is written q_T, $\bar{Z}_T$. No hypothesis or conclusion changed.

(c) Final proof, verbatim: body of PROOF_v3.tex

The proof below is the body of `PROOF_v3.tex` from `\begin{document}` to `\end{document}` without the title block. Its preamble definitions are local to this chapter; labels carry the prefix `T14:`.

1 Setting and statement

Everything is conditional on the data and on a successful calibration $\mathcal{C}$; $\mathbb{E}$ denotes expectation over the final-run randomness given (data, $\mathcal{C}$). By EX-S and CAL the final islands use streams independent of $\mathcal{C}$, so given (data, $\mathcal{C}$) each island evolves as described below; by EX-A every quantity is computed exactly.

Bridge. Ω is a finite set. For $S \in \Omega$, $\gamma_0(S) > 0$ and $d(S) \in \mathbb{R}$; for real φ, $\gamma_\varphi(S) = \gamma_0(S)e^{\varphi d(S)}$, $\bar{Z}_\varphi = \sum_{S\in\Omega} \gamma_\varphi(S)$ and $q_\varphi = \gamma_\varphi/\bar{Z}_\varphi$. Thus $q_\varphi(S) > 0$ for all $S \in \Omega$. The target evidence is $Z = m(\emptyset) \sum_{S\in\Omega} \gamma_0(S)e^{d(S)}$ (the bridge normaliser at temperature $\varphi = 1$) with $m(\emptyset) > 0$.

Grid. $0 = \varphi_0 < \varphi_1 < \cdots < \varphi_T = 1$ (so $T \geq 1$), $\Delta_t = \varphi_t - \varphi_{t-1}$, $q_t = q_{\varphi_t}$, $\bar{Z}_t = \bar{Z}_{\varphi_t}$, and $G_t(S) = e^{\Delta_t d(S)} > 0$ for $t = 1, \ldots, T$. Since $\varphi_T = 1$, $Z = m(\emptyset)\bar{Z}_T$. From here on a numerical subscript on q or $\bar{Z}$ is always a stage index ($q_t = q_{\varphi_t}$), never a temperature.

One final island ($N \geq 1$ particles).

(a) $\xi_0^1, \ldots, \xi_0^N$ are iid with law q_0.

(b) For $t = 1, \ldots, T$: conditionally on $\mathcal{F}_{t-1}$ (defined below), $\zeta_t^1, \ldots, \zeta_t^N$ are iid with law

$$\mu_t := \frac{\sum_{i=1}^N G_t(\xi_{t-1}^i)\, \delta_{\xi_{t-1}^i}}{\sum_{i=1}^N G_t(\xi_{t-1}^i)}$$

(multinomial resampling); conditionally on $\mathcal{F}_{t-1}$ and $\zeta_t = (\zeta_t^1, \ldots, \zeta_t^N)$, the ξ_t^j are independent with $\xi_t^j \sim K_t(\zeta_t^j, \cdot)$, where K_t is a Markov kernel on Ω with $q_t K_t = q_t$ (one move round at φ_t). If these laws are given conditionally on a larger past (for instance one that also contains other islands), they hold in particular given $\mathcal{F}_{t-1}$, respectively given $(\mathcal{F}_{t-1}, \zeta_t)$, by the tower property, because μ_t is $\mathcal{F}_{t-1}$-measurable and the law of ξ_t^j depends only on ζ_t^j.

(c) $\mathcal{F}_t = \sigma(\xi_0, \zeta_1, \xi_1, \ldots, \zeta_t, \xi_t)$, $\mathcal{F}_0 = \sigma(\xi_0)$, and $\widehat{q}_t = N^{-1} \sum_{i=1}^N \delta_{\xi_t^i}$, so $\widehat{q}_t(f) = N^{-1} \sum_i f(\xi_t^i)$.

(d) $\widehat{Z}_r = m(\emptyset)\bar{Z}_0 \prod_{t=1}^T \widehat{q}_{t-1}(G_t)$. The pooled evidence is $\widehat{Z} = \frac{1}{4} \sum_{r=1}^4 \widehat{Z}_r$ over four islands, each of the above form with the same grid, kernels and N.

Only $\widehat{q}_0, \ldots, \widehat{q}_{T-1}$ enter $\widehat{Z}_r$; what the island does after stage T (resampling, burn-in and measured rounds, local search) is irrelevant here. Since Ω is finite, every function $f : \Omega \to \mathbb{R}$ is bounded and all expectations below are of bounded random variables.

Reference model. $S^\circ \in \Omega$ is fixed given (data, $\mathcal{C}$). Put $\eta_t = 1 - q_t(S^\circ)$ $(0 \le t \le T)$, $\varepsilon_* = \max_{0\le t\le T} \eta_t$, $R_t = G_t/G_t(S^\circ)$ $(1 \le t \le T)$, $A_t = \prod_{v=1}^{t} \widehat{q}_{v-1}(R_v)$ with $A_0 = 1$, and $b_t = q_0(S^\circ)/q_t(S^\circ)$ with $b_0 = 1$. Note $R_t(S^\circ) = 1$, $R_t > 0$, $A_t > 0$, $b_t > 0$, and $\varepsilon_* < 1$ because $q_t(S^\circ) > 0$ for every t.

Theorem 1 (T14(i))**.** *For every $N \ge 1$:*

(i-a) $\widehat{Z}_r/Z = A_T/b_T$ *surely;*

(i-b)

$$\mathbb{E}\Big|\frac{\widehat{Z}_r}{Z} - 1\Big| \;\le\; \frac{\sum_{t=1}^{T}\big(b_{t-1}\eta_{t-1} + b_t\eta_t\big) + |1 - b_T|}{b_T} \;\le\; \frac{2}{1-\varepsilon_*}\sum_{t=0}^{T}\eta_t;$$

(i-c) the same two bounds hold with the pooled $\widehat{Z}$ in place of $\widehat{Z}_r$;

(i-d) the endpoint term cannot be dropped in general: there is an instance of the setting above with $\mathbb{E}|\widehat{Z}_r/Z - 1| > \sum_{t=1}^{T}(b_{t-1}\eta_{t-1} + b_t\eta_t)/b_T$.

2 Three exact identities

Lemma 2 (change of temperature)**.** *For $1 \le t \le T$ and every $h : \Omega \to \mathbb{R}$,*

$$q_{t-1}(G_t h) = \frac{\bar{Z}_t}{\bar{Z}_{t-1}}\, q_t(h).$$

Proof. For $S \in \Omega$, $q_{t-1}(S)G_t(S) = \gamma_0(S)e^{\varphi_{t-1}d(S)}e^{\Delta_t d(S)}/\bar{Z}_{t-1} = \gamma_{\varphi_t}(S)/\bar{Z}_{t-1} = (\bar{Z}_t/\bar{Z}_{t-1})\, q_t(S)$. Multiply by $h(S)$ and sum over Ω. □

Lemma 3 (ratio of the b's)**.** *For $1 \le t \le T$, $b_t/b_{t-1} = \dfrac{\bar{Z}_t/\bar{Z}_{t-1}}{G_t(S^\circ)}$.*

Proof. $b_t/b_{t-1} = q_{t-1}(S^\circ)/q_t(S^\circ)$, and by Lemma 2 with $h = \mathbf{1}_{\{S^\circ\}}$, $q_{t-1}(S^\circ)G_t(S^\circ) = (\bar{Z}_t/\bar{Z}_{t-1})q_t(S^\circ)$. Divide by $G_t(S^\circ)q_t(S^\circ) > 0$. □

Lemma 4 (one stage in conditional mean)**.** *For $1 \le t \le T$ and every $f : \Omega \to \mathbb{R}$,*

$$\mathbb{E}\big[\widehat{q}_t(f) \;\big|\; \mathcal{F}_{t-1}\big] = \frac{\widehat{q}_{t-1}(G_t\, K_t f)}{\widehat{q}_{t-1}(G_t)}, \qquad K_t f(x) := \sum_{y\in\Omega} K_t(x,y) f(y).$$

Proof. Fix j. Given $\mathcal{F}_{t-1}$ and ζ_t, $\xi_t^j \sim K_t(\zeta_t^j, \cdot)$, so $\mathbb{E}[f(\xi_t^j) \mid \mathcal{F}_{t-1}, \zeta_t] = K_t f(\zeta_t^j)$. Given $\mathcal{F}_{t-1}$, $\zeta_t^j \sim \mu_t$, so by the tower property

$$\mathbb{E}[f(\xi_t^j) \mid \mathcal{F}_{t-1}] = \mu_t(K_t f) = \frac{\sum_i G_t(\xi_{t-1}^i) K_t f(\xi_{t-1}^i)}{\sum_i G_t(\xi_{t-1}^i)} = \frac{\widehat{q}_{t-1}(G_t K_t f)}{\widehat{q}_{t-1}(G_t)}.$$

Average over $j = 1, \dots, N$. □

(Only the conditional marginal laws of the ζ_t^j and ξ_t^j were used; the independence across particles is not needed for first moments.)

3 The weighted mean identity

Proposition 5. *For $0 \le t \le T-1$ and every $f : \Omega \to \mathbb{R}$,*

$$\mathbb{E}\big[A_t\, \widehat{q}_t(f)\big] = b_t\, q_t(f).$$

Proof. Induction on t, for all f at once. For $t = 0$: $A_0 = b_0 = 1$ and $\mathbb{E}\widehat{q}_0(f) = q_0(f)$ because the ξ_0^i have law q_0.

Let $1 \le t \le T-1$ and assume the claim at $t-1$ for all functions. $A_t = A_{t-1}\widehat{q}_{t-1}(R_t)$ is $\mathcal{F}_{t-1}$-measurable, so by Lemma 4,

$$\mathbb{E}[A_t\widehat{q}_t(f)] = \mathbb{E}\Big[A_{t-1}\,\widehat{q}_{t-1}(R_t)\,\frac{\widehat{q}_{t-1}(G_tK_tf)}{\widehat{q}_{t-1}(G_t)}\Big] = \mathbb{E}\big[A_{t-1}\,\widehat{q}_{t-1}(R_tK_tf)\big],$$

since $R_t = G_t/G_t(S^\circ)$ gives $\widehat{q}_{t-1}(R_t)/\widehat{q}_{t-1}(G_t) = 1/G_t(S^\circ)$ and $\widehat{q}_{t-1}(G_tK_tf)/G_t(S^\circ) = \widehat{q}_{t-1}(R_tK_tf)$. The induction hypothesis with the function R_tK_tf, then Lemma 2 with $h = K_tf$, invariance $q_t(K_tf) = q_t(f)$, and Lemma 3 give

$$\mathbb{E}[A_t\widehat{q}_t(f)] = b_{t-1}\,q_{t-1}(R_tK_tf) = \frac{b_{t-1}}{G_t(S^\circ)}\cdot\frac{\bar{Z}_t}{\bar{Z}_{t-1}}\,q_t(K_tf) = b_t\,q_t(f). \qquad \square$$

4 Proof of Theorem 1

(i-a). Since $G_t = G_t(S^\circ)R_t$, $\widehat{q}_{t-1}(G_t) = G_t(S^\circ)\widehat{q}_{t-1}(R_t)$, and $\prod_{t=1}^T G_t(S^\circ) = e^{d(S^\circ)\sum_t \Delta_t} = e^{d(S^\circ)}$ because $\varphi_T - \varphi_0 = 1$. Hence

$$\frac{\widehat{Z}_r}{Z} = \frac{\bar{Z}_0}{\bar{Z}_T}\,e^{d(S^\circ)}\,A_T.$$

Now, since $\varphi_T = 1$, $q_T(S^\circ)/q_0(S^\circ) = \big(\gamma_0(S^\circ)e^{d(S^\circ)}/\bar{Z}_T\big)/\big(\gamma_0(S^\circ)/\bar{Z}_0\big) = e^{d(S^\circ)}\bar{Z}_0/\bar{Z}_T$, so $\widehat{Z}_r/Z = A_T\,q_T(S^\circ)/q_0(S^\circ) = A_T/b_T$.

One telescoping increment. Fix $1 \le t \le T$. Since $R_t(S^\circ) = 1$,

$$\widehat{q}_{t-1}(R_t) - 1 = \widehat{q}_{t-1}(R_t - 1) = \widehat{q}_{t-1}\big((R_t-1)\mathbf{1}_{\Omega\setminus\{S^\circ\}}\big),$$

and, because $R_t > 0$, $|\widehat{q}_{t-1}(R_t) - 1| \le \widehat{q}_{t-1}(R_t\mathbf{1}_{\Omega\setminus\{S^\circ\}}) + \widehat{q}_{t-1}(\mathbf{1}_{\Omega\setminus\{S^\circ\}})$. Multiplying by $A_{t-1} > 0$,

$$|A_t - A_{t-1}| = A_{t-1}|\widehat{q}_{t-1}(R_t) - 1| \le A_{t-1}\widehat{q}_{t-1}\big(R_t\mathbf{1}_{\Omega\setminus\{S^\circ\}}\big) + A_{t-1}\widehat{q}_{t-1}\big(\mathbf{1}_{\Omega\setminus\{S^\circ\}}\big).$$

Take expectations and apply Proposition 5 at $t-1 \in \{0,\dots,T-1\}$:

$$\mathbb{E}|A_t - A_{t-1}| \le b_{t-1}\,q_{t-1}\big(R_t\mathbf{1}_{\Omega\setminus\{S^\circ\}}\big) + b_{t-1}\eta_{t-1}.$$

By Lemma 2 with $h = \mathbf{1}_{\Omega\setminus\{S^\circ\}}$ and then Lemma 3,

$$b_{t-1}\,q_{t-1}\big(R_t\mathbf{1}_{\Omega\setminus\{S^\circ\}}\big) = \frac{b_{t-1}}{G_t(S^\circ)}\cdot\frac{\bar{Z}_t}{\bar{Z}_{t-1}}\,q_t(\Omega\setminus\{S^\circ\}) = b_t\eta_t.$$

Therefore

$$\mathbb{E}|A_t - A_{t-1}| \le b_{t-1}\eta_{t-1} + b_t\eta_t \qquad (1 \le t \le T). \tag{1}$$

(i-b), first inequality. By (i-a),

$$\frac{\widehat{Z}_r}{Z} - 1 = \frac{A_T - 1}{b_T} + \frac{1 - b_T}{b_T}, \qquad |A_T - 1| = |A_T - A_0| \le \sum_{t=1}^{T} |A_t - A_{t-1}|.$$

Taking expectations and using (1),

$$\mathbb{E}\Big|\frac{\widehat{Z}_r}{Z} - 1\Big| \le \frac{\sum_{t=1}^{T}(b_{t-1}\eta_{t-1} + b_t\eta_t) + |1 - b_T|}{b_T}.$$

(i-b), second inequality. For $0 \le s \le T$,

$$\frac{b_s}{b_T} = \frac{q_T(S^\circ)}{q_s(S^\circ)} \le \frac{1}{q_s(S^\circ)} = \frac{1}{1 - \eta_s} \le \frac{1}{1 - \varepsilon_*}.$$

Hence

$$\frac{\sum_{t=1}^{T}(b_{t-1}\eta_{t-1} + b_t\eta_t)}{b_T} \le \frac{\sum_{t=1}^{T}(\eta_{t-1} + \eta_t)}{1 - \varepsilon_*} = \frac{2\sum_{t=0}^{T}\eta_t - \eta_0 - \eta_T}{1 - \varepsilon_*}.$$

For the endpoint term, $b_T = q_0(S^\circ)/q_T(S^\circ)$ gives

$$\frac{|1 - b_T|}{b_T} = \frac{|q_T(S^\circ) - q_0(S^\circ)|}{q_0(S^\circ)} = \frac{|\eta_0 - \eta_T|}{1 - \eta_0} \le \frac{\eta_0 + \eta_T}{1 - \varepsilon_*},$$

using $|\eta_0 - \eta_T| \le \max(\eta_0, \eta_T) \le \eta_0 + \eta_T$ (both are in $[0, 1)$). Adding the two displays gives the bound $2(1 - \varepsilon_*)^{-1}\sum_{t=0}^{T}\eta_t$.

(i-c). Both right-hand sides of (i-b) depend only on $(q_t)_{0 \le t \le T}$ and S°, which are fixed by the data and $\mathcal{C}$; they are therefore the same for every island. Call the middle expression M. By the triangle inequality, $|\widehat{Z}/Z - 1| = \big|\frac{1}{4}\sum_r(\widehat{Z}_r/Z - 1)\big| \le \frac{1}{4}\sum_r|\widehat{Z}_r/Z - 1|$, so $\mathbb{E}|\widehat{Z}/Z - 1| \le \frac{1}{4}\sum_r \mathbb{E}|\widehat{Z}_r/Z - 1| \le M \le 2(1 - \varepsilon_*)^{-1}\sum_t \eta_t$. No independence between islands is used.

(i-d). Take $\Omega = \{S^\circ, x\}$, $T = 1$, $N = 1$, $m(\emptyset) = 1$, K_1 the identity kernel (it leaves every law invariant), and fix $a = 0.01$, $c = 0.1$. Put $\gamma_0(S^\circ) = 1 - a$, $\gamma_0(x) = a$, $d(S^\circ) = 0$, $d(x) = \log\frac{c(1-a)}{a(1-c)}$. Then $q_0 = (1 - a, a)$, and $\gamma_1 = (1 - a,\ c(1 - a)/(1 - c)) = \frac{1-a}{1-c}(1 - c, c)$, so $q_1 = (1 - c, c)$. With $T = N = 1$ (so stage 1 is temperature 1), $\widehat{Z}_r/Z = \bar{Z}_0 G_1(\xi_0^1)/\bar{Z}_1 = q_1(\xi_0^1)/q_0(\xi_0^1)$ (Lemma 2 with $h = \mathbf{1}_{\{y\}}$ gives $\bar{Z}_0 G_1(y)/\bar{Z}_1 = q_1(y)/q_0(y)$), and $\xi_0^1 \sim q_0$, so

$$\mathbb{E}\Big|\frac{\widehat{Z}_r}{Z} - 1\Big| = \sum_{y \in \Omega}|q_1(y) - q_0(y)| = 2|c - a| = 0.18.$$

Here $\eta_0 = a$, $\eta_1 = c$, $b_0 = 1$, $b_1 = (1 - a)/(1 - c) = 1.1$, and

$$\frac{b_0\eta_0 + b_1\eta_1}{b_1} = \frac{0.01 + 0.11}{1.1} = 0.1090\ldots < 0.18.$$

So the bound without the endpoint term fails, while the full middle expression of (i-b) equals $0.1090\ldots + |1 - 1.1|/1.1 = 0.2$ and the right-hand side equals $(2/0.9)(0.11) = 0.2444\ldots$, both ≥ 0.18 as the theorem requires. □

Remark 6. Proposition 5 is equivalent to the frozen-run identity $\mathbb{E}[\widehat{Z}_{r,t}\widehat{q}_t(f)] = Z_t q_t(f)$ (the form in which the project's conjecture T12(i) states it; that file is not needed here) rewritten through $\widehat{Z}_{r,t} = m(\emptyset)\bar{Z}_0 e^{\varphi_t d(S^\circ)}A_t$ and $Z_t = m(\emptyset)\bar{Z}_t$; it is proved above directly, so the argument uses nothing from T12. Nothing about the internal structure of the move round is used beyond $q_t K_t = q_t$, and nothing about EXP/EXP′ or the statistical model enters.

(d) Review record

FINAL_REPORT.md: statement changes

3. Statement changes

There were no mathematical changes. The statement analysis concluded it was correct as stated (proof.md step 1(a)).

- **Notation (after Astra #1 finding 1).** "$q_1 = \pi$, $Z = m(\emptyset)\bar{Z}_1$" clashed with stage indexing. It was rewritten so that a numerical subscript is always a stage index ($Z = m(\emptyset)\bar{Z}_T$). This is logged in STATEMENT.md §4.
- **Reading of an informal sentence.** "The endpoint term $|1 - b_T|$ is essential" is read as: in the abstract setting, dropping the term breaks the bound for some instance. This reading is stated as (i-d), and it is not claimed that such an instance comes from a real SMC-BVS data set. All three reviewers judged this reading fair.

FINAL_REPORT.md: review history

4. Review history

Stage	Reviewer	Input	Verdict	Findings
A	Astra #1 (gpt-6-astra, high), session 01a16418-6bd3-78f1-ad35-0348b23c7968	PROOF_v1	verified	1 exposition (stage/temperature subscript clash): accepted, fixed in v2
A	Astra #1, same session (resume)	PROOF_v2 + log	verified	agrees with the fix; no new findings
B	Astra #2 (gpt-6-astra, high), new session 01a1641c-418a-7463-a810-25b59b68de23	PROOF_v2	verified	none
B	Opus 5.5 xhigh (Claude Personal, T3 delegate_task)	PROOF_v2	verified	3 exposition: conditioning on F_{t−1}; (i-c) wording; remark citing T12. All accepted, unified into v3
B	Astra #2 (resume)	PROOF_v3 + log	verified	agrees with all answers; none
B	Opus 5.5 xhigh (new round task)	PROOF_v3 + log	verified	agrees with all three answers; none

- Fatal or gap findings: none, in any round. CLAIMS_OF_FATAL_FLAWS.md records this and logs the exposition findings with the answers.
- Astra judge: not used.
- All replies are saved in `review/`. Stage B Opus scratch scripts are in `review/stageB_opus_scratch/`.
- Temp packet folders were deleted at the end; a copy of the Stage A packet is kept in `review/packet_stageA_copy/`.

CLAIMS_OF_FATAL_FLAWS.md (every entry, verbatim)

CLAIMS_OF_FATAL_FLAWS — T14(i)

Reviewer: Astra (gpt-6-astra, reasoning high), session 01a16418-6bd3-78f1-ad35-0348b23c7968, round 1 on PROOF_v1 (2026-10-22). Verdict: verified.

Fatal or gap findings

None. Astra raised no fatal and no gap finding.

Exposition findings (recorded for completeness)

Finding 1 (exposition), verbatim claim: "The notation $q_1, \bar{Z}_1$ is used both for temperature 1 and for stage 1. These meanings generally differ when $T > 1$. In particular, the assertion "$q_T = q_1$" requires the temperature interpretation of the right-hand side."

Lead's answer: accepted and fixed. PROOF_v2.tex defines Z = m(∅)Σγ$_0$e^{d} (the bridge normaliser at φ = 1), states that a numerical subscript on q or $\bar{Z}$ is always a stage index, and writes Z = m(∅)$\bar{Z}$_T and q_T(S°)/q$_0$(S°) = e^{d(S°)}$\bar{Z}_0/\bar{Z}$_T in the proof of (i-a). The (i-d) example keeps q$_1$, $\bar{Z}_1$ because there T = 1, as now noted. STATEMENT.md §1 was reworded the same way (notation only; change log §4).

Stage B (Protocol Amendment 1), round 1 on PROOF_v2, 2026-10-22

Reviewers: Astra #2 (gpt-6-astra, high, new session 01a1641c-418a-7463-a810-25b59b68de23): verdict verified, no findings. Opus 5.5 (xhigh, Claude Personal, clientRequestId T14i-stageB-opus-personal-r1): verdict verified, three exposition findings.

Fatal or gap findings

None. Neither Stage B reviewer raised a fatal or gap finding.

Exposition findings (Opus), all accepted and fixed in the unified PROOF_v3.tex

Opus finding 1, verbatim claim: "the statement gives the resampling and move laws "conditionally on the past". The proof conditions on the smaller σ-field F_{t−1} = σ(ξ$_0$, ζ$_1$, …, ξ_{t−1}) without saying why that is allowed." **Lead's answer: accepted.** §1 item (b) now says that laws given on a larger past hold in particular given F_{t−1} (resp. (F_{t−1}, ζ_t)) by the tower property, since μ_t is F_{t−1}-measurable and the law of ξ_t^j depends only on ζ_t^j.

Opus finding 2, verbatim claim: "the proof says the islands "share the grid, the kernels, N and S°". Only the grid, γ$_0$, d and S° are actually needed." **Lead's answer: accepted.** (i-c) now says both right-hand sides depend only on (q_t)_{0≤t≤T} and S°, which the data and $\boldsymbol{\mathcal{C}}$ fix.

Opus finding 3, verbatim claim: "it cites T12(i) (E[$\hat{Z}$_{r,t} q̂_t(f)] = Z_t q_t(f)), which is not in the packet, so I cannot check it against its source." **Lead's answer: accepted.** The remark now says Proposition 5 is equivalent to the identity in the form in which the project's conjecture T12(i) states it, that the file is not needed, and that the argument uses nothing from T12. (The remark is not used by the proof.)

Chapter 6

T15 (L-OCC): occupancy from an evidence lower tail

Final proof: `proofs/T15_LOCC_occupancy/proof/PROOF_v4.tex`. *Sources of parts (a), (b), (d):* `theorems/T15_LOCC_occupancy.md`, `STATEMENT.md`, `FINAL_REPORT.md`, `CLAIMS_OF_FATAL_FLAWS.md`.

(a) Original conjecture text (theorem file, verbatim)

T15 (L-OCC): Occupancy from an evidence lower tail

Status: conjectured, not proven. Methodology SMC-BVS v3.0 (2026-10-19), SHA-256 93043758058a6d422cca6ca4b6be8f3a9e8149b6f6de2e1bab1ce31c44b8019c. Read with `00_COMMON.md`.

Category / rank: algorithmic convergence (source of BASIN's δ); **core** (lemma).

Definitions used

Island evidence $\hat{Z}_r$; $f_r(\mathfrak{A}^c)$ = the fraction of island r's particles outside a set $\mathfrak{A}$ (fixed given $\boldsymbol{\mathcal{C}}$) at stage T or at a fixed terminal recording prefix (as in T12(i)); $W := \hat{Z}_r/Z$.

Assumptions

EX-A, EX-S, CAL; prefixes as specified in T12.

Precise statement

Conditional on a successful $\boldsymbol{\mathcal{C}}$, for every $c, u \in (0, 1)$:

$E\, f_r(\mathfrak{A}^c) \le \min\{1, \pi(\mathfrak{A}^c)/c + P(\hat{Z}_r/Z < c)\}$, $P(f_r(\mathfrak{A}^c) > u) \le \pi(\mathfrak{A}^c)/(cu) + P(\hat{Z}_r/Z < c)$. If $E|\hat{Z}_r/Z - 1| \le B$ then the lower-tail term is $\le B/(1-c)$. These are calibration-conditional bounds; conditional on an already realised population its actual outside fraction must be used. The lower tail must be islandwise; a pooled-evidence lower tail does not substitute.

Dependencies

Builds on: T12 (and optionally T14 or T25 for the lower tail). Used by: T23, T24, T26.

Numerical evidence (provisional)

provisional evidence (Sol check J6): exact finite-particle check of E[Wf] = π(𝔄ᶜ) (weighted fraction 0.01980 = target outside 0.01980) and of the linear bound. Files: judge/sol2/J6_FIRST_MOMENTS.md.

Proof-route hint (optional)

T12(i) with f = 1_{𝔄ᶜ} gives E[Wf_r] = π(𝔄ᶜ); split at W = c using 0 ≤ f_r ≤ 1; Markov for the probability bound. No substantive hardest step; the content is the lower-tail input. *When proof.md is run on this file, its lead considers the statement both with and without these hints; the hints never reach the reviewers.*

(b) STATEMENT.md (verbatim)

STATEMENT — T15 (L-OCC): occupancy from an evidence lower tail

Source: `theorems/T15_LOCC_occupancy.md` (SMC-BVS v3.0, METHODOLOGY_v3.md SHA-256 93043758058a6d422cca6ca4b6be8f3a9e8149b6f6de2e1bab1ce31c44b8019c), read with `theorems/00_COMMON.md`. Parts to prove: all.

1. Setting (compact; the method file is authoritative)

- **Target.** Ω = {S ⊆ [p] : |S| ≤ K} is a finite set of models. Each S ∈ Ω has a strictly positive unnormalised posterior weight $\gamma_1(S) = \varpi(S)m(S)$ (prior × marginal likelihood). Z := Σ_{S∈Ω} $\gamma_1(S)$ ∈ (0, ∞) is the evidence and π := γ_1/Z the posterior on Ω. For A ⊆ Ω, π(A) = Σ_{S∈A} π(S).
- **Calibration. $\mathcal{C}$** is the frozen output of the calibration phase (temperature grid 0 = $\varphi_0 < \varphi_1 < \ldots <$ φ_T = 1, budgets, N = N_e, and all other settings). It is measurable with respect to calibration random streams 0–9 and the data. "Successful $\mathcal{C}$" means the calibration ended with a constructed grid and the final run is launched. P_$\mathcal{C}$ and E_$\mathcal{C}$ denote probability and expectation over the final-run randomness (streams 10–13) given $\mathcal{C}$ and the data.
- **Final run, island r ∈ {1, 2, 3, 4}.** N particles drawn i.i.d. from the start law π_0; for t = 1, …, T: reweight by G_t(S) = exp{(φ_t − φ_{t−1})d(S)}, multinomial resampling, one move round at temperature φ_t; then L burn-in rounds and 4 measured rounds at φ = 1 (no reweighting in these terminal rounds). A move round is a list of elementary steps applied, for each particle, in a uniformly random order drawn independently of the particle's state.
- **Island evidence.** $\hat{Z}_r := m(\emptyset)\bar{Z}_0 \prod_{t=1}^{T} (1/N) \Sigma_{i=1}^{N} G_t(S_i^{(t-1)})$, the product over all stages (the terminal rounds contribute no factor). Here m(∅) > 0 is the marginal likelihood of the empty model, $\pi_0 = \gamma_0/\bar{Z}_0$ is the composite start law with normaliser $\bar{Z}_0 = \Sigma_\Omega \gamma_0 > 0$ ($\gamma_0 > 0$), and d(S) is the real-valued discrepancy (all defined in `theorems/00_COMMON.md` §§1–2). Since every factor is a finite positive number (each $G_t(S) = e^{(\varphi_t-\varphi_{t-1})d(S)} \in (0, \infty)$), $\hat{Z}_r$ is a finite, strictly positive random variable under EX-A. The proof uses only W_r ≥ 0. Write **$W_r := \hat{Z}_r / Z$**. Pooled evidence: $\hat{Z} := \frac{1}{4} \Sigma_r \hat{Z}_r$, and $\bar{W} := \hat{Z}/Z = \frac{1}{4} \Sigma_r W_r$.
- **Recording positions.** $\mathcal{T}$ denotes the set of positions τ in island r's terminal part at which T12(i) is stated: (a) τ = "end of stage T" (after the stage-T resampling and move round); (b) any fixed elementary prefix within the L + 4 terminal rounds; (c) "immediately before a specified Gibbs label" in a terminal round, under the state-independent random order. $\hat{q}_{r,\tau}$ is the empirical

measure (1/N)Σ_i δ_{S_i(τ)} of island r's particle states at position τ.

- **Outside set and outside fraction.** $\mathfrak{A} \subseteq \Omega$ is a set that is fixed given $\boldsymbol{\mathcal{C}}$ ($\boldsymbol{\mathcal{C}}$- and data-measurable, independent of streams 10–13). f_r := q̂_{r,τ}(1_{$\mathfrak{A}^c$}) ∈ [0, 1], the fraction of island r's particles outside $\mathfrak{A}$ at a fixed $\tau \in \mathcal{T}$.

2. Hypotheses

- **EX-A** (exact arithmetic), **EX-S** (independent ideal uniform streams, with the method's stream separation), **CAL** ($\boldsymbol{\mathcal{C}}$ uses streams 0–9 only; streams 10–13 are fresh; escalation is decided before any final randomness). These are standing assumptions of the setting.
- **(H-T12) Frozen-run identity** (the conclusion of the sibling conjecture T12(i), used here only as an explicit hypothesis): conditional on a successful $\boldsymbol{\mathcal{C}}$, for island r, each $\tau \in \mathcal{T}$, and every bounded function f on Ω that is fixed given $\boldsymbol{\mathcal{C}}$,

  $E_{\boldsymbol{\mathcal{C}}}[\, \hat{Z}_r \cdot \hat{q}_{r,\tau}(f)\,] = Z \cdot \pi(f)$.

  (At these positions the target is $q_T = q_1 = \pi$ and $Z_T = Z$.) Parts (i)–(ii) use only the instance $f = 1_{\{\mathfrak{A}^c\}}$, and only the inequality $E_{\boldsymbol{\mathcal{C}}}[W_r f_r] \le \pi(\mathfrak{A}^c)$. The first claim of (iii), $P_{\boldsymbol{\mathcal{C}}}(W_r < c) \le B/(1 - c)$, uses only its own L^1 hypothesis; its "consequently" clause rests on (i)–(ii) and hence on the same instance of (H-T12). Part (iv) is a statement about abstract laws and uses no hypothesis; it shows that even the identity of (H-T12) for every test function does not yield the pooled form.

3. Conclusions

Fix a successful $\boldsymbol{\mathcal{C}}$, an island r, a position $\tau \in \mathcal{T}$, and a set $\mathfrak{A}$ fixed given $\boldsymbol{\mathcal{C}}$. Let $f_r = \hat{q}_{r,\tau}(1_{\{\mathfrak{A}^c\}})$ and $W_r = \hat{Z}_r/Z$.

(i) Mean bound. For every $c \in (0, 1)$:
$E_{\boldsymbol{\mathcal{C}}}\, f_r \le \min\{\, 1, \pi(\mathfrak{A}^c)/c + P_{\boldsymbol{\mathcal{C}}}(W_r < c)\, \}$.

(ii) Tail bound. For every $c, u \in (0, 1)$:
$P_{\boldsymbol{\mathcal{C}}}(f_r > u) \le \pi(\mathfrak{A}^c)/(cu) + P_{\boldsymbol{\mathcal{C}}}(W_r < c)$.

(iii) Lower tail from an L^1 bound. If $E_{\boldsymbol{\mathcal{C}}}|W_r - 1| \le B$ for a constant $B \ge 0$, then for every $c \in (0, 1)$, $P_{\boldsymbol{\mathcal{C}}}(W_r < c) \le B/(1 - c)$. Consequently (i) and (ii) hold with $P_{\boldsymbol{\mathcal{C}}}(W_r < c)$ replaced by $B/(1 - c)$.

(iv) The lower tail must be islandwise. The bounds (i)–(ii) with the pooled lower tail $P_{\boldsymbol{\mathcal{C}}}(\bar{W} < c)$ in place of the islandwise $P_{\boldsymbol{\mathcal{C}}}(W_r < c)$ do not follow from the structure used, even with the identity of (H-T12) for every test function. (This is a non-implication about direct substitution; it does not assert that every conceivable occupancy bound must use islandwise evidence.) Precisely, fix $c, u \in (0, 1)$ and $a \in (0, cu/4]$.

(iv-a) There are four i.i.d. random pairs (W_r, f_r), $r = 1, \ldots, 4$, with $W_r > 0$ (strictly), $f_r \in \{0, 1\}$, $E\, W_r = 1$ and $E[W_r f_r] = a$, such that, with $\bar{W} = \tfrac{1}{4} \Sigma_r W_r$,

$E\, f_1 > a/c + P(\bar{W} < c)$ and $P(f_1 > u) > a/(cu) + P(\bar{W} < c)$; in particular, since $\min\{1, x\} \le x$, also $E\, f_1 > \min\{1, a/c + P(\bar{W} < c)\}$.

(iv-b) Moreover, let Ω' be any finite set, π' any probability on Ω', $\mathfrak{A}' \subseteq \Omega'$ with $\pi'(\mathfrak{A}'^c) = a$, and $N \ge 1$. (For example, the method's Ω and π whenever some $\mathfrak{A} \subseteq \Omega$ has $\pi(\mathfrak{A}^c) = a$; such an $\mathfrak{A}$ exists for $a :=$ the smallest posterior mass of a single model whenever $|\Omega| \ge 4/(cu)$, since that mass lies in $(0, 1/|\Omega|] \subseteq (0, cu/4]$.) Then the pairs of (iv-a) can be realised jointly with random populations $(S_{r,1}, \ldots, S_{r,N}) \in \Omega'^N$, $r = 1, \ldots, 4$, such that the four pairs $(W_r, (S_{r,i})_{i \le N})$ are i.i.d., $f_r = \hat{q}_r(1_{\{\mathfrak{A}'^c\}})$ for

the empirical measure $\hat{q}_r = (1/N)\,\Sigma_i\,\delta_{S_{r,i}}$, and

$E[\,W_r \cdot \hat{q}_r(g)\,] = \pi'(g)$ for every function $g : \Omega' \to \mathbb{R}$,

which is the identity of (H-T12) with $\hat{Z}_r := Z'\,W_r$ for any constant $Z' > 0$, for every test function g and not only for $g \in \{1, 1_{\mathfrak{A}'^{c}}\}$. Each island has one population; using it at every recording position ($\hat{q}_{r,\tau} := \hat{q}_r$ for all $\tau \in \mathcal{T}$) gives the identity at every position as well. The four islands of this example are i.i.d.; that property only strengthens the example and is not asserted here as a property of the method.

4. Scope remarks (from the theorem file; interpretations, not separate claims)

- (i)–(iii) are statements about the law of the final run given $\boldsymbol{\mathcal{C}}$ (calibration-conditional). They give no bound on the outside fraction of an already realised population; for a realised population its actual f_r must be used.
- Part (iv) formalises the theorem file's sentence "a pooled-evidence lower tail does not substitute" as a non-implication from the properties listed ($W_r > 0$, f_r a fraction in [0,1], i.i.d. islands, and the identity of (H-T12) for every test function and every recording position). It does not claim that the SMC final run, for a particular data set and calibration, realises the counterexample law.

5. Changes relative to the theorem file

- None to the inequalities, their constants or their quantifiers.
- The theorem file's dependency on T12 ("Builds on: T12"; "prefixes as specified in T12") is stated as the explicit hypothesis (H-T12), as the run rules require for sibling conjectures. The positions $\mathcal{T}$ are exactly those named in T12(i).
- The sentence on pooled lower tails is made precise as part (iv) (a non-implication), and the sentence on realised populations is kept as a scope remark (§4).
- Assumption EXP (and its replacement EXP′) is not used.
- v2 (after Astra round 1, exposition finding 1): in (iv) the example pairs are required to have $W_r > 0$ strictly (v1 allowed $W_r \geq 0$), matching the strict positivity of actual island evidence, and a sentence fixes the scope of the heading. The inequalities of (i)–(iii) are unchanged.
- v3 (after Stage B, Opus exposition findings 1–5; no fatal or gap findings in any review): (iv) now holds for every $a \in (0, cu/4]$ (v2: $a = cu/4$ only) and adds (iv-b), realising the example with particle populations that satisfy the full identity (H-T12) for every test function; the $\min\{1,\cdot\}$ form is refuted explicitly; the sentence presenting island i.i.d. as a method property is removed; $\hat{Z}_r$'s symbols and positivity are spelled out; the hypothesis-use note in §2 is made exact. (i)–(iii) unchanged.
- v4 (after Stage B round 2; both reviewers verified v3 with exposition findings only): wording only — the hypothesis-use note for (iii) separates its two claims; "full identity" replaced by "identity for every test function", with the single-population/every-position remark; the method's-Ω example states when a set of mass a exists. No mathematical content changed.
- Post-review alignment (after Stage B round 3, Opus finding 1, marked optional): the last "full identity" in the §4 scope remark replaced by the v4 wording. Wording only, in a scope remark; not re-sent for review.

(c) Final proof, verbatim: body of PROOF_v4.tex

The proof below is the body of `PROOF_v4.tex` *from* `\begin{document}` *to* `\end{document}` *without the title block. Its preamble definitions are local to this chapter; labels carry the prefix* `T15:`.

1 Setting and hypotheses

We use the notation of STATEMENT.md. Ω is the finite model space, $\pi = \gamma_1/Z$ the posterior with evidence $Z \in (0,\infty)$, and $\mathcal{C}$ a successful frozen calibration; $\mathbb{P}_{\mathcal{C}}$ and $\mathbb{E}_{\mathcal{C}}$ are the probability and expectation over the final-run randomness given $\mathcal{C}$ and the data. For an island r, $\hat{Z}_r$ is its evidence estimate, which under EX-A is a finite, strictly positive random variable, and

$$W_r := \hat{Z}_r/Z \in (0,\infty).$$

Fix a recording position $\tau \in \mathcal{T}$ and a set $\mathfrak{A} \subseteq \Omega$ fixed given $\mathcal{C}$, and let

$$f_r := \hat{q}_{r,\tau}(1_{\mathfrak{A}^c}) = \frac{1}{N}\sum_{i=1}^{N} 1\{S_i(\tau) \notin \mathfrak{A}\} \in [0,1].$$

Since $\mathfrak{A}$ is fixed given $\mathcal{C}$, the function $f = 1_{\mathfrak{A}^c}$ is bounded and fixed given $\mathcal{C}$, so hypothesis (H-T12) applies to it and gives

$$\mathbb{E}_{\mathcal{C}}[W_r f_r] = \frac{\mathbb{E}_{\mathcal{C}}[\hat{Z}_r\, \hat{q}_{r,\tau}(1_{\mathfrak{A}^c})]}{Z} = \frac{Z\,\pi(1_{\mathfrak{A}^c})}{Z} = \pi(\mathfrak{A}^c). \tag{1}$$

For parts (i)–(iii) only the inequality $\mathbb{E}_{\mathcal{C}}[W_r f_r] \le \pi(\mathfrak{A}^c)$ is used. The random variable $W_r f_r$ is nonnegative, and by (1) it is integrable.

Throughout, r, τ and $\mathfrak{A}$ are fixed, and we drop the subscript r: $W = W_r$, $f = f_r$.

2 Two elementary facts

Lemma 1 (Splitting at c)**.** *Let $W \ge 0$ and $f \in [0,1]$ be random variables on a common probability space, with $\mathbb{E}[Wf] \le a < \infty$. Then for every $c > 0$,*

$$\mathbb{E}\big[f\, 1\{W \ge c\}\big] \le a/c.$$

Proof. On the event $\{W \ge c\}$ we have $W/c \ge 1$ and $f \ge 0$, hence $f \le fW/c$. Off this event, $f\,1\{W \ge c\} = 0 \le fW/c$ because $f, W \ge 0$. So pointwise $f\,1\{W \ge c\} \le fW/c$, and taking expectations gives $\mathbb{E}[f1\{W \ge c\}] \le \mathbb{E}[fW]/c \le a/c$. □

Lemma 2 (Markov)**.** *If $V \ge 0$ is a random variable and $s > 0$, then $\mathbb{P}(V \ge s) \le \mathbb{E}V/s$.*

Proof. $s\,1\{V \ge s\} \le V$ pointwise; take expectations. □

3 Proof of (i): the mean bound

Fix $c \in (0,1)$. Since $0 \le f \le 1$, we have $\mathbb{E}_{\mathcal{C}} f \le 1$. Write $f = f\,1\{W \ge c\} + f\,1\{W < c\}$. By Lemma 1 with $a = \pi(\mathfrak{A}^c)$ (justified by (1)), $\mathbb{E}_{\mathcal{C}}[f1\{W \ge c\}] \le \pi(\mathfrak{A}^c)/c$. Since $f \le 1$, $\mathbb{E}_{\mathcal{C}}[f1\{W < c\}] \le \mathbb{P}_{\mathcal{C}}(W < c)$. Adding,

$$\mathbb{E}_{\mathcal{C}} f \le \pi(\mathfrak{A}^c)/c + \mathbb{P}_{\mathcal{C}}(W < c).$$

Together with $\mathbb{E}_{\mathcal{C}} f \le 1$ this is $\mathbb{E}_{\mathcal{C}} f \le \min\{1,\ \pi(\mathfrak{A}^c)/c + \mathbb{P}_{\mathcal{C}}(W < c)\}$. □

4 Proof of (ii): the tail bound

Fix $c, u \in (0,1)$. Decompose

$$\{f > u\} \subseteq \big(\{f > u\} \cap \{W \geq c\}\big) \cup \{W < c\}.$$

On $\{f > u\} \cap \{W \geq c\}$ we have $fW \geq fc > uc$, using $W \geq c > 0$ and $f > u > 0$. Hence $\{f > u\} \cap \{W \geq c\} \subseteq \{fW \geq cu\}$. By Lemma 2 applied to $V = fW \geq 0$ and $s = cu > 0$, and by (1),

$$\mathbb{P}_{\mathcal{C}}(fW \geq cu) \leq \mathbb{E}_{\mathcal{C}}[fW]/(cu) \leq \pi(\mathfrak{A}^c)/(cu).$$

By the union bound,

$$\mathbb{P}_{\mathcal{C}}(f > u) \leq \pi(\mathfrak{A}^c)/(cu) + \mathbb{P}_{\mathcal{C}}(W < c). \qquad \square$$

5 Proof of (iii): the lower tail from an L^1 bound

Assume $\mathbb{E}_{\mathcal{C}}|W - 1| \leq B$, and fix $c \in (0,1)$. On $\{W < c\}$, $|W - 1| \geq 1 - W > 1 - c > 0$. Hence $\{W < c\} \subseteq \{|W - 1| \geq 1 - c\}$, and Lemma 2 with $V = |W - 1|$ and $s = 1 - c > 0$ gives

$$\mathbb{P}_{\mathcal{C}}(W < c) \leq \mathbb{P}_{\mathcal{C}}(|W - 1| \geq 1 - c) \leq \frac{\mathbb{E}_{\mathcal{C}}|W - 1|}{1 - c} \leq \frac{B}{1 - c}.$$

Substituting this bound into (i) and (ii) gives, for every $c, u \in (0,1)$,

$$\mathbb{E}_{\mathcal{C}} f \leq \min\Big\{1,\ \frac{\pi(\mathfrak{A}^c)}{c} + \frac{B}{1-c}\Big\}, \qquad \mathbb{P}_{\mathcal{C}}(f > u) \leq \frac{\pi(\mathfrak{A}^c)}{cu} + \frac{B}{1-c}. \qquad \square$$

6 Proof of (iv): the lower tail must be islandwise

Fix $c, u \in (0,1)$ and $a \in (0, cu/4]$. Note $a < 1/4$.

6.1 Proof of (iv-a)

Set

$$\varepsilon := \frac{1}{2}\min\Big\{c,\ \frac{4(1-c)}{3},\ \frac{4a}{3}\Big\} > 0, \qquad h := 4 - 3\varepsilon, \qquad b := \frac{a - 3\varepsilon/4}{h}.$$

Elementary bounds on the constants. Since $\varepsilon \leq 2a/3 \leq cu/6 < 1/6$, we have $h \in (7/2, 4)$. Since $3\varepsilon/4 \leq a/2 < a$, we have $b > 0$; and $b < a/h < a/3 < 1/12$. Moreover $\varepsilon \leq c/2 < c$, and $\varepsilon \leq 2(1-c)/3$ gives $3\varepsilon/4 \leq (1-c)/2$, hence

$$h/4 = 1 - 3\varepsilon/4 \ \geq\ (1+c)/2 \ >\ c. \tag{2}$$

The law. Let (W_r, f_r), $r = 1, \ldots, 4$, be i.i.d. copies of a pair (W, f) with the three-point law

$$\mathbb{P}\big((W,f) = (\varepsilon, 1)\big) = \tfrac{3}{4}, \qquad \mathbb{P}\big((W,f) = (h,1)\big) = b, \qquad \mathbb{P}\big((W,f) = (h,0)\big) = \tfrac{1}{4} - b.$$

The probabilities are positive (as $0 < b < 1/12$) and sum to 1. Clearly $W \in \{\varepsilon, h\} \subset (0, \infty)$, so $W > 0$ strictly, and $f \in \{0,1\}$. Also $\mathbb{P}(W = h) = 1/4$ and $\mathbb{P}(f = 1) = 3/4 + b$.

The required moment properties. $\mathbb{E}W = \frac{3}{4}\varepsilon + \frac{1}{4}h = \frac{3}{4}\varepsilon + 1 - \frac{3}{4}\varepsilon = 1$, and $\mathbb{E}[Wf] = \frac{3}{4}\varepsilon + hb = \frac{3}{4}\varepsilon + a - \frac{3}{4}\varepsilon = a$.

The pooled lower tail. Let $K := \#\{r : W_r = h\} \sim \mathrm{Bin}(4, 1/4)$. Then $\bar{W} = \frac{1}{4}\big(Kh + (4-K)\varepsilon\big)$. If $K = 0$, then $\bar{W} = \varepsilon < c$. If $K \geq 1$, then $\bar{W} \geq h/4 > c$ by (2). Hence $\{\bar{W} < c\} = \{K = 0\}$ and $\mathbb{P}(\bar{W} < c) = (3/4)^4 = 81/256$.

The two left-hand sides. $\mathbb{E}f_1 = \mathbb{P}(f_1 = 1) = \frac{3}{4} + b > \frac{3}{4}$. Since $f_1 \in \{0, 1\}$ and $u < 1$, $\{f_1 > u\} = \{f_1 = 1\}$, so $\mathbb{P}(f_1 > u) = \frac{3}{4} + b > \frac{3}{4} = \frac{192}{256}$.

The two right-hand sides. Since $a \leq cu/4$, $a/c \leq u/4 < 1/4$ and $a/(cu) \leq 1/4$. Hence

$$\frac{a}{c} + \mathbb{P}(\bar{W} < c) < \frac{64}{256} + \frac{81}{256} = \frac{145}{256}, \qquad \frac{a}{cu} + \mathbb{P}(\bar{W} < c) \leq \frac{64}{256} + \frac{81}{256} = \frac{145}{256}.$$

Since $145/256 < 192/256$, both

$$\mathbb{E}f_1 > a/c + \mathbb{P}(\bar{W} < c) \qquad \text{and} \qquad \mathbb{P}(f_1 > u) > a/(cu) + \mathbb{P}(\bar{W} < c)$$

hold. As $\min\{1, x\} \leq x$ for every real x, also $\mathbb{E}f_1 > \min\{1,\ a/c + \mathbb{P}(\bar{W} < c)\}$. The pairs are independent and identically distributed and satisfy $W_r > 0$, $f_r \in \{0, 1\}$, $\mathbb{E}W_r = 1$ and $\mathbb{E}[W_r f_r] = a$, which proves (iv-a). □

6.2 Proof of (iv-b)

Let Ω' be a finite set, π' a probability on Ω', $\mathfrak{A}' \subseteq \Omega'$ with $\pi'(\mathfrak{A}'^c) = a$, and $N \geq 1$. Since $a \in (0, 1)$, both $\pi'(\mathfrak{A}'^c) = a > 0$ and $\pi'(\mathfrak{A}') = 1 - a > 0$, so the conditional laws

$$\pi'_{\mathrm{out}} := \pi'(\,\cdot \mid \mathfrak{A}'^c), \qquad \pi'_{\mathrm{in}} := \pi'(\,\cdot \mid \mathfrak{A}')$$

are well defined; π'_{out} is carried by $\mathfrak{A}'^c$ and π'_{in} by $\mathfrak{A}'$.

Construction. Take the i.i.d. pairs (W_r, f_r) of (iv-a). Independently across r, and given (W_r, f_r), draw $S_{r,1}, \ldots, S_{r,N}$ i.i.d. from π'_{out} if $f_r = 1$ and from π'_{in} if $f_r = 0$. (Formally: let $(U_{r,i})$ and $(V_{r,i})$, $r \leq 4$, $i \leq N$, be independent families of i.i.d. variables with laws π'_{out} and π'_{in}, independent of the pairs, and set $S_{r,i} := U_{r,i}$ if $f_r = 1$ and $S_{r,i} := V_{r,i}$ if $f_r = 0$.) The four quadruples $\big(W_r, f_r, (U_{r,i})_i, (V_{r,i})_i\big)$ are i.i.d.: the pairs (W_r, f_r) are i.i.d., the families $(U_{r,i})$ and $(V_{r,i})$ are independent of each other and of the pairs, and their laws do not depend on r. Since $(W_r, (S_{r,i})_i)$ is the same measurable function of each quadruple, the four pairs $(W_r, (S_{r,i})_{i \leq N})$ are i.i.d.

The outside fraction. If $f_r = 1$, every $S_{r,i} \in \mathfrak{A}'^c$, so $\hat{q}_r(1_{\mathfrak{A}'^c}) = 1$; if $f_r = 0$, every $S_{r,i} \in \mathfrak{A}'$, so $\hat{q}_r(1_{\mathfrak{A}'^c}) = 0$. Hence $f_r = \hat{q}_r(1_{\mathfrak{A}'^c})$.

The identity. Fix $g : \Omega' \to \mathbb{R}$ (bounded, as Ω' is finite). Since $(U_{r,i})$ and $(V_{r,i})$ are independent of (W_r, f_r),

$$\mathbb{E}\big[W_r\, \hat{q}_r(g) \,\big|\, W_r, f_r\big] = W_r\Big(f_r\, \pi'_{\mathrm{out}}(g) + (1 - f_r)\, \pi'_{\mathrm{in}}(g)\Big),$$

with $\pi'_{\mathrm{out}}(g) = \pi'(g 1_{\mathfrak{A}'^c})/a$ and $\pi'_{\mathrm{in}}(g) = \pi'(g 1_{\mathfrak{A}'})/(1 - a)$. All variables involved are bounded, so taking expectations and using $\mathbb{E}[W_r f_r] = a$ and $\mathbb{E}[W_r(1 - f_r)] = \mathbb{E}W_r - \mathbb{E}[W_r f_r] = 1 - a$,

$$\mathbb{E}\big[W_r\, \hat{q}_r(g)\big] = a \cdot \frac{\pi'(g 1_{\mathfrak{A}'^c})}{a} + (1 - a) \cdot \frac{\pi'(g 1_{\mathfrak{A}'})}{1 - a} = \pi'(g 1_{\mathfrak{A}'^c}) + \pi'(g 1_{\mathfrak{A}'}) = \pi'(g).$$

Multiplying by any constant $Z' > 0$ gives $\mathbb{E}[(Z'W_r)\, \hat{q}_r(g)] = Z'\pi'(g)$, the identity of (H-T12) with $\hat{Z}_r := Z'W_r$, for every test function g. Each island has a single population; using it at every recording position ($\hat{q}_{r,\tau} := \hat{q}_r$ for all $\tau \in \mathcal{T}$) gives the identity at every position. This proves (iv-b). □

Remark 1 (Consistency with (i)–(iii)). In the example of (iv-a), the islandwise bounds hold, as they must. Since $\varepsilon < c < h$, $\mathbb{P}(W_1 < c) = \mathbb{P}(W_1 = \varepsilon) = 3/4$, so $a/c + \mathbb{P}(W_1 < c) \geq 3/4 + b = \mathbb{E}f_1$ because $b < a/3 < a/c$; and $a/(cu) + \mathbb{P}(W_1 < c) \geq 3/4 + b = \mathbb{P}(f_1 > u)$ because $b < a/3 < a/(cu)$. The pooled version fails because of the event that island 1 has small evidence $W_1 = \varepsilon$ and $f_1 = 1$ while another island has large evidence h, which keeps $\bar{W}$ above c. Part (iv) is a non-implication about substituting the pooled lower tail into (i)–(ii); it does not claim that the SMC final run, for a particular data set and calibration, realises this law.

Remark 2 (Scope and use of hypotheses). Parts (i)–(iii) bound the law of f_r under $\mathbb{P}_{\mathcal{C}}$; they give no bound on the outside fraction of a realised population. Parts (i) and (ii) use only $W_r \geq 0$, $0 \leq f_r \leq 1$ and the instance $f = 1_{\mathfrak{A}^c}$ of (H-T12) (in fact only $\mathbb{E}_{\mathcal{C}}[W_r f_r] \leq \pi(\mathfrak{A}^c)$). The first claim of part (iii), $\mathbb{P}_{\mathcal{C}}(W_r < c) \leq B/(1-c)$, uses only its own hypothesis $\mathbb{E}_{\mathcal{C}}|W_r - 1| \leq B$ (which also makes W_r integrable) and Markov's inequality, and not (H-T12); its "consequently" clause substitutes this bound into (i) and (ii), and so rests on the same instance of (H-T12) as they do. Part (iv) concerns explicitly constructed laws and uses no property of the final run. Assumption EXP (or EXP′) is not used anywhere.

(d) Review record

FINAL_REPORT.md: statement changes

2. Statement changes and why

The inequalities, constants and quantifiers of the theorem file are unchanged. The other changes:

- **T12 dependency made explicit.** The theorem file says "Builds on T12". STATEMENT.md states T12(i) as the hypothesis (H-T12), as the run rules require. No statement-checker call was needed, because this follows the run rules and is not a new assumption.
- **Pooled-tail sentence made precise.** The theorem file's sentence "a pooled-evidence lower tail does not substitute" became part (iv), a non-implication with an explicit example. It was strengthened during review:
  - v2: strictly positive evidence (Astra #1);
  - v3: every outside mass a ∈ (0, cu/4], plus (iv-b), which realises the example with particle populations satisfying the T12 identity for every test function (suggested by the Opus reviewer);
  - v4: wording only.
- **Realised-population sentence kept as a scope remark.** The theorem file's sentence on realised populations is a scope remark: the bounds concern the law of the run, not a realised population.
- **EXP / EXP′ not needed.** No decision from Decision Point 2 affects T15.

FINAL_REPORT.md: review history

3. Review history

Stage	Reviewer	Version	Verdict	Findings
A	Astra #1 (session 01a16418-957c-7832-97e0-ccb8b8f10081), r1	v1	verified	1 exposition (example used W = 0) → fixed in v2
A	Astra #1, r2	v2	verified	none
B	Astra #2 (new session 01a1641c-660b-7822-a3a2-f4a9f9c67d7b), r1	v2	verified	none
B	Opus 5.5 xhigh (Claude Personal), r1	v2	verified	5 exposition → fixed in v3
B	Astra #2, r2 (ruling on v3 + log)	v3	verified	2 exposition → fixed in v4
B	Opus, r2 (ruling on v3 + log)	v3	verified	4 exposition (2 shared with Astra #2) → fixed in v4
B	Astra #2, r3	v4	verified	none of any severity
B	Opus, r3	v4	verified	1 optional wording note → aligned in STATEMENT §4

- **Fatal or gap findings:** none, from any reviewer, at any round. CLAIMS_OF_FATAL_FLAWS.md records this, with every exposition finding and the lead's answer.
- **Rulings on the log:** both Stage B reviewers ruled on it and agree that nothing is outstanding.
- **Astra judge:** not used.
- **Lead's judgment of the findings:** all were accepted as correct. None changed parts (i)–(iii), which are word for word the same as in v1.

CLAIMS_OF_FATAL_FLAWS.md (every entry, verbatim)

CLAIMS_OF_FATAL_FLAWS — T15 (L-OCC)

Fatal or gap findings

None. Eight reviews, and none raised a fatal or gap finding:

- Astra #1, round 1, on PROOF_v1: verified; 1 exposition finding.
- Astra #1, round 2, on PROOF_v2: verified; no new findings.
- Stage B Astra #2, round 1, on PROOF_v2: verified; no findings.
- Stage B Opus 5.5 xhigh, round 1, on PROOF_v2: verified; 5 exposition findings.
- Stage B Astra #2, round 2, on PROOF_v3 + this log: verified; 2 exposition findings. Agrees that no fatal or gap finding is outstanding. Rulings: A1-1, O-1, O-2, O-3, O-5 resolved; O-4 partially resolved.
- Stage B Opus, round 2, on PROOF_v3 + this log: verified; 4 exposition findings. Agrees that no fatal or gap finding is outstanding. Rulings: O-1 to O-5 resolved.
- Stage B Astra #2, round 3, on PROOF_v4: verified; no findings of any severity; B2-1 to B2-4 resolved.
- Stage B Opus, round 3, on PROOF_v4: verified; B2-1 to B2-4 resolved; one optional exposition note (a leftover "full identity" in the STATEMENT §4 scope remark, which it judged accurate) was aligned afterwards (wording only).

Exposition findings and the lead's answers (kept for the record)

#	Reviewer	Finding (short)	Answer	Change
A1-1	Astra #1 r1	The (iv) example has W = 0 with probability 3/4, but actual island evidence is strictly positive. Heading scope needs care.	Accepted.	v2: example uses ε > 0; scope sentence added to (iv).
O-1	Opus B r1	(H-T12) gives the identity for every test function g, but the example checks only g ∈ {1, 1_{$\mathfrak{A}^c$}}. Opus suggested a populations-based extension.	Accepted; adopted as a strengthening.	v3: new (iv-b) realises the example with particle populations satisfying E[W $\hat{q}$(g)] = π'(g) for all g, on any finite Ω' (including the method's Ω, π) with $\pi'(\mathfrak{A}'^c)$ = a. (iv) now holds for every a ∈ (0, cu/4].
O-2	Opus B r1	Island i.i.d. given $\boldsymbol{\mathcal{C}}$ is presented as a method property, but it is not among the hypotheses.	Accepted.	v3: the claim about the method is removed; the example's i.i.d. property is described as only strengthening it.
O-3	Opus B r1	$\bar{Z}_0$, d, m(∅), π_0 are undefined, and the positivity of $\hat{Z}$_r is asserted, not derived.	Accepted.	v3: the symbols are defined briefly with a pointer to 00_COMMON; positivity is derived from positive factors. The proof uses only W_r ≥ 0.
O-4	Opus B r1	The Scope remark misdescribes the hypothesis use of (iii) and (iv).	Accepted.	v3: the remark is rewritten part by part.
O-5	Opus B r1	(iv) refutes the pooled form of (i) without the min{1,·} and does not state that the min form is refuted too.	Accepted.	v3: the one-line step min{1,x} ≤ x is added to (iv-a) and its proof.
B2-1	Astra #2 r2 = Opus r2 #1	The i.i.d. justification in (iv-b) cites (W_r, U, V) and leaves out f_r.	Accepted (both reviewers raised it independently).	v4: the argument now uses the quadruples (W_r, f_r, U_r, V_r).
B2-2	Astra #2 r2 = Opus r2 #2	"Part (iii) does not use (H-T12)" holds only for its first claim; the "consequently" clause rests on (i)–(ii).	Accepted.	v4: the note in STATEMENT §2 and the proof's final remark separate the two claims.
B2-3	Opus r2 #3	(iv-b) gives the identity at one position, while "full identity" suggests all of $\mathcal{T}$.	Accepted.	v4: "full identity" is replaced by "identity for every test function", plus a remark that reusing the population at every τ gives every position.
B2-4	Opus r2 #4	"The method's Ω and π" example needs some set of posterior mass exactly a.	Accepted.	v4: the example states when such a set exists (smallest atom, if #Ω ≥ 4/(cu)).

Unified version after Stage B round 1: v3. After Stage B round 2 (wording only): STATEMENT.md (v4 bullet in §5) and proof/PROOF_v4.tex/.pdf. (i)–(iii) are unchanged in statement and proof since v1.

Chapter 7

T20 (T15), clauses (i)–(ii): local-search semantics

Final proof: `proofs/T20_T15_local_search/proof/PROOF_v3.tex`. *Sources of parts (a), (b), (d):* `theorems/T20_T15_local_search.md`, `STATEMENT.md`, `FINAL_REPORT.md`, `CLAIMS_OF_FATAL_FLAWS.md`.

(a) Original conjecture text (theorem file, verbatim)

T20 (T15): Local-search semantics and conditional repair

Status: conjectured, not proven. Methodology SMC-BVS v3.0 (2026-10-19), SHA-256 93043758058a6d422cca6ca4b6be8f3a9e8149b6f6de2e1bab1ce31c44b8019c. Read with `00_COMMON.md`.

Category / rank: deterministic component validity and statistical consistency. Clauses (i)–(ii): **core**; (iii) ORTH repair: **secondary**; (iv) COH extension: **optional**. v2's SE-only "every best move is a true move" clause is withdrawn.

Definitions used

The local search: from a centre S_c, each scan scores every feasible flip and swap of the centre; best-improvement moves, at most ℓ moves, then one extra scan; `local_certified` flag; running all-scored HPM $\hat{S}$; E_n := {S_0 is the strict global maximiser of F} (00_COMMON §§1–2; THEORY_BRIEF §2).

Assumptions

EX-A and execution of the relevant scans for (i)–(ii); G, R, SE and BM-C (through T03) for the statistical reading; ORTH and BM-∞ for (iii); COH and BM-∞ for (iv).

Precise statement

(i) [certificate] `local_certified` certifies exactly that the returned model was itself scanned as a centre and had no strictly improving feasible flip or swap within the cap; equal-score neighbours are allowed; weak local maximality only. All scanned candidates, including rejected ones, enter the running HPM comparison; a model first discovered by the final extra scan can be the reported HPM

without a certificate. (ii) [one neighbour] On E_n, every completed scan from a feasible flip-or-swap neighbour of S_0 scores S_0; if a move remains the search moves to S_0; if the subsequent scan executes it certifies S_0. Under T03, $P_0(E_n) \to 1$, and a sufficient global margin is $\min_{S \neq S_0}\{F(S_0) - F(S)\} \geq \log[\pi(S_0)/(1-\pi(S_0))] \geq \tfrac{1}{2} \log n - \tfrac{1}{2} \log \log n - O_{P_0}(1)$. (iii) [ORTH repair] Under G, R, ORTH, BM-∞: with probability → 1, simultaneously for every feasible starting centre with $|S_c \Delta S_0| \leq \ell$, each selected best improvement reduces $|S \Delta S_0|$ (by 1 for a true addition or noise deletion, by 2 for a true-for-noise swap); the search reaches S_0 within $|S_c \Delta S_0|$ moves and the extra scan certifies it. Equal magnitudes are unnecessary. (iv) [COH extension] The same conclusion under G, R, COH, BM-∞, with the obligation of proving monotonicity of the selected best move uniformly over the centres with the integrated-variance score. (v) [scope] Posterior concentration does not imply the entry condition of (iii), nor bound an arbitrary centre's size by m_n; the move limit can bind; a later search repeats an earlier ascent only if its centre and comparisons are unchanged. Under SE alone a signal-bearing proxy can win the first improvement from ∅ (correlation matrix [[1,0,.65],[0,1,.65],[.65,.65,1]], $S_0 = \{1,2\}$: eigenvalue floor 0.081 > 0, proxy population fit gain 1.69 vs 1 per true singleton).

Dependencies

Builds on: T03, T05, T11. Used by: T21, T24, T31.

Numerical evidence (provisional)

provisional evidence (Sol check J3): on the design above with θ = (1,1,0), best-improvement paths from ∅ at n = 64, 512, 2048 select the proxy first and recover S_0 afterwards; at n = 128 the path stops at a certified wrong local maximum although S_0 is globally best; one-neighbour scoring/certification semantics confirmed; a truth first hit on the final scan is scored but not certified. Files: judge/sol1/J3_LOCAL_SEARCH.md.

Proof-route hint (optional)

(i)–(ii): frozen-centre scan rule and the all-scored ledger (T11). (iii): the gain statistics G_l do not depend on the centre under ORTH; one no-lucky event $\{\max_l G_l < x_k \text{ for all } k \leq s_0+\ell\}$ of probability → 1 separates true-coordinate gains from noise gains in the exact integrated odds. (iv): control residualised gains uniformly over admissible centres (deviation $\leq 4c_n|T| \log p/\varphi_0^2(1+o(1))$ from ORTH). Hardest step: best-move ordering after integrating variance. *When proof.md is run on this file, its lead considers the statement both with and without these hints; the hints never reach the reviewers.*

(b) STATEMENT.md (verbatim)

STATEMENT.md — T20 (T15), clauses (i)–(ii): local-search semantics

Source: `Theory_architecture_v3/theorems/T20_T15_local_search.md` (clauses (i)–(ii) only; (iii)–(v) are out of scope). Method: SMC-BVS v3.0, `Methodology/SMC_BVS_v3_20261019/METHODOLOGY_v3.md`, SHA-256 93043758058a6d422cca6ca4b6be8f3a9e8149b6f6de2e1bab1ce31c44b8019c (checked 2026-10-22). Statement analysis outcome: **(a) correct as stated**; no hypothesis was changed or added. Section 5 lists the readings used to make the informal wording precise.

0. Original clauses (verbatim)

(i) [certificate] `local_certified` certifies exactly that the returned model was itself scanned as a centre and had no strictly improving feasible flip or swap within the cap; equal-score neighbours are allowed; weak local maximality only. All scanned candidates, including rejected ones, enter the running HPM comparison; a model first discovered by the final extra scan can be the reported HPM without a certificate.

(ii) [one neighbour] On E_n, every completed scan from a feasible flip-or-swap neighbour of S_0 scores S_0; if a move remains the search moves to S_0; if the subsequent scan executes it certifies S_0. Under T03, $P_0(E_n) \to 1$, and a sufficient global margin is $\min_{S \neq S_0}\{F(S_0) - F(S)\} \geq \log[\pi(S_0)/(1-\pi(S_0))] \geq \tfrac{1}{2} \log n - \tfrac{1}{2} \log \log n - O_{P_0}(1)$.

Assumptions named by the source for (i)–(ii): EX-A (exact scores and comparisons) and execution of the relevant scans; for the statistical reading, G, R, SE and BM-C or BM-∞ through T03. The source defines (its "Definitions used"): "E_n := {S_0 is the strict global maximiser of F}".

Method text formalised in §2 (METHODOLOGY_v3 §6.10, verbatim): "**Local search:** from the current best scored model (the centre), scan every feasible flip and swap (additions beyond K excluded) with the centre frozen during the scan; move to the best strict improvement (lexicographic ties); at most ℓ moves, then one further scan; stop earlier when a scan finds no improvement. Every evaluated model updates the global best. If the last scan still improves, set `local_limit` (no restart). `local_certified` only when the returned model is a scanned centre with no improving neighbour. The search never alters particles." and "**HPM:** the strict maximum of $\log \varpi(S) + \log m(S)$ over every model ever scored (tables, pilot, calibration, final rounds, rejected proposals, independence proposals, local searches), kept as a running best that survives cache eviction. Only exactly equal scores invoke the tie rule: smallest |S|, then lexicographically smallest index set." Execution order (§6.1/§7): "… final local search; final checks (flags only); report" — nothing is scored after the final local search. $\ell = \lceil \log_2(n + p + 1) \rceil$ (§3).

1. Setting

- $p \geq 1$ predictors, $n \geq 3$, $K = \min(p, \lfloor n/2 \rfloor) \geq 1$, state space $\Omega = \{S \subseteq [p] : |S| \leq K\}$.
- **Score.** $F : \Omega \to \mathbb{R}$. Clause (i) and the deterministic part of clause (ii) are proved for an **arbitrary** $F : \Omega \to \mathbb{R}$; only (ii-B)–(ii-C) use the method's score $F(S) = \log[\varpi(S)\, m(S)]$, $\pi(S) = e^{F(S)}/\Sigma_{T \in \Omega} e^{F(T)}$ for $S \in \Omega$, and the convention $\pi(S) := 0$ for $S \notin \Omega$. The method's score (00_COMMON §1; METHODOLOGY_v3 §5), on standardised data $X \in \mathbb{R}^{n \times p}$, $y \in \mathbb{R}^n$ ($y \neq 0$; columns and y centred, $\|x_j\|^2 = \|y\|^2 = n - 1$): for $S \in \Omega$ with $k = |S|$, X_S the columns in S, $G_S = X_S^T X_S$, $z_S = X_S^T y$, $H_S = (n+1)G_S + I_k$, $D_S = \det H_S / \det(G_S + I_k)$, $Q_S = y^T y - n\, z_S^T H_S^{-1} z_S$ (for $S = \emptyset$: $D_\emptyset = 1$, $Q_\emptyset = y^T y$); $a_0 = b_0 = 0.01$, $\nu = a_0 + n/2$; $m(S) = (2b_0)^{a_0}\, \Gamma(\nu) / (\pi^{n/2}\, \Gamma(a_0)) \cdot D_S^{-1/2}\, (2b_0 + Q_S)^{-\nu}$; $w(k) = \tfrac{1}{2} / C(2p-1, k)$, $W_K = \Sigma_{j \leq K} C(p, j)\, w(j)$, $\varpi(S) = w(|S|)/W_K$. (This is the marginal likelihood of the model $y \mid \beta_S, \sigma^2 \sim N(X_S \beta_S, \sigma^2 I)$, $\beta_S \mid \sigma^2 \sim N(0, n\sigma^2 (G_S + I)^{-1})$, $\sigma^2 \sim IG(a_0, b_0)$, times the beta-binomial $BB(1, p)$ prior capped at K; the proof uses only the displayed formulas.)
- **EX-A.** All scores are exact real numbers and all comparisons are exact.
- **HPM order** $\succ$ (method §6.10): $S \succ T$ iff $F(S) > F(T)$; or $F(S) = F(T)$ and $|S| < |T|$; or $F(S) = F(T)$, $|S| = |T|$ and the increasingly sorted index list of S is lexicographically smaller than that of T. This is a strict total order on Ω. For a finite nonempty $\mathcal{A} \subseteq \Omega$, best($\mathcal{A}$) is its $\succ$-maximum. The running HPM

after any step is best($\mathcal{S}$) where $\mathcal{S}$ is the set of all models scored so far in the run (tables, pilot, calibration, final rounds, accepted and rejected proposals including independence proposals, local scans).

- **Move order $\triangleright$.** A fixed strict total order on Ω with $F(S) > F(T) \Rightarrow S \triangleright T$ (maximise F, exact ties broken by a fixed deterministic rule; the method says "lexicographic ties"). Every result holds for every such rule.
- **Feasible flip-or-swap neighbourhood** of $c \in \Omega$: $\mathcal{N}(c) = \{c \cup \{j\} : j \notin c, |c| < K\} \cup \{c \setminus \{i\} : i \in c\} \cup \{(c \setminus \{i\}) \cup \{j\} : i \in c, j \notin c\} \subseteq \Omega$. (All swaps, not only neighbour-graph swaps; additions beyond K excluded.) The relation $N \in \mathcal{N}(c)$ is symmetric on Ω.

2. The local search (method §6.10), as formalised

Input: the finite nonempty set $\mathcal{S}^- \subseteq \Omega$ of models scored in the run before the search starts; the move cap $\ell \geq 0$ (the method has $\ell = \lceil \log_2(n+p+1) \rceil$).

```
c_0 := best(𝒮⁻);  𝒮 := 𝒮⁻;  k := 0
loop:
   scan k (centre c_k frozen): 𝒮 := 𝒮 ∪ 𝒩(c_k)          # every neighbour scored; running HPM updated
   I_k := {N ∈ 𝒩(c_k) : F(N) > F(c_k)}                   # strict improvements over the centre
   if I_k = ∅:  stop; outcome NOIMP; m := k
   elif k < ℓ:  c_{k+1} := the ▷-maximum of I_k; k := k + 1  # a move
   else (k = ℓ): stop; outcome LIMIT; m := ℓ                # the "one further scan" still improved
```

Outputs: centres $\mathcal{C} = \{c_0, \ldots, c_m\}$; final scored set $\mathcal{S}$_end; **returned model R := best($\mathcal{S}$_end)** (the running HPM when the search ends); `local_limit` := 1 iff outcome LIMIT; **`local_certified` := 1 iff $R \in \mathcal{C}$ and there is no $N \in \mathcal{N}(R)$ with $F(N) > F(R)$** (method: "`local_certified` only when the returned model is a scanned centre with no improving neighbour").

The run may contain several local searches (after the pilot, after each set of diagnostic islands, after the final run); between them the scored set may grow. In the method nothing is scored after the final local search, so for the final search R equals the reported HPM $\hat{S}$.

Execution hypothesis (EXEC). Every scan that the procedure calls is completed, i.e. all of $\mathcal{N}(c_k)$ is scored.

3. Clause (i): certificate semantics (EX-A, EXEC; arbitrary F)

For any execution of one local search:

- **(i-a) Characterisation.** `local_certified` = 1 iff the outcome is NOIMP and R = c_m. In that case $F(R) = \max_{\mathcal{S}\text{_end}} F$ and $F(R) \geq F(N)$ for every $N \in \mathcal{N}(R)$ (weak local maximality over the feasible flips and swaps; neighbours with $F(N) = F(R)$ are permitted, and every such neighbour N satisfies $R \succ N$, i.e. $|N| > |R|$, or $|N| = |R|$ and N is lexicographically later). Moreover $R \in \mathcal{C}$ holds iff R was a scan centre in any local search of the run up to and including this one.
- **(i-b) Only weak local information.** The whole execution (centres, outcome, $\mathcal{S}$_end, R, both flags) is unchanged if F is replaced by any $F' : \Omega \to \mathbb{R}$ with $F' = F$ on $\mathcal{S}$_end, the F'-run using the same input $\mathcal{S}^-$, the same ℓ, the HPM order $\succ$ computed from F', and the move order computed from F' with the same fixed tie-break rule. Hence `local_certified` = 1 carries no information about F on $\Omega \setminus \mathcal{S}$_end; in particular, when $\Omega \setminus \mathcal{S}$_end $\neq \emptyset$ it does not certify global maximality (when $\mathcal{S}$_end = Ω, R is the global maximiser of F trivially).

- **(i-c) Ledger.** Every model scored by any scan of the search, whether or not it was selected as a move, belongs to $\mathcal{S}$_end, so R ⪰ N (R = N or R ≻ N) for each of them; in particular F(R) ≥ F(N). For the final search R = $\hat{S}$.
- **(i-d) Final extra scan.** On outcome LIMIT: R ∈ $\mathcal{N}$(c_ℓ), F(R) > F(c_ℓ), R was first scored by scan ℓ (R ∉ the scored set before scan ℓ), R was never a scan centre of any local search of the run up to this one, and `local_certified` = 0. For the final search this R is the reported HPM $\hat{S}$.
- **(i-e) Generic case.** If exactly one model of $\mathcal{S}$_end attains max_{$\mathcal{S}$_end} F, then `local_certified` = 1 ⟺ outcome NOIMP ⟺ `local_limit` = 0.

4. Clause (ii): one neighbour of S_0

Let $S_0 \subseteq [p]$ be the true support and **$E_n := \{S_0 \in \Omega$ and $F(S_0) > F(S)$ for every $S \in \Omega \setminus \{S_0\}\}$**.

(ii-A) Deterministic (EX-A, EXEC; arbitrary F). On E_n, for every local search of the run and every scan k of it whose centre c_k is a feasible flip-or-swap neighbour of S_0 (c_k ∈ $\mathcal{N}(S_0)$): 1. scan k scores S_0 ($S_0 \in \mathcal{N}$(c_k)); 2. $S_0 \in$ I_k and S_0 is the ▷-maximum of I_k; if k < ℓ (a move remains), c_{k+1} = S_0; 3. if k < ℓ, scan k+1 (with centre S_0) is called by the procedure; it finds I_{k+1} = ∅, the search stops with outcome NOIMP, R = S_0 and `local_certified` = 1; 4. if k = ℓ (no move remains), the search stops with outcome LIMIT, R = S_0 and `local_certified` = 0. In both cases S_0 is the running HPM from the moment scan k scores it; for the final search $\hat{S} = S_0$.

(ii-B) Margin inequality (method's F; deterministic). If $S_0 \in \Omega$ then $0 < \pi(S_0) < 1$ and
$\Delta_n := \min_{S \in \Omega \setminus \{S_0\}} [F(S_0) - F(S)] \geq \lambda_n := \log[\pi(S_0)/(1 - \pi(S_0))]$. Consequently $\{\pi(S_0) > ½\} \subseteq$ E_n.

(ii-C) Statistical (method's F). Regime: a sequence of problems indexed by $n \to \infty$ (design fixed, P_0 = law of the Gaussian errors), $r_n := \sqrt{(\log n / n)}$. Hypothesis **H03** (the conclusion of T03(i), conjectured there under R1 with G, R, SE and BM-C or BM-∞; it is used here only as a hypothesis): there is n_0 with $S_0 \in \Omega$ for all $n \geq n_0$, and $1 - \pi(S_0) = O_{P_0}(r_n)$. Under H03: 1. $P_0(E_n) \to 1$; 2. $(½ \log n - ½ \log\log n - \lambda_n)^+ = O_{P_0}(1)$ (over $n \geq n_0$), i.e. $\Delta_n \geq \lambda_n \geq ½ \log n - ½ \log\log n - O_{P_0}(1)$; 3. under EX-A and EXEC, for any joint law of data and algorithm randomness whose data marginal is P_0, the probability that conclusions (ii-A)1–4 hold simultaneously for every scan of every local search of the run is at least $P_0(E_n)$, hence tends to 1.

5. Readings used (the informal text made precise)

1. "Returned model" = the running HPM when the search ends (R above); this matches the source's "a model first discovered by the final extra scan can be the reported HPM without a certificate". If one instead read it as the last centre c_m: on outcome LIMIT, R ≠ c_m (by (i-d), F(R) > F(c_m)), but the flag is 0 under both readings (c_m has a strictly improving neighbour); on outcome NOIMP, R = c_m unless another model of $\mathcal{S}$_end has exactly the score F(c_m) and precedes c_m in the HPM order. So the two readings give different flag values only on NOIMP with such an exact tie.
2. "Scanned as a centre" = R ∈ $\mathcal{C}$; (i-a) shows this is the same as having been a centre in any local search of the run up to and including the current one.
3. "Within the cap" = among feasible models, |S| ≤ K. (Reading it as the move cap ℓ gives nothing different: certification implies outcome NOIMP at some m ≤ ℓ.) The method's "`local_certified` only when …" is read as "iff", as the source's "certifies exactly that" requires.

4. "Weak local maximality only" is made precise by (i-a) (what is implied) and (i-b) (nothing about unscored models is implied).
5. "Can be the reported HPM without a certificate" is made precise by (i-d): on outcome LIMIT the reported model is such a model and is uncertified. Whether outcome LIMIT has positive probability for given data is not asserted. (A second, tie-only route to an uncertified reported model also exists: on NOIMP, if some scored model has exactly the score F(c_m) and precedes c_m in the HPM order, then R is the HPM-first of all such models (it may be a neighbour of c_m or a model scored earlier), R ≠ c_m, and both flags are 0; (i-a) covers it.)
6. "If the subsequent scan executes" = EXEC; in the procedure a scan always follows a move (a regular scan or the one further scan).
7. "Under T03" = hypothesis H03; "a sufficient global margin is … ≥ …" = the chain Δ_n ≥ λ_n ≥ ½ log n − ½ log log n − O_{P_0}(1) of (ii-B)–(ii-C), with Δ_n > 0 equivalent to E_n when $S_0 \in \Omega$.

6. Change log

- v3 → v3.1 (2026-10-22, after the Stage B ruling round; text of §1 and §5 only, proof unchanged): §5 item 5 tie-route sentence made precise (Opus N1 = Astra #2 exposition finding); §1 list of HPM sources now names independence proposals (Opus N2).
- v2 → v3 (2026-10-22, after the Stage B reviews; exposition only, no hypothesis added or changed): verbatim source text for E_n and for METHODOLOGY_v3 §6.10 added to §0; (i-a) states that equal-score neighbours of a certified R come after R in the HPM order; (i-b) says which move order the F′-run uses and qualifies "does not certify global maximality" by $\Omega \setminus \mathcal{S}$_end ≠ ∅; (ii-A) closing sentence "from the moment scan k scores it"; (ii-C)3 names EX-A and EXEC; §5 items 3 and 5 extended.
- v1 → v2 (2026-10-22, after Astra #1 round 1; no hypothesis or claim of §§3–4 changed): §1 now writes out the method's score formulas (Q_S, D_S, m, ϖ) in full, so the statement is self-contained; §5 item 1 corrected (the two readings of "returned model" also differ on outcome LIMIT, where both give flag 0); §5 item 2 wording corrected ("up to and including the current one").

(c) Final proof, verbatim: body of PROOF_v3.tex

The proof below is the body of `PROOF_v3.tex` *from* `\begin{document}` *to* `\end{document}` *without the title block. Its preamble definitions are local to this chapter; labels carry the prefix* `T20:`.

All notation, the procedure and the claims are those of `STATEMENT.md`; section numbers §1–§4 below refer to it. We recall what is used.

1 Setting and conventions

$\Omega = \{S \subseteq [p] : |S| \le K\}$ with $p \ge 1$, $n \ge 3$, $K = \min(p, \lfloor n/2 \rfloor) \ge 1$. A score is a function $F : \Omega \to \mathbb{R}$. All comparisons are exact (EX-A).

Orders. The HPM order $\succ$: $S \succ T$ iff $F(S) > F(T)$, or $F(S) = F(T)$ and $|S| < |T|$, or $F(S) = F(T)$, $|S| = |T|$ and the increasingly sorted index list of S is lexicographically smaller than that of T. Distinct sets of equal size have distinct sorted lists, and lexicographic order on lists of equal length is a strict total order; hence $\succ$ is a strict total order on Ω (irreflexive, transitive, and

any two distinct elements are comparable). We write $S \succeq T$ for ($S \succ T$ or $S = T$). For finite nonempty $\mathcal{A} \subseteq \Omega$, best($\mathcal{A}$) is the $\succ$-maximum. Since $F(S) > F(T) \Rightarrow S \succ T$,

$$F(\text{best}(\mathcal{A})) = \max_{\mathcal{A}} F. \tag{1}$$

The move order: fix a strict total order τ on Ω (the tie-break rule; it does not depend on F). Put $S \rhd_F T$ iff $F(S) > F(T)$, or $F(S) = F(T)$ and $S \tau T$. This is a strict total order with $F(S) > F(T) \Rightarrow S \rhd_F T$. Every strict total order $\rhd$ on Ω with $F(S) > F(T) \Rightarrow S \rhd T$ (the move order of §1) equals $\rhd_F$ for $\tau := \rhd$; so nothing is lost by writing the move order as $\rhd_F$. Again, for finite nonempty $\mathcal{A}$ the $\rhd_F$-maximum of $\mathcal{A}$ has score $\max_{\mathcal{A}} F$.

Neighbourhoods. For $c \in \Omega$, $\mathcal{N}(c) = \{c \cup \{j\} : j \notin c,\ |c| < K\} \cup \{c \setminus \{i\} : i \in c\} \cup \{(c \setminus \{i\}) \cup \{j\} : i \in c,\ j \notin c\}$. Every element of $\mathcal{N}(c)$ has size at most K, so $\mathcal{N}(c) \subseteq \Omega$; and $c \notin \mathcal{N}(c)$ (a flip changes the size, a swap removes $i \in c$ and inserts $j \notin c$, $j \neq i$). Moreover $\mathcal{N}(c) \neq \emptyset$: if $c = \emptyset$ the addition $\{1\}$ is feasible ($K \geq 1$), and if $c \neq \emptyset$ any deletion is in $\mathcal{N}(c)$. So maxima of F over $\mathcal{N}(c)$ are maxima over finite nonempty sets.

Lemma 1 (symmetry)**.** *For $c, S \in \Omega$: $S \in \mathcal{N}(c) \iff c \in \mathcal{N}(S)$.*

Proof. By the symmetry of the roles it suffices to prove $\Rightarrow$. (a) $S = c \cup \{j\}$, $j \notin c$: then $j \in S$ and $c = S \setminus \{j\}$ is a deletion from S. (b) $S = c \setminus \{i\}$, $i \in c$: then $i \notin S$, $|S| = |c| - 1 < |c| \leq K$, so the addition $S \cup \{i\} = c$ is feasible and lies in $\mathcal{N}(S)$. (c) $S = (c \setminus \{i\}) \cup \{j\}$ with $i \in c$, $j \notin c$: then $j \neq i$, $j \in S$, $i \notin S$, and $c = (S \setminus \{j\}) \cup \{i\}$ is a swap of S. □

The procedure (§2). Input: finite nonempty $\mathcal{S}^- \subseteq \Omega$ (all models scored in the run before the search) and $\ell \geq 0$. Put $c_0 = \text{best}(\mathcal{S}^-)$, $\mathcal{S}_{<0} = \mathcal{S}^-$. For $k = 0, 1, \ldots$: scan k scores $\mathcal{N}(c_k)$, so the scored set becomes $\mathcal{S}_{<k+1} := \mathcal{S}_{<k} \cup \mathcal{N}(c_k)$; let $I_k = \{N \in \mathcal{N}(c_k) : F(N) > F(c_k)\}$. If $I_k = \emptyset$, stop with outcome NOIMP and $m = k$. Else, if $k < \ell$, let c_{k+1} be the $\rhd_F$-maximum of I_k and continue with $k + 1$. Else ($k = \ell$) stop with outcome LIMIT and $m = \ell$. Nothing else is scored during the search. Outputs: $\mathcal{C} = \{c_0, \ldots, c_m\}$, $\mathcal{S}_{\text{end}} = \mathcal{S}_{<m+1}$, $R = \text{best}(\mathcal{S}_{\text{end}})$, `local_limit` $= 1$ iff LIMIT, and `local_certified` $= 1$ iff $R \in \mathcal{C}$ and no $N \in \mathcal{N}(R)$ has $F(N) > F(R)$. (EXEC: each scan scores all of $\mathcal{N}(c_k)$.)

2 The basic invariant

Lemma 2. *The procedure stops with $m \leq \ell$. For every $k \leq m$:*

(a) $c_k \in \mathcal{S}_{<k}$ (each centre has been scored before its scan);

(b) $F(c_k) = \max_{\mathcal{S}_{<k}} F$;

(c) if $k < m$ then $F(c_{k+1}) = \max_{\mathcal{N}(c_k)} F > F(c_k)$; in particular $F(c_0) < F(c_1) < \cdots < F(c_m)$ and the centres are distinct.

Moreover:

(d) on NOIMP, $\max_{\mathcal{S}_{\text{end}}} F = F(c_m)$;

(e) on LIMIT, $\max_{\mathcal{S}_{\text{end}}} F = \max_{\mathcal{N}(c_\ell)} F > F(c_\ell) = \max_{\mathcal{S}_{<\ell}} F$, so every maximiser of F on $\mathcal{S}_{\text{end}}$ lies in $\mathcal{N}(c_\ell) \setminus \mathcal{S}_{<\ell}$.

Proof. A move happens only from a scan with index $k < \ell$, and each move raises the index by one; so at most ℓ moves occur and the procedure stops at some $m \le \ell$ (each scan is a finite computation).

(a) $c_0 = \mathrm{best}(\mathcal{S}^-) \in \mathcal{S}^- = \mathcal{S}_{<0}$; and for $k < m$, $c_{k+1} \in I_k \subseteq \mathcal{N}(c_k) \subseteq \mathcal{S}_{<k+1}$.

(b), (c) by induction. For $k = 0$, (b) is (1). Assume (b) for some $k < m$. Since $k < m$, scan k led to a move, so $I_k \neq \emptyset$. Let $M = \max_{I_k} F$. Elements of $\mathcal{N}(c_k) \setminus I_k$ have $F \le F(c_k) < M$, so $M = \max_{\mathcal{N}(c_k)} F$. The $\rhd_F$-maximum c_{k+1} of I_k has $F(c_{k+1}) = M$. Hence $F(c_{k+1}) = \max_{\mathcal{N}(c_k)} F > F(c_k)$, which is (c), and $\max_{\mathcal{S}_{<k+1}} F = \max\{\max_{\mathcal{S}_{<k}} F, \max_{\mathcal{N}(c_k)} F\} = \max\{F(c_k), M\} = M = F(c_{k+1})$, which is (b) for $k+1$.

(d) On NOIMP, $I_m = \emptyset$, i.e. $F(N) \le F(c_m)$ for all $N \in \mathcal{N}(c_m)$; with (b), $\max_{\mathcal{S}_{\mathrm{end}}} F = \max\{\max_{\mathcal{S}_{<m}} F, \max_{\mathcal{N}(c_m)} F\} = F(c_m)$.

(e) On LIMIT, $m = \ell$ and $I_\ell \neq \emptyset$, so $\max_{\mathcal{N}(c_\ell)} F > F(c_\ell) = \max_{\mathcal{S}_{<\ell}} F$ by (b); hence $\max_{\mathcal{S}_{\mathrm{end}}} F = \max_{\mathcal{N}(c_\ell)} F$, and any maximiser T on $\mathcal{S}_{\mathrm{end}}$ has $F(T) > \max_{\mathcal{S}_{<\ell}} F$, so $T \notin \mathcal{S}_{<\ell}$ and $T \in \mathcal{N}(c_\ell)$. □

Lemma 3 (earlier scores)**.** *If $R \in \mathcal{S}^-$ then $R = c_0$.*

Proof. $c_0 = \mathrm{best}(\mathcal{S}^-)$ and $R \in \mathcal{S}^-$ give $c_0 \succeq R$. $R = \mathrm{best}(\mathcal{S}_{\mathrm{end}})$ and $c_0 \in \mathcal{S}^- \subseteq \mathcal{S}_{\mathrm{end}}$ give $R \succeq c_0$. If $R \neq c_0$ we would have $c_0 \succ R \succ c_0$, hence $c_0 \succ c_0$ by transitivity, contradicting irreflexivity. □

3 Clause (i)

(i-a). Suppose `local_certified` $= 1$. Then $R = c_j$ for some $j \le m$, and $\mathcal{N}(c_j)$ contains no N with $F(N) > F(c_j)$, i.e. $I_j = \emptyset$. If $j < m$ the procedure moved after scan j, which requires $I_j \neq \emptyset$; so $j = m$. On LIMIT, $I_m = I_\ell \neq \emptyset$; so the outcome is NOIMP. Conversely, if the outcome is NOIMP and $R = c_m$, then $R \in \mathcal{C}$ and $I_m = \emptyset$ says that no $N \in \mathcal{N}(R)$ has $F(N) > F(R)$, so `local_certified` $= 1$.

In that case $F(R) = F(c_m) = \max_{\mathcal{S}_{\mathrm{end}}} F$ by Lemma 2(d), and $F(R) \ge F(N)$ for all $N \in \mathcal{N}(R)$ because $I_m = \emptyset$. Equality $F(N) = F(R)$ is not excluded by the definition (only strict improvements are), so equal-score neighbours are compatible with `local_certified` $= 1$; for such N, since $N \in \mathcal{S}_{\mathrm{end}}$ and $R = \mathrm{best}(\mathcal{S}_{\mathrm{end}})$, the tie rule gives $R \succ N$.

Finally, $R \in \mathcal{C}$ trivially implies that R was a scan centre in some local search of the run up to this one. Conversely, suppose R was a scan centre of an earlier local search. By Lemma 2(a) applied to that search, R was scored then, so $R \in \mathcal{S}^-$; by Lemma 3, $R = c_0 \in \mathcal{C}$.

(i-b). Let $F' : \Omega \to \mathbb{R}$ with $F' = F$ on $\mathcal{S}_{\mathrm{end}}$, and run the procedure with F' in place of F (same $\mathcal{S}^-$, ℓ, tie rules $\succ$ via sizes and indices, and τ). Write primes for the F'-run. We show by induction on k that the two runs coincide step by step. The restriction of $\succ$ to a set $\mathcal{A}$ depends on F only through $F|_{\mathcal{A}}$ (and on sizes and indices); since $\mathcal{S}^- \subseteq \mathcal{S}_{\mathrm{end}}$, $c'_0 = \mathrm{best}'(\mathcal{S}^-) = \mathrm{best}(\mathcal{S}^-) = c_0$ and $\mathcal{S}'_{<0} = \mathcal{S}_{<0}$. Suppose $c'_j = c_j$ and $\mathcal{S}'_{<j} = \mathcal{S}_{<j}$ for all $j \le k$, where $k \le m$. Scan k scores the same set $\mathcal{N}(c_k)$ (it depends only on c_k and K), so $\mathcal{S}'_{<k+1} = \mathcal{S}_{<k+1}$. Since $\{c_k\} \cup \mathcal{N}(c_k) \subseteq \mathcal{S}_{\mathrm{end}}$, F' and F agree on it, so $I'_k = I_k$; hence the stop/move decision at scan k is the same (it depends on whether I_k is empty and on k versus ℓ). If a move occurs, $\rhd_{F'}$ and $\rhd_F$ agree on $I_k \subseteq \mathcal{S}_{\mathrm{end}}$ (they are defined by the values of F', resp. F, there and the fixed τ), so $c'_{k+1} = c_{k+1}$. Hence $m' = m$, the outcomes coincide, $\mathcal{C}' = \mathcal{C}$, $\mathcal{S}'_{\mathrm{end}} = \mathcal{S}_{\mathrm{end}}$, and $R' = \mathrm{best}'(\mathcal{S}_{\mathrm{end}}) = R$ because $\succ'$ and $\succ$ agree on $\mathcal{S}_{\mathrm{end}}$. The flag `local_limit` coincides. For `local_certified`: if $R \notin \mathcal{C}$ both flags are 0; if $R \in \mathcal{C}$ then $\mathcal{N}(R) \subseteq \mathcal{S}_{\mathrm{end}}$ (it was scanned), so the condition "no $N \in \mathcal{N}(R)$ with $F'(N) > F'(R)$" coincides with the one for F.

Consequently the value of `local_certified` is the same for every assignment of scores to the models of $\Omega \setminus \mathcal{S}_{\mathrm{end}}$. In particular, if $\Omega \setminus \mathcal{S}_{\mathrm{end}} \neq \emptyset$, the choice $F' = F$ on $\mathcal{S}_{\mathrm{end}}$ and $F' = F(R) + 1$

elsewhere leaves `local_certified` unchanged while R is not a global maximiser of F'. So, as a property of the procedure, `local_certified` $= 1$ certifies nothing beyond the facts in (i-a) about scored models; when some model is unscored it is not a certificate of global maximality. (If $\mathcal{S}_{\text{end}} = \Omega$, then $R = \text{best}(\Omega)$ is a global maximiser of F by (1), whatever the flag.) (No claim is made here about which score functions arise from data.)

(i-c). Every model scored by scan k is in $\mathcal{N}(c_k) \subseteq \mathcal{S}_{\text{end}}$, whether or not it is c_{k+1}. As $R = \text{best}(\mathcal{S}_{\text{end}})$, $R \succeq N$ and, by (1), $F(R) \geq F(N)$ for each such N. In the method the reported HPM $\hat{S}$ is the running best over every model scored in the run, and nothing is scored after the final local search (STATEMENT.md §2); so for the final search $\hat{S} = \text{best}(\mathcal{S}_{\text{end}}) = R$.

(i-d). On LIMIT, R maximises F on $\mathcal{S}_{\text{end}}$ by (1); Lemma 2(e) gives $R \in \mathcal{N}(c_\ell) \setminus \mathcal{S}_{<\ell}$ and $F(R) > F(c_\ell)$. Thus R was first scored by scan ℓ. Every centre c_k of this search lies in $\mathcal{S}_{<k} \subseteq \mathcal{S}_{<\ell}$ (Lemma 2(a)), and every centre of an earlier local search was scored in that search, hence lies in $\mathcal{S}^- \subseteq \mathcal{S}_{<\ell}$ (the scored set is cumulative: it records every model ever scored in the run). So R was never a scan centre of any local search of the run up to this one; in particular $R \notin \mathcal{C}$ and `local_certified` $= 0$. For the final search, $\hat{S} = R$ by (i-c).

(i-e). Assume exactly one model of $\mathcal{S}_{\text{end}}$ attains $\max_{\mathcal{S}_{\text{end}}} F$. On NOIMP, $c_m \in \mathcal{S}_{\text{end}}$ attains it (Lemma 2(d)) and so does R by (1); by uniqueness $R = c_m$, and (i-a) gives `local_certified` $=$ 1. On LIMIT, `local_certified` $= 0$ by (i-d). Since `local_limit` $= 1$ exactly on LIMIT, `local_certified` $= 1 \iff$ NOIMP $\iff$ `local_limit` $= 0$. □

4 Clause (ii), deterministic part (ii-A)

Assume E_n: $S_0 \in \Omega$ and $F(S_0) > F(S)$ for all $S \in \Omega \setminus \{S_0\}$. Consider any local search of the run and any scan k of it with $c_k \in \mathcal{N}(S_0)$.

1. By Lemma 1, $S_0 \in \mathcal{N}(c_k)$, and by EXEC scan k scores it.

2. $c_k \neq S_0$ (as $S_0 \notin \mathcal{N}(S_0)$), so $F(S_0) > F(c_k)$ and $S_0 \in I_k$. For $N \in I_k \setminus \{S_0\}$, $F(S_0) > F(N)$, hence $S_0 \rhd_F N$. So S_0 is the $\rhd_F$-maximum of I_k, and if $k < \ell$ the procedure sets $c_{k+1} = S_0$.

3. If $k < \ell$, the procedure continues to scan $k+1$ with centre S_0 (after any move the loop performs another scan: a regular one if $k+1 < \ell$, the one further scan if $k+1 = \ell$). Every $N \in \mathcal{N}(S_0)$ differs from S_0, so $F(N) < F(S_0)$ and $I_{k+1} = \emptyset$: the search stops with NOIMP, $m = k+1$, $c_m = S_0$. Since $S_0 \in \mathcal{S}_{\text{end}}$ and $F(S_0) > F(T)$ for every other T, $R = \text{best}(\mathcal{S}_{\text{end}}) = S_0 = c_m$, and (i-a) gives `local_certified` $= 1$.

4. If $k = \ell$, then $I_\ell \ni S_0$ is nonempty and the search stops with LIMIT; $R = S_0$ as in 3; `local_certified` $= 0$ by (i-d) (and `local_limit` $= 1$).

From the moment scan k scores S_0, it is in the scored set and strictly beats every other model, so the running HPM, being the $\succ$-maximum of a scored set containing S_0, equals S_0 for the rest of the run; for the final search, $\hat{S} = S_0$. □

Corollary 4. *On E_n, if $S_0 \in \mathcal{S}^-$ at the start of some local search, that search has $c_0 = S_0$, stops after scan 0 with NOIMP, and returns $R = S_0$ with `local_certified` $= 1$.*

Proof. $c_0 = \text{best}(\mathcal{S}^-) = S_0$ since S_0 has the strictly largest score; $I_0 = \emptyset$ as in step 3; $R = S_0 = c_0$ and (i-a) applies. □

5 Clause (ii), margin and statistical parts

From here on $F(S) = \log \varpi(S) + \log m(S)$ is the method's score, with ϖ, m, G_S, z_S, H_S, D_S and $Q_S = y^\top y - n\, z_S^\top H_S^{-1} z_S$ exactly as defined in STATEMENT.md §1, $\pi(S) = e^{F(S)} / \sum_{T \in \Omega} e^{F(T)}$ for $S \in \Omega$ and $\pi(S) := 0$ for $S \notin \Omega$.

Lemma 5. *For every $S \in \Omega$, $0 < \varpi(S) m(S) < \infty$. Hence F is real-valued on Ω and $0 < \pi(S) < 1$ for every $S \in \Omega$.*

Proof. $\varpi(S) = w(|S|)/W_K$ with $w(k) = \frac{1}{2} / \binom{2p-1}{k} > 0$ for $0 \le k \le K \le p \le 2p-1$, and W_K a finite positive sum. In $m(S) = \frac{(2b_0)^{a_0}\Gamma(\nu)}{\pi^{n/2}\Gamma(a_0)} D_S^{-1/2} (2b_0 + Q_S)^{-\nu}$ the constant is positive and finite ($a_0 = b_0 = 0.01$, $\nu = a_0 + n/2$). For $S = \emptyset$ the conventions are $D_\emptyset = 1$ and $Q_\emptyset = y^\top y$. For $k = |S| \ge 1$: $G_S = X_S^\top X_S \succeq 0$, so $G_S + I$ and $H_S = (n+1)G_S + I$ are positive definite and $D_S = \det H_S / \det(G_S + I) \in (0, \infty)$. Since $k \le K \le n$, take a thin SVD $X_S = U \Sigma V^\top$ with $U \in \mathbb{R}^{n \times k}$, $U^\top U = I_k$, V orthogonal and $\Sigma = \mathrm{diag}(\sigma_1, \dots, \sigma_k) \ge 0$. Then $H_S^{-1} = V((n+1)\Sigma^2 + I)^{-1} V^\top$ and $X_S H_S^{-1} X_S^\top = U\, \mathrm{diag}\big(\sigma_i^2 / ((n+1)\sigma_i^2 + 1)\big) U^\top \preceq \frac{1}{n+1} U U^\top \preceq \frac{1}{n+1} I_n$. With $z_S = X_S^\top y$, $n z_S^\top H_S^{-1} z_S = n\, y^\top X_S H_S^{-1} X_S^\top y \le \frac{n}{n+1} y^\top y$, so, by the definition $Q_S = y^\top y - n z_S^\top H_S^{-1} z_S$, $Q_S \ge y^\top y / (n+1) \ge 0$ and $2b_0 + Q_S > 0$. Hence $m(S) \in (0, \infty)$. Finally $\emptyset$ and $\{1\}$ belong to Ω ($K \ge 1$), so $|\Omega| \ge 2$, and each $\pi(S)$ is a positive term of a sum of at least two positive terms divided by that sum: $0 < \pi(S) < 1$. □

(ii-B). Let $S_0 \in \Omega$. By Lemma 5, $\lambda_n = \log[\pi(S_0)/(1 - \pi(S_0))]$ is a real number and $\Omega \setminus \{S_0\}$ is nonempty and finite; choose $S^* \in \Omega \setminus \{S_0\}$ with $F(S^*) = \max_{S \ne S_0} F(S)$, so $\Delta_n = F(S_0) - F(S^*)$. Then

$$1 - \pi(S_0) = \sum_{S \in \Omega \setminus \{S_0\}} \pi(S) \;\ge\; \pi(S^*), \qquad \text{so} \qquad \frac{\pi(S_0)}{1 - \pi(S_0)} \le \frac{\pi(S_0)}{\pi(S^*)} = e^{F(S_0) - F(S^*)} = e^{\Delta_n}.$$

Taking logarithms, $\lambda_n \le \Delta_n$. If $\pi(S_0) > \frac{1}{2}$ then $S_0 \in \Omega$ (by the convention $\pi = 0$ off Ω) and $\lambda_n > 0$, so $\Delta_n > 0$, i.e. $F(S_0) > F(S)$ for every $S \in \Omega \setminus \{S_0\}$: E_n holds. □

(ii-C). We use the standard definitions: real random variables Y_n are $O_{P_0}(1)$ if for every $\varepsilon > 0$ there are $M < \infty$ and N with $P_0(|Y_n| > M) < \varepsilon$ for all $n \ge N$; and $X_n = O_{P_0}(a_n)$ means $X_n / a_n = O_{P_0}(1)$. (If one prefers "for all n" in place of "for all $n \ge N$", the two agree for a.s. finite variables, by enlarging M to cover the finitely many $n < N$.) Note $r_n = \sqrt{\log n / n} \to 0$ and, for $n \ge 3$, $\log \log n$ is defined and $\log r_n = \frac{1}{2} \log \log n - \frac{1}{2} \log n$.

Assume H03 and fix $\varepsilon > 0$. There are $C \in (0, \infty)$ and N_1 (enlarging C if necessary, we may take $C \ge 1$, so $\log(2C) > 0$) with $P_0(1 - \pi(S_0) > C r_n) < \varepsilon$ for $n \ge N_1$. Let $N_2 \ge \max(n_0, N_1, 3)$ be such that $C r_n < \frac{1}{2}$ for $n \ge N_2$. Fix $n \ge N_2$ and work on the event $B_n = \{1 - \pi(S_0) \le C r_n\}$, which has $P_0(B_n) > 1 - \varepsilon$. On B_n, $\pi(S_0) > \frac{1}{2}$; we also know $1 - \pi(S_0) > 0$ (Lemma 5, $S_0 \in \Omega$ as $n \ge n_0$).

1. On B_n, $\pi(S_0) > \frac{1}{2}$, so E_n holds by (ii-B). Hence $P_0(E_n^c) \le P_0(B_n^c) < \varepsilon$ for all $n \ge N_2$. As ε was arbitrary, $P_0(E_n) \to 1$.

2. On B_n (with $\pi(S_0) > \frac{1}{2}$),

$$\lambda_n = \log \pi(S_0) - \log(1 - \pi(S_0)) \;\ge\; -\log 2 - \log C - \log r_n = \tfrac{1}{2} \log n - \tfrac{1}{2} \log \log n - \log(2C).$$

Let $Y_n = (\frac{1}{2} \log n - \frac{1}{2} \log \log n - \lambda_n)^+$, a finite random variable for $n \ge n_0$. On B_n, $Y_n \le \log(2C) =: M$. Hence $P_0(Y_n > M) < \varepsilon$ for all $n \ge N_2$, i.e. $Y_n = O_{P_0}(1)$. Together with (ii-B), for $n \ge n_0$, $\Delta_n \ge \lambda_n \ge \frac{1}{2} \log n - \frac{1}{2} \log \log n - Y_n$ with $Y_n = O_{P_0}(1)$.

3. Assume EX-A and EXEC, as in (ii-A). The conclusions (ii-A)1–4 were derived on E_n for every local search, every scan with a centre in $\mathcal{N}(S_0)$, every input $\mathcal{S}^-$ and every ℓ; they use no property of the algorithm's randomness. Hence, for any joint law of data and algorithm randomness whose data marginal is P_0, the set A_n of outcomes on which these conclusions hold for every scan of every local search of the run contains the event E_n, which depends on the data only. Therefore $P(A_n) \geq P_0(E_n) \to 1$ (read $P(A_n)$ as an inner probability if the measurability of A_n is not established; E_n is measurable, being defined by finitely many strict inequalities between continuous functions of y). □

(d) Review record

FINAL_REPORT.md: statement changes

2. Changes to the statement

No hypothesis was changed or added. The step 1 outcome was (a), correct as stated.

To make the informal clauses precise, STATEMENT.md:

- formalises the method's §6.10 search as a short algorithm (§2), with the source text quoted verbatim in §0;
- uses T03 only through the explicit hypothesis H03;
- names the "scans execute" assumption EXEC;
- records seven readings in §5. Two examples:
  - "returned model" means the running HPM at the end of the search;
  - "can be the reported HPM without a certificate" is proved as a conditional ("on LIMIT, ..."). No claim is made that LIMIT actually happens for real data.

Reviewer-driven text corrections (change log in STATEMENT.md §6):

- v1 → v2:
  - an explanatory remark in §5 was wrong and was fixed (A1-1);
  - the score formulas were written out in full (A1-2).
- v2 → v3:
  - (i-b)'s "does not certify global maximality" is now qualified by "when some model is unscored". A reviewer noted it was false if every model had been scored;
  - other edits are clarifications only.
- v3 → v3.1: readings and setting wording only.

FINAL_REPORT.md: review history

3. Review history

Stage A: lead + Astra #1 (GPT-6 Astra, high, session 01a1641e-caed-7313-8c23-8d32e3a413f3).

- Round 1 on v1: *not verified*.
  - One "fatal" finding, in an explanatory reading note rather than a claim. The reviewer said it does not affect §§3–4.
  - One gap: Q_S was not defined in the packet.
  - One exposition finding.
  - All three were accepted and fixed.
- Round 2 on v2: *verified*. Astra agreed with both answers.

Stage B: two fresh independent reviews of v2, started together.

- **Astra #2** (new session 01a16424-4661-7c72-9494-a9613c43944d): *verified*. No fatal or gap findings; one cosmetic remark.
- **Opus 5.5 xhigh** (native Agent, read-only Plan type, Claude Work account): *verified*. Ten exposition or scope notes (F1–F10); seven applied in v3.
- **Ruling round on v3 and the claims log:** both *verified*, agreeing with every answer.
  - Both raised the same remaining exposition point, fixed in STATEMENT v3.1.
  - One harmless note (N3, on measurability) was left unapplied so the reviewed proof text stays final.

No disputes; the Astra judge was not used. The full log, with every fatal and gap finding verbatim and the lead's answers, is in `CLAIMS_OF_FATAL_FLAWS.md`. All replies are in `review/`.

Numerical sanity check. A Sonnet assistant ran a single-seed local check (`helpers/numcheck/NUMCHECK.md`).

- Scope: 72 small problems ($p \le 8$), where the whole model space can be scored, and 8,640 random search runs.
- Result: 0 failures across all checks of (i)–(ii), including tie canaries.
- This is a sanity check, not part of the proof.

CLAIMS_OF_FATAL_FLAWS.md (every entry, verbatim)

CLAIMS_OF_FATAL_FLAWS.md — T20 (T15) (i)–(ii)

Every fatal or gap finding, verbatim, with the lead's answer.

Stage A — Astra #1 (session 01a1641e-caed-7313-8c23-8d32e3a413f3), round 1, on PROOF_v1 + STATEMENT v1

A1-1 (fatal, per the reviewer: "it does not invalidate the formal conclusions in §§3–4")

1. **Location:** STATEMENT.md §5, item 1, beginning “If one instead reads it as the last centre…”.
   Severity: fatal — this explanatory claim is false; it does not invalidate the formal conclusions in §§3–4.

Claim: The running HPM R and the last centre c_m do not differ only in the presence of score ties. Unique attainment of the largest scored value does not ensure that they coincide.

Evidence: Take $p = 1$, $n = 3$, hence $K = 1$, with

$$\mathcal{S}^- = \{\varnothing\}, \qquad \ell = 0, \qquad F(\varnothing) = 0, \quad F(\{1\}) = 1.$$

Scan 0 scores $\{1\}$ and stops with LIMIT. The last centre is $c_m = \varnothing$, whereas $R = \{1\}$. There are no score ties, and the scored maximum is unique. More generally, (i-d) itself establishes $F(R) > F(c_m)$ on every LIMIT outcome. Thus the explanatory sentence contradicts a formal claim that the proof correctly establishes.

Lead's answer: accepted and fixed. The sentence in the readings section was wrong: on LIMIT, R ≠ c_m always. It was an explanatory remark, not one of the claims of §§3–4, and the proof never used it. STATEMENT.md §5 item 1 now reads: on LIMIT, R ≠ c_m but the flag is 0 under both readings (c_m has a strictly improving neighbour); on NOIMP, R = c_m unless another scored model has exactly the score F(c_m) and precedes c_m in the HPM order; so the two readings give different flag values only on NOIMP with such an exact tie. (Check: under the "last centre" reading the flag is 1 iff NOIMP; under the adopted reading it is 1 iff NOIMP and R = c_m, by (i-a).)

A1-2 (gap)

2. **Location:** PROOF_v1.tex, “Clause (ii), margin and statistical parts,” lemma labelled `lem:finite`, proof for nonempty S.

   Severity: gap.

   Claim: The asserted lower bound on Q_S is not justified by definitions supplied in the permitted files.

   Evidence: The proof establishes

   $$n z_S^\top H_S^{-1} z_S \le \frac{n}{n+1} y^\top y$$

   and immediately concludes

   $$Q_S \ge \frac{y^\top y}{n+1} \ge 0.$$

   However, neither supplied text defines Q_S for nonempty S, or states an identity connecting it to the preceding quadratic form. Only $Q_\varnothing$ is defined. Consequently this implication, and therefore this particular derivation of positivity of the displayed marginal likelihood, cannot be verified from the permitted files. The identification of those detailed formulas with the method’s marginal likelihood also relies on an external source whose contents are not supplied.

   This is an auxiliary derivational gap: the stated setting already assumes a real-valued score, and the subsequent posterior-odds calculation is correct.

Lead's answer: accepted and fixed. The packet lacked the definition. STATEMENT.md §1 now writes out the method's score in full, copied from 00_COMMON §1 and METHODOLOGY_v3 §5: G_S, z_S, H_S, D_S, Q_S = $y^T y - n\, z_S^T H_S^{-1} z_S$, ν, m(S), w, W_K, ϖ. It also says the proof uses only these displayed formulas. PROOF_v2's Lemma (finite) now cites that definition explicitly at the step Q_S ≥ $y^T y/(n+1)$. The identification with a Bayesian marginal likelihood is not used anywhere in the proof; the formulas are the definition of the score.

Astra #1 round 2 (same session, on PROOF_v2 + STATEMENT v2 + this log): agrees with both answers ("I agree with the answer and confirm that the fix works" for A1-1 and A1-2); no new findings; verdict **verified** (review/astra1_r2.md). End of Stage A; proof candidate = PROOF_v2.

Stage B — Astra #2 (fresh session 01a16424-4661-7c72-9494-a9613c43944d) and Opus 5.5 xhigh reviewer, round 1, on PROOF_v2 + STATEMENT v2 (change log removed from the packet copy)

No fatal or gap findings. Astra #2: verdict verified; one remark (the inner-probability parenthetical in (ii-C)3 is unnecessary; kept, harmless). Opus: verdict verified; findings F1–F10, all exposition or scope (review/astra2_r1.md, review/opus_r1.md).

Lead's handling of the exposition/scope findings (unified revision PROOF_v3 + STATEMENT v3; no hypothesis or claim content changed):

- F1 accepted: (ii-C)3 and proof step 3 now name EX-A and EXEC.
- F2 accepted: (i-b) now says the F′-run uses $\succ$ and the move order computed from F′ with the same fixed tie-break rule (the proof already did this via τ).
- F3 accepted: (i-b) "does not certify global maximality" now qualified by $\Omega \setminus \mathcal{S}_\text{end} \neq \emptyset$, with the trivial $\mathcal{S}_\text{end} = \Omega$ case noted (statement and proof).
- F4 accepted: (i-a) now states that every equal-score neighbour N of a certified R satisfies $R \succ N$.
- F5 accepted: proof §1 shows $\mathcal{N}(c) \neq \emptyset$.
- F6 accepted: proof (ii-C) takes $C \geq 1$ (so $\log(2C) > 0$).
- F7 accepted: "from the moment scan k scores S_0" (statement and proof).
- F8 noted (scope): (i-c) holds by construction of the formal model; the link to the method is now quoted verbatim in STATEMENT §0 (METHODOLOGY_v3 §6.10 and §6.1).
- F9 partly addressed: the source's E_n definition and the method's §6.10 text are now quoted verbatim in STATEMENT §0; H03 remains an explicit hypothesis (T03 is a conjecture being proved in a sibling thread), as the statement says.
- F10 accepted: §5 item 5 now mentions the tie-only route (NOIMP with an equal-score neighbour preceding c_m), covered by (i-a).

Stage B ruling round (on PROOF_v3 + STATEMENT v3 + this log)

- **Opus reviewer (r2): verified.** Agrees with every answer in the log; each change works. New findings N1–N3, exposition only (review/opus_r2.md).
- **Astra #2 (r2, same session): verified.** Agrees with every answer (table of rulings); one exposition finding, the same point as Opus N1 (review/astra2_r2.md).
- Lead: N1 (= Astra #2's point) and N2 accepted, fixed in STATEMENT v3.1 (§5 item 5 and §1 text only; proof unchanged). N3 (measurability of E_n via a.s. continuity of standardisation) noted; harmless, both reviewers agree; not applied, to keep the reviewed proof text final.
- No fatal or gap finding is outstanding. No dispute; Astra judge not used.

Chapter 8

T22 (T16), parts (i), (ii), (iv): aggregate start bounds and endpoint propagation

Final proof: `proofs/T22_T16_start_aggregation/proof/PROOF_v3.tex`. *Sources of parts (a), (b), (d):* `theorems/T22_T16_start_aggregation.md`, `STATEMENT.md`, `FINAL_REPORT.md`, `CLAIMS_OF_FATAL_FLAWS.md`.

(a) Original conjecture text (theorem file, verbatim)

T22 (T16): Aggregate start bounds and conditional concentration for the actual blocks

Status: conjectured, not proven. Methodology SMC-BVS v3.0 (2026-10-19), SHA-256 93043758058a6d422cca6ca4b6be8f3a9e8149b6f6de2e1bab1ce31c44b8019c. Read with `00_COMMON.md`.

Category / rank: deterministic start analysis (i)–(ii), (iv): **core** (lemma); statistical concentration (iii), (v): **secondary** (theorem).

Definitions used

Tables r_b, γ_0, π_0, w(k) (00_COMMON §2); AGG's q_n, e_n for a feasible reference S° (s = |S°|, U_b* = S°∩C_b) with the stated conventions; u_n := (p−s)$^{-1}$Σ_{l∉S°} max_{S°∩C_{b(l)}⊆T⊆C_{b(l)}\l} r_{b(l)}(T∪l)/r_{b(l)}(T); uncapped weight γ₀^{unc}(S) = w(|S|)∏_b r_b(S∩C_b); δ_n^{comp} = sup_{S⊉S°}γ₀^{unc}(S)/γ₀^{unc}(S∪S°); r_k (T10); R_i = 1/q_i(S_0) − 1 for i ∈ {0,1}; leakage quantities E_{T,l}, v_T; inventory codes VIS, NL_avg(κ), EXP, BO.

Assumptions

EX-A for (i), (ii), (iv); G, R, within-block SE, VIS, NL_avg(κ) and EXP for (iii); T03's hypotheses for the full-endpoint conclusion in (iv); BO, SE, bounded signal energy and BM-∞ for (v).

Precise statement

(i) [deterministic aggregation] If q_n < 1: $\pi_0(S°) \geq (1-q_n)^{s+1}/(1+e_n)^s$ and $1 - \pi_0(S°) \leq (s+1)q_n/(1-q_n) + se_n$. Moreover $q_n \leq u_n$, so $u_n < 1$ permits its substitution. A further candidate bound is $\pi_0(S°) \geq (1-u_n)^{s+1}/[1 + (2^s - 1)\delta_n^{comp}]$. Conventions: empty maxima are 0; q = u = 0 if s = p; e = δ = 0 if s = 0. (ii) [reference-to-truth odds] $\pi_0(S_0) = \pi_0(S°)\cdot[w(s_0)/w(s)]\cdot\prod_b r_b(S_0 \cap C_b)/r_b(S° \cap C_b)$ exactly; if $S° \subseteq S_0$, successive feasible additions give $\pi_0(S_0)/\pi_0(S°) \geq \prod_{j=0}^{s_0-s-1} r_{s+j}$. Normalised reference mass remains essential. (iii) [statistical, with EXP] For fixed κ > 0: $q_n \leq u_n = O_{P_0}(2^{B\sqrt{(\log n)}n^{-\kappa}})$ when EXP's remainder is $O_{P_0}(1)$ (exponent form $2^{B\sqrt{(\log n)}n^{-\kappa+o_{P_0}(1)}}$ with an $o_{P_0}(\log n)$ remainder); EXP's visibility lower bound with $M_n^V \to \infty$ and B ≤ 16 gives $e_n = o_{P_0}(n^{-m})$ for every fixed m; hence $1 - \pi_0(S_0) = O_{P_0}(2^{B\sqrt{(\log n)}n^{-\kappa}} + e_n)$. Exact zero leakage permits κ = ½; NL_η supplies the admissible κ of the inventory. Neither substitutes for EXP. (iv) [endpoint propagation] For $0 \leq \varphi \leq 1$: $1/q_\varphi(S_0) - 1 \leq R_0^{1-\varphi}R_1^\varphi \leq \max(R_0, R_1)$; hence $\inf_{\varphi\in[0,1]}q_\varphi(S_0) \geq \min\{\pi_0(S_0), \pi(S_0)\}$. With T03 and (iii) this gives uniform concentration on [0,1]. (v) [BO] Under BO, completing the true variables in a block removes its signal leakage into the remaining coordinates of that block; with SE, bounded signal energy and BM-∞ the projection argument supplies VIS; the score conclusion still requires EXP. No over-tempered clause is retained. Start concentration implies neither schedule completion nor evidence accuracy, nor that every initial particle equals S_0.

Dependencies

Builds on: T01, T03, T10. Used by: T21, T26, T27.

Numerical evidence (provisional)

provisional evidence (Sol check J5): 468 table cases (adversarial and seeded), 376 applicable bounds, 0 failures, including empty, full and binding-cap references; actual partition at n = 64, p = 9, B = 3; observed EXP remainder range [−0.10, 0.17] on 20 additions (cannot establish its stochastic order); two-branch bound verified algebraically. Files: judge/sol1/J5_COMPOSITE_START.md, J5_tables.json.

Proof-route hint (optional)

Dominate the completed-table generating function coefficientwise by $\exp\{z(p-s)q_n\}$; prior ratio $w(s+j)/w(s) \leq (s+j)!/(s!(p-s)^j)$; $\Sigma_j C(s+j, j)q_n^j = (1-q_n)^{-(s+1)}$; charge incomplete configurations to their completions (≤ e_n per missing coordinate, $(1+e_n)^s$); the uncapped sum dominates the capped normaliser. (iii): truncate $|Z| \leq h_n$ uniformly over the ≤ $p2^B$ table comparisons; $a_l \leq C(n+1)^{-1/2}\exp\{(\delta+\sigma^*Z)^2/(2v_T)\}$ by EXP; the two-branch bound gives $E[\cdot; |Z| \leq h_n] \leq (2h_n/\sqrt{(2\pi)})n^{-1/2}e^{E_{T,l}}$; linearity and Markov, no independence. (iv): Hölder on the total odds $\Sigma_{S\neq S_0}[\gamma_0(S)/\gamma_0(S_0)]^{1-\varphi}[\gamma_1(S)/\gamma_1(S_0)]^\varphi \leq R_0^{1-\varphi}R_1^\varphi$. Hardest step: deriving EXP from primitive assumptions (the uniform block-score expansion over $O(p2^B)$ entries). *When proof.md is run on this file, its lead considers the statement both with and without these hints; the hints never reach the reviewers.*

(b) STATEMENT.md (verbatim)

STATEMENT — T22 (T16), parts (i), (ii), (iv)

Source: `Theory_architecture_v3/theorems/T22_T16_start_aggregation.md` (SMC-BVS

v3.0, method SHA-256 93043758…8019c). Only parts (i), (ii), (iv) are stated here; parts (iii) and (v) are out of scope for this run. Parts (i), (ii), (iv-a), (iv-b) are deterministic: they hold for every realisation of the data; (iv-c) is a conditional probabilistic consequence of (iv-b). Exact arithmetic is assumed (idealisation EX-A): π_0, q_φ and all tables are the exact mathematical objects defined below.

Status: statement approved with corrections by Astra (step 1b), 2026-10-22; see "Changes to the statement" at the end.

1. Setting and definitions

Data and model space. Integers n ≥ 3 and p ≥ 1. X ∈ ℝ^{n×p} (columns x_1, …, x_p) and y ∈ $\mathbb{R}^n$ are fixed (in the method they are the standardised design and response, $\|x_j\|^2 = \|y\|^2 = n - 1$; the statements below use only that they are real). K = min(p, ⌊n/2⌋). Ω = {S ⊆ [p] : |S| ≤ K} ("feasible models"). For S ⊆ [p] (any size), with X_S the columns in S:

- G_S = $X_S^T X_S$, z_S = $X_S^T y$, H_S = (n+1)G_S + I_{|S|}, D_S = det H_S / det(G_S + I_{|S|}),
- t_S = n $z_S^T H_S^{-1} z_S$, Q_S = $y^T y$ − t_S, ν = a_0 + n/2 with $a_0 = b_0 = 0.01$,
- m(S) = $c_n \cdot D_S^{-1/2} (2b_0 + Q_S)^{-\nu}$, where $c_n = (2b_0)^{a_0} \Gamma(\nu) / (\pi^{n/2} \Gamma(a_0)) > 0$ (for S = ∅: D_∅ = 1, Q_∅ = $y^T y$),
- BF(S) = m(S)/m(∅).
- Size prior: w(k) = ½ / C(2p−1, k) for 0 ≤ k ≤ p; W_K = Σ_{k≤K} C(p,k)w(k); ϖ(S) = w(|S|)/W_K for S ∈ Ω.
- Posterior: π(S) = ϖ(S)m(S)/Z for S ∈ Ω, Z = Σ_{S∈Ω} ϖ(S)m(S).

Blocks and tables. C_1, …, C_J is a partition of [p] into nonempty blocks with |C_b| ≤ B ≤ K (in the method: the deterministic complete-linkage/cut/next-fit partition with B = min(⌊√n⌋, ⌊√p⌋, 16); parts (i) and (iv) use only that it is a partition and that tables are positive). b(l) denotes the block containing l. Tables: r_b(U) = BF(U) for U ⊆ C_b (so r_b(∅) = 1 and r_b(U) > 0).

Start law and bridge. For S ∈ Ω: $\gamma_0(S) = \varpi(S) \prod_b r_b(S \cap C_b)$, $\bar{Z}_0 = \Sigma_{S\in\Omega} \gamma_0(S)$, $\pi_0(S) = \gamma_0(S)/\bar{Z}_0$. Uncapped weight, for every S ⊆ [p]: $\gamma_0^{unc}(S) = w(|S|) \prod_b r_b(S \cap C_b)$. Discrepancy d(S) = log BF(S) − Σ_b log r_b(S ∩ C_b). For φ ∈ [0,1] and S ∈ Ω: $\gamma_\varphi(S) = \gamma_0(S) e^{\varphi\, d(S)}$, $\bar{Z}_\varphi = \Sigma_{S\in\Omega} \gamma_\varphi(S)$, $q_\varphi(S) = \gamma_\varphi(S)/\bar{Z}_\varphi$. (Thus $q_0 = \pi_0$; that $q_1 = \pi$ is a fact to be checked in the proof.) Masses of sets outside Ω are 0.

Reference quantities (AGG). Fix a reference S° ∈ Ω, s = |S°|, U_b* = S° ∩ C_b.

- For each block b and integer j ≥ 0: $a_{bj} = \Sigma_{U : U_b^* \subseteq U \subseteq C_b, |U \setminus U_b^*| = j} r_b(U)/r_b(U_b^*)$ (so $a_{b0} = 1$, and $a_{bj} = 0$ for j > |C_b \ U_b*|).
- $D_b = \max_{1 \le j \le |C_b \setminus U_b^*|} (j!\, a_{bj})^{1/j}$, with D_b = 0 when C_b = U_b* (empty maximum).
- $q = (p-s)^{-1} \Sigma_b D_b$ if s < p; q = 0 if s = p.
- $e = 2p \cdot \max\{ [r_b(U)/r_b(U \cup U_b^*)]^{1/a} : b, U \subseteq C_b, a := |U_b^* \setminus U| > 0 \}$; e = 0 when the set is empty (in particular when s = 0).
- $u = (p-s)^{-1} \Sigma_{l \notin S^\circ} \max\{ r_{b(l)}(T \cup \{l\})/r_{b(l)}(T) : U^*_{b(l)} \subseteq T \subseteq C_{b(l)} \setminus \{l\} \}$ if s < p; u = 0 if s = p.
- $\delta^{comp} = \sup\{ \gamma_0^{unc}(S)/\gamma_0^{unc}(S \cup S^\circ) : S \in \Omega, S \not\supseteq S^\circ \}$, with $\delta^{comp} = 0$ when s = 0. (The proof may use this sup over feasible S; the bound then also holds for the larger sup over all S ⊆ [p] with S ⊉ S°.)

Addition floors. For $0 \le k \le p - 1$: $r_k = (k+1)/((2p-k-1)\sqrt{(n+1)})$.

Truth. $S_0 \subseteq [p]$ is a fixed set (the support of the true coefficient vector), $s_0 = |S_0|$. For $i \in \{0, 1\}$, when $S_0 \in \Omega$: $R_i = 1/q_i(S_0) - 1$.

2. Claims

(i) Deterministic aggregation. For every partition and positive tables as above and every $S^\circ \in \Omega$:

- (i-a) If $q < 1$, then $\pi_0(S^\circ) \ge (1-q)^{s+1}/(1+e)^s$ and $1 - \pi_0(S^\circ) \le (s+1)q/(1-q) + s \cdot e$.
- (i-b) $q \le u$. Hence, if $u < 1$, both bounds of (i-a) hold with u in place of q.
- (i-c) If $u < 1$, then $\pi_0(S^\circ) \ge (1-u)^{s+1}/[1 + (2^s - 1)\delta^{comp}]$.

(ii) Reference-to-truth odds. Assume $S_0 \in \Omega$ (i.e. $s_0 \le K$).

- (ii-a) $\pi_0(S_0) = \pi_0(S^\circ) \cdot [w(s_0)/w(s)] \cdot \prod_b r_b(S_0 \cap C_b)/r_b(S^\circ \cap C_b)$, exactly.
- (ii-b) If moreover $S^\circ \subseteq S_0$, then $\pi_0(S_0)/\pi_0(S^\circ) \ge \prod_{j=0}^{s_0-s-1} r_{s+j}$ (empty product = 1).
- Remark (interpretive, not a separate mathematical claim): (ii) controls $\pi_0(S_0)$ only relative to $\pi_0(S^\circ)$; a lower bound on the normalised reference mass $\pi_0(S^\circ)$ (e.g. from (i)) is still required.

(iv) Endpoint propagation. Assume $S_0 \in \Omega$.

- (iv-a) For every $\varphi \in [0,1]$: $1/q_\varphi(S_0) - 1 \le R_0^{1-\varphi} R_1^{\varphi} \le \max(R_0, R_1)$, with the convention $0^0 = 1$.
- (iv-b) Hence $\inf_{\varphi \in [0,1]} q_\varphi(S_0) \ge \min\{\pi_0(S_0), \pi(S_0)\}$; equivalently $\sup_{\varphi \in [0,1]} [1 - q_\varphi(S_0)] \le \max\{1 - \pi_0(S_0), 1 - \pi(S_0)\}$.
- (iv-c) Consequence (conditional). Consider a sequence of data sets indexed by $n \to \infty$ under a probability P_0. If $P_0(S_0 \in \Omega) \to 1$, $1 - \pi(S_0) \to 0$ in P_0-probability, and $1 - \pi_0(S_0) \to 0$ in P_0-probability, then $\sup_{\varphi \in [0,1]} [1 - q_\varphi(S_0)] \to 0$ in P_0-probability (uniform concentration of the bridge on $[0,1]$). The two convergence hypotheses are the conclusions of T03 (posterior concentration) and of part (iii) of T22; here they are explicit hypotheses, not results of this run.

Changes to the statement

Statement check by Astra (session 01a16418-acfd-7cb3-8783-b91d3752db30, reply in review/astra_01_statement_check.log): **APPROVED WITH CORRECTIONS**, 2026-10-22. The corrections are applied below.

Approved changes:

- C1 (added hypothesis): (ii) and (iv) assume $S_0 \in \Omega$ ($s_0 \le K$). Reason: π_0, q_φ and π are distributions on Ω; if $s_0 > K$ then $\pi_0(S_0) = 0$ while the right side of (ii-a) is positive, so (ii-a) is false (instance: $n = 3$, $p = 2$, $x_1 = x_2 = y = (-1, 0, 1)^T$, singleton blocks, $S_0 = \{1,2\}$, $S^\circ = \emptyset$), and R_i is undefined in (iv). The method's regime R assumes "eventually $s_0 \le K$", so nothing is lost asymptotically. Astra: necessary and minimal (not needed for (iv-b), which is trivial outside Ω).
- C6 (generalisation, recorded at Astra's request): the source's setting is the standardised design and response and the method's complete-linkage/cut/next-fit partition. STATEMENT.md quantifies over all real X, y and all partitions of $[p]$ into blocks with $|C_b| \le K$ (and, for (i), all positive tables). This is a strictly more general statement; the source setting is a special case.

- C7 (scope reading): (i-c) is stated under u < 1, the condition the source attaches to substituting u ("u_n < 1 permits its substitution"); the source's "candidate bound" sentence does not repeat it.

Accepted readings:

- C2: the final sentence of the source's (iv), "With T03 and (iii) this gives uniform concentration on [0,1]", is stated as (iv-c) with the conclusions of T03 and (iii) as explicit hypotheses. (Astra notes that the hypothesis $P_0(S_0 \in \Omega) \to 1$ is redundant given either concentration hypothesis; it is kept, harmlessly.) (iv-c) is a conditional probabilistic consequence of the deterministic (iv-b); "all three parts are deterministic" refers to (i), (ii), (iv-a), (iv-b).
- C3: δ^{comp}'s supremum is taken over feasible $S \not\supseteq S^\circ$ (a smaller sup, hence a stronger result that implies the reading over all $S \subseteq [p]$).
- C4: "Normalised reference mass remains essential" is recorded as an interpretive remark, not a claim.
- C5 (provenance, corrected): r_k = (k+1)/((2p−k−1)√(n+1)) is copied verbatim from `theorems/T10_T11_addition_hazard.md`, "Definitions used" ("r_k := (k+1)/((2p−k−1)√(n+1))"); the lead checked this. Astra objected only because T10 was not in its packet; the formula is used here as an explicit definition. All other definitions are transcriptions of the source and 00_COMMON.md, confirmed by Astra.
- No EXP / EXP′ hypothesis is needed for (i), (ii), (iv) (confirmed by Astra).

(c) Final proof, verbatim: body of PROOF_v3.tex

The proof below is the body of `PROOF_v3.tex` from `\begin{document}` to `\end{document}` without the title block. Its preamble definitions are local to this chapter; labels carry the prefix `T22:`.

All notation is that of `STATEMENT.md`. Every claim except (iv-c) is deterministic (and (iv-c) is a direct consequence of (iv-b)): the data (X, y), the partition $C_1, \dots, C_J$ and the reference $S^\circ \in \Omega$ are fixed, and all quantities are exact. We write $S + l := S \cup \{l\}$, $\varrho := n + 1$, and, for $S \subseteq [p]$, $M_S := X_S X_S^\top \in \mathbb{R}^{n \times n}$ (with $M_\emptyset = 0$). Throughout, $A \succeq B$ is the Loewner order. The letter e always denotes the AGG quantity of the statement (never Euler's number), e^k is its k-th power, and exponentials are written $\exp(\cdot)$. We use $x^0 := 1$ for every $x \geq 0$ (in particular $e^0 = 1$, and $D_b^0 = 1$ when $D_b = 0$).

1 Preliminaries on the score

Lemma 1 (Positivity and a closed form)**.** *For every $S \subseteq [p]$,*

$$t_S = \frac{n}{\varrho}\, y^\top \big(I_n - (I_n + \varrho M_S)^{-1}\big) y, \qquad D_S = \frac{\det(I_n + \varrho M_S)}{\det(I_n + M_S)}.$$

Consequently $0 \leq t_S \leq \frac{n}{n+1} y^\top y$, $Q_S \geq y^\top y/(n+1) \geq 0$, $D_S \geq 1$, and $m(S) \in (0, \infty)$. In particular every table entry $r_b(U)$, every $w(k)$ $(0 \leq k \leq p)$, every $\gamma_0(S)$, $\gamma_0^{\mathrm{unc}}(S)$, $\gamma_\varphi(S)$ is strictly positive and finite.

Proof. For $S = \emptyset$ both formulas give $t_\emptyset = 0$, $D_\emptyset = 1$, matching the definitions. Let $S \neq \emptyset$, $k = |S|$, $A = X_S$. The matrices $I_k + \varrho A^\top A$ and $I_n + \varrho A A^\top$ are positive definite, and $(I_n + \varrho A A^\top)A =$

$A(I_k + \varrho A^\top A)$, so $A(I_k + \varrho A^\top A)^{-1} = (I_n + \varrho AA^\top)^{-1}A$. Hence, since $H_S = \varrho A^\top A + I_k$,

$$AH_S^{-1}A^\top = (I_n + \varrho AA^\top)^{-1}AA^\top = \tfrac{1}{\varrho}(I_n + \varrho AA^\top)^{-1}\big((I_n + \varrho AA^\top) - I_n\big) = \tfrac{1}{\varrho}\big(I_n - (I_n + \varrho M_S)^{-1}\big),$$

and $t_S = n\, y^\top AH_S^{-1}A^\top y$ gives the formula for t_S. Since $0 \prec (I_n + \varrho M_S)^{-1} \preceq I_n$, the matrix $I_n - (I_n + \varrho M_S)^{-1}$ has eigenvalues in $[0, 1)$, so $0 \le t_S \le \frac{n}{\varrho} y^\top y$ and $Q_S = y^\top y - t_S \ge y^\top y/\varrho \ge 0$. By Sylvester's determinant identity $\det(I_k + \tau A^\top A) = \det(I_n + \tau AA^\top)$ for every real τ, applied with $\tau = \varrho$ and $\tau = 1$, we get the formula for D_S; if $\lambda_1, \dots, \lambda_n \ge 0$ are the eigenvalues of $M_S \succeq 0$, then $D_S = \prod_{i=1}^n (1 + \varrho\lambda_i)/(1 + \lambda_i) \ge 1$ because $\varrho \ge 1$. Finally $2b_0 + Q_S \ge 2b_0 > 0$, $c_n > 0$, so $m(S) = c_n D_S^{-1/2}(2b_0 + Q_S)^{-\nu} \in (0, \infty)$. Positivity of the remaining objects follows: $w(k) = \frac{1}{2}/\binom{2p-1}{k} > 0$ for $0 \le k \le p \le 2p - 1$, $W_K \ge w(0) > 0$, $r_b(U) = m(U)/m(\emptyset) > 0$, and $\exp(\varphi d(S)) \in (0, \infty)$. □

Lemma 2 (One addition)**.** *For every $S \subseteq [p]$ and $l \notin S$: $Q_{S+l} \le Q_S$, $D_{S+l}/D_S \le n+1$, and hence*

$$\frac{m(S + l)}{m(S)} \;\ge\; (n + 1)^{-1/2}.$$

Proof. The matrix M_S does not depend on the order of the columns, and $M_{S+l} = M_S + x_l x_l^\top \succeq M_S$. Then $I_n + \varrho M_{S+l} \succeq I_n + \varrho M_S \succ 0$, and inversion reverses the Loewner order on positive definite matrices, so $(I_n + \varrho M_{S+l})^{-1} \preceq (I_n + \varrho M_S)^{-1}$; by Lemma 1, $t_{S+l} \ge t_S$, i.e. $Q_{S+l} \le Q_S$.

For the determinant ratio, the matrix determinant lemma $\det(P + xx^\top) = \det P\,(1 + x^\top P^{-1}x)$ ($P \succ 0$) and Lemma 1 give

$$\frac{D_{S+l}}{D_S} = \frac{\det(I_n + \varrho M_S + \varrho\, x_l x_l^\top)/\det(I_n + \varrho M_S)}{\det(I_n + M_S + x_l x_l^\top)/\det(I_n + M_S)} = \frac{1 + \varrho\,\xi}{1 + \zeta},$$

with $\xi = x_l^\top (I_n + \varrho M_S)^{-1} x_l$ and $\zeta = x_l^\top (I_n + M_S)^{-1} x_l$. Since $I_n + \varrho M_S \succeq I_n + M_S \succ 0$, $0 \le \xi \le \zeta$. Thus $1 + \varrho\xi \le 1 + \varrho\zeta \le \varrho(1 + \zeta)$ (as $\varrho \ge 1$), i.e. $D_{S+l}/D_S \le \varrho$.

Therefore, with $\nu > 0$ and $2b_0 + Q_{S+l} \le 2b_0 + Q_S$ (both positive),

$$\frac{m(S + l)}{m(S)} = \Big(\frac{D_S}{D_{S+l}}\Big)^{1/2}\Big(\frac{2b_0 + Q_S}{2b_0 + Q_{S+l}}\Big)^{\nu} \;\ge\; \varrho^{-1/2} \cdot 1. \qquad \square$$

Lemma 3 (Prior ratios)**.** *For $0 \le k \le p - 1$: $w(k + 1)/w(k) = (k + 1)/(2p - k - 1)$, and $w(k)/w(k + 1) \le 2p - 1 < 2p$. If $0 \le s < p$ and $1 \le j \le p - s$, then*

$$\frac{w(s + j)}{w(s)} \;\le\; \frac{(s + j)!}{s!\,(p - s)^j}.$$

Proof. With $N = 2p - 1$, $w(k+1)/w(k) = \binom{N}{k}/\binom{N}{k+1} = (k+1)/(N-k) = (k+1)/(2p-k-1)$; its inverse is $(2p-k-1)/(k+1) \le 2p-1$. For the product, $w(s+j)/w(s) = \prod_{i=0}^{j-1}(s+i+1)/(2p-s-i-1)$, and for $0 \le i \le j-1 \le p-s-1$ we have $2p - s - i - 1 \ge 2p - s - (p - s - 1) - 1 = p \ge p - s > 0$. Hence the product is at most $\prod_{i=0}^{j-1}(s + i + 1)/(p - s) = (s + j)!/(s!(p - s)^j)$. □

2 Part (i)

Because $\gamma_0(S) = \gamma_0^{\mathrm{unc}}(S)/W_K$ for $S \in \Omega$ and $S^\circ \in \Omega$,

$$\pi_0(S^\circ) = \frac{\gamma_0^{\mathrm{unc}}(S^\circ)}{\sum_{S \in \Omega} \gamma_0^{\mathrm{unc}}(S)}. \tag{1}$$

Completion bijection. Let $\mathcal{T} = \{T : S^\circ \subseteq T \subseteq [p]\}$. The map $\Phi(S) = (S \cup S^\circ,\ S^\circ \setminus S)$ is a bijection from the power set of $[p]$ onto $\mathcal{T} \times 2^{S^\circ}$, with inverse $(T, M) \mapsto T \setminus M$. Indeed, $(S \cup S^\circ) \setminus (S^\circ \setminus S) = S$; and for $(T, M) \in \mathcal{T} \times 2^{S^\circ}$, $(T \setminus M) \cup S^\circ = T$ (since $M \subseteq S^\circ \subseteq T$) and $S^\circ \setminus (T \setminus M) = S^\circ \cap M = M$. Note $|T \setminus M| = |T| - |M|$.

Lemma 4 (Completed sum)**.** *If $q < 1$, then $\sum_{T \in \mathcal{T}} \gamma_0^{\mathrm{unc}}(T) \le (1-q)^{-(s+1)}\, \gamma_0^{\mathrm{unc}}(S^\circ)$.*

Proof. If $s = p$, then $\mathcal{T} = \{S^\circ\}$ and the claim is $\gamma_0^{\mathrm{unc}}(S^\circ) \le \gamma_0^{\mathrm{unc}}(S^\circ)$ (here $q = 0$). Let $s < p$. For $T \in \mathcal{T}$ put $V_b = (T \cap C_b) \setminus U_b^* \subseteq C_b \setminus U_b^*$. Since the blocks partition $[p]$, $T \mapsto (V_b)_b$ is a bijection from $\mathcal{T}$ onto $\prod_b 2^{C_b \setminus U_b^*}$, with $T \cap C_b = U_b^* \cup V_b$ and $|T| = s + \sum_b |V_b|$. Hence

$$\frac{\gamma_0^{\mathrm{unc}}(T)}{\gamma_0^{\mathrm{unc}}(S^\circ)} = \frac{w(|T|)}{w(s)} \prod_b \frac{r_b(U_b^* \cup V_b)}{r_b(U_b^*)},$$

and grouping the T by $j = |T| - s$ and expanding the product of sums,

$$\sum_{T \in \mathcal{T}} \frac{\gamma_0^{\mathrm{unc}}(T)}{\gamma_0^{\mathrm{unc}}(S^\circ)} = \sum_{j=0}^{p-s} \frac{w(s+j)}{w(s)}\, c_j, \qquad c_j := \sum_{(j_b)_b:\ \sum_b j_b = j} \prod_b a_{b j_b},$$

where j_b ranges over nonnegative integers ($a_{b j_b} = 0$ when $j_b > |C_b \setminus U_b^*|$).

Coefficient bound. For every b and $j \ge 0$, $a_{bj} \le D_b^j / j!$: for $j = 0$ both sides equal 1 ($D_b^0 = 1$); for $1 \le j \le |C_b \setminus U_b^*|$ it is the definition of D_b as a maximum; for larger j, $a_{bj} = 0$. By the multinomial theorem,

$$c_j \le \sum_{\sum_b j_b = j} \prod_b \frac{D_b^{j_b}}{j_b!} = \frac{(\sum_b D_b)^j}{j!} = \frac{((p-s)q)^j}{j!}.$$

Prior bound. By Lemma 3 (and trivially for $j = 0$), $w(s+j)/w(s) \le (s+j)!/(s!(p-s)^j)$ for $0 \le j \le p - s$. Therefore

$$\sum_{T \in \mathcal{T}} \frac{\gamma_0^{\mathrm{unc}}(T)}{\gamma_0^{\mathrm{unc}}(S^\circ)} \le \sum_{j=0}^{p-s} \frac{(s+j)!}{s!\,(p-s)^j} \cdot \frac{(p-s)^j q^j}{j!} = \sum_{j=0}^{p-s} \binom{s+j}{j} q^j \le \sum_{j=0}^{\infty} \binom{s+j}{j} q^j = (1-q)^{-(s+1)},$$

using $q \ge 0$ and the negative binomial series $\sum_{j \ge 0} \binom{s+j}{j} x^j = (1-x)^{-(s+1)}$ for $|x| < 1$ (obtained by differentiating the geometric series s times and dividing by $s!$). □

Lemma 5 (Charging incomplete sets to completions)**.** *For every $S \subseteq [p]$ with $\Phi(S) = (T, M)$: $\gamma_0^{\mathrm{unc}}(S) \le e^{|M|}\, \gamma_0^{\mathrm{unc}}(T)$.*

Proof. If $M = \emptyset$ then $S = T$. Let $M \ne \emptyset$; then $s \ge 1$, the set over which e is a maximum is nonempty (it contains $(b, \emptyset)$ for any b with $U_b^* \ne \emptyset$), and $\epsilon := e/(2p) > 0$ satisfies $r_b(U)/r_b(U \cup U_b^*) \le \epsilon^a$ whenever $U \subseteq C_b$ and $a = |U_b^* \setminus U| > 0$. Fix b and put $U = S \cap C_b$. Since $T = S \cup S^\circ$, $T \cap C_b = U \cup U_b^*$, and $a_b := |U_b^* \setminus U| = |M \cap C_b|$. If $a_b > 0$, $r_b(S \cap C_b)/r_b(T \cap C_b) \le \epsilon^{a_b}$; if $a_b = 0$ then $U \supseteq U_b^*$ and the ratio is $1 = \epsilon^0$. Multiplying over b (the $M \cap C_b$ partition M): $\prod_b r_b(S \cap C_b)/r_b(T \cap C_b) \le \epsilon^{|M|}$. For the prior, $|S| = |T| - |M|$ and, by Lemma 3 applied $|M|$ times (each intermediate size k satisfies $k \le p - 1$), $w(|S|)/w(|T|) \le (2p)^{|M|}$. Hence $\gamma_0^{\mathrm{unc}}(S)/\gamma_0^{\mathrm{unc}}(T) \le (2p\epsilon)^{|M|} = e^{|M|}$. □

Proposition 1 ((i-a))**.** *If $q < 1$, then $\pi_0(S^\circ) \ge (1-q)^{s+1}/(1+e)^s$ and $1 - \pi_0(S^\circ) \le (s+1)q/(1-q) + s\,e$.*

Proof. All terms are positive, so dropping the feasibility restriction and using the bijection Φ, then Lemmas 5 and 4,

$$\begin{aligned}\sum_{S\in\Omega}\gamma_0^{\mathrm{unc}}(S) &\le \sum_{S\subseteq[p]}\gamma_0^{\mathrm{unc}}(S) = \sum_{T\in\mathcal{T}}\sum_{M\subseteq S^\circ}\gamma_0^{\mathrm{unc}}(T\setminus M)\\ &\le \sum_{T\in\mathcal{T}}\gamma_0^{\mathrm{unc}}(T)\sum_{M\subseteq S^\circ}e^{|M|} = (1+e)^s\sum_{T\in\mathcal{T}}\gamma_0^{\mathrm{unc}}(T)\\ &\le \frac{(1+e)^s}{(1-q)^{s+1}}\,\gamma_0^{\mathrm{unc}}(S^\circ).\end{aligned}$$

With (1) this is the first bound. (The completions T may have $|T|>K$; their uncapped weights are used only in this upper bound.) For the second, put $x=(1-q)^{s+1}\in(0,1]$ and $z=(1+e)^{-s}\in(0,1]$. Then $1-\pi_0(S^\circ)\le 1-xz=(1-x)+x(1-z)\le(1-x)+(1-z)$. Bernoulli's inequality gives $1-x\le(s+1)q\le(s+1)q/(1-q)$. The function $g(x)=1-(1+x)^{-s}$ satisfies $g(0)=0$ and $g'(x)=s(1+x)^{-s-1}\le s$ for $x\ge 0$, so $1-z=g(e)\le se$. □

Proposition 2 ((i-b))**.** *$q\le u$. Hence, if $u<1$, both bounds of (i-a) hold with u in place of q.*

Proof. If $s=p$, $q=u=0$. Let $s<p$. For $l\notin S^\circ$, with $b=b(l)$, put $\mu_l=\max\{r_b(T+l)/r_b(T): U_b^*\subseteq T\subseteq C_b\setminus\{l\}\}$; the index set is nonempty because $l\notin U_b^*$ (so $T=U_b^*$ is admissible), and $u=(p-s)^{-1}\sum_{l\notin S^\circ}\mu_l$. Fix b and $V\subseteq C_b\setminus U_b^*$ with $|V|=j\ge 1$, and list $V=\{v_1,\dots,v_j\}$ in any order. With $T_i=U_b^*\cup\{v_1,\dots,v_i\}$ we have $U_b^*\subseteq T_{i-1}\subseteq C_b\setminus\{v_i\}$, so telescoping,

$$\frac{r_b(U_b^*\cup V)}{r_b(U_b^*)}=\prod_{i=1}^{j}\frac{r_b(T_{i-1}+v_i)}{r_b(T_{i-1})}\le\prod_{l\in V}\mu_l.$$

Summing over V, $j!\,a_{bj}\le j!\sum_{V\subseteq C_b\setminus U_b^*,|V|=j}\prod_{l\in V}\mu_l\le\big(\sum_{l\in C_b\setminus U_b^*}\mu_l\big)^j$, because expanding the j-th power of a sum of nonnegative terms produces each product over a j-subset exactly $j!$ times (once per ordering), plus further nonnegative terms. Taking j-th roots and the maximum over $1\le j\le|C_b\setminus U_b^*|$, $D_b\le\sum_{l\in C_b\setminus U_b^*}\mu_l$ (also when $C_b=U_b^*$, both sides being 0). The sets $C_b\setminus U_b^*$ partition $[p]\setminus S^\circ$, so $(p-s)q=\sum_b D_b\le\sum_{l\notin S^\circ}\mu_l=(p-s)u$.

If $u<1$ then $q\le u<1$, so (i-a) applies; since $x\mapsto(1-x)^{s+1}$ is nonincreasing and $x\mapsto(s+1)x/(1-x)$ is nondecreasing on $[0,1)$, the bounds with q imply those with u. □

Proposition 3 ((i-c))**.** *If $u<1$, then $\pi_0(S^\circ)\ge(1-u)^{s+1}/[1+(2^s-1)\delta^{\mathrm{comp}}]$.*

Proof. By the bijection Φ, $\sum_{S\in\Omega}\gamma_0^{\mathrm{unc}}(S)=\sum_{T\in\mathcal{T}}\sum_{M\subseteq S^\circ}\mathbf{1}\{T\setminus M\in\Omega\}\gamma_0^{\mathrm{unc}}(T\setminus M)$. Fix $T\in\mathcal{T}$. The term $M=\emptyset$ is at most $\gamma_0^{\mathrm{unc}}(T)$. For $\emptyset\ne M\subseteq S^\circ$ with $S:=T\setminus M\in\Omega$: $S\not\supseteq S^\circ$ (it misses $M\ne\emptyset$) and $S\cup S^\circ=T$, so $\gamma_0^{\mathrm{unc}}(S)\le\delta^{\mathrm{comp}}\gamma_0^{\mathrm{unc}}(T)$ by definition of δ^{comp} (a maximum over a finite set). There are at most 2^s-1 such M (the nonempty subsets of S°, some of which may be excluded by feasibility; none when $s=0$). Hence $\sum_{S\in\Omega}\gamma_0^{\mathrm{unc}}(S)\le[1+(2^s-1)\delta^{\mathrm{comp}}]\sum_{T\in\mathcal{T}}\gamma_0^{\mathrm{unc}}(T)$. Since $q\le u<1$ (part (i-b)), Lemma 4 and monotonicity give $\sum_T\gamma_0^{\mathrm{unc}}(T)\le(1-q)^{-(s+1)}\gamma_0^{\mathrm{unc}}(S^\circ)\le(1-u)^{-(s+1)}\gamma_0^{\mathrm{unc}}(S^\circ)$, and (1) concludes. The argument used the supremum over feasible S only, so the bound also holds with the (larger) supremum over all $S\subseteq[p]$, $S\not\supseteq S^\circ$. □

3 Part (ii)

Assume $S_0\in\Omega$.

Proposition 4 ((ii-a)). $\pi_0(S_0) = \pi_0(S^\circ)\,[w(s_0)/w(s)]\,\prod_b r_b(S_0 \cap C_b)/r_b(S^\circ \cap C_b)$.

Proof. For $S \in \Omega$, $\pi_0(S) = w(|S|)\prod_b r_b(S \cap C_b)/(W_K \bar{Z}_0)$, and $\pi_0(S^\circ) > 0$ (Lemma 1). Divide the expression for S_0 by that for S°. □

Lemma 6 (One feasible addition under π_0). *If $S \in \Omega$, $k = |S| < K$ and $l \notin S$, then $S + l \in \Omega$ and $\pi_0(S + l) \geq r_k\,\pi_0(S)$.*

Proof. $|S + l| = k + 1 \leq K$. Let $b = b(l)$ and $U = S \cap C_b$; only block b changes, $(S + l) \cap C_b = U + l$. By (ii-a)'s computation and Lemmas 3 and 2 (note $k \leq K - 1 \leq p - 1$),

$$\frac{\pi_0(S + l)}{\pi_0(S)} = \frac{w(k + 1)}{w(k)} \cdot \frac{r_b(U + l)}{r_b(U)} = \frac{k + 1}{2p - k - 1} \cdot \frac{m(U + l)}{m(U)} \geq \frac{k + 1}{(2p - k - 1)\sqrt{n + 1}} = r_k. \qquad \square$$

Proposition 5 ((ii-b)). *If $S^\circ \subseteq S_0$, then $\pi_0(S_0)/\pi_0(S^\circ) \geq \prod_{j=0}^{s_0 - s - 1} r_{s+j}$.*

Proof. If $s_0 = s$ then $S_0 = S^\circ$ and both sides equal 1. Otherwise list $S_0 \setminus S^\circ = \{l_1, \ldots, l_{s_0 - s}\}$ and put $S^{(i)} = S^\circ \cup \{l_1, \ldots, l_i\}$, so $|S^{(i)}| = s + i$. For $0 \leq i \leq s_0 - s - 1$, $|S^{(i)}| = s + i < s_0 \leq K$, so Lemma 6 gives $\pi_0(S^{(i+1)}) \geq r_{s+i}\pi_0(S^{(i)})$. Multiplying these $s_0 - s$ inequalities (all terms positive) gives the claim, since $S^{(0)} = S^\circ$ and $S^{(s_0 - s)} = S_0$. □

Remark 1. (ii) bounds $\pi_0(S_0)$ only relative to $\pi_0(S^\circ)$; an absolute bound needs a lower bound on the normalised mass $\pi_0(S^\circ)$, such as part (i).

4 Part (iv)

Assume $S_0 \in \Omega$.

Lemma 7 (Endpoints and geometric interpolation). *$q_1 = \pi$, and for every $\varphi \in [0, 1]$ and $S \in \Omega$, $\gamma_\varphi(S) = \gamma_0(S)^{1-\varphi}\gamma_1(S)^\varphi$.*

Proof. By Lemma 1, $d(S)$ is finite and

$$\gamma_1(S) = \gamma_0(S)\exp(d(S)) = \varpi(S)\prod_b r_b(S \cap C_b) \cdot \frac{\mathrm{BF}(S)}{\prod_b r_b(S \cap C_b)} = \frac{\varpi(S)m(S)}{m(\emptyset)}.$$

Hence $q_1(S) = \varpi(S)m(S)/\sum_{S' \in \Omega} \varpi(S')m(S') = \pi(S)$. Also $\gamma_0 \exp(\varphi d) = \gamma_0^{1-\varphi}(\gamma_0 \exp(d))^\varphi$. □

Proposition 6 ((iv-a), (iv-b)). *For every $\varphi \in [0, 1]$, $1/q_\varphi(S_0) - 1 \leq R_0^{1-\varphi}R_1^\varphi \leq \max(R_0, R_1)$ (with $0^0 = 1$). Hence $\inf_{\varphi \in [0,1]} q_\varphi(S_0) \geq \min\{\pi_0(S_0), \pi(S_0)\}$, equivalently $\sup_{\varphi \in [0,1]}[1 - q_\varphi(S_0)] \leq \max\{1 - \pi_0(S_0), 1 - \pi(S_0)\}$.*

Proof. Since $\gamma_\varphi > 0$ on Ω, $1/q_\varphi(S_0) - 1 = \sum_{S \in \Omega \setminus \{S_0\}} \gamma_\varphi(S)/\gamma_\varphi(S_0)$. With $\alpha_S = \gamma_0(S)/\gamma_0(S_0)$ and $\beta_S = \gamma_1(S)/\gamma_1(S_0)$ (positive), Lemma 7 gives $\gamma_\varphi(S)/\gamma_\varphi(S_0) = \alpha_S^{1-\varphi}\beta_S^\varphi$, and $R_0 = \sum_{S \neq S_0} \alpha_S$, $R_1 = \sum_{S \neq S_0} \beta_S$ (sums over $\Omega \setminus \{S_0\}$), using $q_0 = \pi_0$, $q_1 = \pi$.

Since $n \geq 3$ and $p \geq 1$, $K \geq 1$, so $\emptyset$ and $\{1\}$ both lie in Ω and $\Omega \setminus \{S_0\} \neq \emptyset$; hence $R_0, R_1 > 0$ (the convention $0^0 = 1$ in the statement is never needed). For $\varphi \in \{0, 1\}$ the first inequality is an equality. For $\varphi \in (0, 1)$, Hölder's inequality with conjugate exponents $1/(1 - \varphi)$ and $1/\varphi$ gives

$$\sum_{S \neq S_0} \alpha_S^{1-\varphi}\beta_S^\varphi \leq \Big(\sum_{S \neq S_0} \alpha_S\Big)^{1-\varphi}\Big(\sum_{S \neq S_0} \beta_S\Big)^\varphi = R_0^{1-\varphi}R_1^\varphi.$$

Finally $R_0^{1-\varphi} R_1^{\varphi} \le \max(R_0, R_1)^{1-\varphi} \max(R_0, R_1)^{\varphi} = \max(R_0, R_1)$.

For (iv-b): $q_\varphi(S_0) = 1/(1 + [1/q_\varphi(S_0) - 1]) \ge 1/(1 + \max(R_0, R_1)) = \min\{1/(1 + R_0), 1/(1 + R_1)\} = \min\{q_0(S_0), q_1(S_0)\} = \min\{\pi_0(S_0), \pi(S_0)\}$, for every $\varphi \in [0, 1]$. Taking the infimum gives the claim; the supremum form is the same statement. □

Proposition 7 ((iv-c))**.** *Along a sequence of data sets with* $P_0(S_0 \in \Omega) \to 1$, $1 - \pi(S_0) \to 0$ *and* $1 - \pi_0(S_0) \to 0$ *in* P_0*-probability,* $\sup_{\varphi \in [0,1]} [1 - q_\varphi(S_0)] \to 0$ *in* P_0*-probability.*

Proof. (Masses of S_0 are 0 when $S_0 \notin \Omega$.) Let $\Sigma_n = \sup_{\varphi \in [0,1]} [1 - q_\varphi(S_0)]$. On $E_n = \{S_0 \in \Omega\}$, (iv-b) gives $\Sigma_n \le \max\{1 - \pi_0(S_0), 1 - \pi(S_0)\} \le [1 - \pi_0(S_0)] + [1 - \pi(S_0)]$. For $\eta > 0$,

$$P_0(\Sigma_n > \eta) \le P_0(E_n^c) + P_0\big(1 - \pi_0(S_0) > \eta/2\big) + P_0\big(1 - \pi(S_0) > \eta/2\big) \to 0.$$

Measurability: $\varphi \mapsto q_\varphi(S_0)$ is continuous on $[0, 1]$ (a ratio of finite sums of positive functions $\gamma_0(S) \exp(\varphi d(S))$), so Σ_n equals the supremum over rational φ and is measurable whenever each $q_\varphi(S_0)$ is; in any case the displayed bound holds with outer probability. □

Remark 2 (Scope). No assumption on the noise, the design, or the score expansion (EXP or EXP′) was used: (i), (ii) and (iv) are identities and inequalities valid for every data set, every partition with positive tables (for (ii-b): tables equal to Bayes factors), and every feasible reference. Statistical content enters only through (iv-c)'s hypotheses.

(d) Review record

FINAL_REPORT.md: statement changes

Changes to the statement

All changes were approved by Astra #1 as statement checker (step 1b), with the verdict "APPROVED WITH CORRECTIONS". Details are in STATEMENT.md, "Changes to the statement".

- **C1, added hypothesis:** (ii) and (iv) assume S_0 is a feasible model, $|S_0| \le K = \min(p, \lfloor n/2 \rfloor)$.
  - Without it, the identity in (ii) is false: $\pi_0(S_0) = 0$ while the right side is positive.
  - Astra confirmed this with a concrete instance ($n = 3$, $p = 2$, $S_0 = \{1,2\}$).
  - Asymptotically nothing is lost: the regime R already assumes "eventually $s_0 \le K$".
- **C6, generalisation:** the proof holds for any real X, y and any block partition, and (i) holds for any positive tables. The method's setting is a special case.
- **C7, scope:** (i-c) is stated under $u < 1$.
- **Readings:**
  - "With T03 and (iii) ..." in (iv) became a conditional corollary, (iv-c). The conclusions of T03 and of part (iii) enter only as explicit hypotheses; nothing about T03 or (iii) is proved here.
  - δ^comp's supremum is taken over feasible models. This makes the bound stronger and implies the other reading.
  - "Normalised reference mass remains essential" is treated as a remark, not a claim.
- **EXP / EXP′ is not needed** for (i), (ii) or (iv). Astra confirmed this. Part (iii), which needs EXP′, was left alone as instructed, and so was part (v).

- **The proof-route hint** never entered STATEMENT.md or any packet.

FINAL_REPORT.md: review history

Review history

Stage A: lead and Astra #1 (session 01a16418-acfd-7cb3-8783-b91d3752db30)

- **Statement check:** approved with corrections (`review/astra_01_statement_check.log`).
- **Review of PROOF_v1:** VERDICT verified, one exposition finding (`review/astra_02_review_A1_v1.log`).
  - The finding: "2^s − 1 such M" should read "at most 2^s − 1".
  - I accepted it and fixed it in PROOF_v2, the proof candidate.

Stage B: two independent reviews of PROOF_v2, started together with fresh context and given only the packet

- **Astra #2** (new session 01a1641f-ea40-7570-bde0-1de23647a8bc): no findings, VERDICT verified (`review/astra2_B_review_v2.log`).
- **Opus 5.5, xhigh:** five exposition findings, VERDICT verified (`review/opus_B_review_v2.md`).
  - The findings were: a display overflowing the margin, an unstated step for D_$S \geq 1$, symbol clashes, an impossible case $\Omega = \{S_0\}$, and the 0^0 convention.
  - I accepted all five and fixed them in the unified PROOF_v3 (`review/CHANGES_v2_to_v3.md`, `review/REVIEW_PROCESSING.md`).
- **Round 2:** PROOF_v3 and the claims log went back to both reviewers.
  - Astra #2: agrees with the log, VERDICT verified (`review/astra2_B_round2_v3.log`).
  - Opus: agrees with the log, confirms all five fixes, VERDICT verified (`review/opus_B_round2_v3.md`).
  - Opus raised one new cosmetic point, N1: the letter x is reused as a dummy variable in one paragraph of (i-a).
  - I accepted N1 but deferred it, so that the final file is exactly the version every reviewer verified.
- **CLAIMS_OF_FATAL_FLAWS.md:** no fatal or gap finding in any review. Both Stage B reviewers agreed with the log.
- **Disputes:** none, and the Astra judge was not used.

Numerical sanity check (not a review, not evidence of proof)

- A Sonnet assistant wrote an independent brute-force implementation from STATEMENT.md and ran it with one seed, locally (`helpers/numcheck/REPORT.md`).
- Scale: 10,000 small instances ($p \leq 9$, the cap often binding, adversarial tables for (i)). In total about 1.1M checks of part (i), 3.1M of (ii-a) and 0.54M φ-points of (iv).

- 0 failures.
- All 4 planted false variants were detected.

CLAIMS_OF_FATAL_FLAWS.md (every entry, verbatim)

CLAIMS_OF_FATAL_FLAWS.md — T22 (T16) parts (i), (ii), (iv)

Every fatal or gap finding from every review, verbatim, with the lead's answer.

Stage A (Astra #1, review of PROOF_v1; review/astra_02_review_A1_v1.log)

No fatal or gap findings (one exposition finding; VERDICT: verified).

Stage B (Astra #2 and Opus xhigh, reviews of PROOF_v2; review/astra2_B_review_v2.log, review/opus_B_review_v2.md)

No fatal or gap findings. Astra #2: no findings, VERDICT: verified. Opus: five exposition findings, VERDICT: verified.

The log is therefore empty of fatal and gap claims. Exposition findings and their handling are in review/REVIEW_PROCESSING.md.

Stage B round 2 — rulings on this log (PROOF_v3)

- Astra #2 (review/astra2_B_round2_v3.log): "I agree with the log concerning my review: Round 1 raised no fatal or gap finding." No findings in PROOF_v3. VERDICT: verified.
- Opus xhigh (review/opus_B_round2_v3.md): "The log records no fatal or gap claims from any review. I agree with that ruling." All five exposition fixes confirmed. One new exposition finding (N1: dummy variable x reused in the (i-a) paragraph); accepted by the lead, deferred (cosmetic; PROOF_v3 kept as the verified final). VERDICT: verified. No dispute; the Astra judge was not needed.

Chapter 9

T23(ii): pathwise Rao–Blackwell PIP bound and its basin consequence

Final proof: `proofs/T23_T18_basin_prefix/proof/PROOF_v5.tex`. *Sources of parts (a), (b), (d):* `theorems/T23_T18_basin_prefix.md`, `STATEMENT.md`, `FINAL_REPORT.md`, `CLAIMS_OF_FATAL_FLAWS.md`.

(a) Original conjecture text (theorem file, verbatim)

T23 (T18): Basin-to-output guarantee with recording-prefix control

Status: conjectured, not proven. Methodology SMC-BVS v3.0 (2026-10-19), SHA-256 93043758058a6d422cca6ca4b6be8f3a9e8149b6f6de2e1bab1ce31c44b8019c. Read with `00_COMMON.md`.

Category / rank: conditional algorithmic convergence; abstract theorem (i)–(iv): **core**; conditional instances (v): **secondary**.

Definitions used

Final-run terminal rounds (L burn-in + 4 measured) at $\varphi = 1$; f_H(S_0^c) = fraction of an island's particles outside S_0 after H terminal rounds; B_{i,h} = indicator that particle i is outside S_0 at some point during measured round h (including its start); $\bar{B}_r = (4N_e)^{-1}\Sigma_{h=1}^{4}\Sigma_i B_{i,h}$; $\hat{PIP}_{rj}$, primary average; $u_\pi = \delta_1/\pi(S_0)$; $\rho_* = \min\{1,(B+3)u_\pi\}$; BASIN($\mathcal{R}$, r_0, a, η, ρ, δ) (inventory); $\mathfrak{P} = \{\tilde{S}_0, S_0\}$.

Assumptions

EX-A, EX-S, CAL, BASIN; T03's hypotheses supply statistical truth stability; the instances in (v) need their own exit, gain and initial-population hypotheses.

Precise statement

(i) For one island, after H terminal rounds: $E_{\mathcal{C}}$ f_H(S_0^c) ≤ δ + (1−a)^{⌊H/r_0⌋} + H(η + ρ). Conditioning on the realised initial population replaces δ by its actual outside-basin fraction. (ii) [pathwise RB bound] $\max_j|\hat{PIP}_{rj} - 1\{j \in S_0\}| \le \bar{B}_r + \max_j|o_j(S_0,1) - 1\{j \in S_0\}| \le \bar{B}_r + u_\pi$. Hence for the primary average, $E_{\mathcal{C}}\max_j|\hat{PIP}_j^{primary} - 1\{j\in S_0\}| \le \delta + (1-a)^{\lfloor L/r_0\rfloor} + (L+4)(\eta+\rho) + u_\pi =:$

$\mathcal{B}$_n; no particle independence and no union over p coordinates is needed; the conditional MPM error probability is ≤ 2$\mathcal{B}$_n. (iii) [truth stability] For the actual hybrid round, ρ ≤ ρ_*: the Gibbs first-departure probabilities sum over distinct one-flip neighbours, and each of the B+2 other labels contributes ≤ u_π by reversibility. (iv) [initial occupancy] T15 supplies δ ≤ π($\mathcal{R}^c$)/c + P_$\mathcal{C}$(Ẑ_r/Z < c), 0 < c < 1, with a per-island lower tail. (v) [conditional instances] For either single-flip form of ONE, if the actual one-round first-exit probability from $\mathfrak{P}$ is uniformly ≤ η_$\mathfrak{P}$, reflection and T18 give the candidate entrance bound a ≥ π(S_0|$\mathfrak{P}$) − η_$\mathfrak{P}$; the pair is useful when (L+4)η_$\mathfrak{P}$ → 0 and the initial-population condition holds. For a proxy swap, the natural basin {proxy state, S_0, S_0 ∪ proxy} needs verified entrance and exit bounds, or T24's $\mathcal{R}_1$. ONE and TL_avg alone establish neither basin claim.

Dependencies

Builds on: T03, T09, T12, T15, T18. Used by: T24, T26, T31. T25 may supply the per-island lower tail.

Numerical evidence (provisional)

provisional evidence (Sol check J4): labelled-round enumeration (n = 8, p = 4) confirms the first-exit partition (residual $2.2\cdot10^{-16}$) and that end-of-round fractions alone do not give the pathwise RB bound (max visit-minus-endpoint 0.578). provisional evidence (Sol check J6): occupancy identity. Files: judge/sol2/J4_REPAIR_BASIN.md.

Proof-route hint (optional)

Blockwise entrance trials over ⌊H/r_0⌋ blocks; separate basin exits from truth departures; bound measured rewards by whole-round bad-path indicators (a pre-coordinate fraction is bounded in expectation by its round-start fraction plus truth's prefix-departure probability); T12 and T15 for the initial occupancy. Hardest step: deriving a useful basin for the actual mixed-label kernel. *When proof.md is run on this file, its lead considers the statement both with and without these hints; the hints never reach the reviewers.*

(b) STATEMENT.md (verbatim)

STATEMENT — T23(ii): pathwise Rao–Blackwell PIP bound and its basin consequence

Source: `theorems/T23_T18_basin_prefix.md`, part (ii) only (SMC-BVS v3.0, METHODOLOGY_v3.md SHA-256 93043758058a6d422cca6ca4b6be8f3a9e8149b6f6de2e1bab1ce31c44b8019c), read with `theorems/00_COMMON.md`. Parts (i), (iii), (iv), (v) are not claimed here.

Statement analysis outcome: **(a) correct as stated.** No assumption is changed or added. EXP / EXP′ is not needed. No other v3 conjecture is used as a hypothesis: the expectation bound is proved from the BASIN conditions directly (it does not cite part (i)). Section 5 lists the readings fixed here.

1. Setting (compact; the method file is authoritative)

- **Models and target.** Ω = {S ⊆ [p] : |S| ≤ K}, K = min(p, ⌊n/2⌋). Each S ∈ Ω has weight γ_1(S) = ϖ(S)m(S) > 0; Z = Σ_Ω γ_1 and π = γ_1/Z on Ω. The data (X, y) are fixed throughout.
- **Gibbs conditional.** For S ∈ Ω and j ∈ [p], o_j(S, 1) is the inclusion probability that the Gibbs step

for coordinate j uses at state S (at φ = 1):

- if j ∈ S, or j ∉ S and |S| < K: o_j(S,1) = $\gamma_1(S \cup \{j\}) / [\gamma_1(S \cup \{j\}) + \gamma_1(S \setminus \{j\})]$;
- if j ∉ S and |S| = K (addition excluded): o_j(S,1) = 0.

- **Truth and stability constant.** $S_0 \subseteq [p]$ is a fixed set (the true support). $\delta_1 = 1 - \pi(S_0)$, $u_\pi = \delta_1/\pi(S_0)$ ($u_\pi = +\infty$ if $\pi(S_0) = 0$).
- **Round at φ = 1.** One round of a particle is m = p + B + 2 labelled elementary steps (p Gibbs coordinate steps, one uniform swap, B neighbour swaps, one independence proposal from π_0), applied in a uniformly random order drawn independently of the particle's state, afresh for every particle and every round. Each Gibbs step for coordinate j records o_j(S,1) at the state S held immediately before its draw. A round started at x produces a path $x = X^{(0)}, X^{(1)}, \ldots, X^{(m)}$ ($X^{(k)}$ = state after k steps). K_1 is the one-round transition kernel (law of $X^{(m)}$ given $X^{(0)} = x$); K_1^r its r-fold power.
- **Calibration. $\mathcal{C}$** is the frozen calibration output (grid, T, L, N_e, all settings), measurable with respect to streams 0–9 and the data; we fix a $\mathcal{C}$ for which the final run is executed (successful grid construction). P_$\mathcal{C}$, E_$\mathcal{C}$ are probability and expectation over the final-run randomness (streams 10–13) given $\mathcal{C}$.
- **Final run, island r ∈ {1,2,3,4}.** N = N_e particles; T bridge stages (reweight, multinomial resampling, one round per stage); then the **terminal rounds**: L burn-in rounds and 4 measured rounds, all with the kernel K_1 and with no reweighting or resampling. For particle i, X_{i,s} denotes its state after s terminal rounds (s = 0: after the stage-T resampling and the stage-T round). Measured round h ∈ {1,2,3,4} is terminal round L + h; it starts at X_{i,L+h−1}.
- **Estimators.** For island r, $\hat{PI}_{rj} = (4N)^{-1} \Sigma_{h=1}^{4} \Sigma_{i=1}^{N}$ o_j(S_{i,h,j}, 1), where S_{i,h,j} is particle i's state immediately before its Gibbs step for j in measured round h. Primary $\hat{PI}_j = ¼ \Sigma_r \hat{PI}_{rj}$. $\hat{MPM} = \{j : \hat{PI}_j > ½\}$.
- **Bad-round indicators.** B_{i,h} = 1 if particle i's path in measured round h, $X^{(0)}, \ldots, X^{(m)}$ (including its start $X^{(0)} = X_{i,L+h-1}$), contains a state ≠ S_0; else 0. $\bar{B}_r = (4N)^{-1} \Sigma_{h=1}^{4} \Sigma_{i=1}^{N}$ B_{i,h}.

2. Hypotheses

- **EX-A** (exact scores and probabilities, so o_j and π_0 draws are exact), **EX-S** (independent ideal uniform streams with the method's stream separation), **CAL** ($\mathcal{C}$ uses streams 0–9 only; streams 10–13 are fresh).
- **BASIN($\mathcal{R}$, r_0, a, η, ρ, δ)**, given $\mathcal{C}$: $\mathcal{R} \subseteq \Omega$ with $S_0 \in \mathcal{R}$; an integer $r_0 \geq 1$; numbers a, η, ρ ∈ [0,1], δ ≥ 0, such that
  - (entrance) $K_1^{r_0}(x, \{S_0\}) \geq a$ for every $x \in \mathcal{R}$;
  - (basin exit) for every $x \in \mathcal{R}$, the probability that a round started at x has some state outside $\mathcal{R}$ on its path $X^{(1)}, \ldots, X^{(m)}$ is ≤ η;
  - (truth departure) the probability that a round started at S_0 has some state ≠ S_0 on its path $X^{(1)}, \ldots, X^{(m)}$ is ≤ ρ;
  - (initial occupancy) for every island r, $E_{\mathcal{C}}[\, N^{-1} \Sigma_i 1\{X_{i,0} \notin \mathcal{R}\} \,] \leq \delta$.

Part (ii)'s first inequality (pathwise) uses only the setting and EX-A; the second uses $S_0 \in \Omega$; the

expectation and MPM bounds use EX-S, CAL and BASIN.

3. Conclusions

(ii-a) **Pathwise RB bound.** On every realisation of the final run, for each island r,
max_j |PIP̂_{rj} − 1{j ∈ S_0}| ≤ B̄_r + max_j |o_j(S_0,1) − 1{j ∈ S_0}|.
If $S_0 \notin \Omega$, the method does not define o_j(S_0, 1); it may then be given any real value, and the inequality holds for every such choice, because every path state lies in Ω and so B̄_r = 1.

(ii-b) **Truth term.** If $S_0 \in \Omega$, then max_j |o_j(S_0,1) − 1{j ∈ S_0}| ≤ u_π. (If $S_0 \notin \Omega$, u_π = +∞ and the bound is trivial.)

(ii-c) **Expectation bound for the primary PIP.** Under BASIN,
E_$\mathcal{C}$ max_j |PIP̂_j − 1{j ∈ S_0}| ≤ δ + (1−a)^{⌊L/r_0⌋} + (L+4)(η+ρ) + u_π =: $\mathcal{B}$_n.

(ii-d) **MPM.** P_$\mathcal{C}$(MPM̂ ≠ S_0) ≤ 2$\mathcal{B}$_n.

The bounds use no independence between particles and no union bound over the p coordinates.

4. Scope

- All conclusions are given the data and a fixed successful $\mathcal{C}$ (calibration-conditional). They say nothing about whether BASIN holds; statistical instances are parts (iii)–(v), not claimed here.

5. Readings fixed here (precisions, not changes)

1. **ρ is the whole-round departure probability from S_0.** The theorem file says "truth prefix-departure ≤ ρ" and, in (iii), bounds ρ by summing first-departure probabilities over all p + B + 2 labels of a round. B_{i,h} counts a departure anywhere in the round. The whole-round reading is the one consistent with both.
2. **η is the one-round first-exit probability from $\mathcal{R}$,** maximised over starts in $\mathcal{R}$ (path version). The proof uses only its consequence $K_1(x, \mathcal{R}^c) \le \eta$ for $x \in \mathcal{R}$, so the end-of-round version would also suffice.
3. **Terminal rounds have no resampling** (the method resamples after every reweighting; there is no reweighting at φ = 1 after stage T). Particle identities persist through the L + 4 terminal rounds.
4. **δ is per island** (islands are i.i.d. given $\mathcal{C}$ under EX-S and CAL, so a bound for one island holds for all).
5. "MPM error" means the event MPM̂ ≠ S_0.

(c) Final proof, verbatim: body of PROOF_v5.tex

The proof below is the body of `PROOF_v5.tex` *from* `\begin{document}` *to* `\end{document}` *without the title block. Its preamble definitions are local to this chapter; labels carry the prefix* `T23:`.

All notation, the setting and the hypotheses are those of `STATEMENT.md`. We recall the objects used. The data are fixed throughout.

1 Setting

Models. $\Omega = \{S \subseteq [p] : |S| \le K\}$; $\gamma_1(S) > 0$ for $S \in \Omega$; $\pi = \gamma_1/Z$ on Ω. For $S \in \Omega$ and $j \in [p]$ the Gibbs inclusion probability at $\varphi = 1$ is

$$o_j(S) = \begin{cases} \dfrac{\gamma_1(S \cup \{j\})}{\gamma_1(S \cup \{j\}) + \gamma_1(S \setminus \{j\})}, & j \in S, \text{ or } j \notin S \text{ and } |S| < K, \\ 0, & j \notin S \text{ and } |S| = K. \end{cases}$$

(We write $o_j(S)$ for $o_j(S, 1)$.) In both cases $o_j(S) \in [0, 1]$, and both sets in the first case belong to Ω. $S_0 \subseteq [p]$ is fixed. We extend π by $\pi(S) = 0$ for $S \notin \Omega$; then $\delta_1 = 1 - \pi(S_0)$, $u_\pi = \delta_1/\pi(S_0)$, with $u_\pi = +\infty$ when $\pi(S_0) = 0$, i.e. exactly when $S_0 \notin \Omega$. When $S_0 \notin \Omega$ the quantity $o_j(S_0)$ is not defined by the method; in that case we give it an arbitrary real value c_j^* (the results below hold for every choice).

Rounds. A round started at $x \in \Omega$ applies the $m = p + B + 2$ labelled elementary steps in a uniformly random order and produces a path $x = X^{(0)}, X^{(1)}, \dots, X^{(m)}$, all in Ω. The Gibbs step for coordinate j occurs exactly once in the round; if it is the k-th step ($1 \le k \le m$), it records $o_j(X^{(k-1)})$. $K_1(x, \cdot)$ is the law of $X^{(m)}$ and K_1^r its r-fold power. For $x \in \Omega$ and $A \subseteq \Omega$ write

$$\mathrm{Path}_x(A^c) = \Pr\big(X^{(k)} \notin A \text{ for some } 1 \le k \le m \;\big|\; X^{(0)} = x\big),$$

the probability that a round started at x visits A^c.

Terminal rounds of island r. $N = N_e$ particles. $X_{i,s}$ is particle i's state after s terminal rounds ($s = 0$: after the stage-T resampling and the stage-T round). The terminal rounds are L burn-in and 4 measured rounds, all with kernel K_1, with no reweighting and no resampling. Measured round $h \in \{1, 2, 3, 4\}$ is terminal round $L + h$; its path for particle i is $X_{i,h}^{(0)} = X_{i,L+h-1}, X_{i,h}^{(1)}, \dots, X_{i,h}^{(m)} = X_{i,L+h}$. Let $k_{i,h}(j)$ be the position of the Gibbs step for j in that round and $S_{i,h,j} = X_{i,h}^{(k_{i,h}(j)-1)}$ the state immediately before it. Then

$$\widehat{\mathrm{PIP}}_{rj} = \frac{1}{4N} \sum_{h=1}^{4} \sum_{i=1}^{N} o_j(S_{i,h,j}),$$

$$B_{i,h} = \mathbf{1}\big\{X_{i,h}^{(k)} \ne S_0 \text{ for some } 0 \le k \le m\big\}, \qquad \overline{B}_r = \frac{1}{4N} \sum_{h=1}^{4} \sum_{i=1}^{N} B_{i,h}.$$

These objects belong to island r; we suppress the index r in $X_{i,s}$, $S_{i,h,j}$, $B_{i,h}$, $k_{i,h}(j)$ and $\mathcal{F}_s^i$ below, and keep it in $\widehat{\mathrm{PIP}}_{rj}$ and $\overline{B}_r$. The primary estimator is $\widehat{\mathrm{PIP}}_j = \frac{1}{4}\sum_{r=1}^{4} \widehat{\mathrm{PIP}}_{rj}$ and $\widehat{\mathrm{MPM}} = \{j : \widehat{\mathrm{PIP}}_j > \frac{1}{2}\}$.

BASIN$(\mathcal{R}, r_0, a, \eta, \rho, \delta)$**.** $\mathcal{R} \subseteq \Omega$, $S_0 \in \mathcal{R}$, $r_0 \ge 1$ an integer, $a, \eta, \rho \in [0, 1]$, $\delta \ge 0$, and

(B1) $K_1^{r_0}(x, \{S_0\}) \ge a$ for every $x \in \mathcal{R}$;

(B2) $\mathrm{Path}_x(\mathcal{R}^c) \le \eta$ for every $x \in \mathcal{R}$;

(B3) $\mathrm{Path}_{S_0}(\{S_0\}^c) \le \rho$;

(B4) $E_{\mathcal{C}}\big[N^{-1}\sum_i \mathbf{1}\{X_{i,0}\notin\mathcal{R}\}\big]\le\delta$ for every island r.

Since the event "$X^{(m)}\notin A$" is contained in the event "$X^{(k)}\notin A$ for some $1\le k\le m$",

$$K_1(x,\mathcal{R}^c)\le\eta\ (x\in\mathcal{R}),\qquad K_1(S_0,\{S_0\}^c)\le\rho. \tag{1}$$

2 The Markov structure of the terminal rounds

Throughout, $\mathcal{C}$ is fixed (a successful calibration) and all probabilities are $P_{\mathcal{C}}$. Fix an island r.

Elementary steps. For a label e (one of the p Gibbs coordinates, the uniform swap, the B neighbour swaps, the independence proposal) let $K_e(x,\cdot)$ be the law of the state after step e is applied at state x, as prescribed by the method at $\varphi=1$. By `STATEMENT.md` §1, every terminal round uses the same round, with the same kernel K_1, fixed by the data and $\mathcal{C}$. So the K_e do not depend on the round, the particle or the particle population. By the definition of a round, a round started at x_0 has path law

$$\Pr\big(X^{(1)}=x_1,\dots,X^{(m)}=x_m\ \big|\ X^{(0)}=x_0\big)=\frac{1}{m!}\sum_{\sigma}\prod_{t=1}^{m}K_{\sigma(t)}(x_{t-1},\{x_t\}), \tag{2}$$

where σ runs over the orderings of the m labels and $\sigma(t)$ is the label at position t.

The form of EX-S used. Each random choice of the algorithm (an order of the labels, or the outcome of one elementary step) is made from uniform variates. EX-S (ideal independent uniform streams), together with CAL (island r's streams are among 10–13, independent of streams 0–9 and hence of $\mathcal{C}$), is used in the following form:

(F) For each random choice c of island r's final run, let $\mathcal{D}_c$ be the σ-field generated by all variates of island r used by the random choices made before c. Conditionally on $\mathcal{D}_c$, the fresh variates from which c is made (a sequence, if c uses a random number of them) are i.i.d. uniform. Hence, given $\mathcal{D}_c$, the choice c has the law that the method prescribes for it given the current state.

Concretely: the order of a round is uniform over the $m!$ orderings given $\mathcal{D}_c$ (the method draws it "independently of the particle's state, afresh for every particle and every round"); and the outcome of step e applied at the current state x has law $K_e(x,\cdot)$ given $\mathcal{D}_c$. Under EX-A these laws are realised exactly. Variates are attributed to the choice that uses them, whenever they were generated. So (F) does not depend on whether particles are processed round by round across the population or one particle at a time, on how the variates are split into sub-streams, or on whether variates are generated in batches ahead of use. It is the idealisation EX-S itself, stated for the choices made by the algorithm.

Per-particle σ-fields. For a particle i and $0\le s\le L+3$, let $\mathcal{F}_s^i$ be the σ-field generated by all variates of island r used by the random choices made before particle i's terminal round $s+1$ begins. Then $\mathcal{F}_s^i\subseteq\mathcal{F}_{s+1}^i$, because every choice made before round $s+1$ of particle i begins is also made before its round $s+2$ begins. Also $X_{i,0},\dots,X_{i,s}$ are $\mathcal{F}_s^i$-measurable, because they are computed from the initial draws, the bridge stages and particle i's own terminal rounds $1,\dots,s$, all of which precede its round $s+1$.

Lemma 1. *Assume EX-A, EX-S (in the form (F)) and CAL. For every particle i, every $0 \le s \le L+3$ and every set $\Gamma \subseteq \Omega^{m+1}$ of paths,*

$$P_{\mathcal{C}}\big((Z^{(0)}_{i,s+1},\dots,Z^{(m)}_{i,s+1}) \in \Gamma \;\big|\; \mathcal{F}^i_s\big) = \Pr\big((X^{(0)},\dots,X^{(m)}) \in \Gamma \;\big|\; X^{(0)} = x\big)\Big|_{x=X_{i,s}} \quad a.s.,$$

where $(Z^{(0)}_{i,s+1},\dots,Z^{(m)}_{i,s+1})$ is particle i's path in terminal round $s+1$, so $Z^{(0)}_{i,s+1} = X_{i,s}$ (no resampling between terminal rounds). In the notation of §1, the path of measured round h is $X^{(k)}_{i,h} = Z^{(k)}_{i,L+h}$. In particular, $P_{\mathcal{C}}(X_{i,s+1} \in A \mid \mathcal{F}^i_s) = K_1(X_{i,s}, A)$. Moreover, for $0 \le s < s+q \le L+4$ and $A \subseteq \Omega$, $P_{\mathcal{C}}(X_{i,s+q} \in A \mid \mathcal{F}^i_s) = K_1^q(X_{i,s}, A)$.

Proof. Write $Y_t = Z^{(t)}_{i,s+1}$ for $0 \le t \le m$, and let Σ be the order drawn for this round. Let $\mathcal{D}_0$ be the σ-field of all variates used by the random choices made before Σ is chosen, and, for $1 \le t \le m$, $\mathcal{D}_t$ the σ-field of all variates used by the random choices made before the outcome of the t-th step of this round is chosen. Then $\mathcal{F}^i_s \subseteq \mathcal{D}_0 \subseteq \mathcal{D}_1 \subseteq \cdots \subseteq \mathcal{D}_m$. $Y_0 = X_{i,s}$ is $\mathcal{D}_0$-measurable, and $\Sigma, Y_0, \dots, Y_{t-1}$ are $\mathcal{D}_t$-measurable. By (F), $P_{\mathcal{C}}(\Sigma = \sigma \mid \mathcal{D}_0) = 1/m!$ for every ordering σ, and $P_{\mathcal{C}}(Y_t = y \mid \mathcal{D}_t) = K_{\Sigma(t)}(Y_{t-1}, \{y\})$ for $1 \le t \le m$. Fix $x_0, \dots, x_m \in \Omega$ and an ordering σ. Conditioning successively on $\mathcal{D}_m, \mathcal{D}_{m-1}, \dots, \mathcal{D}_1, \mathcal{D}_0$ (tower property; all spaces are finite) gives

$$P_{\mathcal{C}}\big(\Sigma = \sigma,\ Y_0 = x_0, \dots, Y_m = x_m \;\big|\; \mathcal{D}_0\big) = \mathbf{1}\{X_{i,s} = x_0\}\,\frac{1}{m!}\prod_{t=1}^{m} K_{\sigma(t)}(x_{t-1}, \{x_t\}).$$

Indeed, at stage t the factor $\mathbf{1}\{\Sigma = \sigma, Y_0 = x_0, \dots, Y_{t-1} = x_{t-1}\}$ is $\mathcal{D}_t$-measurable, and on that event $P_{\mathcal{C}}(Y_t = x_t \mid \mathcal{D}_t) = K_{\sigma(t)}(x_{t-1}, \{x_t\})$, a constant. Sum over σ and condition on $\mathcal{F}^i_s \subseteq \mathcal{D}_0$. The right side is $\mathcal{F}^i_s$-measurable, so by (2) the conditional law of the path given $\mathcal{F}^i_s$ is the round path law started at $X_{i,s}$. Summing over the paths in Γ gives the first claim. The second claim is the case $\Gamma = \{\text{paths with } x_m \in A\}$.

For the q-round claim, induct on q. The case $q = 1$ is the second claim. If it holds for q, then, since $\mathcal{F}^i_s \subseteq \mathcal{F}^i_{s+q}$ and $X_{i,s+q}$ is $\mathcal{F}^i_{s+q}$-measurable (here $s+q \le L+3$),

$$P_{\mathcal{C}}(X_{i,s+q+1} \in A \mid \mathcal{F}^i_s) = E_{\mathcal{C}}\big[K_1(X_{i,s+q}, A) \;\big|\; \mathcal{F}^i_s\big] = \sum_y K_1^q(X_{i,s}, \{y\}) K_1(y, A) = K_1^{q+1}(X_{i,s}, A). \quad \square$$

3 The pathwise bound

Lemma 2 (pathwise RB bound)**.** *Assume EX-A (the recorded value is the exact o_j at the pre-draw state). On every realisation, for every island r and every $j \in [p]$,*

$$|\widehat{\mathrm{PIP}}_{rj} - \mathbf{1}\{j \in S_0\}| \le \overline{B}_r + (1 - \overline{B}_r)\,|o_j(S_0) - \mathbf{1}\{j \in S_0\}| \le \overline{B}_r + \max_{l \in [p]} |o_l(S_0) - \mathbf{1}\{l \in S_0\}|.$$

Proof. If $S_0 \notin \Omega$, then no path state (all lie in Ω) equals S_0, so $B_{i,h} = 1$ for all (i,h) and $\overline{B}_r = 1$; since $o_j \in [0,1]$, the left side is $\le 1 \le 1 + |c^*_l - \mathbf{1}\{l \in S_0\}|$ for any value c^*_l, and both inequalities hold. Assume now $S_0 \in \Omega$. Write $c_j = \mathbf{1}\{j \in S_0\}$. By the triangle inequality,

$$|\widehat{\mathrm{PIP}}_{rj} - c_j| = \Big|\frac{1}{4N}\sum_{h,i}\big(o_j(S_{i,h,j}) - c_j\big)\Big| \le \frac{1}{4N}\sum_{h,i} |o_j(S_{i,h,j}) - c_j|.$$

Fix (i,h). The state $S_{i,h,j} = X^{(k_{i,h}(j)-1)}_{i,h}$ is one of the states $X^{(0)}_{i,h}, \dots, X^{(m-1)}_{i,h}$ of the path of measured round h. If $B_{i,h} = 0$, every state of that path equals S_0 (so $S_0 \in \Omega$), hence $S_{i,h,j} = S_0$

and $|o_j(S_{i,h,j}) - c_j| = |o_j(S_0) - c_j|$. If $B_{i,h} = 1$, then $|o_j(S_{i,h,j}) - c_j| \le 1$ because $o_j \in [0,1]$ and $c_j \in \{0,1\}$. Thus $|o_j(S_{i,h,j}) - c_j| \le B_{i,h} + (1 - B_{i,h})|o_j(S_0) - c_j|$. Averaging over (i,h) gives the first inequality, and $(1 - \overline{B}_r) \in [0,1]$ gives the second. The bound holds for each j, hence for $\max_j$. □

Lemma 3 (truth term)**.** *If $S_0 \in \Omega$, then* $\max_{j \in [p]} |o_j(S_0) - \mathbf{1}\{j \in S_0\}| \le u_\pi$.

Proof. $\pi(S_0) > 0$ since $\gamma_1 > 0$ on Ω. Let $j \in S_0$. Then $S_0 \setminus \{j\} \in \Omega$, $S_0 \setminus \{j\} \neq S_0$, and

$$1 - o_j(S_0) = \frac{\gamma_1(S_0 \setminus \{j\})}{\gamma_1(S_0) + \gamma_1(S_0 \setminus \{j\})} \le \frac{\gamma_1(S_0 \setminus \{j\})}{\gamma_1(S_0)} = \frac{\pi(S_0 \setminus \{j\})}{\pi(S_0)} \le \frac{1 - \pi(S_0)}{\pi(S_0)} = u_\pi.$$

Let $j \notin S_0$. If $|S_0| = K$, then $o_j(S_0) = 0$ and the bound is trivial. If $|S_0| < K$, then $S_0 \cup \{j\} \in \Omega$, $S_0 \cup \{j\} \neq S_0$, and

$$o_j(S_0) = \frac{\gamma_1(S_0 \cup \{j\})}{\gamma_1(S_0 \cup \{j\}) + \gamma_1(S_0)} \le \frac{\pi(S_0 \cup \{j\})}{\pi(S_0)} \le \frac{1 - \pi(S_0)}{\pi(S_0)} = u_\pi.$$

In both displays we used $\pi(S') \le \pi(\Omega \setminus \{S_0\}) = 1 - \pi(S_0)$ for $S' \in \Omega \setminus \{S_0\}$. □

4 Expected bad-round fraction

Fix an island r and a particle i; write $X_s = X_{i,s}$ and $\mathcal{F}_s = \mathcal{F}_s^i$ $(0 \le s \le L+3)$, and $P_{\mathcal{C}}$ for the conditional law given $\mathcal{C}$. All conditioning below is on $\mathcal{F}_s$ with $s \le L+3$, as required by Lemma 1. Put $\bar{\eta} = \max_{x \in \mathcal{R}} K_1(x, \mathcal{R}^c)$ and $\bar{\rho} = K_1(S_0, \{S_0\}^c)$; by (1), $\bar{\eta} \le \eta$ and $\bar{\rho} \le \rho$.

Lemma 4 (occupancy at round ends)**.** *Under EX-A, EX-S, CAL and (B1), for every integer $0 \le H \le L+4$,*

$$P_{\mathcal{C}}(X_H \neq S_0) \le P_{\mathcal{C}}(X_0 \notin \mathcal{R}) + (1-a)^{\lfloor H/r_0 \rfloor} + H(\bar{\eta} + \bar{\rho}).$$

Proof. Let $k = \lfloor H/r_0 \rfloor$, so $kr_0 \le H$. Define the events

$$\begin{aligned}
E_1 &= \{X_0 \notin \mathcal{R}\},\\
E_2 &= \{X_s \in \mathcal{R},\ X_{s+1} \notin \mathcal{R} \text{ for some } 0 \le s \le H-1\},\\
E_3 &= \{X_s = S_0,\ X_{s+1} \neq S_0 \text{ for some } 0 \le s \le H-1\},\\
E_4 &= \{X_{jr_0} \in \mathcal{R} \text{ for } 0 \le j \le k-1\} \cap \{X_{jr_0} \neq S_0 \text{ for } 1 \le j \le k\}.
\end{aligned}$$

(For $k = 0$, E_4 is the whole space.)

Inclusion: $\{X_H \neq S_0\} \subseteq E_1 \cup E_2 \cup E_3 \cup E_4$. Suppose $X_H \neq S_0$ and none of E_1, E_2, E_3 occurs. Since E_1 fails, $X_0 \in \mathcal{R}$; since E_2 fails, induction on s gives $X_s \in \mathcal{R}$ for all $0 \le s \le H$. If $X_s = S_0$ for some $s \le H$, then, since E_3 fails, induction gives $X_{s'} = S_0$ for all $s \le s' \le H$, contradicting $X_H \neq S_0$. So $X_s \in \mathcal{R} \setminus \{S_0\}$ for all $0 \le s \le H$; as $jr_0 \le H$ for $0 \le j \le k$, E_4 occurs.

Bound on E_2 and E_3. By the union bound and Lemma 1,

$$\begin{aligned}
P_{\mathcal{C}}(E_2) &\le \sum_{s=0}^{H-1} P_{\mathcal{C}}(X_s \in \mathcal{R},\ X_{s+1} \notin \mathcal{R})\\
&= \sum_{s=0}^{H-1} E_{\mathcal{C}}\big[\mathbf{1}\{X_s \in \mathcal{R}\} K_1(X_s, \mathcal{R}^c)\big] \le H\bar{\eta},\\
P_{\mathcal{C}}(E_3) &\le \sum_{s=0}^{H-1} P_{\mathcal{C}}(X_s = S_0,\ X_{s+1} \neq S_0)\\
&= \sum_{s=0}^{H-1} E_{\mathcal{C}}\big[\mathbf{1}\{X_s = S_0\} K_1(S_0, \{S_0\}^c)\big] \le H\bar{\rho}.
\end{aligned}$$

Bound on E_4. For $1 \le l \le k$ let $G_l = \{X_{jr_0} \in \mathcal{R} \text{ for } 0 \le j \le l-1\} \cap \{X_{jr_0} \neq S_0 \text{ for } 1 \le j \le l\}$, and G_0 the whole space; so $E_4 = G_k$. For $1 \le l \le k$, $G_l = G_{l-1} \cap \{X_{(l-1)r_0} \in \mathcal{R}\} \cap \{X_{lr_0} \neq S_0\}$, and $G_{l-1} \cap \{X_{(l-1)r_0} \in \mathcal{R}\} \in \mathcal{F}_{(l-1)r_0}$. Since $(l-1)r_0 \le H - r_0 \le L+3$ and $lr_0 \le H \le L+4$, the q-round part of Lemma 1 with $q = r_0$ gives $P_{\mathcal{C}}(X_{lr_0} = S_0 \mid \mathcal{F}_{(l-1)r_0}) = K_1^{r_0}(X_{(l-1)r_0}, \{S_0\})$, which is $\ge a$ on $\{X_{(l-1)r_0} \in \mathcal{R}\}$ by (B1). Hence

$$P_{\mathcal{C}}(G_l) = E_{\mathcal{C}}\Big[\mathbf{1}_{G_{l-1}\cap\{X_{(l-1)r_0}\in\mathcal{R}\}}\big(1 - K_1^{r_0}(X_{(l-1)r_0}, \{S_0\})\big)\Big] \le (1-a)P_{\mathcal{C}}(G_{l-1}),$$

and $P_{\mathcal{C}}(E_4) = P_{\mathcal{C}}(G_k) \le (1-a)^k$.

Summing the four bounds proves the lemma. □

Lemma 5 (one measured round)**.** *Under EX-A, EX-S, CAL and (B1)–(B3), for every particle i and $h \in \{1,2,3,4\}$,*

$$P_{\mathcal{C}}(B_{i,h} = 1) \le P_{\mathcal{C}}(X_{i,0} \notin \mathcal{R}) + (1-a)^{\lfloor L/r_0 \rfloor} + (L+3)\eta + (L+4)\rho.$$

Proof. Let $H = L + h - 1$, the number of terminal rounds before measured round h; $X_{i,h}^{(0)} = X_{i,H}$. If $B_{i,h} = 1$ then either $X_{i,H} \neq S_0$, or $X_{i,H} = S_0$ and some state $X_{i,h}^{(k)}$, $1 \le k \le m$, differs from S_0. By Lemma 1 with $s = H$ (the path of measured round h is $Z_{i,H+1}^{(\cdot)}$) and (B3), the second event has probability $E_{\mathcal{C}}[\mathbf{1}\{X_{i,H} = S_0\}\mathrm{Path}_{S_0}(\{S_0\}^c)] \le \rho$. By Lemma 4,

$$P_{\mathcal{C}}(B_{i,h} = 1) \le P_{\mathcal{C}}(X_{i,0} \notin \mathcal{R}) + (1-a)^{\lfloor H/r_0 \rfloor} + H(\bar\eta + \bar\rho) + \rho.$$

Now $H \ge L$ and $0 \le 1-a \le 1$ give $(1-a)^{\lfloor H/r_0 \rfloor} \le (1-a)^{\lfloor L/r_0 \rfloor}$; and $H \le L+3$, $\bar\eta \le \eta$, $\bar\rho \le \rho$ give $H(\bar\eta + \bar\rho) + \rho \le (L+3)\eta + (L+4)\rho$. □

5 The theorem

Theorem 6 (T23(ii))**.** *(a) Under EX-A, on every realisation, for every island r:* $\max_j |\widehat{\mathrm{PIP}}_{rj} - \mathbf{1}\{j \in S_0\}| \le \overline{B}_r + \max_j |o_j(S_0) - \mathbf{1}\{j \in S_0\}|$. *If $S_0 \notin \Omega$, the method does not define $o_j(S_0, 1)$; it may then be given any real value, and the inequality holds for every such choice, because every path state lies in Ω and so $\overline{B}_r = 1$.*

(b) If $S_0 \in \Omega$: $\max_j |o_j(S_0) - \mathbf{1}\{j \in S_0\}| \le u_\pi$.

(c) Under EX-A, EX-S, CAL and BASIN$(\mathcal{R}, r_0, a, \eta, \rho, \delta)$,

$$E_{\mathcal{C}} \max_j |\widehat{\mathrm{PIP}}_j - \mathbf{1}\{j \in S_0\}| \le \delta + (1-a)^{\lfloor L/r_0 \rfloor} + (L+4)(\eta + \rho) + u_\pi =: \mathcal{B}_n.$$

(d) Under the same hypotheses, $P_{\mathcal{C}}(\widehat{\mathrm{MPM}} \neq S_0) \le 2\mathcal{B}_n$.

Proof. (a) is Lemma 2 and (b) is Lemma 3.

(c) BASIN gives $S_0 \in \mathcal{R} \subseteq \Omega$, so (a) and (b) give, for each island r, on every realisation, $\max_j |\widehat{\mathrm{PIP}}_{rj} - \mathbf{1}\{j \in S_0\}| \le \overline{B}_r + u_\pi$. For the primary estimator, by the triangle inequality,

$$\begin{aligned}\max_j |\widehat{\mathrm{PIP}}_j - \mathbf{1}\{j \in S_0\}| &= \max_j \Big| \frac{1}{4}\sum_{r=1}^{4} \big(\widehat{\mathrm{PIP}}_{rj} - \mathbf{1}\{j \in S_0\}\big) \Big| \\ &\le \frac{1}{4}\sum_{r=1}^{4} \max_j |\widehat{\mathrm{PIP}}_{rj} - \mathbf{1}\{j \in S_0\}| \\ &\le \frac{1}{4}\sum_{r=1}^{4} \overline{B}_r + u_\pi.\end{aligned}$$

For each island, linearity of expectation and Lemma 5 give

$$E_{\mathcal{C}}\overline{B}_r = \frac{1}{4N}\sum_{h=1}^{4}\sum_{i=1}^{N} P_{\mathcal{C}}(B_{i,h}=1) \le E_{\mathcal{C}}\Big[\frac{1}{N}\sum_{i=1}^{N}\mathbf{1}\{X_{i,0} \notin \mathcal{R}\}\Big] + (1-a)^{\lfloor L/r_0 \rfloor} + (L+3)\eta + (L+4)\rho,$$

which is $\le \delta + (1-a)^{\lfloor L/r_0 \rfloor} + (L+4)(\eta+\rho)$ by (B4). Taking $E_{\mathcal{C}}$ in the previous display proves (c). (When $u_\pi = +\infty$ the claim is trivial; under BASIN, $u_\pi < \infty$.)

(d) If $\widehat{\mathrm{MPM}} \neq S_0$, there is j with either $j \in S_0$ and $\widehat{\mathrm{PIP}}_j \le \frac{1}{2}$, so $|\widehat{\mathrm{PIP}}_j - 1| \ge \frac{1}{2}$; or $j \notin S_0$ and $\widehat{\mathrm{PIP}}_j > \frac{1}{2}$, so $|\widehat{\mathrm{PIP}}_j - 0| > \frac{1}{2}$. Hence $\{\widehat{\mathrm{MPM}} \neq S_0\} \subseteq \{\max_j |\widehat{\mathrm{PIP}}_j - \mathbf{1}\{j \in S_0\}| \ge \frac{1}{2}\}$, and Markov's inequality and (c) give $P_{\mathcal{C}}(\widehat{\mathrm{MPM}} \neq S_0) \le 2\mathcal{B}_n$. □

Remark 7. No independence between particles is used: Lemma 5 is a statement about one particle, and (c) uses only linearity of expectation over particles. No union bound over coordinates is used: the bad-round indicators $B_{i,h}$ do not depend on j, so one bound serves all p coordinates at once.

Remark 8. Lemma 4 uses only (B1), with the end-of-round quantities $\bar{\eta}, \bar{\rho}$; (B2)–(B3) enter Lemma 5 through $\bar{\eta} \le \eta$ and $\bar{\rho} \le \rho$ (1). The path form of (B3) is needed once, in Lemma 5, because the Gibbs recordings of a measured round are taken at intermediate states of the round, not at its end.

(d) Review record

FINAL_REPORT.md: statement changes

Statement changes

None. Step-1 outcome: (a), correct as stated.

- EXP/EXP′ is not needed.
- No sibling conjecture is used as a hypothesis; the occupancy step is proved directly from BASIN, without citing part (i).
- STATEMENT.md §5 records five readings. These are precisions, not changes: 1. ρ is the whole-round departure probability from S_0. 2. η is the one-round first-exit probability from the basin $\boldsymbol{\mathcal{R}}$. 3. There is no resampling in the L + 4 terminal rounds. 4. δ applies to every island. 5. "MPM error" means the event $\widehat{\mathrm{MPM}} \neq S_0$.
- The proof states EX-S in the form it uses, (F): each random choice of the algorithm, given everything used by earlier choices, has the law the method prescribes. Both Stage B reviewers ruled (F) a faithful reading of EX-S and CAL, not an added assumption.

FINAL_REPORT.md: review history

Review history

Stage A (lead + Astra #1). Astra #1: gpt-6-astra, reasoning high, session 01a16419-ed12-7492-9b6f-8736751fe274.

- PROOF_v1 → "No findings ... VERDICT: verified".

Stage B, round 1 (independent, started together; each reviewer had only STATEMENT.md, the proof and the report format).

- **Astra #2** (new session 01a1641c-c05f-7d81-9ac6-6bcc512573e3, high): no findings, verified.
- **Opus 5.5 reviewer** (effort xhigh, read-only Plan agent, own account): verified, with 6 exposition findings, all accepted → PROOF_v2. The findings:
  - Lemma 1's stopping-time step was not written out.
  - The kernel being fixed given $\boldsymbol{\mathcal{C}}$ was not cited to the statement.
  - The case $S_0 \notin \Omega$ needed a convention.
  - Lemma 4's range of H was not bounded.
  - Lemma 4's hypothesis list was not minimal.
  - EX-A and the island index were not stated explicitly.

Round 2 (rulings on v2).

- Both verified and agreed with all six changes.
- They raised 7 more exposition points, all accepted → PROOF_v3. The main ones:
  - Lemma 1 cited method details that STATEMENT.md does not contain (both reviewers).
  - Lemma 1 assumed the population is processed round by round.
- The fix rewrote Lemma 1 with per-particle σ-fields and a step-by-step conditioning proof.

Round 3 (rulings on v3).

- Astra #2: no findings, verified; it ruled (F) faithful.
- Opus: verified, with 2 exposition points (define the σ-fields by variates *used*, not *drawn*; a clash of path notation). Both accepted → PROOF_v4, wording only.

Round 4 (confirmation of v4). Both verified.

- The Opus reviewer added one cosmetic note: the path symbol Z reuses the evidence symbol Z. It said this "needs no action". It is deferred, not changed, so the final proof is exactly the text both reviewers confirmed.

Overall.

- No fatal or gap finding in any round, from any reviewer (CLAIMS_OF_FATAL_FLAWS.md).
- No dispute, so no Astra judge.

- The lead holds the proof correct. Astra #1 verified v1; Astra #2 and the Opus reviewer verified v2, v3 and v4. The v1→v4 changes are exposition plus a more careful proof of Lemma 1. Astra #1 was not re-consulted after Stage A.

CLAIMS_OF_FATAL_FLAWS.md (every entry, verbatim)

CLAIMS_OF_FATAL_FLAWS — T23(ii)

Fatal or gap findings, verbatim, with the lead's answer.

Stage A (Astra #1, session 01a16419-ed12-7492-9b6f-8736751fe274)

None. Reply: "No findings." / "VERDICT: verified" (review/r1_ASTRA_REPLY.md).

Stage B, round 1

- Astra #2 (session 01a1641c-c05f-7d81-9ac6-6bcc512573e3): none. Reply: "No findings." / "VERDICT: verified" (review/B_astra2_r1_REPLY.md).
- Opus 5.5 xhigh reviewer: none. Its six findings are all graded "exposition" by the reviewer. Its summary says "I found no fatal error and no gap" and ends "VERDICT: verified" (review/B_opus_r1_REPLY.md).

There are no fatal or gap findings to log. The exposition findings and the changes made for them are in review/B_CHANGES_v1_to_v2.md; the unified proof is proof/PROOF_v2.tex.

Stage B, round 2 (rulings on PROOF_v2)

- Astra #2: "I found no fatal error or gap, and one exposition issue"; agrees with all six changes; "VERDICT: verified" (review/B_astra2_r2_REPLY.md).
- Opus reviewer: six new findings, all exposition; "I add no fatal or gap entry for round 2"; agrees with the log entry describing its round-1 report; "VERDICT: verified" (review/B_opus_r2_REPLY.md).

Still no fatal or gap findings to log. The round-2 exposition findings and the changes made for them are in review/B_CHANGES_v2_to_v3.md; the unified proof is proof/PROOF_v3.tex.

Stage B, round 3 (rulings on PROOF_v3)

- Astra #2: "No findings."; agrees with every v2→v3 change; finds (F) "a faithful reading of EX-S and CAL, rather than an added assumption"; "VERDICT: verified" (review/B_astra2_r3_REPLY.md).
- Opus reviewer: two findings, both exposition; "I add no fatal or gap entry"; finds (F) a faithful reading provided "drawn before" is read as "used by earlier choices"; "VERDICT: verified" (review/B_opus_r3_REPLY.md).

No fatal or gap finding was raised in any round. The two round-3 exposition points are fixed in proof/PROOF_v4.tex (review/B_CHANGES_v3_to_v4.md).

Stage B, round 4 (confirmation of PROOF_v4)

- Astra #2: "No findings."; the changes "preserve every argument, hypothesis, conclusion, and constant"; "VERDICT: verified" (review/B_astra2_r4_REPLY.md).
- Opus reviewer: one cosmetic exposition note (the symbol Z for terminal-round paths reuses the evidence symbol Z), with "This needs no action for the verdict"; "VERDICT: verified" (review/B_opus_r4_REPLY.md). Lead: deferred, not changed, so that the final proof is exactly the text both reviewers confirmed.

Final: no fatal or gap finding in any round (Stage A or B); no dispute; no Astra judge.

Chapter 10

T31 (T22), part (i): Markov coverage of $0.05\,\widehat{Z}$ and what it certifies

Final proof: `proofs/T31_T22_certificates/proof/PROOF_v3.tex`. *Sources of parts (a), (b), (d):* `theorems/T31_T22_certificates.md`, `STATEMENT.md`, `FINAL_REPORT.md`, `CLAIMS_OF_FATAL_FLAWS.md`.

(a) Original conjecture text (theorem file, verbatim)

T31 (T22): What the reported checks and certificates can establish

Status: conjectured, not proven. Methodology SMC-BVS v3.0 (2026-10-19), SHA-256 93043758058a6d422cca6ca4b6be8f3a9e8149b6f6de2e1bab1ce31c44b8019c. Read with `00_COMMON.md`.

Category / rank: validity and certification limits; **core** (proposition).

Definitions used

Reported quantities: $\hat{Z}$, the bound $0.05\hat{Z}$, island SD of $\log \hat{Z}_r$ (E2), E1, E3, low_empirical_ESS, local_certified, global_certified; $Z_W := Z\pi(W)$ (distinct from the comparator mean r_{WZ}); $Z_V := \Sigma_{S\in V}\varpi(S)m(S)$; Q_f; thresholds t fixed before final randomness; weight comparability ratio C_{eq}.

Assumptions

EX-A, EX-S, CAL; MISS-E/CPL-E for (iii)–(iv); the usability, threshold-measurability or external-bound assumptions stated in the clauses.

Precise statement

(i) Under T12: $P_{\mathcal{C}}(Z \ge \alpha\hat{Z}) \ge 1-\alpha$ and $E_{\mathcal{C}} \log \hat{Z} \le \log Z$; so $0.05\hat{Z}$ has coverage at least 95% under the ideal final-run assumptions; it is not a relative-error interval; conditioning afterwards on final flags does not preserve this coverage automatically. (ii) [estimator-class example, not a regression construction] For $0 < a, \varepsilon < 1$, an unbiased nonnegative estimator equal to aZ with probability $1-\varepsilon$ and to $Z\{1-(1-\varepsilon)a\}/\varepsilon$ otherwise has four independent copies all equal to aZ with probability $(1-\varepsilon)^4$: zero log dispersion with arbitrarily severe underestimation. Independence and agreement give no uniformly informative multiplicative-accuracy certificate over this class. (iii) Under T28, diagnostics

measurable from the evidence coupling have the comparator's law up to h_n^E; score/search diagnostics need T29's information control; passing transfers only if the comparator passes with high probability; calibration flags are fixed given $\mathcal{C}$, recomputed final E2 is not. (iv) [upper bounds] If U uses only coupled information and P_{ref,$\mathcal{C}$}(U^W ≤ CZ_W) ≥ 1 − α′ with CZ_W < Z, then P_$\mathcal{C}$(Z ≤ U) ≤ h_n^E + α′ on that loss instance; not an unconditional impossibility theorem. (v) [discovery after selection] For a threshold t fixed before final randomness: P_$\mathcal{C}${∃S : π_0(S) ≥ t, S missed by the iid draws} ≤ min{1, ⌊1/t⌋e^{−Q_ft}}; and with probability ≥ 1 − δ every model with Q_fπ_0(S) ≥ log[1/(δπ_0(S))] has been scored (usable after selection). Substituting $\pi_0(\hat{S})$ into a fixed-model miss formula is invalid as a competitor certificate; neither statement shows all better-posterior models have large start mass. (vi) local_certified has exactly T20(i)'s meaning; global_certified is restricted to completed validated enumeration; a genuine external U ≥ Z would certify the best scored model globally whenever U − Z_V < max_{S∈V}ϖ(S)m(S) (v3 supplies no such U). (vii) Restricted ESS controls only T13's restricted moment; grid equality across rebuilds certifies neither omitted mass nor mixing; if M models have unnormalised weight ratio ≤ C_eq at a temperature, V captures at most C_eq|V|/M of their mass; "near-equivalent" without a quantitative condition supplies no bound.

Dependencies

Builds on: T12, T13, T11, T20, T28, T29 (and T05–T07, T27 for interpretation). Used by: nothing (terminal interpretation).

Numerical evidence (provisional)

provisional evidence (Sol check J7): exact finite law p = (0.4, 0.3, 0.2, 0.1), 3 draws: fixed-model miss 0.216, all-above-0.25 miss 0.532, selected-threshold competitor miss 0.54 vs naive 0.545 (conditional on the selected label: 1 vs 0.74); fixed-threshold union bound 0.559. provisional evidence (Sol check J8): information-transfer corrections.

Proof-route hint (optional)

Markov and Jensen; the explicit rare-tail law; information-preserving coupling (T17); at most ⌊1/t⌋ atoms of mass ≥ t for the union bound; allocate error δπ_0(S) per atom for the simultaneous statement; T13 for (vii). Hardest step: none in the algebra; a verified production loss instance is the difficult premise. *When proof.md is run on this file, its lead considers the statement both with and without these hints; the hints never reach the reviewers.*

(b) STATEMENT.md (verbatim)

STATEMENT — T31 (T22), part (i) only

Method: SMC-BVS v3.0 (METHODOLOGY_v3.md, SHA-256 93043758058a6d422cca6ca4b6be8f3a9e8149b6f6de2e1bab1ce31c44b8019c).

Source text (verbatim, clause (i) of the theorem file)

> (i) Under T12: P_$\mathcal{C}$(Z ≥ α$\hat{Z}$) ≥ 1 − α and E_$\mathcal{C}$ log $\hat{Z}$ ≤ log Z; so 0.05$\hat{Z}$ has coverage at least 95% under the ideal final-run assumptions; it is not a relative-error interval; conditioning afterwards on final flags does not preserve this coverage automatically.

Assumptions listed for the whole file: EX-A, EX-S, CAL. Clause (i) uses T12 (a sibling conjecture), which enters below only as the explicit hypothesis (H-T12).

1. Setting and definitions

Fixed data (X, y), already standardised as in the method (§4): $n \geq 3$, $\|y\|^2 = n - 1 > 0$, $p \geq 1$ retained columns. All objects below are deterministic functions of the data unless stated otherwise.

- State space $\Omega = \{S \subseteq [p] : |S| \leq K\}$, $K = \min(p, \lfloor n/2 \rfloor)$ (finite).
- $G_S = X_S^T X_S$, $z_S = X_S^T y$, $H_S = (n+1)G_S + I$, $D_S = \det H_S / \det(G_S + I)$, $Q_S = y^T y - n\, z_S^T H_S^{-1} z_S$, $\nu = 0.01 + n/2$, $a_0 = b_0 = 0.01$; $m(S) = (2b_0)^{a_0}\, \Gamma(\nu) / (\pi^{n/2}\, \Gamma(a_0)) \cdot D_S^{-1/2}\, (2b_0 + Q_S)^{-\nu}$ (for $S = \emptyset$: $D_\emptyset = 1$, $Q_\emptyset = y^T y$).
- $w(k) = \tfrac{1}{2} / C(2p-1, k)$; $W_K = \sum_{k \leq K} C(p,k)\, w(k)$; $\varpi(S) = w(|S|)/W_K$.
- Evidence $Z = \sum_{S \in \Omega} \varpi(S)\, m(S)$. $BF(S) = m(S)/m(\emptyset)$.
- Blocks $C_1, \ldots, C_J$ (a fixed partition of $[p]$ determined by X); tables $r_b(A) = BF(A)$, $A \subseteq C_b$. $\gamma_0(S) = \varpi(S) \prod_b r_b(S \cap C_b)$; $\bar{Z}_0 = \sum_{S \in \Omega} \gamma_0(S)$; discrepancy $d(S) = \log BF(S) - \sum_b \log r_b(S \cap C_b)$.
- Calibration $\boldsymbol{\mathcal{C}}$: the random object produced by the calibration phase (grid $0 = \varphi_0 < \varphi_1 < \ldots < \varphi_T = 1$, T, L, N_e, tier, candidate set and all other frozen settings). A calibration value c is **successful** if it yields a frozen grid, so that the method runs its final phase.
- **Final run given c.** For a successful c: 4 islands $r = 1, \ldots, 4$, each with N_e particles in Ω, run with the frozen grid of c and the final streams 10–13 (island r uses stream $9 + r$). Writing $S_{r,i}^{(t-1)} \in \Omega$ for particle i of island r at the moment stage t's reweighting is applied ($t = 1, \ldots, T$), the island evidence is (method §6.12)
  $\hat{Z}_r = m(\emptyset)\, \bar{Z}_0 \prod_{t=1}^{T} (1/N_e) \sum_{i=1}^{N_e} \exp\{(\varphi_t - \varphi_{t-1})\, d(S_{r,i}^{(t-1)})\}$,
  and the pooled evidence is $\hat{Z} = \tfrac{1}{4} \sum_{r=1}^{4} \hat{Z}_r$. The reported lower bound is $0.05 \cdot \hat{Z}$.
- P_c, E_c: probability and expectation over the final streams 10–13, with the calibration fixed at c. Under CAL and EX-S (and joint measurability of the final run in (c, streams)), these give a version of the conditional law and expectation given $\boldsymbol{\mathcal{C}}$, on the event of a successful calibration; the source text writes them $P_{\boldsymbol{\mathcal{C}}}$, $E_{\boldsymbol{\mathcal{C}}}$. All conclusions below are stated for P_c, for every successful c.
- **Final E2 flag** (method §6.9, recomputed after the final run as a flag only): E2_final is raised iff the standard deviation of the four numbers $\log \hat{Z}_1, \ldots, \log \hat{Z}_4$ exceeds 1 nat. The method does not fix the divisor; both the population SD (divisor 4) and the sample SD (divisor 3) are covered below.

2. Assumptions

- **EX-A** (declared idealisation): exact scores, sums, products, probabilities and comparisons.
- **EX-S** (declared idealisation): independent ideal uniform random streams with the method's stream separation; the algorithm's outputs are measurable functions of the streams.
- **CAL** (method construction): $\boldsymbol{\mathcal{C}}$ uses streams 0–9 only; streams 10–13 are fresh (independent of $\boldsymbol{\mathcal{C}}$); escalation is decided before any final randomness.
- **(H-T12)** [taken from the sibling conjecture T12 (iii), used here only as a hypothesis]: for every successful calibration value c, $E_c[\hat{Z}] = Z$. (T12 (i) states $E_c[\hat{Z}_r] = Z$ for each island, which implies this.)

3. Conclusions to prove

Fix any successful calibration value c. Under EX-A, EX-S, CAL and (H-T12):

(a) Coverage. For every $\alpha \in (0, 1)$: $P_c(Z \geq \alpha\hat{Z}) \geq 1 - \alpha$. In particular $P_c(Z \geq 0.05 \cdot \hat{Z}) \geq 0.95$.

(b) Log bias. $\log \hat{Z}$ is a well-defined, integrable random variable under P_c, and $E_c[\log \hat{Z}] \leq \log Z$.

The remaining two conclusions make precise "it is not a relative-error interval" and "conditioning afterwards on final flags does not preserve this coverage automatically". They are **non-implication statements**: they show that the properties of $\hat{Z}$ that yield (a) and (b) do not yield more. Let $\mathcal{L}(Z)$ be the set of joint laws of a vector $(\hat{Z}_1, \ldots, \hat{Z}_4)$ such that the $\hat{Z}_r$ are independent and identically distributed, each takes finitely many values, all in $(0, \infty)$, and $E[\hat{Z}_r] = Z$; set $\hat{Z} = \frac{1}{4} \Sigma_r \hat{Z}_r$ and define E2_final from $(\hat{Z}_r)$ as above. Every law in $\mathcal{L}(Z)$ satisfies (a) and (b) (by the same proof) and has every property of $\hat{Z}$ that the proof of (a)–(b) uses (positive, finitely many values, mean Z). The extra requirements in $\mathcal{L}(Z)$ (i.i.d. islands with islandwise mean Z) are not hypotheses about the actual final run; they only make the non-implications stronger. These clauses do not assert that any particular data set realises the laws exhibited, nor that they are not realised. This non-implication reading is the intended one; the per-data-set reading about the actual estimator would be false: for fixed data the estimator satisfies $e^{-\Delta}Z \leq \hat{Z} \leq e^{\Delta}Z$ surely, with $\Delta = \max_\Omega d - \min_\Omega d$, a data-dependent constant that is not among the reported quantities listed above and that no property of $\mathcal{L}(Z)$ supplies (proof, Remark at the end).

(c) One-sided, not a relative-error interval. For every $c_0 > 0$ and every $\eta \in (0, 1)$ there is a law in $\mathcal{L}(Z)$ with $P(\hat{Z} < c_0 Z) \geq 1 - \eta$. Consequently, for no constants $0 < c_1 \leq 1 \leq c_2 < \infty$ and $\gamma > 0$ does the bound $P(c_1 Z \leq \hat{Z} \leq c_2 Z) \geq \gamma$ follow from the properties defining $\mathcal{L}(Z)$ (in particular not from those used in (a)–(b)), even though (a) holds for every such law; the event $\{Z \geq \alpha\hat{Z}\}$ places no upper limit on $Z/\hat{Z}$.

(d) Conditioning on a final flag. For every $\alpha \in (0, 1)$:

- (d1) there is a law in $\mathcal{L}(Z)$ with $P(\text{E2_final raised}) > 0$ and $P(Z \geq \alpha\hat{Z} \mid \text{E2_final raised}) = 0$;
- (d2) for every $\gamma > 0$ there is a law in $\mathcal{L}(Z)$ with $P(\text{E2_final not raised}) > 0$ and $P(Z \geq \alpha\hat{Z} \mid \text{E2_final not raised}) < \gamma$.

Both hold for either SD divisor. The source's "final flags" is read existentially: one final flag (E2_final) whose conditioning breaks the coverage suffices to show that conditioning on final flags does not preserve it automatically (a flag independent of $\hat{Z}$ would preserve it). Consequently, coverage at level $1 - \alpha$ conditional on either value of the final E2 flag does not follow from the properties defining $\mathcal{L}(Z)$, although (a) holds unconditionally for every such law.

Approved changes to the statement

None to hypotheses or conclusions. Exposition only (after Stage B review round 1, Opus reviewer findings 1, 2, 5): the paragraph before (c) and the sentence after (d2) were reworded to state why the non-implication reading is the intended one, that the extra requirements of $\mathcal{L}(Z)$ are not hypotheses about the actual run, and that "final flags" is read existentially. Round 2 (both Stage B reviewers): "the only true one" replaced by "the intended one; the per-data-set reading would be false"; "the method does not report" replaced by "not among the reported quantities listed above"; the $P_c \leftrightarrow$ conditional-law sentence now carries the joint-measurability proviso. Versions reviewed: review/STATEMENT_as_reviewed_r1.md (round 1), review/STATEMENT_as_reviewed_r2.md (round 2).

(c) Final proof, verbatim: body of PROOF_v3.tex

The proof below is the body of `PROOF_v3.tex` from `\begin{document}` to `\end{document}` without the title block. Its preamble definitions are local to this chapter; labels carry the prefix `T31:`.

All notation is that of STATEMENT.md. Throughout, c is a fixed successful calibration value; $\mathbb{P}_c, \mathbb{E}_c$ are the probability and expectation over the final streams 10–13 with the calibration fixed at c.

1 Positivity and finiteness

Lemma 1. *Under EX-A and EX-S:*

(i) $m(S) \in (0,\infty)$ *for every* $S \in \Omega$*; hence* $Z \in (0,\infty)$.

(ii) $\bar{Z}_0 \in (0,\infty)$ *and* $d(S) \in \mathbb{R}$ *for every* $S \in \Omega$.

(iii) For each r, $\widehat{Z}_r$ *is a random variable under* $\mathbb{P}_c$ *taking finitely many values, all in* $(0,\infty)$*. Hence* $\widehat{Z} = \frac{1}{4}\sum_r \widehat{Z}_r$ *and* $\log \widehat{Z}$ *are random variables taking finitely many values,* $\widehat{Z} > 0$*, and* $\log \widehat{Z}$ *is bounded, in particular integrable.*

Proof. (i) For $S = \emptyset$, $D_\emptyset = 1$ and $Q_\emptyset = y^\top y = n - 1 > 0$. Let $S \neq \emptyset$, $k = |S|$, and take a full singular value decomposition $X_S = U\Sigma V^\top$ with $U \in \mathbb{R}^{n\times n}$, $V \in \mathbb{R}^{k\times k}$ orthogonal and $\Sigma \in \mathbb{R}^{n\times k}$ with diagonal entries $\sigma_1, \ldots, \sigma_{\min(n,k)} \geq 0$ and zeros elsewhere. Since $G_S = X_S^\top X_S \succeq 0$, both $H_S = (n+1)G_S + I$ and $G_S + I$ are positive definite, so $D_S = \det H_S / \det(G_S + I) > 0$. Further $H_S = V\{(n+1)\Sigma^\top\Sigma + I\}V^\top$, so

$$nX_S H_S^{-1} X_S^\top = U\left[n\Sigma\{(n+1)\Sigma^\top\Sigma + I\}^{-1}\Sigma^\top\right]U^\top,$$

and the bracket is diagonal with entries $n\sigma_j^2/\{(n+1)\sigma_j^2 + 1\} \in [0, n/(n+1))$ and zeros. Hence $I_n - nX_S H_S^{-1} X_S^\top$ has all eigenvalues at least $1/(n+1)$, and

$$Q_S = y^\top y - nz_S^\top H_S^{-1} z_S = y^\top\big(I_n - nX_S H_S^{-1} X_S^\top\big)y \;\geq\; \frac{y^\top y}{n+1} > 0.$$

With $a_0 = b_0 = 0.01$ and $\nu = 0.01 + n/2$, the constants $(2b_0)^{a_0}$, $\Gamma(\nu)$, $\Gamma(a_0)$, $\pi^{n/2}$ are finite and positive, $D_S^{-1/2} \in (0,\infty)$ and $(2b_0 + Q_S)^{-\nu} \in (0,\infty)$; so $m(S) \in (0,\infty)$. Since $1 \leq p$ and $K \leq p \leq 2p-1$, each $\binom{2p-1}{k}$ with $k \leq K$ is a positive integer, so $w(k) > 0$, $W_K \in (0,\infty)$ and $\varpi(S) > 0$. As Ω is finite, $Z = \sum_\Omega \varpi(S)m(S) \in (0,\infty)$.

(ii) By (i), $\mathrm{BF}(S) = m(S)/m(\emptyset) \in (0,\infty)$ for $S \in \Omega$. The table entries that enter γ_0 and d are $r_b(S \cap C_b)$ with $S \in \Omega$; since $|S \cap C_b| \leq |S| \leq K$, $S \cap C_b \in \Omega$ and so $r_b(S \cap C_b) = \mathrm{BF}(S \cap C_b) \in (0,\infty)$. Hence $\gamma_0(S) \in (0,\infty)$, $\bar{Z}_0 = \sum_\Omega \gamma_0(S) \in (0,\infty)$ (Ω finite and nonempty), and $d(S) = \log \mathrm{BF}(S) - \sum_b \log r_b(S \cap C_b)$ is a finite real number.

(iii) Given c, the number of stages T and the particle count N_e are fixed positive integers ($T \geq 1$ because $\phi_0 = 0 < \phi_T = 1$; $N_e \geq 1$ as a particle count) and the grid $\phi_0, \ldots, \phi_T$ is fixed. By STATEMENT.md §1 every particle state $S_{r,i}^{(t-1)}$ lies in the finite set Ω, so the array $A_r = (S_{r,i}^{(t-1)})_{1\leq i\leq N_e,\, 1\leq t\leq T}$ takes values in the finite set $\Omega^{N_e T}$. By the formula of STATEMENT.md §1, $\widehat{Z}_r = g(A_r)$ with the deterministic map

$$g(a) = m(\emptyset)\,\bar{Z}_0 \prod_{t=1}^{T} \frac{1}{N_e}\sum_{i=1}^{N_e} \exp\{(\phi_t - \phi_{t-1})\, d(a_{i,t})\}.$$

By (i)–(ii) every factor is finite and positive (an average of finitely many finite positive numbers), so g maps the finite set $\Omega^{N_e T}$ into $(0,\infty)$; under EX-A this value is computed exactly. Hence, pointwise on the sample space, $\widehat{Z}_r \in g(\Omega^{N_e T})$, a finite subset of $(0,\infty)$. Each $\widehat{Z}_r$ is an output of the algorithm, hence a random variable under $\mathbb{P}_c$ by EX-S. The same then holds for $\widehat{Z} = \frac{1}{4}\sum_r \widehat{Z}_r$, and $\log \widehat{Z}$ is a random variable taking finitely many real values, so it is bounded. □

2 Coverage and log bias: conclusions (a) and (b)

The next lemma isolates the only properties of $\widehat{Z}$ that (a) and (b) use.

Lemma 2. *Let W be a random variable on some probability space, taking finitely many values, all in $(0,\infty)$, with $\mathbb{E}W = Z \in (0,\infty)$. Then for every $\alpha \in (0,1)$, $\mathbb{P}(Z \geq \alpha W) \geq 1-\alpha$, and $\mathbb{E}\log W \leq \log Z$.*

Proof. Since $W \geq 0$, pointwise $W \geq (Z/\alpha)\mathbf{1}\{W > Z/\alpha\}$; taking expectations, $Z = \mathbb{E}W \geq (Z/\alpha)\,\mathbb{P}(W > Z/\alpha)$, so $\mathbb{P}(\alpha W > Z) \leq \alpha$ (Markov's inequality), and $\mathbb{P}(Z \geq \alpha W) = 1 - \mathbb{P}(\alpha W > Z) \geq 1-\alpha$.

Let $v_1, \ldots, v_m \in (0,\infty)$ be the distinct values of W and $p_i = \mathbb{P}(W = v_i)$, so $\sum_i p_i = 1$ and $\sum_i p_i v_i = Z$. By concavity of log on $(0,\infty)$ (finite Jensen inequality), $\mathbb{E}\log W = \sum_i p_i \log v_i \leq \log \sum_i p_i v_i = \log Z$. □

Theorem 3 (Conclusions (a) and (b))**.** *Under EX-A, EX-S, CAL and (H-T12), for every successful calibration value c:*

(a) $\mathbb{P}_c(Z \geq \alpha\widehat{Z}) \geq 1-\alpha$ for every $\alpha \in (0,1)$; in particular $\mathbb{P}_c(Z \geq 0.05\,\widehat{Z}) \geq 0.95$.

(b) $\log \widehat{Z}$ is a bounded (hence integrable) random variable under $\mathbb{P}_c$ and $\mathbb{E}_c \log \widehat{Z} \leq \log Z$.

Proof. By Lemma 1, under $\mathbb{P}_c$ the variable $W = \widehat{Z}$ takes finitely many values, all in $(0,\infty)$, $\log \widehat{Z}$ is bounded, and $Z \in (0,\infty)$. By (H-T12), $\mathbb{E}_c\widehat{Z} = Z$. Lemma 2 gives (a) and the inequality in (b); the case $\alpha = 0.05$ gives $\mathbb{P}_c(Z \geq 0.05\,\widehat{Z}) \geq 0.95$. □

Remark 4 ($\mathbb{P}_c$ and conditioning on $\mathcal{C}$). STATEMENT.md defines $\mathbb{P}_c, \mathbb{E}_c$ as the law of the final run with the calibration frozen at c, and all conclusions are proved for $\mathbb{P}_c$ for every successful c. Under CAL and EX-S the final outputs are a function $F(c,U)$ of the calibration and the final streams U, which are independent of $\mathcal{C}$; by the standard substitution (freezing) property of conditional expectation, on the event of a successful calibration, $\mathbb{P}(\cdot \mid \mathcal{C}) = \mathbb{P}_{\mathcal{C}}$ almost surely (for jointly measurable F). This is how the source's $\mathbb{P}_{\mathcal{C}}, \mathbb{E}_{\mathcal{C}}$ are read; CAL is used only here.

Remark 5. Let $\mathcal{L}(Z)$ be as in STATEMENT.md §3: joint laws of $(\widehat{Z}_1, \ldots, \widehat{Z}_4)$ with i.i.d. coordinates, each taking finitely many values in $(0,\infty)$, with $\mathbb{E}\widehat{Z}_r = Z$. For every law in $\mathcal{L}(Z)$, $\widehat{Z} = \frac{1}{4}\sum_r \widehat{Z}_r$ takes finitely many values in $(0,\infty)$ and $\mathbb{E}\widehat{Z} = Z$; so Lemma 2 applies and (a), (b) hold for it. The proof of Theorem 3 uses only these properties of $\widehat{Z}$ (positivity, finitely many values, mean Z). The extra properties in the definition of $\mathcal{L}(Z)$ (i.i.d. islands, islandwise mean Z) are not hypotheses about the actual final run; adding them to the class only makes the non-implications (c) and (d) stronger.

3 The bound is one-sided: conclusion (c)

Lemma 6 (rare-tail law)**.** *Let $c_0 > 0$ and $\eta \in (0,1)$. Put $c' = \min(c_0, 1)/2 \in (0, \frac{1}{2}]$, $\varepsilon = \eta/4 \in (0, \frac{1}{4})$ and $B = \{1-(1-\varepsilon)c'\}/\varepsilon$. Let $\widehat{Z}_1, \dots, \widehat{Z}_4$ be i.i.d. with $\mathbb{P}(\widehat{Z}_r = c'Z) = 1-\varepsilon$ and $\mathbb{P}(\widehat{Z}_r = BZ) = \varepsilon$. This law belongs to $\mathcal{L}(Z)$ and $\mathbb{P}(\widehat{Z} < c_0 Z) \geq 1-\eta$.*

Proof. $(1-\varepsilon)c' < 1$, so $B > 0$; and $B - c' = (1-c')/\varepsilon > 0$, so the two values are distinct and positive. $\mathbb{E}\widehat{Z}_r = Z\{(1-\varepsilon)c' + \varepsilon B\} = Z$. So the law is in $\mathcal{L}(Z)$. On the event $A = \{\widehat{Z}_1 = \cdots = \widehat{Z}_4 = c'Z\}$ we have $\widehat{Z} = c'Z < c_0 Z$ (as $c' \leq c_0/2$), and by independence and Bernoulli's inequality $\mathbb{P}(A) = (1-\varepsilon)^4 \geq 1 - 4\varepsilon = 1 - \eta$. □

Corollary 7 (Conclusion (c))**.** *For every $c_0 > 0$ and $\eta \in (0,1)$ some law in $\mathcal{L}(Z)$ has $\mathbb{P}(\widehat{Z} < c_0 Z) \geq 1-\eta$. Consequently, for no constants $0 < c_1 \leq 1 \leq c_2 < \infty$ and $\gamma > 0$ does $\mathbb{P}(c_1 Z \leq \widehat{Z} \leq c_2 Z) \geq \gamma$ hold for every law in $\mathcal{L}(Z)$, although every law in $\mathcal{L}(Z)$ satisfies (a).*

Proof. The first sentence is Lemma 6. For the second, given c_1, c_2, γ we may take $\gamma \leq 1$ (otherwise the bound fails trivially). Apply Lemma 6 with $c_0 = c_1$ and $\eta = \gamma/2 \in (0, \frac{1}{2}]$: then $\mathbb{P}(c_1 Z \leq \widehat{Z} \leq c_2 Z) \leq \mathbb{P}(\widehat{Z} \geq c_1 Z) \leq \eta < \gamma$. The last clause is Remark 5. □

Remark 8. On the event A of Lemma 6, $Z/\widehat{Z} = 1/c'$, which is arbitrarily large as $c_0 \downarrow 0$, while the coverage event $\{Z \geq \alpha\widehat{Z}\}$ holds on A for every $\alpha \in (0,1)$. So the reported $0.05\,\widehat{Z}$ is a one-sided lower confidence bound for Z: the coverage event places no upper limit on $Z/\widehat{Z}$, and $[0.05\,\widehat{Z}, \infty)$ is not a relative-error interval.

4 Conditioning on the final E2 flag: conclusion (d)

Lemma 9 (SD and range)**.** *Let $x \in \mathbb{R}^4$ with range $R = \max_i x_i - \min_i x_i$ and $SS = \sum_i (x_i - \bar{x})^2$. Then $SS \geq R^2/2$, so the population SD $\sqrt{SS/4}$ and the sample SD $\sqrt{SS/3}$ are both at least $R/(2\sqrt{2})$. In particular, if $R \geq 3$ both exceed 1, and if $R = 0$ both equal 0.*

Proof. If $R = 0$ all coordinates are equal and $SS = 0$. Otherwise let $u = \max_i x_i$, $v = \min_i x_i$, attained at distinct indices. For any real m, $(u-m)^2 + (v-m)^2 = \frac{1}{2}(u-v)^2 + 2\{\frac{1}{2}(u+v) - m\}^2 \geq \frac{1}{2}R^2$; with $m = \bar{x}$ this gives $SS \geq R^2/2$. Then $SS/4 \geq R^2/8$ and $SS/3 \geq R^2/6 \geq R^2/8$. If $R \geq 3$, both SDs are at least $3/(2\sqrt{2}) = \sqrt{9/8} > 1$. □

Corollary 10. *Suppose each $\widehat{Z}_r/Z$ takes values in a finite set $\mathcal{A} \subset (0, \infty)$ such that $|\log a - \log a'| \geq 3$ for all distinct $a, a' \in \mathcal{A}$. Then, for either SD divisor, $\mathrm{E2}_{\mathrm{final}}$ is not raised if and only if $\widehat{Z}_1 = \widehat{Z}_2 = \widehat{Z}_3 = \widehat{Z}_4$.*

Proof. $\log \widehat{Z}_r = \log Z + \log(\widehat{Z}_r/Z)$, and adding a constant does not change the SD. If all four are equal, the SD is $0 \leq 1$. If not, two of them take distinct values in $\mathcal{A}$, so the range of $(\log \widehat{Z}_r)_r$ is at least 3 and, by Lemma 9, the SD exceeds 1. □

Theorem 11 (Conclusion (d1))**.** *Let $\alpha \in (0,1)$, $a = \frac{1}{4}$, $b = 8/\alpha$, $\varepsilon = 3\alpha/(32-\alpha)$, and let $\widehat{Z}_1, \dots, \widehat{Z}_4$ be i.i.d. with $\mathbb{P}(\widehat{Z}_r = aZ) = 1-\varepsilon$, $\mathbb{P}(\widehat{Z}_r = bZ) = \varepsilon$. This law is in $\mathcal{L}(Z)$, $\mathbb{P}(\mathrm{E2}_{\mathrm{final}}$ raised$) > 0$, and $\mathbb{P}(Z \geq \alpha\widehat{Z} \mid \mathrm{E2}_{\mathrm{final}}$ raised$) = 0$, for either SD divisor.*

Proof. $0 < 3\alpha < 32 - \alpha$, so $\varepsilon \in (0,1)$. $\mathbb{E}\widehat{Z}_r/Z = a + \varepsilon(b-a) = \frac{1}{4} + \frac{3}{4b-1} \cdot \frac{4b-1}{4} = 1$. So the law is in $\mathcal{L}(Z)$. Further $\log(b/a) = \log(32/\alpha) > \log 32 > 3$ (as $e^3 < 20.1$), so Corollary 10 applies with $\mathcal{A} = \{a, b\}$: $\mathrm{E2}_{\mathrm{final}}$ is raised iff the four values are not all equal. Hence $\mathbb{P}(\mathrm{E2}_{\mathrm{final}}$ raised$) =$

$1-(1-\varepsilon)^4-\varepsilon^4$, which is positive because it is at least the probability $\varepsilon(1-\varepsilon)^3>0$ of the outcome (bZ,aZ,aZ,aZ). On the event $\{\mathrm{E2}_{\mathrm{final}}$ raised$\}$ at least one $\widehat{Z}_r$ equals bZ and the others are at least $aZ>0$, so $\widehat{Z}\geq(b+3a)Z/4>bZ/4=2Z/\alpha$, i.e. $\alpha\widehat{Z}>2Z>Z$. Thus the coverage event fails everywhere on $\{\mathrm{E2}_{\mathrm{final}}$ raised$\}$. □

Theorem 12 (Conclusion (d2))**.** *Let $\alpha\in(0,1)$ and $\gamma>0$. Put $h=2/\alpha$ and let M be an integer with $M\geq h$ and $M^3\geq 81/(\alpha^4\gamma)$. Let $\ell_k=he^{-3k}$ for $k=1,\dots,M$, $\bar{\ell}=M^{-1}\sum_{k=1}^{M}\ell_k$, $q=(1-\bar{\ell})/(h-\bar{\ell})$, and let $\widehat{Z}_1,\dots,\widehat{Z}_4$ be i.i.d. with $\mathbb{P}(\widehat{Z}_r=hZ)=q$ and $\mathbb{P}(\widehat{Z}_r=\ell_kZ)=(1-q)/M$ for each k. Then this law is in $\mathcal{L}(Z)$, $\mathbb{P}(\mathrm{E2}_{\mathrm{final}}$ not raised$)>0$, and $\mathbb{P}(Z\geq\alpha\widehat{Z}\mid\mathrm{E2}_{\mathrm{final}}$ not raised$)<\gamma$, for either SD divisor. Taking $\gamma=1-\alpha$, the conditional coverage given that the flag is not raised is below $1-\alpha$.*

Proof. The law is well defined and in $\mathcal{L}(Z)$. Using $M\geq h$,

$$0<\bar{\ell}=\frac{h}{M}\sum_{k=1}^{M}e^{-3k}<\frac{h}{M}\cdot\frac{e^{-3}}{1-e^{-3}}=\frac{h}{M}\cdot\frac{1}{e^3-1}\leq\frac{1}{e^3-1}<\frac{1}{19},$$

since $e^3-1>19$ (as $e>2.718$ gives $e^3>2.718^3>20.07$). As $h=2/\alpha>2$, we have $0<\bar{\ell}<1<h$; so $0<1-\bar{\ell}<h-\bar{\ell}$ and $q\in(0,1)$. Moreover $q>(1-\frac{1}{19})/h=\frac{9}{19}\alpha>\alpha/3$. The values $h>\ell_1>\cdots>\ell_M>0$ are distinct and the probabilities $q,(1-q)/M,\dots$ are positive and sum to 1. Finally $\mathbb{E}\widehat{Z}_r/Z=qh+(1-q)\bar{\ell}=q(h-\bar{\ell})+\bar{\ell}=1$. So the law is in $\mathcal{L}(Z)$.

The flag. Consecutive values in $\mathcal{A}=\{h,\ell_1,\dots,\ell_M\}$ have log-differences exactly 3, so any two distinct values differ in log by at least 3, and Corollary 10 gives $\{\mathrm{E2}_{\mathrm{final}}$ not raised$\}=\{\widehat{Z}_1=\cdots=\widehat{Z}_4\}$. This event is the disjoint union of $H=\{$all four equal $hZ\}$, with $\mathbb{P}(H)=q^4>0$, and $L=\bigcup_k\{$all four equal $\ell_kZ\}$, with $\mathbb{P}(L)=M\{(1-q)/M\}^4=(1-q)^4M^{-3}$.

Coverage on each part. On H, $\widehat{Z}=hZ$ and $\alpha\widehat{Z}=2Z>Z$: coverage fails. On $\{$all four equal $\ell_kZ\}$, $\alpha\widehat{Z}=2e^{-3k}Z<Z$: coverage holds. Hence, with $x=(1-q)^4M^{-3}>0$,

$$\mathbb{P}(Z\geq\alpha\widehat{Z}\mid\mathrm{E2}_{\mathrm{final}}\text{ not raised})=\frac{x}{q^4+x}<\frac{x}{q^4}\leq\frac{M^{-3}}{q^4}<\frac{81}{\alpha^4M^3}\leq\gamma,$$

using $(1-q)^4\leq1$, $q>\alpha/3$ and $M^3\geq81/(\alpha^4\gamma)$. □

Remark 13 (What (c) and (d) do and do not say). By Remark 5, every law used in Lemma 6 and Theorems 11–12 satisfies (a) and (b), has i.i.d. islands with islandwise mean Z, and has $\widehat{Z}$ positive with finitely many values. So none of these properties — in particular, none of the properties of the final-run evidence used to prove (a) and (b) — implies a two-sided relative-error bound, or coverage conditional on either value of the final E2 flag. These are non-implication statements about the properties used; they do not assert that a particular data set and calibration realise the laws exhibited.

This reading is the one intended, and the per-data-set reading about the actual estimator is false, as follows. For fixed data, put $d_{\min}=\min_\Omega d$, $d_{\max}=\max_\Omega d$ and $\Delta=d_{\max}-d_{\min}$, and let $A=m(\emptyset)\bar{Z}_0>0$. The grid is strictly increasing, so $\phi_t-\phi_{t-1}>0$ and each term $\exp\{(\phi_t-\phi_{t-1})d(S)\}$, hence each stage average, lies in $[e^{(\phi_t-\phi_{t-1})d_{\min}},e^{(\phi_t-\phi_{t-1})d_{\max}}]$. Since $\sum_t(\phi_t-\phi_{t-1})=\phi_T-\phi_0=1$, every $\widehat{Z}_r$, and hence their average $\widehat{Z}$, lies in $[Ae^{d_{\min}},Ae^{d_{\max}}]$. By the definitions of γ_0 and d, $\varpi(S)\mathrm{BF}(S)=\gamma_0(S)e^{d(S)}$, so $Z/A=\sum_\Omega\gamma_0(S)e^{d(S)}/\bar{Z}_0$ is a γ_0-weighted average of the numbers $e^{d(S)}$ and Z lies in the same interval. Hence, surely, $e^{-\Delta}Z\leq\widehat{Z}\leq e^{\Delta}Z$. So, for fixed data, the actual estimator does satisfy a two-sided relative-error bound, but with a data-dependent constant e^{Δ} that is not among the reported quantities listed in STATEMENT.md (the $\widehat{Z}_r$, $\widehat{Z}$ and $0.05\,\widehat{Z}$)

and that no property of $\mathcal{L}(Z)$ supplies. Similarly, the four numbers $\log \widehat{Z}_r$ lie in an interval of length at most Δ, so their sum of squared deviations is at most Δ^2 (as $\sum_i (x_i - \bar{x})^2 \le \sum_i (x_i - m)^2$ for the midpoint m of the interval, and each of the four terms is at most $(\Delta/2)^2$), so the population SD is at most $\Delta/2$ and the sample SD at most $\Delta/\sqrt{3}$; thus, if $\Delta \le \sqrt{3}$, $\mathrm{E2}_{\mathrm{final}}$ is never raised, and if $e^{\Delta} \le 1/\alpha$, coverage holds surely. For such data, conditioning on the flag would preserve coverage; (d) says only that coverage is not preserved *automatically*, that is, as a consequence of the properties used in (a)–(b). One witness flag (E2) suffices for this; a flag of positive probability independent of $\widehat{Z}$ would preserve coverage. Whether laws like those of (c)–(d) are realised by the actual final run for some data set is not addressed here.

(d) Review record

FINAL_REPORT.md: statement changes

Statement changes

None to hypotheses or conclusions. The step-1 decision was (a), correct as stated. The source text is quoted verbatim in STATEMENT.md. "Under T12" became the explicit hypothesis (H-T12). The two informal cautions ("not a relative-error interval", "conditioning on final flags does not preserve coverage automatically") were written as precise non-implication statements (c), (d). After review, explanatory wording only was revised twice; the reviewed versions are kept in `review/STATEMENT_as_reviewed_r1.md` and `_r2.md`. Every reviewer confirmed that no hypothesis or conclusion changed and that the statement is a faithful formalisation of clause (i).

FINAL_REPORT.md: review history

Review history

Stage	Reviewer	Version	Verdict	Findings
A	Astra #1 (GPT-6 Astra, high; session 01a1641c-085b-7532-afa0-084ca7a98792)	v1	verified	none
B r1	Astra #2 (GPT-6 Astra, high; new session 01a1641e-28c0-7c51-a27c-31ffb6ff38dc)	v1	verified	none
B r1	Opus 5.5 xhigh (read-only agent)	v1	verified	11 exposition (incl. the sure bound)
B r2	Astra #2 (resumed)	v2 + log	verified	2 exposition ("only true one" overclaim; unverifiable "method does not report")
B r2	Opus (continued)	v2 + log	verified	5 exposition (same overclaim; small-Δ derivation; implicit steps; measurability proviso; file name)
B r3	Astra #2 (resumed)	v3 + log	verified	none
B r3	Opus (continued)	v3 + log	verified	3 exposition (cosmetic; recorded, not applied so the final text is the reviewed v3)

No reviewer, in any round, raised a fatal or gap finding. All exposition findings and the lead's answers are in `CLAIMS_OF_FATAL_FLAWS.md`; the reviewers agreed with every answer. The Astra judge was

not needed. Replies: `review/astra_01_review_r1.log`, `review/stageB_astra2_r{1,2,3}.log`, `review/stageB_opus_r{1,2,3}.md`.

Sanity check (not evidence of proof): a Haiku assistant computed the example laws of (c), (d1) and (d2) exactly for several α, γ, c_0, η. All 104 checks passed and the canary was exact (`helpers/numcheck/check.py`, `results.txt`).

CLAIMS_OF_FATAL_FLAWS.md (every entry, verbatim)

CLAIMS_OF_FATAL_FLAWS — T31 (T22) part (i)

Fatal or gap findings

None. Three independent reviews of PROOF_v1 (Astra #1, Astra #2, Opus 5.5 xhigh) reported no fatal and no gap finding; all three returned **verified**.

Exposition findings and the lead's answers (for the record; Opus reviewer, Stage B round 1)

#	Finding (short)	Answer	Change in v2
1	Non-implication reading of (c)/(d) is the only true one: surely e^{−Δ}Z ≤ Ẑ ≤ e^{Δ}Z with Δ = max d − min d; statement should say so	Accepted (checked: $\varpi \cdot BF = \gamma_0 e^{d}$, stage factors in [e^{Δφ d_min}, e^{Δφ d_max}])	STATEMENT §3 paragraph before (c) reworded; new final paragraph of the last Remark in the proof derives the bound
2	"Final flags" plural vs one witness flag	Accepted: existential reading	STATEMENT sentence after (d2); last Remark
3	Link P_c ↔ conditional law given $\mathcal{C}$ not written out; CAL unused	Accepted	New Remark "P_c and conditioning on $\mathcal{C}$" (substitution lemma; CAL used only there)
4	Fidelity to METHODOLOGY_v3 and T12 not checkable from the packet	Noted (scope of the packet). Lead checked: Ẑ_r formula = METHODOLOGY_v3 §6.12 verbatim; E2 = §6.9, "recomputed as flags only" after the final run; (H-T12) = T12(iii) first identity	None
5	Preamble could be read as asserting iid islands / islandwise unbiasedness of the actual run	Accepted	STATEMENT §3 preamble reworded; Remark on $\mathcal{L}(Z)$ extended
6	π_0 and "every move stays in Ω" not in the packet	Accepted	Lemma 1(iii) now cites STATEMENT §1 (states in Ω) only
7	Measurability via "outputs", N_e ≥ 1, T ≥ 1	Accepted	Lemma 1(iii) reworded
8	Table entries with |A| > K	Accepted	Lemma 1(ii): only A = S ∩ C_b ∈ Ω enter
9	"On this event" ambiguous	Accepted	Theorem (d1) proof
10	$e^3 > 20$ unjustified	Accepted	Theorem (d2) proof: e > 2.718 ⇒ $e^3 > 20.07$

#	Finding (short)	Answer	Change in v2
11	"No upper limit on Z/Ẑ" informal	Accepted as informal remark; no change needed	None

Round 2 (PROOF_v2): rulings and new findings

Both Stage B reviewers returned **verified** on PROOF_v2, with no fatal or gap finding, and agreed with the answers to rows 2–11. Row 1: both agree with the sure bound but disagree with the wording "the only true one" (accepted, see R2-1). Full rulings: review/stageB_opus_r2.md, review/stageB_astra2_r2.log.

#	Finding (short; source)	Answer	Change in v3
R2-1	"Only true reading" overclaims; only the per-data-set reading is refuted (Opus R2-1; Astra #2 finding 1)	Accepted	STATEMENT §3 and last Remark: "the intended one; the per-data-set reading would be false"; added "whether such laws are realised by the actual run for some data set is not addressed"
R2-2	Small-Δ claims lack derivation and assume such data exist (Opus)	Accepted	Last Remark: SD ≤ Δ/2 (pop), ≤ Δ/√3 (sample) derived; stated conditionally ("if Δ ≤ √3", "if e^Δ ≤ 1/α"), no existence claim
R2-3	Implicit steps in the sure bound; "method does not compute/report" unverifiable (Opus; Astra #2 finding 2)	Accepted	Steps written out (positive increments, averaging, γ_0-weighted average); "not among the reported quantities listed in STATEMENT.md"
R2-4	Joint-measurability proviso missing in STATEMENT §1 (Opus)	Accepted	STATEMENT §1 P_c sentence now carries the proviso and says conclusions are stated for P_c
R2-5	Proof cites "STATEMENT.md" while packet file is STATEMENT_v2.md (Opus)	No change: the project file is STATE-MENT.md; packet copies are renamed only to keep versions apart	None

Round 3 (PROOF_v3): confirmation

Astra #2 and the Opus reviewer both returned **verified** on PROOF_v3, agreed with all Round 2 answers (R2-1..R2-5: change works), and reported no fatal or gap finding. Opus added three exposition notes (R3-1: the (d) half of the per-data remark is true for every data set by a one-line argument not on the page; R3-2: symbol A reused; R3-3: wording of "reported quantities"). Lead's answer: all three accepted as correct and cosmetic; **not applied**, so that the final text is exactly the reviewed PROOF_v3. They concern commentary only, not any conclusion.